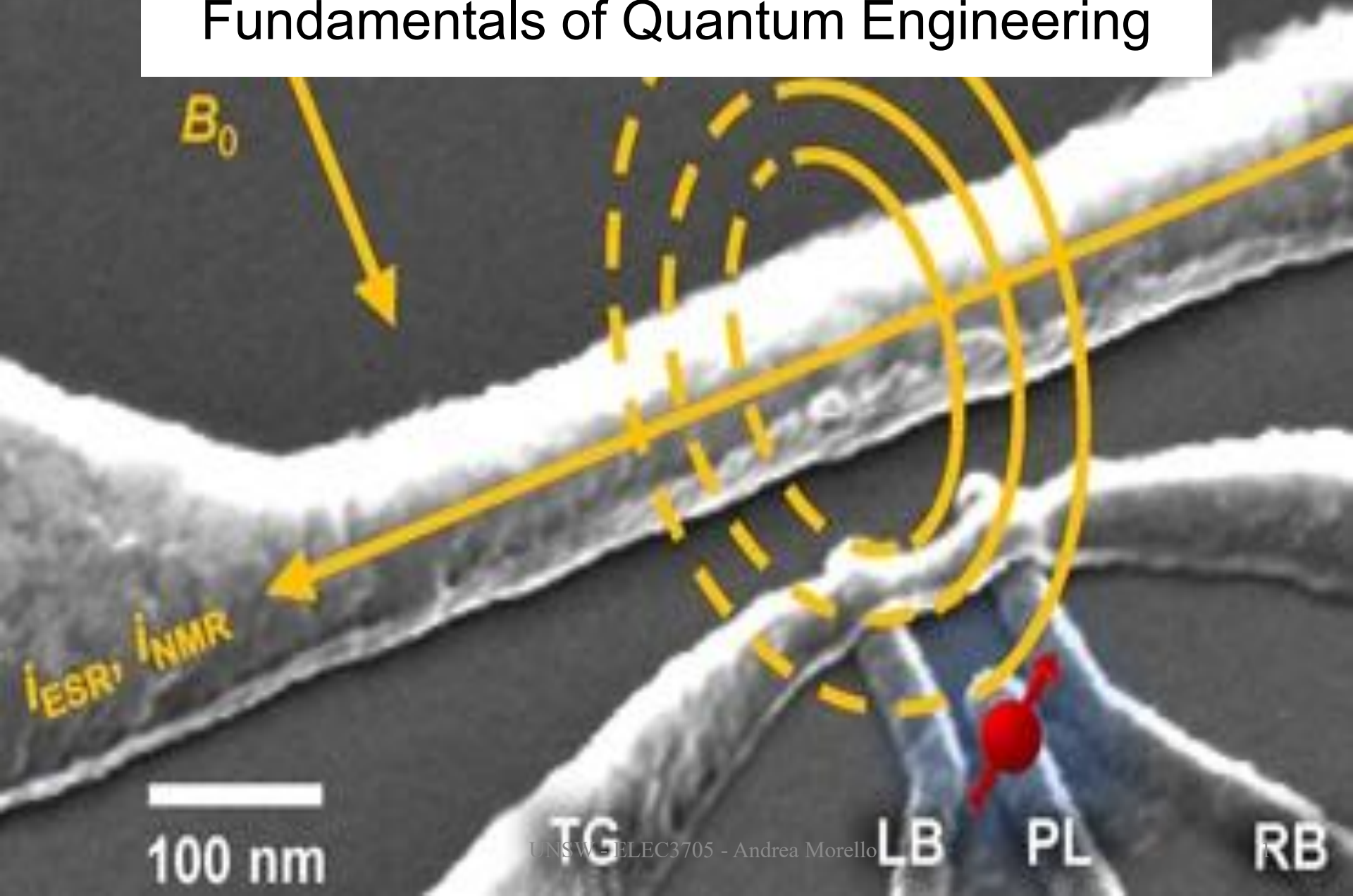


ELEC3705

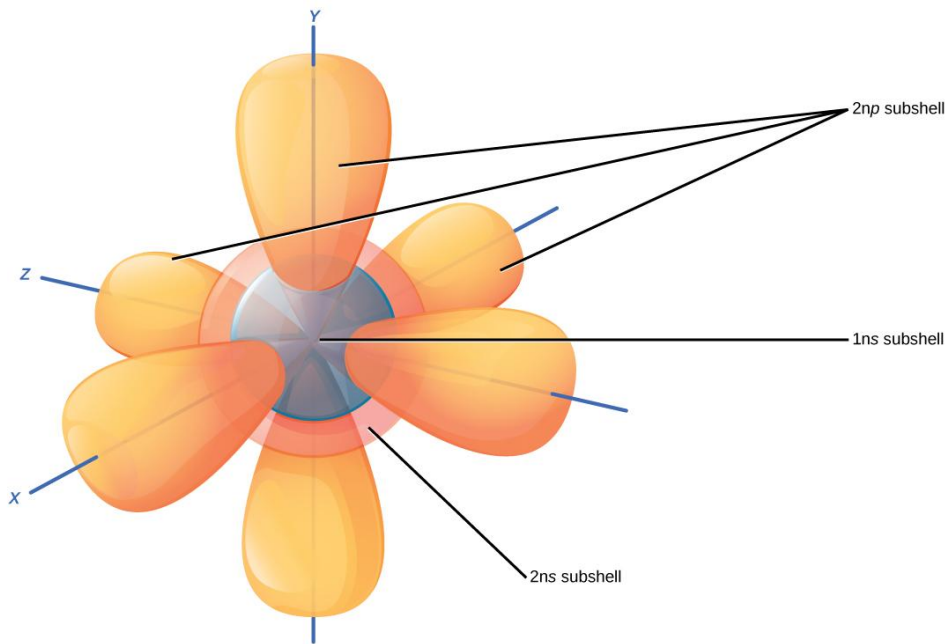
Fundamentals of Quantum Engineering



What is Quantum mechanics?

Old-fashion answer:

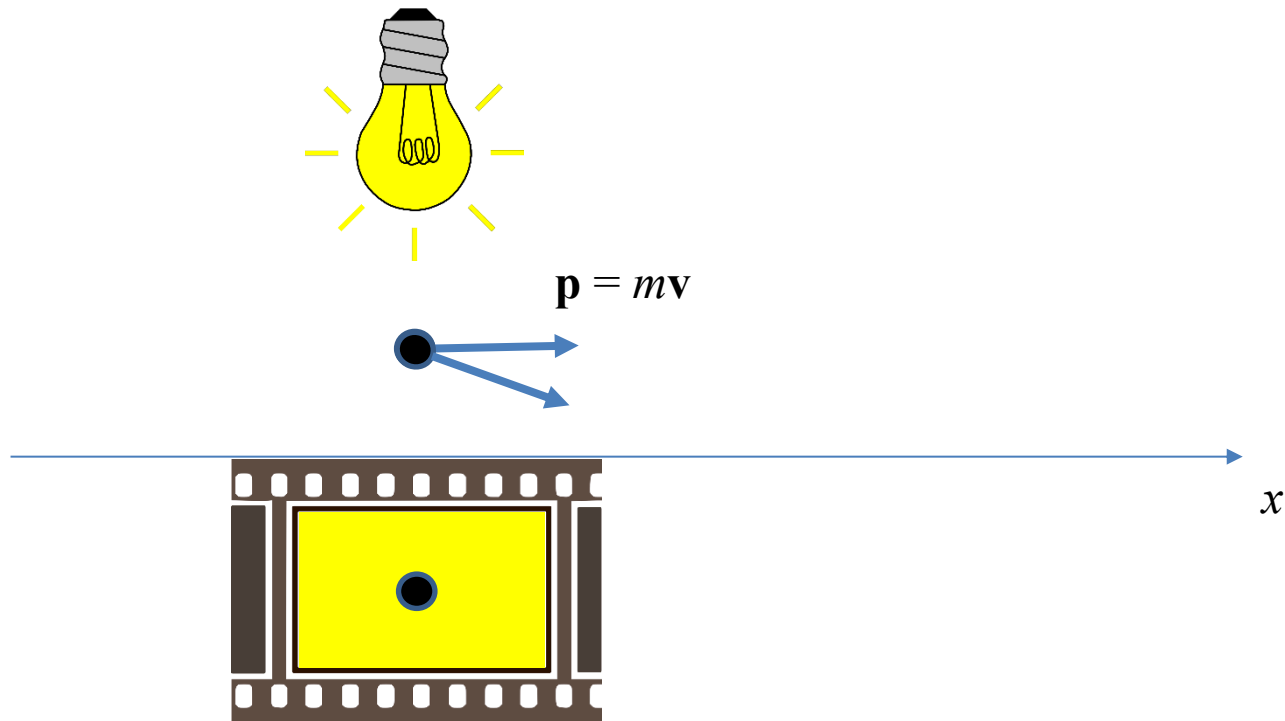
“it’s the study of physics at the microscopic scale, at the level of atoms and electrons”



What is Quantum mechanics?

Modern answer:

“it’s the study of physical systems whose dynamics is governed by variables that cannot be simultaneously observed without disturbance”



Uncertainty principle

Nature poses a *fundamental* limitation to the precision with which we can *simultaneously* measure the position and the momentum of a particle

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}$$

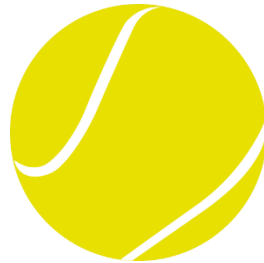
$\sigma_{x,p}$ = standard deviation of the statistical distribution (average uncertainty)

$\hbar = 1.054 \times 10^{-34} \text{ J} \cdot \text{s/rad}$ Planck's constant

Uncertainty principle

Example 1: tennis ball

$$v = 30 \text{ m/s}, \quad m = 0.05 \text{ kg} \quad \Delta v \approx 10^{-16} \text{ m/s} \Rightarrow \Delta x \approx 10^{-17} \text{ m}$$

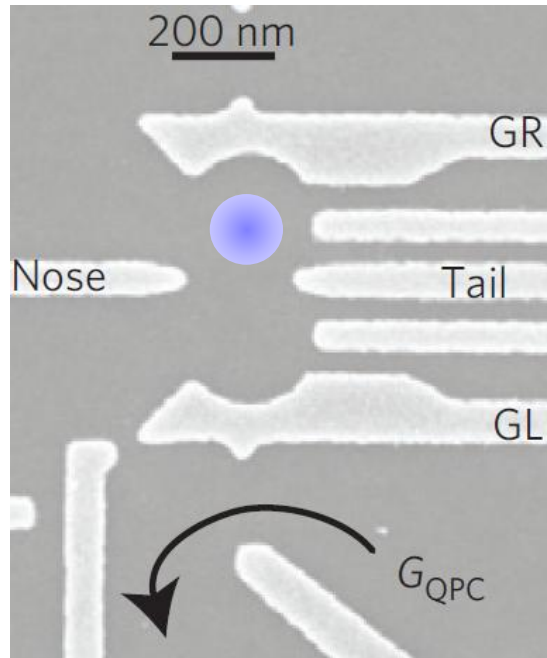


Example 2: electron

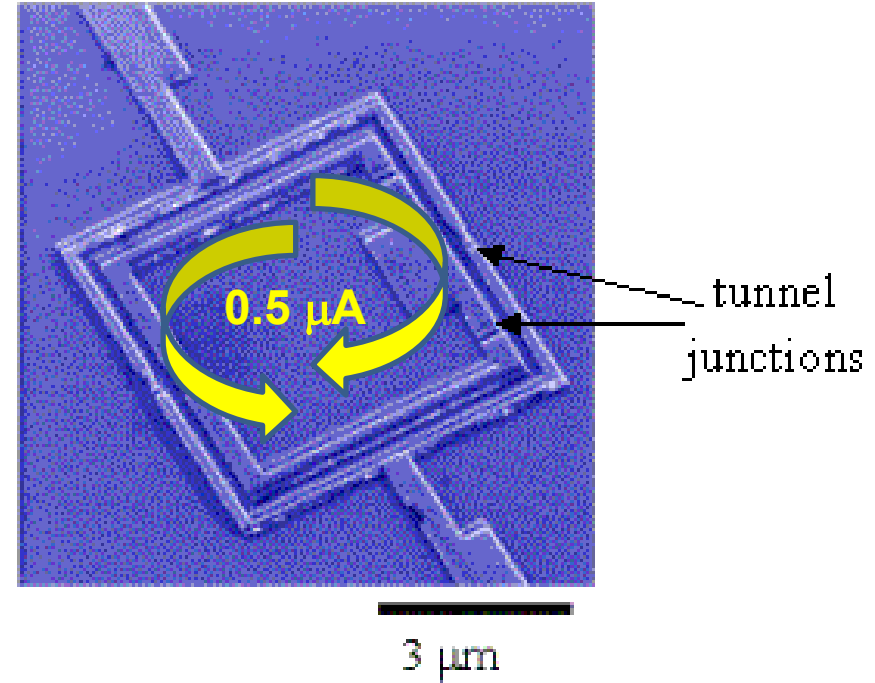
$$v = 30 \text{ m/s}, \quad m = 10^{-30} \text{ kg} \quad \Delta v \approx 10 \text{ m/s} \Rightarrow \Delta x \approx 5 \mu\text{m}$$



Quantum superposition

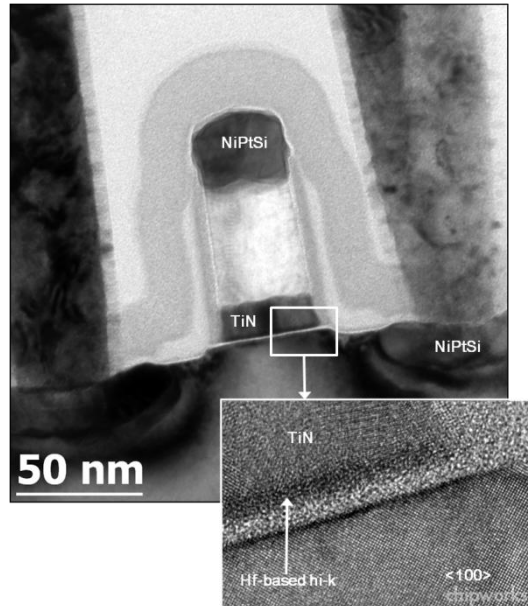


semiconductor
double quantum dot

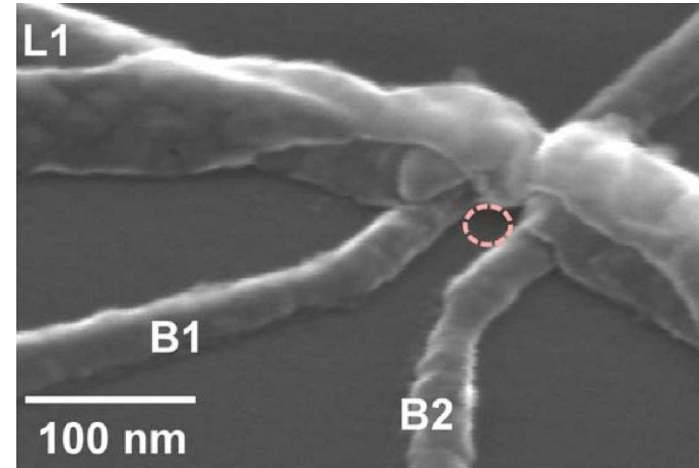


superconducting
quantum interference device

Why it matters to electrical engineers



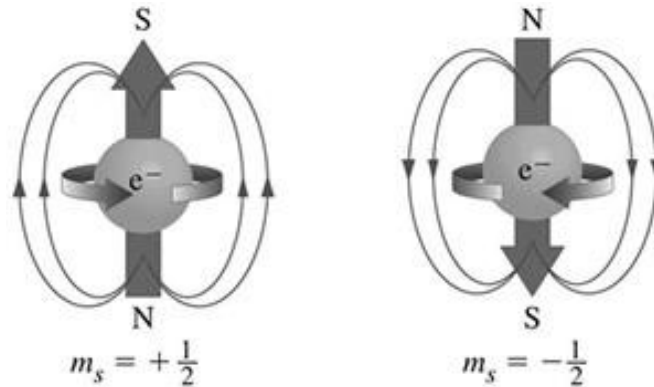
32 nm transistor
“classical” device



30 nm quantum dot
“quantum” device

- Same material
- Same size

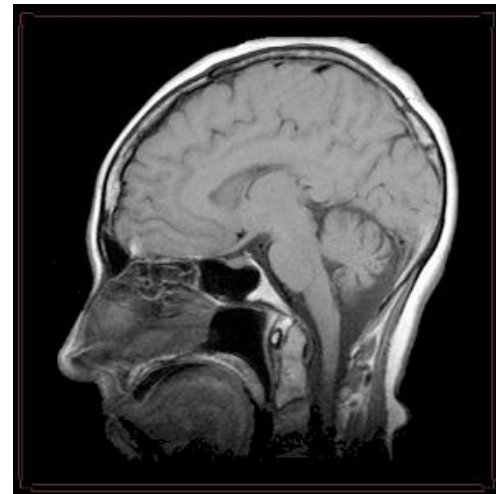
Spin: the simplest quantum system



Spins are everywhere: electrons, protons, neutrons

E.g. each water molecule (70% of your body mass...) has 2 proton spins

→ develop techniques to “see” the spins: Magnetic Resonance Imaging



Quantum Engineering “now”

Atomic clock

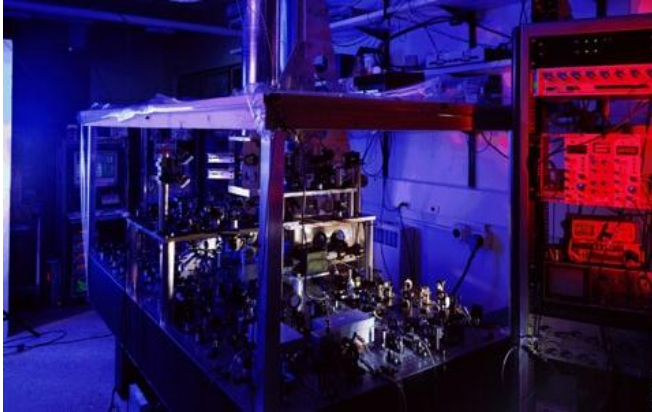


Image: Wikipedia

Quantum random number generator

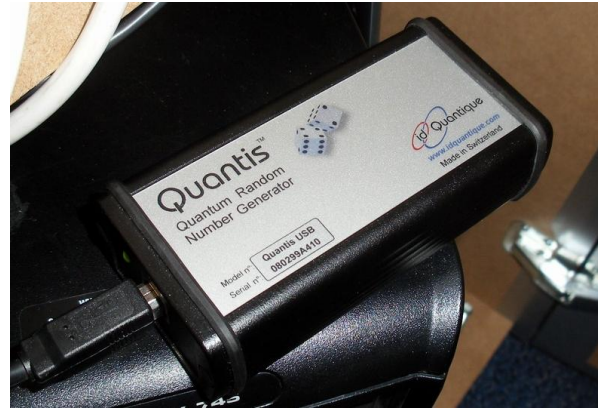


Image: IdQuantique

Quantum sensor for mining



Image: CSIRO

Quantum dot TV



What this course aims to teach

- The origin and formal structure of quantum mechanics;
- How to apply quantum mechanics to simple systems, such as spins and elementary charges;
- Understanding *quantitatively* the complex dynamics of multiple particles and spins, and their uniquely quantum property of entanglement;
- Derive the basic properties of atoms and crystals, including semiconductors;
- Describe the quantum behaviour of nanoscale electronic devices
- Write Python code to calculate the behaviour of quantum systems;
- Prepare for the study of complex, modern, engineered quantum systems: superconducting circuits, quantum computers, quantum-limited amplifiers (→ ELEC4605).

Quantum engineering jobs

Quantum mechanics is no longer just theoretical physics.
It's now a branch of engineering.

There are plenty of jobs looking for people who know all of:

- Electronics
 - Programming
 - Microwave engineering
 - Semiconductor devices
 - Quantum mechanics
 - Quantum information
 - Superconductivity
 - Cryogenics
- “classical” Elec. Eng.
- Quantum Engineering

These people are *rare*, and highly sought-after

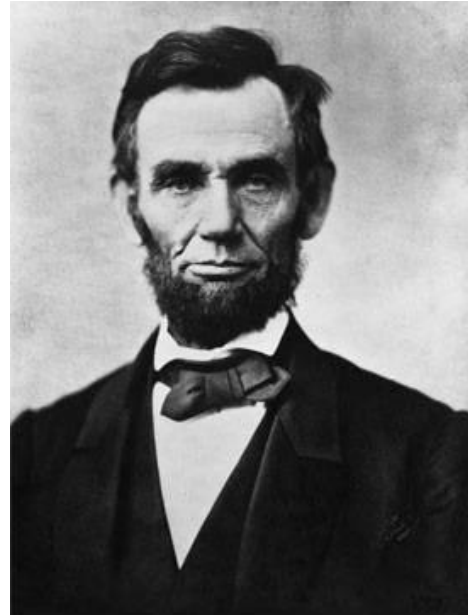


A warning about the internet

The internet is full of content of questionable validity

This is **especially true** for anything that relates to quantum*

This course will teach you how to **do** quantum mechanics – and it is through doing it that you will learn how it truly works



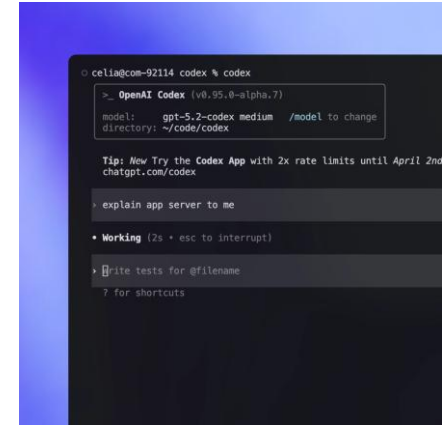
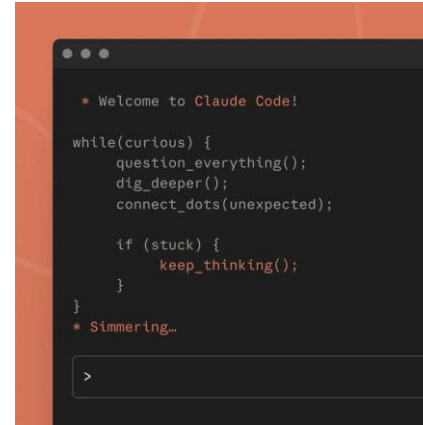
“Don’t believe everything you read on the Internet just because there’s a picture with a quote next to it.”

—Abraham Lincoln

AI use

A *key* component of the course is using Python code to simulate even complicated – and very interesting! – quantum systems

There are very good AI coding tools these days, and they work



This is not a programming course – it's a quantum engineering course. It may be a *good thing* that AI code helpers exist. We can get to interesting quantum simulations without struggling with coding

But, you *must* understand what the code does, and how. You *must* understand how to translate the maths shown in the lectures onto code that does what you need. This is the learning outcome of the course, and we will **examine** you on it

Section 1.1

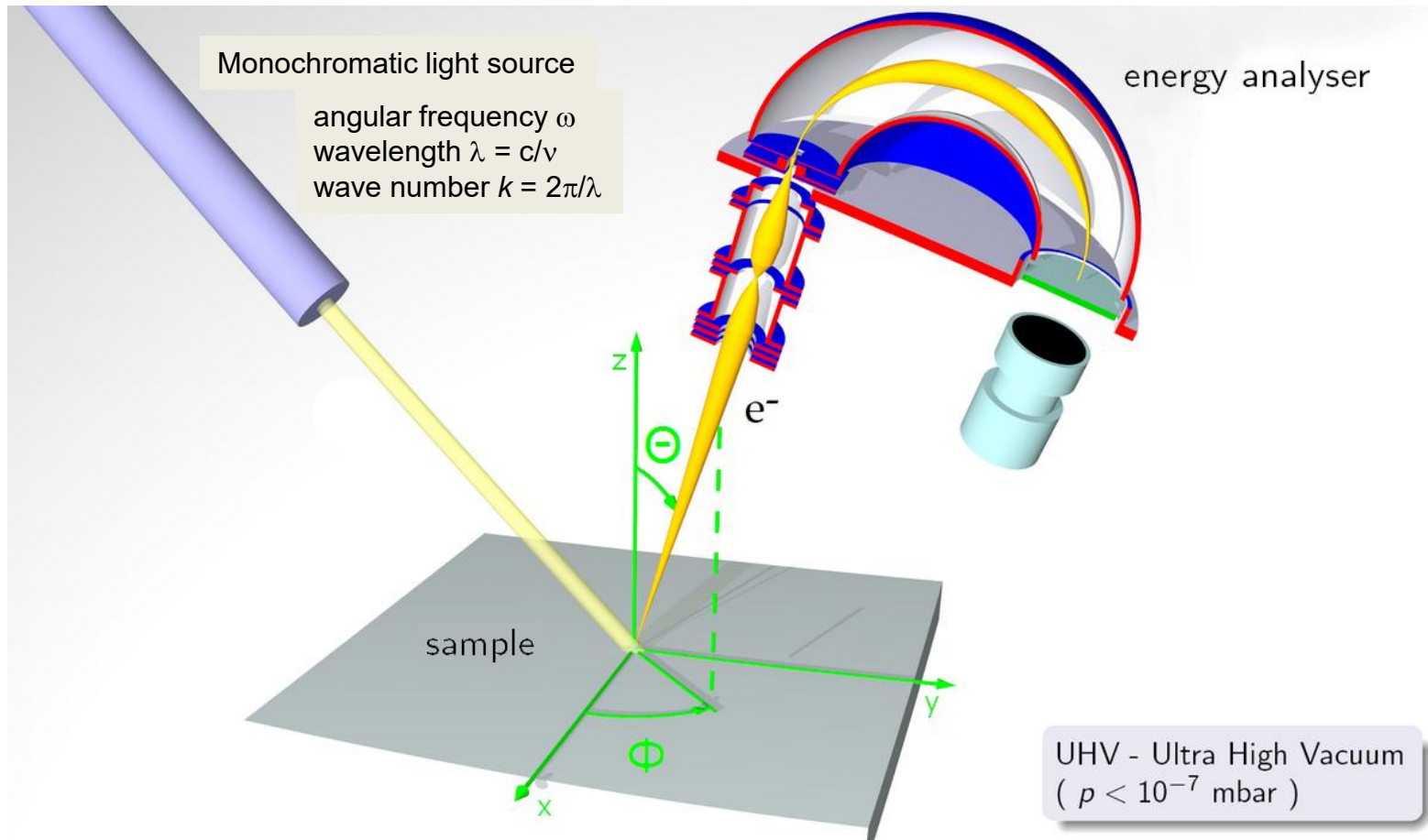
Why Quantum Mechanics?

Learning outcomes:

- A brief history of how Quantum Mechanics was born – out of necessity!
- Understand how wave-particle duality is captured in the equations that define the quantum theory
- An intuitive derivation of the Heisenberg uncertainty principle

Photoelectric effect

Is it possible to extract electrons from a metal using light?
Consider the system below:



Photoelectric effect

Observations:

- 1) Below a certain frequency (color) of the light, no electrons are extracted from the metal. This is independent of the light intensity
- 2) Above a cutoff frequency ω_c , electrons are extracted from the metal, even at very small light intensity
- 3) The energy E of the extracted electrons is proportional to $\omega - \omega_c$
- 4) The momentum, $p = mv = \sqrt{2mE}$, of the extracted electrons is inversely proportional to the wavelength of the light

Planck-Einstein relations

These observations can be explained if one assumes that light comes in discrete packets of energy (**photons**), where:

$$\boxed{\begin{array}{l} E = \hbar\omega \\ \mathbf{p} = \hbar\mathbf{k} \end{array}} \quad \text{or} \quad \boxed{\begin{array}{l} E = h\nu \\ p = h/\lambda \end{array}}$$

$h = 6.626 \times 10^{-34}$ J s is the Planck constant

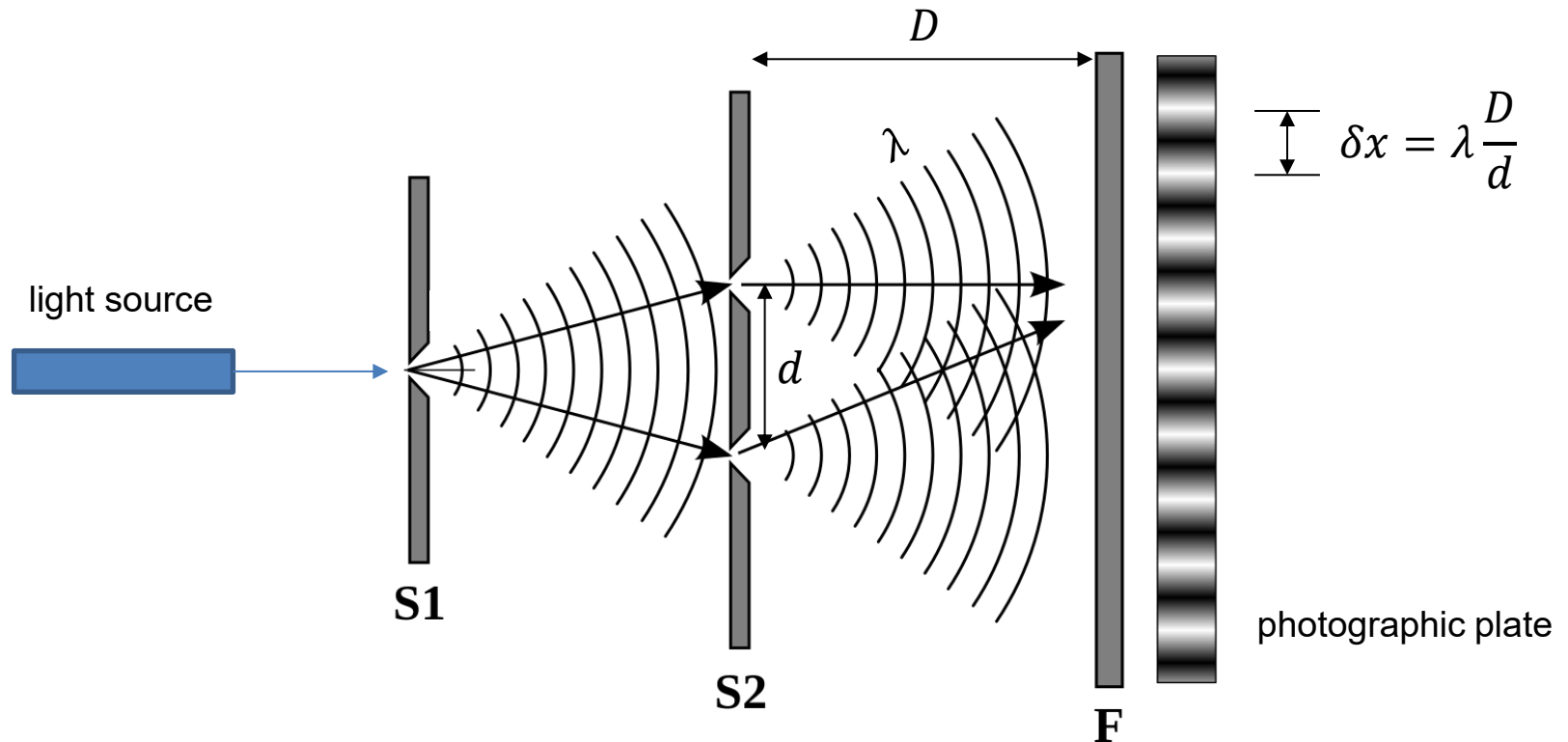
$\hbar = h / 2\pi = 1.054 \times 10^{-34}$ J s / rad is the “reduced” Planck constant

These are the Planck-Einstein relations

They describe light as composed of “particles” of energy $E = h\nu$ determined by the frequency.

But light is also an electromagnetic “wave”!

Double slit experiment



Can be understood by using the linearity of Maxwell's equations

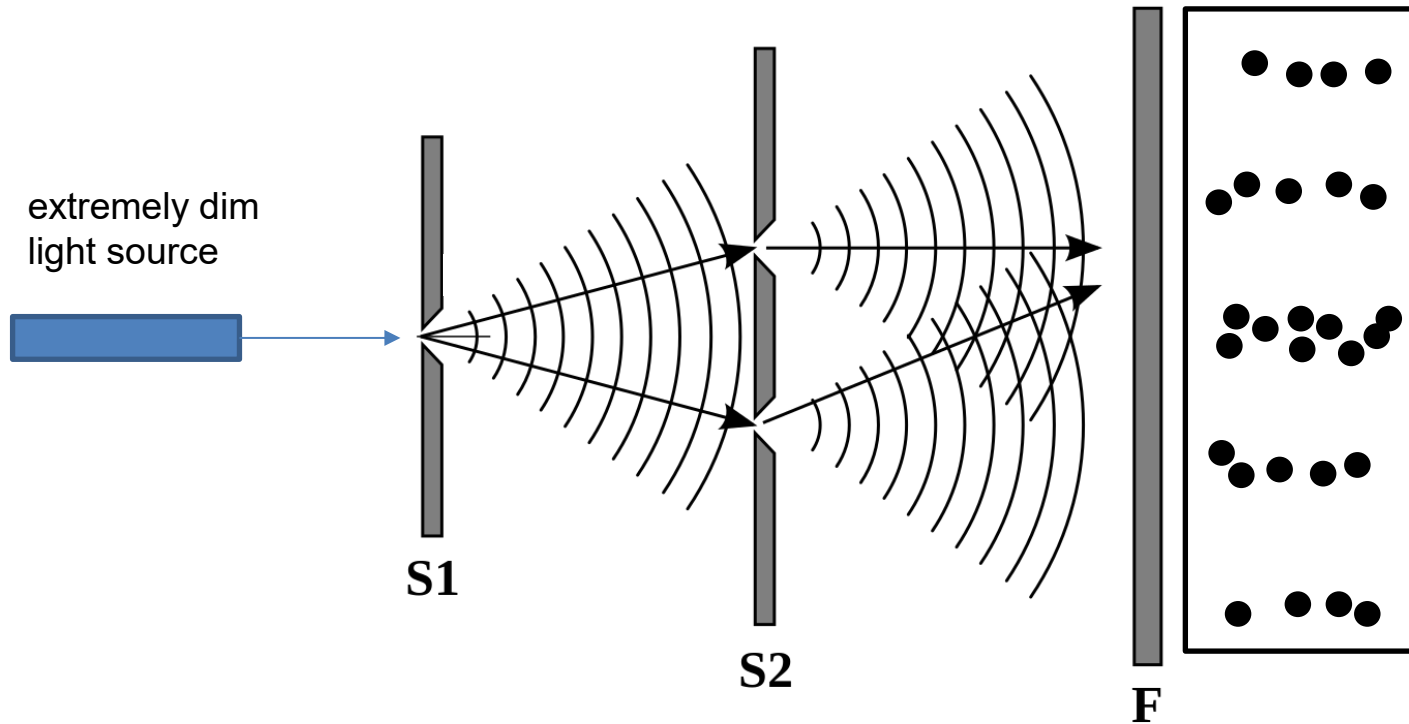
The light intensity is proportional to $|E|^2$

If $\vec{E}(\vec{r}) = \vec{E}_1(\vec{r}) + \vec{E}_2(\vec{r})$ from two sources (the open slits), then

$$I \propto \left| \vec{E}_1(\vec{r}) + \vec{E}_2(\vec{r}) \right|^2 \neq \left| \vec{E}_1(\vec{r}) \right|^2 + \left| \vec{E}_2(\vec{r}) \right|^2$$

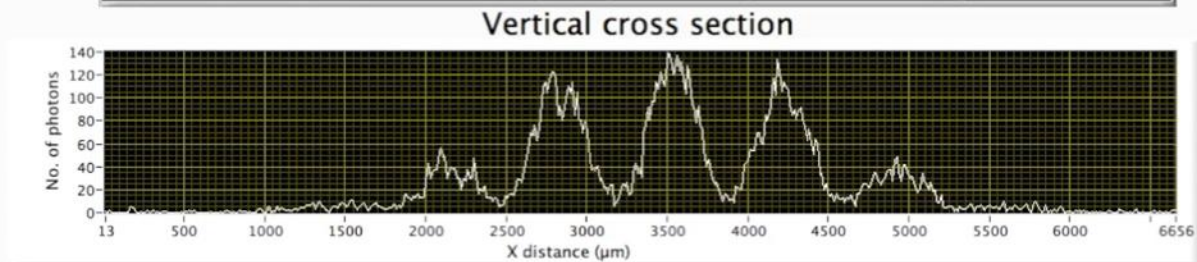
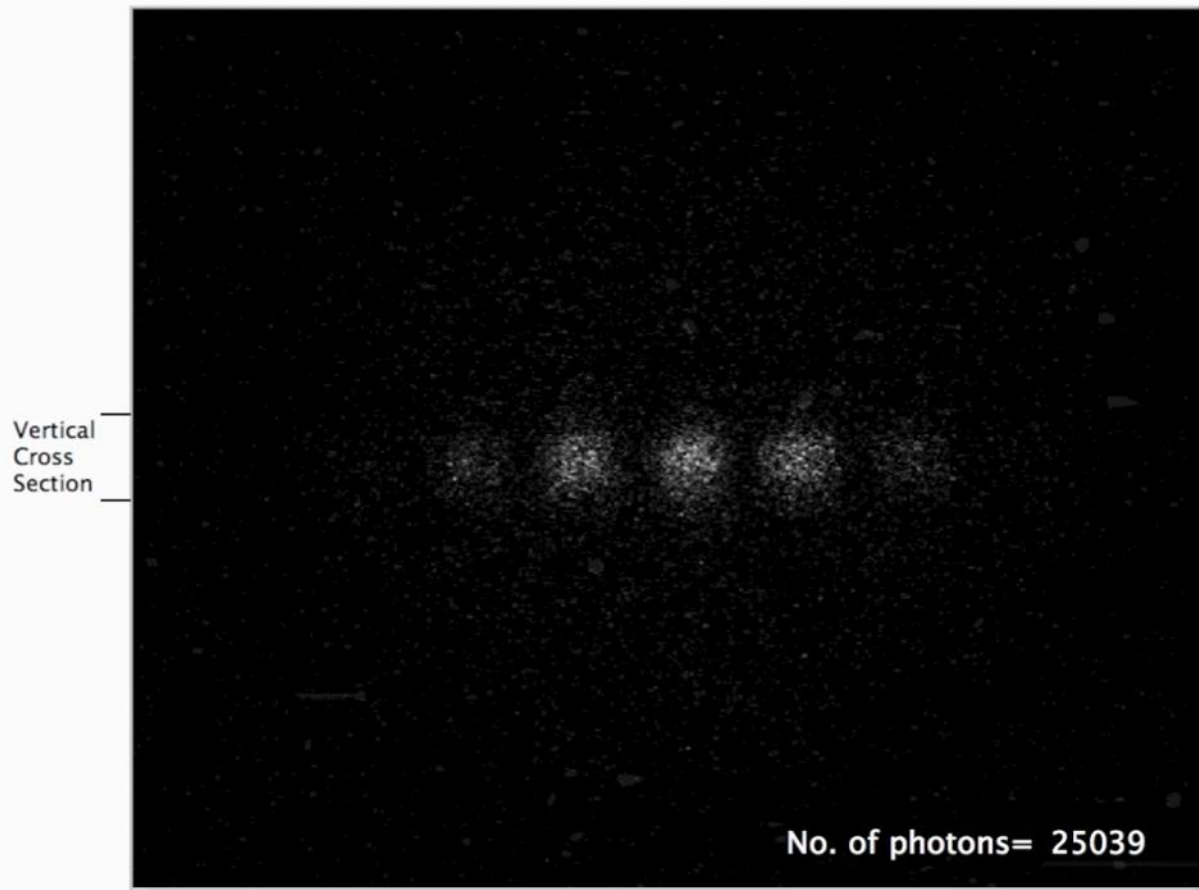
→ shows the wave nature of light

Double slit experiment – low intensity



Individual spots can be seen on the screen → photons are particles, after all.
But, after accumulating many photons, the familiar interference pattern builds up

'Ghost' Young's double slit with a coherent source
photon by photon



Video from:
R.S. Aspden, M.J. Padgett,
G.C. Spalding,
Am. J. Phys. 84, 671 (2016)

Double slit experiment with “particles”

The double slit experiment can be repeated with things that we normally thought of as “particles”, for instance electrons or neutrons.

The results are the same as the double-slit experiment with light!

But, what is the wavelength of a “particle”?

De Broglie hypothesis:

“to every particle is associated a wave, which obeys the same Planck – Einstein relations as applicable for photons”

$$E = \hbar\omega$$

$$\mathbf{p} = \hbar\mathbf{k}$$

Therefore,

$$\lambda = \frac{2\pi}{k} = \frac{h}{p}$$

→ “de Broglie wavelength”

Examples:

Tennis ball $m = 50 \text{ g}$, $v = 30 \text{ m/s} \rightarrow \lambda = 6.6 \times 10^{-34} / (0.05 \times 30) = 4.4 \times 10^{-34} \text{ m}$

Electron $m = 9.1 \times 10^{-31} \text{ kg}$, $v = 30 \text{ m/s} \rightarrow \lambda = 6.6 \times 10^{-34} / (9.1 \times 10^{-31} \times 30) = 24 \text{ } \mu\text{m}$

Again, it seems quantum effects are only significant only for small objects...

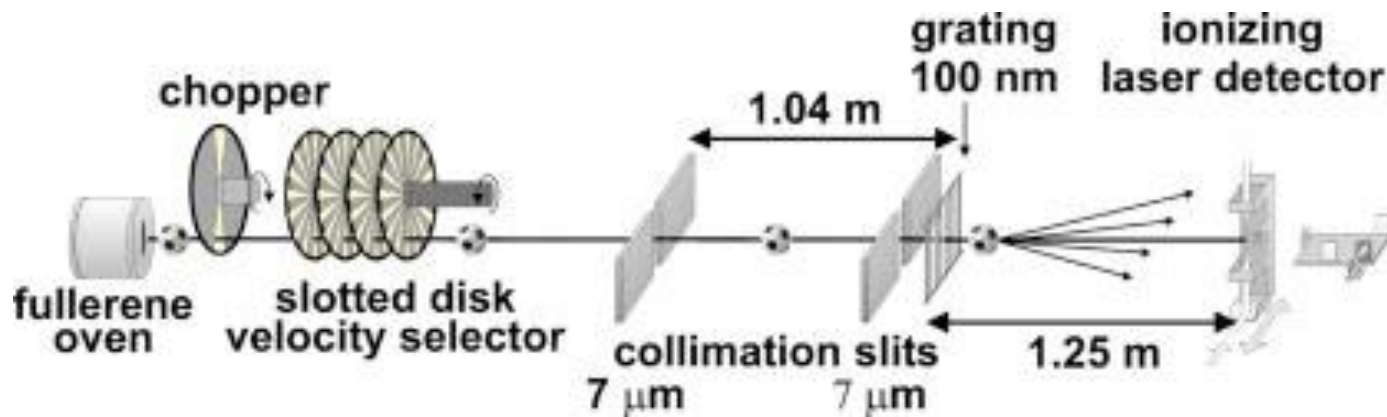
Double slit experiment with “big particles”

The distance between the intensity peaks on the screen is $\delta x = \lambda \frac{D}{d}$, with $\lambda = h/p$

How big a particle can we make show an interference pattern?

- large $\lambda \rightarrow$ small $p \rightarrow$ small velocity (at a given mass)
- small $d \rightarrow$ nanofabricated slits
- large D

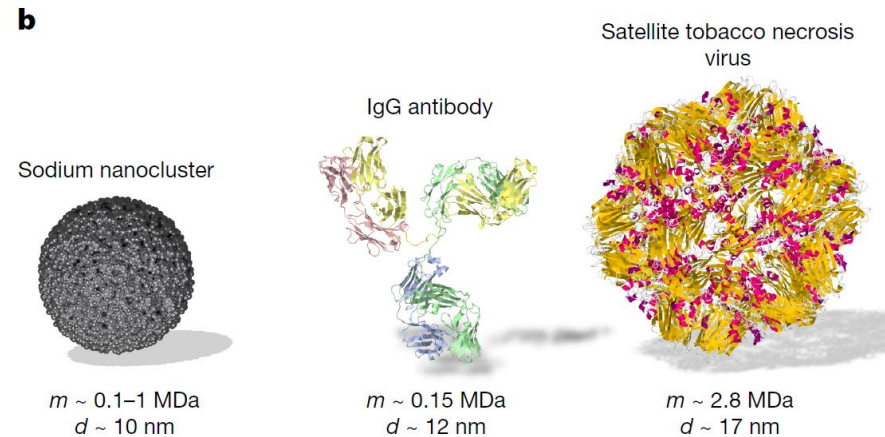
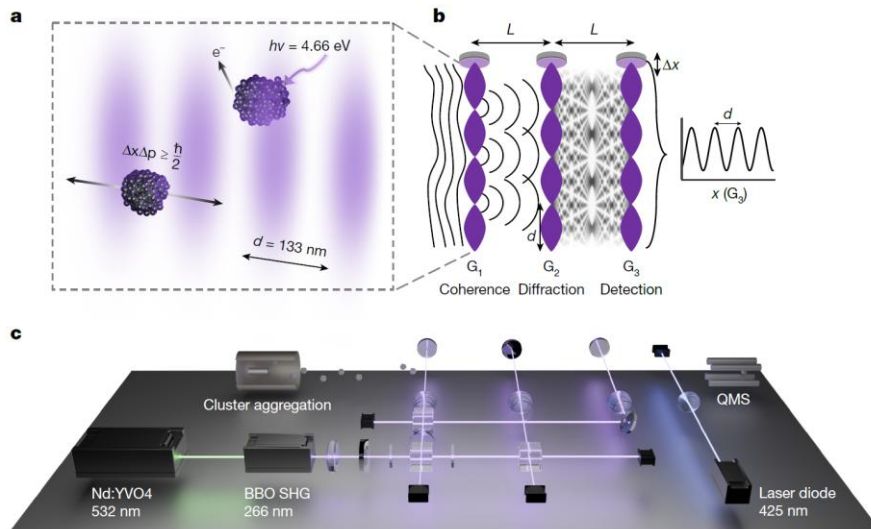
Buckyball (C_{60}) interference experiment



O. Nairz et al., Am. J. Phys. 71, 319 (2003)

Double slit experiment with “big particles”

Biggest objects seen to interfere so far:
sodium clusters with >7000 atoms



Almost as big as a virus...

Note: 1 Da (dalton) = 1/12 of the mass of a ^{12}C atom $\approx$ mass of a proton

S. Pedalino et al., Nature 649, 866 (2026)

Wave-particle duality

The double-slit experiment and the photoelectric effect force us to abandon the view that particles and waves are different things.

Each wave is made of “packets of energy”, to which is associated a wave with wavelength $\lambda = h/p$ and frequency $\nu = E/h$.

In view of the way in which these packets “delocalize” in space and interfere with themselves, the concept of “trajectory” becomes meaningless.

We need a new description of how quantum systems “move”, that properly accounts for their wave nature too.

Wave function

Instead of trying to describe the trajectory of a “particle” with position $\mathbf{r}$ and momentum $\mathbf{p}$ – which would be pointless – we use a wave function $\psi(\mathbf{r}, t)$ which contains all the information we can gather on the particle.

$\psi(\mathbf{r}, t)$ is a probability amplitude, such that

$$C \iiint_V |\psi(\mathbf{r}, t)|^2 d^3\mathbf{r} = P_V(t)$$

where $P_V(t)$ is the probability of finding the particle in the volume V at time t .
 C is a normalization constant chosen such that

$$C \iiint_{-\infty}^{+\infty} |\psi(\mathbf{r}, t)|^2 d^3\mathbf{r} = 1$$

i.e. the particle has to be “somewhere”!

Conceptual shift: instead of trying to calculate a trajectory, we will calculate how the probability of finding the particle in a certain region in space evolves in time.

(Note: the same can be done in classical mechanics, when you have only a statistical description of the initial state of the system)

Wave equation

We know from classical electromagnetism that Maxwell's equations allow solutions in the form of propagating waves (see ELEC3115, Eqs. 35 and 36). Example

$$\mathbf{E}(\mathbf{r}, t) = A e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$$

is a plane wave that solves the wave equation of the form:

$$\nabla^2 \mathbf{E} = \frac{1}{v^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

where $v = \omega/k$.

(Recall that $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the Laplacian operator)

This wave equation contains the partial second derivative of the electric field with respect to both time and space. Taking the second derivative of the plane wave results in the coefficients k^2 and ω^2 which give rise to the $1/v^2$ term in the equation, as appropriate.

Key point: here **v is constant**, independent of ω

Imposing Planck-Einstein relations

The Planck-Einstein relations are the key empirical constraints that any description of a quantum object must fulfill.

Consider a free particle: the only energy is kinetic, thus:

$$E = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

The other Planck-Einstein relation imposes $E = \hbar\omega$, so:

$$\hbar\omega = \frac{\hbar^2 k^2}{2m}$$

Important note: the energy depends **quadratically** on k but **linearly** on ω .

This suggests that an appropriate wave equation for the quantum wavefunction should contain a first derivative w.r.t. time, and the second derivative w.r.t. space

Key point: now **v is not constant** – it depends on the energy of the “particle”, thus on ω

Schroedinger equation

In 1925, Schroedinger postulated (i.e. “guessed”) an equation that describes the time evolution of the wave function of a quantum object:

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}, t) + V(\mathbf{r})\psi(\mathbf{r}, t)$$

This is the **Schroedinger equation**

$V(\mathbf{r})$ is a (position-dependent) potential energy

It's similar (but not identical) to a wave equation. It contains the first derivative w.r.t. time, and the second w.r.t. space. The coefficients are chosen to impose the Planck-Einstein relations.

Because of the first derivative w.r.t. time, the equation *explicitly* contains $i = \sqrt{-1}$ (this is the first time in your life you have encountered $\sqrt{-1}$ in a law of nature!)

Schroedinger equation

Proof: consider a free particle, not subjected to any potentials or forces: $V(\mathbf{r}) = 0$

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}, t)$$

This has a solution of the form:

$$\psi(\mathbf{r}, t) = A \exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)]$$

with the condition:

$$\hbar\omega = \frac{\hbar^2 k^2}{2m}$$

Inserting the Planck-Einstein relations, $E = \hbar\omega$, $\mathbf{p} = \hbar\mathbf{k}$, the above condition means

$$E = \frac{p^2}{2m} = \frac{1}{2}mv^2$$

as expected.

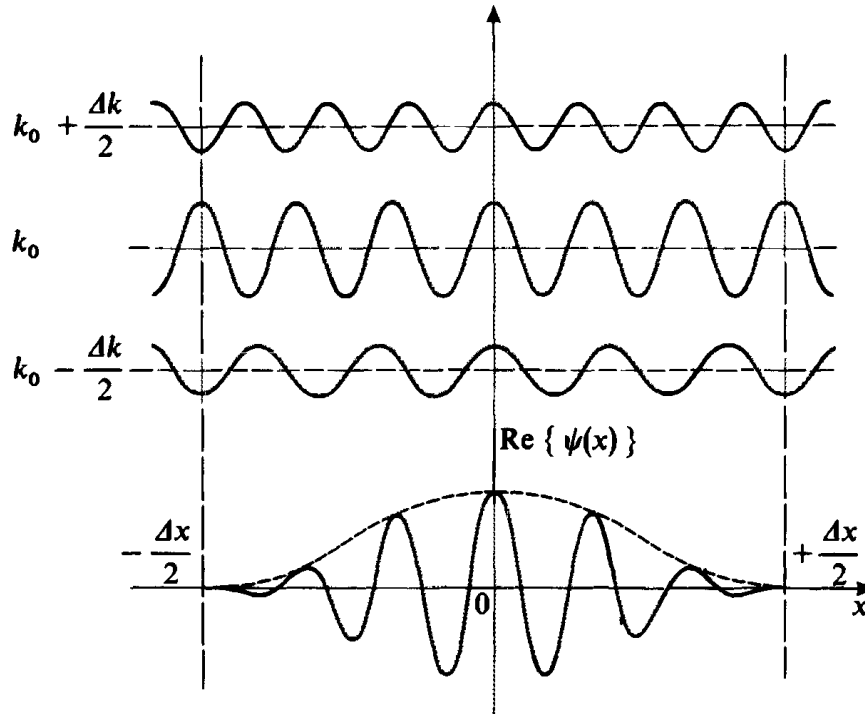
Note: what we wrote is the expression of a monochromatic plane wave. But a plane wave is delocalized over the whole space!!

How do we describe something that is concentrated in some region, resembling a “particle”?

Wave packets

Instead of a single wavelength, let's take 3 different ones (and thus 3 different wave numbers k)

$$\psi(\mathbf{r}, t) = g_1 \exp[i(\mathbf{k}_0 \cdot \mathbf{r} - \omega t)] + \\ + g_2 \exp[i((\mathbf{k}_0 - \delta\mathbf{k}) \cdot \mathbf{r} - \omega t)] + g_3 \exp[i((\mathbf{k}_0 + \delta\mathbf{k}) \cdot \mathbf{r} - \omega t)]$$



The interference between the different wave components causes the wavefunction to no longer be spread out equally in space. But it still has recurrences... How can we push those recurrences away?

Wave packets

The recurrences get farther and farther away the more wave components we add, up to the limit of infinitely many components with different wave number k , each with a coefficient $g(k)$.

Let's take a 1-dimensional problem for simplicity ($\mathbf{r} \rightarrow x$):

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} g(k) \exp[i(kx - \omega t)] dk$$

The meaning of $g(k)$ is found by taking $t = 0$:

$$\psi(x, 0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} g(k) \exp(ikx) dk$$

therefore:

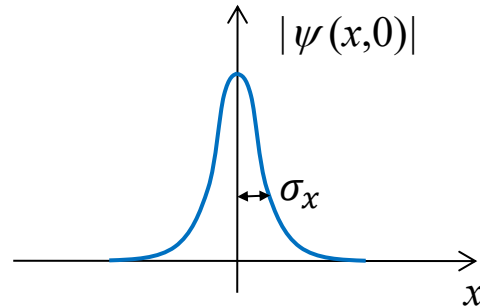
$$g(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \psi(x, 0) \exp(-ikx) dx$$

$g(k)$ is the Fourier transform (in space) of the shape of the wave packet at $t = 0$, i.e., of the probability amplitude of finding the object at point x at $t = 0$.

Heisenberg uncertainty principle

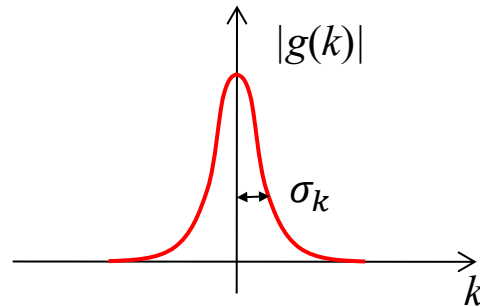
Consider a Gaussian wave packet:

$$\psi(x, 0) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left(-\frac{x^2}{2\sigma_x^2}\right)$$



Its Fourier transform is also a Gaussian:

$$\begin{aligned} g(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \psi(x, 0) \exp(-ikx) dx = \\ &= \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{k^2}{2\sigma_k^2}\right) \end{aligned}$$



From the properties of Fourier transforms, we know that:

$$\sigma_k = 1/\sigma_x \quad \Rightarrow \quad \sigma_k \sigma_x = 1$$

Imposing the Planck-Einstein relation $p = \hbar k \Rightarrow \sigma_p = \hbar \sigma_k$

$$\sigma_p \sigma_x = \hbar$$

The = sign is valid for a Gaussian. For all other shapes, the product of the standard deviations is larger. So, in general, $\sigma_p \sigma_x \geq \hbar$

Heisenberg uncertainty principle

The relation $\sigma_k \sigma_x = 1$, from which we have derived the uncertainty principle, is a simple mathematical property of Fourier transform.

Quantum Mechanics enters the moment you ASSUME the Planck-Einstein relations, i.e. impose (or, rather, formalize the experimental evidence of) wave-particle duality.

Conceptually, you could choose to do the opposite:
if you ASSUME the uncertainty principle as a postulate, then the wave-particle duality is an inescapable consequence of the mathematical properties of Fourier transforms.

The two approaches are perfectly reversible.

Nowadays we take the uncertainty principle as postulate, but historically it probably went the other way, and it's a bit easier to derive.

The “classical limit”

People often say (colloquially) that one obtains the “classical limit” by making $h \rightarrow 0$. Consider the relation:

$$p = \hbar k \Rightarrow \boxed{\lambda \cdot p = h} \text{ (since } k = 2\pi/\lambda \text{)}$$

What does $h = 0$ mean? Two possibilities:

- $\lambda = 0, p \neq 0 \rightarrow$ classical mechanics
(you never thought of the wavelength of a tennis ball...)
- $\lambda \neq 0, p = 0 \rightarrow$ classical wave propagation
(you never thought of the mass of a radio wave...)



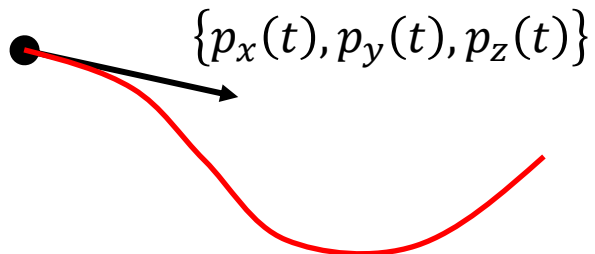
Again we see that quantum mechanics, as DEFINED by an uncertainty principle with $h \neq 0$, is intimately related to the impossibility to “separate” the behavior of particles from that of waves.

Classical vs. quantum description

Classical:

- The particle has a well-defined position and momentum
- The equations of motion predict the trajectory, i.e. the future positions and momenta

$$\{x(t), y(t), z(t)\}$$



$$\{p_x(t), p_y(t), p_z(t)\}$$

Quantum:

- The “object” has a probability of being found in some regions of space, described by a wave function
- The Schroedinger equation predicts the evolution of the wave function in time and space

$$\psi(x, y, z; t)$$



Section 1.2

The mathematical formalism of Quantum Mechanics

Learning outcomes:

- Revise the basic linear algebra necessary to describe quantum systems
- Learn some special properties of matrices that will be reflected in the description of physical systems

Hilbert space

In linear algebra, a function is a vector in an infinite-dimensional vector space. Sometimes it's possible (and convenient) to represent the function using a finite number of “basis functions”, so that the vector space necessary to describe the function has finite dimensions.

This vector space is called **Hilbert space**

A vector space requires a suitable basis.

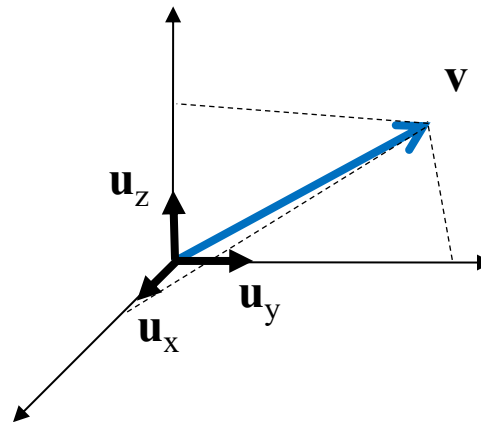
Easy example: 3-dimensional vector space $\mathbb{R}^3$

$$\mathbf{v} = x\mathbf{u}_x + y\mathbf{u}_y + z\mathbf{u}_z$$

$\{\mathbf{u}_x, \mathbf{u}_y, \mathbf{u}_z\}$ are an
orthonormal basis for $\mathbb{R}^3$

$$|\mathbf{u}_\alpha| = 1$$

$$\mathbf{u}_\alpha \cdot \mathbf{u}_\beta = \delta_{\alpha\beta}$$



Row & Column vectors

Once I've chosen the basis of the vector space, I can represent any vector as the array of its coefficients in the chosen basis

Example:

$$\mathbf{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

By default, vectors are represented as columns. I can transpose it to obtain the row:

$$\mathbf{v}^T = (x \quad y \quad z)$$

The scalar (or “inner”, or “dot”) product of vectors is defined as row times column.

Example:

$$\mathbf{a} \cdot \mathbf{b} \rightarrow \mathbf{a}^T \mathbf{b} = (a_1 \quad a_2 \quad a_3) \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \sum_{i=1}^3 a_i b_i$$

Complex numbers

If you are using complex numbers for the coefficients, when you transpose you must also take the complex conjugate.

Example:

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \rightarrow \mathbf{v}^T = (v_1^* \quad v_2^* \quad v_3^*)$$

This is because the inner product of a vector with itself is its modulus squared, which must be a real number:

$$\mathbf{v} \cdot \mathbf{v} \rightarrow \mathbf{v}^T \mathbf{v} = (v_1^* \quad v_2^* \quad v_3^*) \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \sum_{i=1}^3 v_i^* v_i = \sum_{i=1}^3 |v_i|^2$$

Dirac notation

Instead of using “directly” the wavefunction $\psi(x, y, z; t)$, we want to use its coefficients in a suitable basis $\rightarrow \infty$ -dimensional column vector

Example:

$$|\psi\rangle = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \dots \\ \psi_\infty \end{pmatrix} \quad \text{“ket”} \rightarrow \text{column vector}$$

Its transpose is called “bra”

$$\langle\psi| = (\psi_1^* \quad \psi_2^* \quad \dots \quad \psi_\infty^*) \quad \text{“bra”} \rightarrow \text{row vector}$$

They are called like this because, when you take their inner product, you get a “bra(c)ket”

$$\langle\varphi|\psi\rangle = (\varphi_1^* \quad \varphi_2^* \quad \dots \quad \varphi_\infty^*) \begin{pmatrix} \psi_1 \\ \psi_2 \\ \dots \\ \psi_\infty \end{pmatrix}$$

Linear operators

In a vector space, we can define linear operators, i.e. matrices, that transform a vector into another

$$|\psi'\rangle = A|\psi\rangle$$

Linearity means:

$$\begin{array}{ll} \text{if} & |\psi\rangle = \lambda_1|\psi_1\rangle + \lambda_2|\psi_2\rangle \\ \text{then} & |\psi'\rangle = \lambda_1 A|\psi_1\rangle + \lambda_2 A|\psi_2\rangle \end{array}$$

Note:

$\langle\varphi|\psi\rangle$ is a number

$|\varphi\rangle\langle\psi|$ is a matrix

Adjoint operators

Adjoint operators provide the correspondence between bras and kets

$$|\psi'\rangle = A|\psi\rangle \rightarrow \langle\psi'| = \langle\psi|A^\dagger \quad (\text{notice the order!})$$

$A^\dagger$ is the adjoint of A $\dagger$ = “dagger”

It is obtained by transposing and taking the complex conjugate of each element

Example:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1+i & 0 & 5 \\ -i & 3i & 4 \end{pmatrix} \rightarrow A^\dagger = \begin{pmatrix} 1 & 1-i & i \\ 2 & 0 & -3i \\ 3 & 5 & 4 \end{pmatrix}$$

Hermitian operators

Operators for which $A = A^\dagger$ are called Hermitian

They are easy to recognize:

- The elements on opposite side of the diagonal are the complex conjugate of each other
- The elements on the diagonal are real

Example:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1+i & 0 & 5 \\ -i & 3i & i \end{pmatrix} \quad \text{is not Hermitian}$$

$$B = \begin{pmatrix} 1 & 1-i & 3 \\ 1+i & 0 & -3i \\ 3 & 3i & 4 \end{pmatrix} \quad \text{is Hermitian}$$

Basis of a Hilbert space

A set of kets $\{|u_i\rangle\}$ is called orthonormal if

$$\langle u_i | u_j \rangle = \delta_{ij}$$

The set $\{|u_i\rangle\}$ forms a basis of the Hilbert space $\mathcal{E}$ if every $|\psi\rangle$ can be written as:

$$|\psi\rangle = \sum_j c_j |u_j\rangle$$

The coefficients c_j are the inner products of $|\psi\rangle$ with the basis vectors:

$$c_j = \langle u_j | \psi \rangle$$

Important property: closure relation

$$|\psi\rangle = \sum_j c_j |u_j\rangle = \sum_j \langle u_j | \psi \rangle |u_j\rangle$$

Closure relation

Important theorem:

$$|\psi\rangle = \sum_j c_j |u_j\rangle = \sum_j \underbrace{\langle u_j | \psi \rangle}_{\substack{\text{this is a number} \\ \text{it can be multiplied in any order}}} |u_j\rangle$$

$$= \sum_j \underbrace{|u_j\rangle \langle u_j|}_{\text{this is a matrix}}$$

For this to be true, it requires:

$$\sum_j |u_j\rangle \langle u_j| = \mathbb{I}$$

This is called “closure relation”.

The operators $|u_j\rangle \langle u_j|$ projects a vector onto the basis state $|u_j\rangle$

Observables

General linear algebra theorems:

- The eigenvalues of a Hermitian operator are real
- The eigenvectors corresponding to different eigenvalues are orthogonal

A Hermitian operator is called **observable** if its eigenvectors form a basis of the vector space

We call **unitary** an operator for which holds:

$$U^{-1} = U^\dagger$$

The eigenvalues of unitary operators are complex numbers of modulus 1.

Section 2

Spin and the Postulates of Quantum Mechanics

Learning outcomes:

- Describe the behaviour of a simple quantum system using the mathematical tools introduced in the previous section
- Formalize the observed behaviour into fundamental Postulates of Quantum Mechanics
- Recognize how the quantum formalism describes the classical precession of a magnetic dipole

Magnetic moment

In classical electromagnetism, a body with charge q and mass m that revolves around an axis creates a magnetic dipole moment

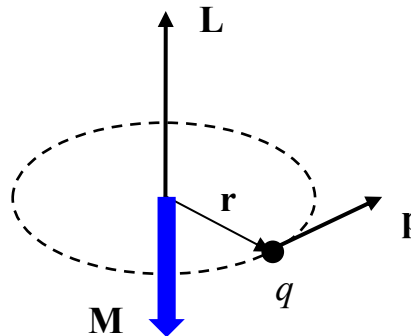
$$\mathbf{M} = \gamma \mathbf{L}$$

where

$\mathbf{L} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times m\mathbf{v}$ is the angular momentum

$\gamma = q/2m$ is the gyromagnetic ratio

Note: γ can be positive or negative, depending on the sign of the charge. For a negative charge (electron), $\mathbf{M}$ and $\mathbf{L}$ are antiparallel.



Magnetic energy

In the presence of a magnetic field $\mathbf{B}$, the magnetic dipole minimizes its energy W by orienting itself along the field (think of the needle of a compass...)

$$W = -\mathbf{M} \cdot \mathbf{B}$$

There is a torque on the angular momentum:

$$\mathbf{T} = \mathbf{M} \times \mathbf{B}$$

In addition, there is a linear force on the centre of mass:

$$\mathbf{F} = -\nabla W = \nabla(\mathbf{M} \cdot \mathbf{B})$$

In a uniform magnetic field, there is no linear force on the dipole.

Assume there is an inhomogeneous magnetic field B_z oriented along $\mathbf{z}$

The force on the dipole is:

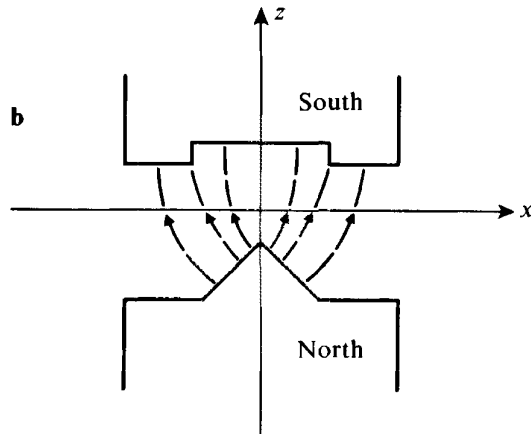
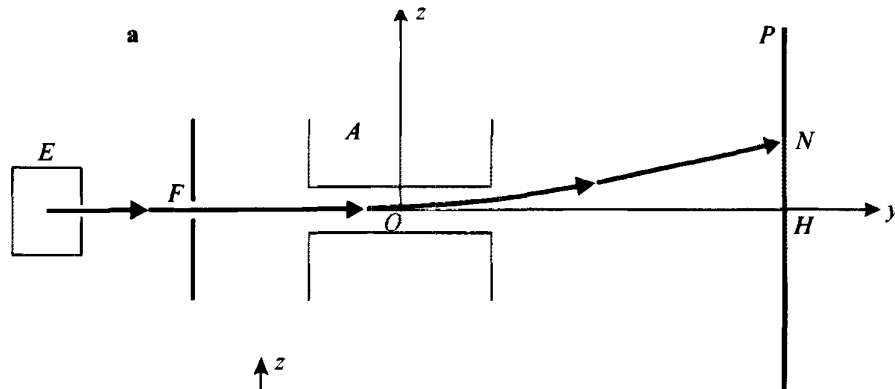
$$F_z = M_z \frac{\partial B_z}{\partial z}$$

The orientation of the force (upwards or downwards) thus depends on whether M_z is positive or negative

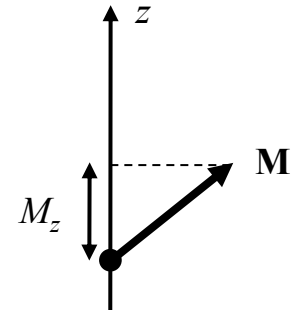
Stern-Gerlach experiment

This famous experiment was conceived to verify if the atoms possess an *intrinsic* magnetic moment, called “**spin**”.

It was done with silver atoms, because they have zero angular momentum – all atomic orbitals are completely filled except for the $5s^1$, which has $L = 0$ anyway.

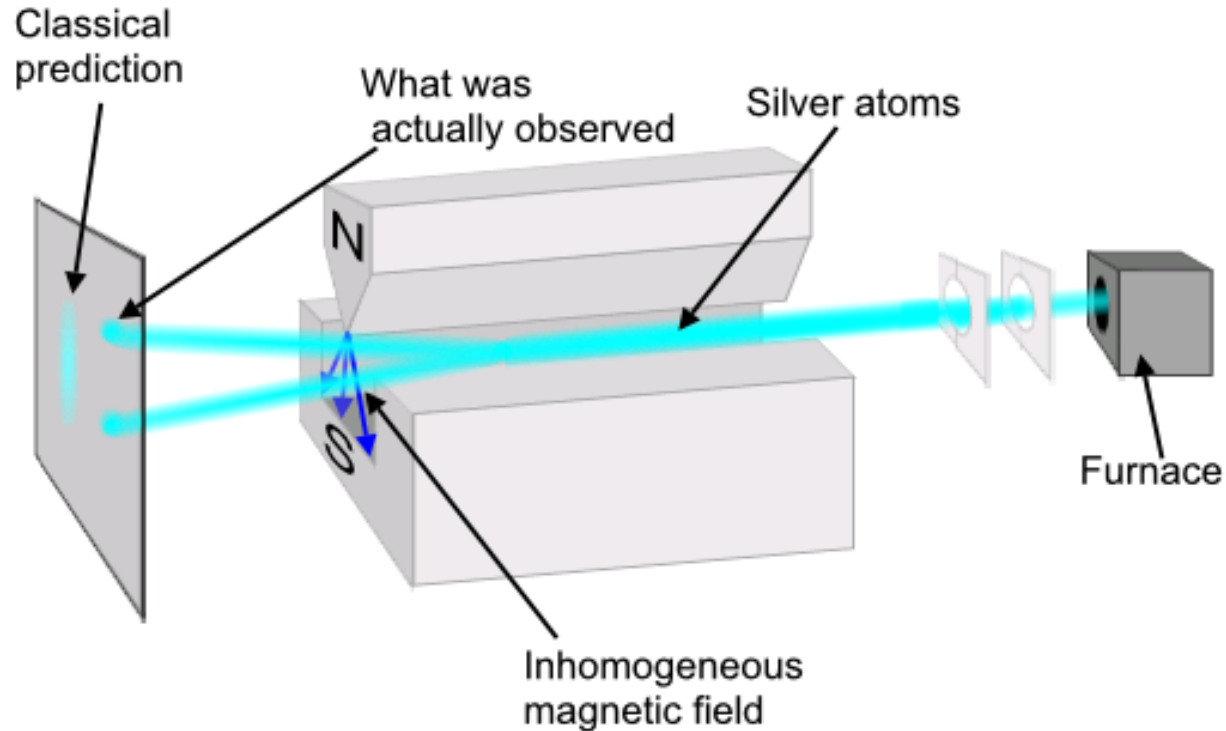


This arrangement of magnets creates a gradient of magnetic field B_z in the z-direction, so that a magnetic dipole will be deflected upwards or downwards depending on the value of M_z when it enters the apparatus.



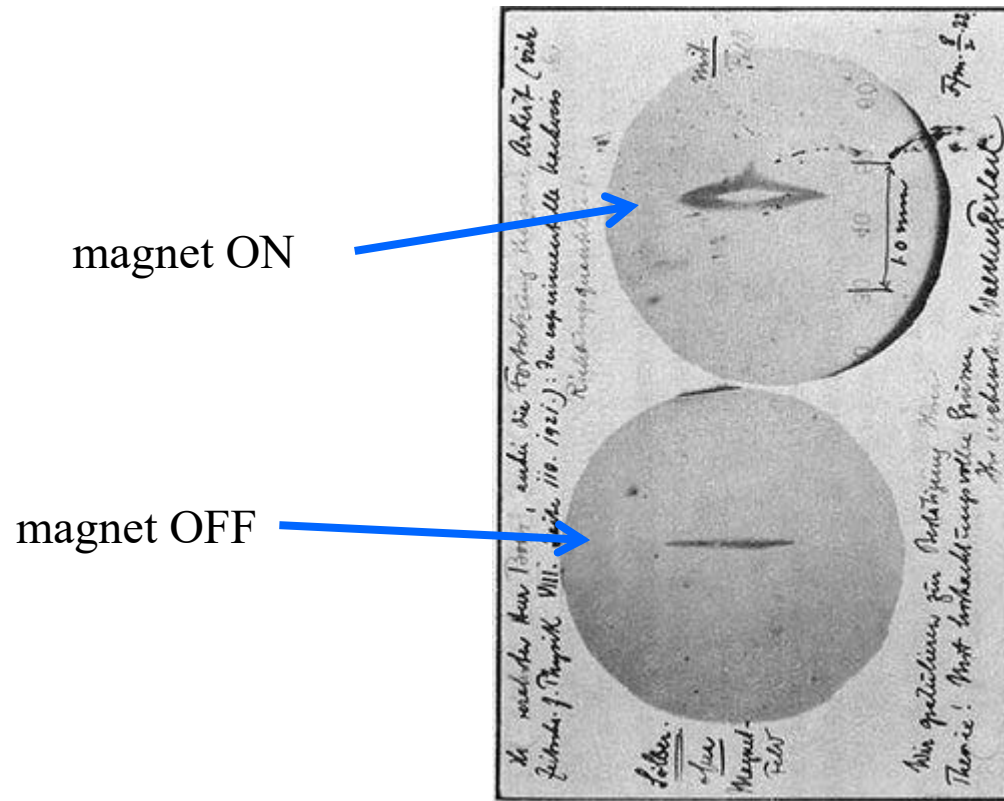
Stern-Gerlach experiment

The beam of silver atoms comes out of a high-temperature furnace, so you would expect that the value of M_z is totally random. Thus, the atoms colliding on the back screen should be distributed over a continuous interval.



Stern-Gerlach experiment

Instead, it was found that the atoms fall on two separate spots, with **no atoms going straight through**



Spin 1/2

The Stern-Gerlach experiment showed three very important things:

- 1) silver atoms have a magnetic moment that does not arise from the orbital angular momentum of its electrons, but is intrinsic to the atom → **spin**.
- 2) this magnetic moment is **quantized**. That is, it behaves like a quantum system with two discrete states. The outcome of the experiment precludes a continuous range of measured values
- 3) The value of the intrinsic angular momentum was measured to be

$$S_z = \pm \frac{\hbar}{2}$$

Notice the dimensions:

$$[\hbar] = \text{J s} = \text{kg m}^2 \text{s}^{-1}$$

S is an angular momentum, like $\mathbf{L} = \mathbf{r} \times \mathbf{p}$

$$[S] = [L] = \text{m kg m s}^{-1} = \text{J s}$$

Postulates of QM: #1

At a time t_0 , the state of a quantum system is specified by a ket $|\psi(t_0)\rangle$ that belongs to a suitable vector space $\mathcal{E}$.

Profound consequence: any ket $|\psi'\rangle = \lambda_1|\psi_1\rangle + \lambda_2|\psi_2\rangle$

is also a legitimate state of the system $\Rightarrow$ **superposition principle**

Practical remark: we will normally assume that the state vectors are normalized:

$$\langle\psi|\psi\rangle = 1$$

Postulate #1 application

Since the Stern-Gerlach experiment showed that there are two states, we need a 2-dimensional vector space.

Orthonormal basis vectors: $|\downarrow\rangle, |\uparrow\rangle$

They must fulfill the following conditions:

$$\langle\downarrow|\downarrow\rangle = \langle\uparrow|\uparrow\rangle = 1 \quad \rightarrow \text{normalized}$$

$$\langle\uparrow|\downarrow\rangle = 0 \quad \rightarrow \text{orthogonal}$$

$$|\downarrow\rangle\langle\downarrow| + |\uparrow\rangle\langle\uparrow| = \mathbb{I} \quad \rightarrow \text{closure relation}$$

Easy choice: $|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Any ket of the form $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$, with $\alpha^2 + \beta^2 = 1$ is a legitimate state of the system

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Postulates of QM: #2

Every measurable physical quantity $\mathcal{A}$ is described by an operator A , which must be an observable.

Profound consequence: you can construct operators in $\mathcal{E}$ that are not observable, i.e. they cannot correspond to measurable physical quantities

Postulate #2 application

The physical quantity measured in the Stern-Gerlach experiment is the z -component of the angular momentum, S_z .

Since it is found to provide only two discrete results, the operator S_z must be an observable in a 2-dimensional Hilbert space, i.e. a 2×2 matrix.

Postulates of QM: #3

When measuring the value of the quantity $\mathcal{A}$, we can only obtain as a result one of the eigenvalues a_n of the observable A .

Profound consequence: for an observable with a discrete set of eigenvalues, the corresponding physical quantity can only be found to have a discrete set of values, not any arbitrary value in between → **quantization**

Postulate #3 application

Putting #2 and #3 together, and recalling that the experimentally observed values of S_z are $\pm\hbar/2$, we deduce that S_z must be a 2×2 observable with eigenvalues $\pm\hbar/2$.

We can *choose* to define S_z as:

$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Spin matrices

Proof that $S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is the correct matrix:

$$S_z |\uparrow\rangle = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} |\uparrow\rangle$$

$$S_z |\downarrow\rangle = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{\hbar}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{\hbar}{2} |\downarrow\rangle$$

i.e. $|\uparrow\rangle, |\downarrow\rangle$ are eigenvectors of S_z with eigenvalues $\pm\hbar/2$

The matrices describing the spin projection along x and y are:

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

with eigenvectors:

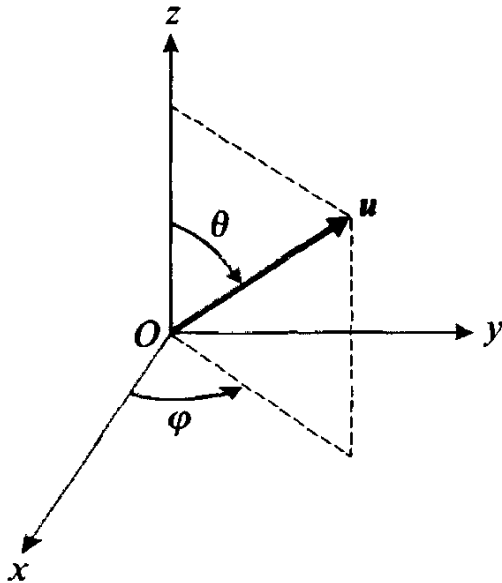
$$|X\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle + |\downarrow\rangle) \quad | - X\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle - |\downarrow\rangle)$$

$$|Y\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle + i|\downarrow\rangle) \quad | - Y\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle - i|\downarrow\rangle)$$

Spin matrices

Consider an arbitrary unit vector $\mathbf{u}$ in 3-dimensional space. The observable corresponding to the spin projection along $\mathbf{u}$ is simply given by:

$$S_u = S_x \sin \theta \cos \phi + S_y \sin \theta \sin \phi + S_z \cos \theta = \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta \cdot e^{-i\phi} \\ \sin \theta \cdot e^{i\phi} & -\cos \theta \end{pmatrix}$$



The eigenvectors of S_u are:

$$|U\rangle = \cos \frac{\theta}{2} e^{-\frac{i\phi}{2}} |\uparrow\rangle + \sin \frac{\theta}{2} e^{\frac{i\phi}{2}} |\downarrow\rangle$$

$$|-U\rangle = -\sin \frac{\theta}{2} e^{-\frac{i\phi}{2}} |\uparrow\rangle + \cos \frac{\theta}{2} e^{\frac{i\phi}{2}} |\downarrow\rangle$$

Postulates of QM: #4

When measuring the quantity $\mathcal{A}$ on a system described by the state $|\psi\rangle$, the probability of obtaining as a result the eigenvalue a_n is given by:

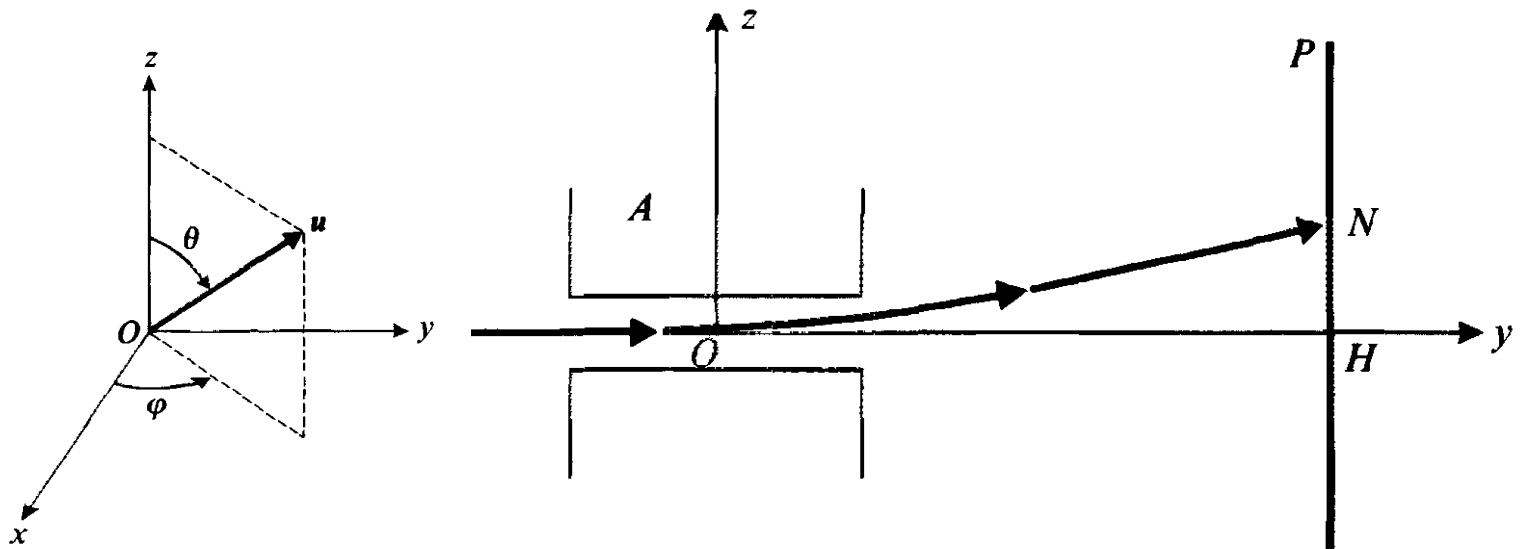
$$\mathcal{P}(a_n) = |\langle u_n | \psi \rangle|^2 \quad \text{“Born’s rule”}$$

where $|u_n\rangle$ is the eigenvector of A with eigenvalue a_n

Profound consequence: the result of the *measurement* of a quantum system is, in general, **probabilistic**

Postulate #4 application

Consider an atom with spin oriented along $\mathbf{u}$, entering the Stern-Gerlach apparatus.



Quantum measurement

The state of the atom just before the measurement is:

$$|\psi(t^-)\rangle = \cos\frac{\theta}{2}e^{-\frac{i\phi}{2}}|\uparrow\rangle + \sin\frac{\theta}{2}e^{\frac{i\phi}{2}}|\downarrow\rangle = \begin{pmatrix} \cos\frac{\theta}{2}e^{-\frac{i\phi}{2}} \\ \sin\frac{\theta}{2}e^{\frac{i\phi}{2}} \end{pmatrix}$$

According to the Born's rule, the probability to find the eigenvalue $+\hbar/2$ corresponding to the eigenvector $|\uparrow\rangle$ is:

$$\mathcal{P}_{\uparrow} = |\langle\uparrow|\psi(t^-)\rangle|^2 = \left| (1 \quad 0) \begin{pmatrix} \cos\frac{\theta}{2}e^{-\frac{i\phi}{2}} \\ \sin\frac{\theta}{2}e^{\frac{i\phi}{2}} \end{pmatrix} \right|^2 = \cos^2\frac{\theta}{2}$$

The result is independent of ϕ , since the apparatus only measures the projection along z .

Example: $\theta = \pi/2 \rightarrow$ spin in the xy plane

$$\mathcal{P}_{\uparrow} = \cos^2\frac{\pi}{4} = \frac{1}{2}$$

i.e. there is a 50% chance of finding the electron pointing “up”

Postulates of QM: #5

If the measurement of $\mathcal{A}$ at time t has given the result a_n , immediately after the measurement the system is found in the normalized projection of the state $|\psi(t^-)\rangle$ before the measurement onto the eigenvector $|u_n\rangle$ corresponding to the measured eigenvalue a_n

$$|\psi(t^-)\rangle \rightarrow |\psi(t^+)\rangle = \frac{P_n |\psi(t^-)\rangle}{\sqrt{\langle \psi(t^-) | P_n | \psi(t^-) \rangle}}$$

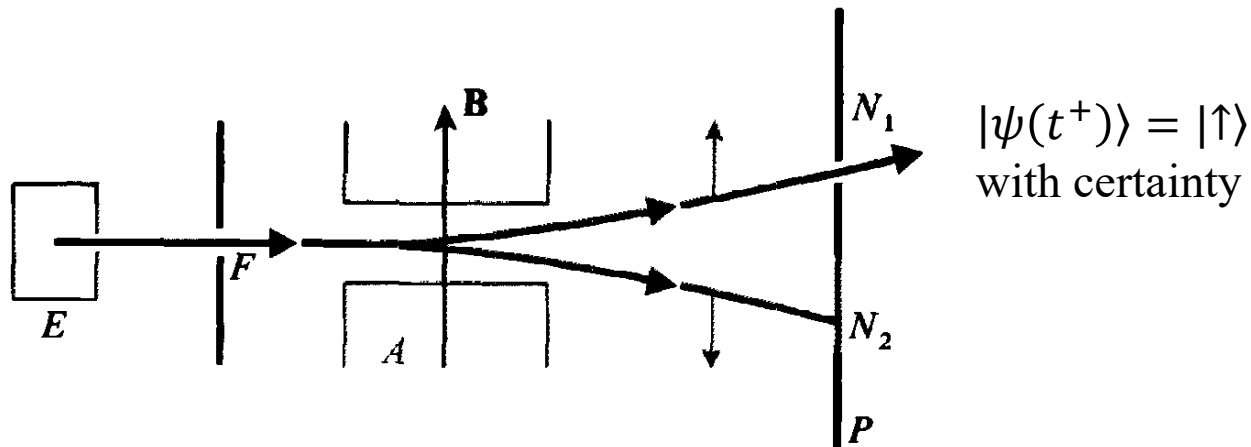
where $P_n = |u_n\rangle\langle u_n|$ is the projection operator onto the eigenstate $|u_n\rangle$
(the denominator serves the purpose of normalizing the wavefunction)

Also known as “collapse of the wavefunction”

Profound consequence: in general, measuring some property of the system *changes its state*

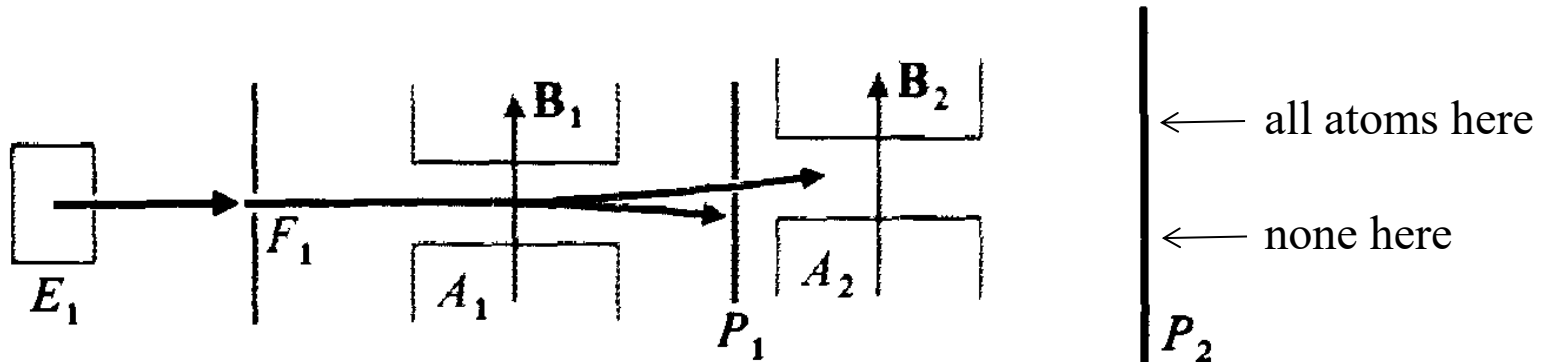
Postulate #5 application

The Stern-Gerlach apparatus can be used to prepare a known quantum state:
if we pierce a hole in the screen where the “up” atoms would collide, then we *know*
that those atoms are in the state $|\psi(t^+)\rangle = |\uparrow\rangle$



State preparation

We can verify that the measurement and the selection of a specific outcome causes a state preparation, by inserting another Stern-Gerlach apparatus after the first one:



Expectation value

Now imagine that the second Stern-Gerlach apparatus was not vertical, but rotated along $\mathbf{u}$. Even though we know with certainty that the atoms coming out of the first apparatus are all in the state $|\uparrow\rangle$, the outcome of the measurement is not certain, because the second apparatus measures $\mathcal{S}_u$, not $\mathcal{S}_z$.

What can we say about the statistics of the second measurement?

We know from Born's rule the probability that each of the eigenvalues may appear as the outcome of the measurement, because we know the quantum state $|\psi\rangle$.

Therefore, the average value (often called “expectation value”) of an observable A given the state $|\psi\rangle$ is:

$$\langle A \rangle_\psi = \sum_n a_n \mathcal{P}(a_n) = \sum_n a_n |\langle u_n | \psi \rangle|^2$$

note: given a complex number c , $|c|^2 = c^* c$
 given two vectors $|\psi\rangle$ and $|\varphi\rangle$, $\langle \varphi | \psi \rangle = \langle \psi | \varphi \rangle^*$

Expectation value

$$\langle A \rangle_\psi = \sum_n a_n |\langle u_n | \psi \rangle|^2 = \sum_n a_n \langle \psi | u_n \rangle \langle u_n | \psi \rangle$$


 $A|u_n\rangle$

note: $a_n |u_n\rangle = A|u_n\rangle$

$$\langle A \rangle_\psi = \sum_n \langle \psi | \underbrace{A|u_n\rangle}_{\text{independent of } n} \underbrace{\langle u_n | \psi \rangle}_{\text{independent of } n} = \langle \psi | A \left[\underbrace{\sum_n |u_n\rangle \langle u_n|}_{= \mathbb{I} \text{ (closure relation)}} \right] | \psi \rangle$$

$$\langle A \rangle_\psi = \langle \psi | A | \psi \rangle$$

In the example given before:

$$\langle S_u \rangle_{|\uparrow\rangle} = \langle \uparrow | S_u | \uparrow \rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta \cdot e^{-i\phi} \\ \sin \theta \cdot e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} \cos \theta$$

Postulates of QM: #6

If a system is in a state $|\psi(t)\rangle$ at time t , its subsequent time evolution is governed by the Schroedinger equation:

$$i\hbar \frac{d|\psi(t)\rangle}{dt} = H(t)|\psi(t)\rangle$$

where $H(t)$ is the Hamiltonian of the system, i.e. the operator that describes its total energy.

Profound consequence: Quantum Mechanics is, for the most part, **deterministic**. Given an initial state, the system evolves in a *predictable* way, according to a linear differential equation called Schrödinger equation. The probabilistic aspects enter when you try to *measure* the state

Postulate #6 application

$$i\hbar \frac{d|\psi(t)\rangle}{dt} = H(t)|\psi(t)\rangle$$

$H(t)$ is a matrix, $|\psi\rangle$ is a vector, so the Schroedinger equation is a system of linear differential equations.

Its formal solution can be written as:

$$|\psi(t)\rangle = U(t, t_0)|\psi(t_0)\rangle$$

with the condition

$$U(t_0, t_0) = \mathbb{I}$$

Time-evolution operator

Substituting $|\psi(t)\rangle = U(t, t_0)|\psi(t_0)\rangle$ into the Schroedinger equation gives:

$$i\hbar \frac{d}{dt} U(t, t_0) |\psi(t_0)\rangle = H(t) U(t, t_0) |\psi(t_0)\rangle$$

since $|\psi(t_0)\rangle$ is arbitrary, $U(t, t_0)$ must satisfy:

$$i\hbar \frac{d}{dt} U(t, t_0) = H(t) U(t, t_0)$$

Assume for simplicity that the Hamiltonian is constant: $H(t) \rightarrow H$
the differential equation above can be formally integrated to yield:

$$U(t, t_0) = \int_{t_0}^t -i \frac{H}{\hbar} U(t', t_0) dt' = \exp \left[-i \frac{H}{\hbar} (t - t_0) \right]$$

Thus, knowing $|\psi(t_0)\rangle$, the quantum state at later times is given by

$$|\psi(t)\rangle = e^{-i \frac{H}{\hbar} (t - t_0)} |\psi(t_0)\rangle$$

where

$$U(t, t_0) = e^{-i \frac{H}{\hbar} (t - t_0)}$$

is the **time evolution operator**.

Exponential of a matrix

Normally, you only know how to do sums and (row by column) multiplications on matrices. How do you calculate some arbitrary function of a matrix, such as e^A ?

Use the Taylor expansion!

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

By analogy:

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!} = \mathbb{I} + A + \frac{A^2}{2} + \frac{A^3}{6} + \dots$$

Note: if A is diagonal, A^n is also diagonal, and every function of A is diagonal too.

$$A = \begin{pmatrix} a_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & a_N \end{pmatrix} \rightarrow e^A = \begin{pmatrix} e^{a_1} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & e^{a_N} \end{pmatrix}$$

In general, if $|\varphi\rangle$ is an eigenvector of A with eigenvalue a , then $|\varphi\rangle$ is also an eigenvector of $F(A)$ with eigenvalue $F(a)$.

Matlab command

Matlab has a command that calculates the exponential of a matrix:

`expm(A)`

Therefore, given an initial state ψ_0 at time $t = 0$, the quantum state ψ_t at a later time t is obtained with **one line of code**:

```
psi_t = expm(-1i * H * t / hbar) * psi_0
```

Python command

Python has a similar command, as part of the `scipy.linalg` package:

```
from scipy.linalg import expm
```

```
psi_t = scipy.linalg.expm(-1j * H * t / hbar) @ psi_0
```

Hamiltonian eigenbasis

Special case:

The Hamiltonian, i.e. the total energy of the system, is an observable physical quantity, so it must be described by a matrix whose eigenvectors form a basis of the vector space.

If we choose to use the eigenvectors of the Hamiltonian as our basis, it means the Hamiltonian appears as a diagonal matrix:

$$H = \begin{pmatrix} E_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & E_N \end{pmatrix}$$

where $E_1, \dots, E_N$ are the eigenvalues (“eigenenergies”).

Therefore, in the Hamiltonian eigenbasis, the time evolution operator is simply:

$$U(t, t_0) = \begin{pmatrix} e^{-i\frac{E_1}{\hbar}(t-t_0)} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & e^{-i\frac{E_N}{\hbar}(t-t_0)} \end{pmatrix}$$

→ If you can describe the system in the Hamiltonian eigenbasis, the time evolution operator can be calculated “by hand”. This is what you’ll find in all textbooks.

Example: spin precession

Classical description:

Consider a magnetic moment $\mathbf{M} = \gamma\mathbf{S}$ placed in a (uniform) magnetic field $\mathbf{B}$. If $\mathbf{S}$ and $\mathbf{B}$ are not parallel, there is a torque $\mathbf{T}$ acting on the angular momentum:

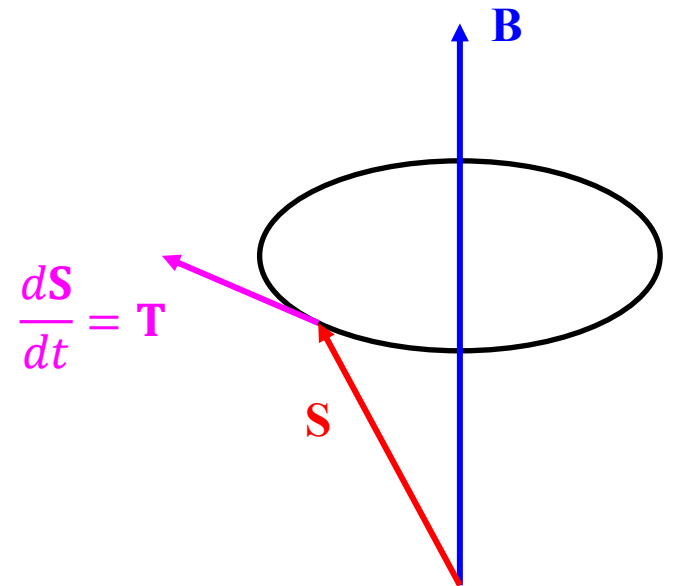
$$\mathbf{T} = \gamma\mathbf{S} \times \mathbf{B}$$

The derivative of the angular momentum is equal to the torque:

$$\frac{d\mathbf{S}}{dt} = \mathbf{T} = \gamma\mathbf{S} \times \mathbf{B}$$

Let us define the quantity:

$$\omega = -\gamma B$$



Example: spin precession

Quantum description:

The energy of a magnetic moment $\mathbf{M} = \gamma\mathbf{S}$ placed in a (uniform) magnetic field $\mathbf{B}$ is

$$W = -\gamma\mathbf{S} \cdot \mathbf{B}$$

Assume that $\mathbf{B} \parallel z$. Since there are no other relevant contributions to the energy of the system, the Hamiltonian is

$$H = -\gamma S_z B = -\gamma B \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \omega \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \frac{\hbar\omega}{2} & 0 \\ 0 & -\frac{\hbar\omega}{2} \end{pmatrix}$$

Note: this Hamiltonian is time-independent if the magnetic field is constant.

It has two eigenenergies:

$$E_1 = \frac{\hbar\omega}{2} \qquad E_2 = -\frac{\hbar\omega}{2}$$

Time evolution

Time evolution operator:

$$U(t, t_0) = e^{-i\frac{H}{\hbar}(t-t_0)} = \begin{pmatrix} e^{-i\frac{E_1}{\hbar}(t-t_0)} & 0 \\ 0 & e^{-i\frac{E_2}{\hbar}(t-t_0)} \end{pmatrix} = \begin{pmatrix} e^{-i\frac{\omega}{2}(t-t_0)} & 0 \\ 0 & e^{i\frac{\omega}{2}(t-t_0)} \end{pmatrix}$$

Easy! Because we used the eigenstates of the Hamiltonian as our basis.

Now let's prepare the spin in the symmetric eigenstate of S_x :

$$|\psi(0)\rangle = |X\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

At a later time t , the state will be:

$$|\psi(t)\rangle = U(t, 0)|\psi(0)\rangle = \begin{pmatrix} e^{-i\frac{\omega}{2}t} & 0 \\ 0 & e^{i\frac{\omega}{2}t} \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\frac{\omega}{2}t} \\ e^{i\frac{\omega}{2}t} \end{pmatrix}$$

Expectation values

To make sense of what this $|\psi(t)\rangle$ means, we need to calculate the expectation values of some physical quantities.

Example:

$$\langle S_z \rangle_{\psi(t)} = \langle \psi(t) | S_z | \psi(t) \rangle = \frac{\hbar}{4} \begin{pmatrix} e^{i\frac{\omega}{2}t} & e^{-i\frac{\omega}{2}t} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} e^{-i\frac{\omega}{2}t} \\ e^{i\frac{\omega}{2}t} \end{pmatrix} = 0$$

$$\begin{aligned} \langle S_x \rangle_{\psi(t)} &= \langle \psi(t) | S_x | \psi(t) \rangle = \frac{\hbar}{4} \begin{pmatrix} e^{i\frac{\omega}{2}t} & e^{-i\frac{\omega}{2}t} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{-i\frac{\omega}{2}t} \\ e^{i\frac{\omega}{2}t} \end{pmatrix} = \\ &= \frac{\hbar}{4} (e^{i\omega t} + e^{-i\omega t}) = \frac{\hbar}{2} \cos(\omega t) \end{aligned}$$

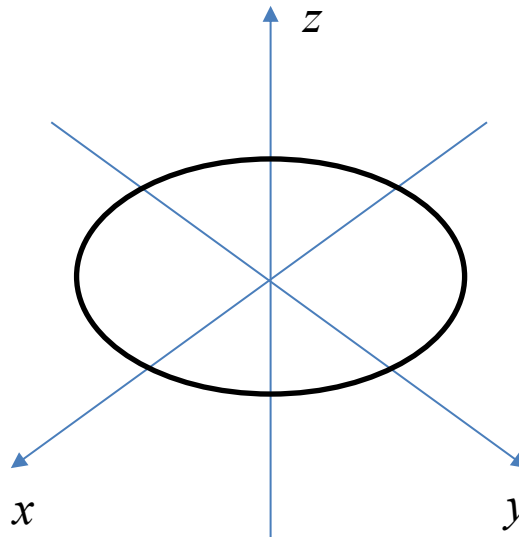
$$\begin{aligned} \langle S_y \rangle_{\psi(t)} &= \langle \psi(t) | S_y | \psi(t) \rangle = \frac{\hbar}{4} \begin{pmatrix} e^{i\frac{\omega}{2}t} & e^{-i\frac{\omega}{2}t} \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} e^{-i\frac{\omega}{2}t} \\ e^{i\frac{\omega}{2}t} \end{pmatrix} = \\ &= \frac{\hbar}{4} (-ie^{i\omega t} + ie^{-i\omega t}) = \frac{\hbar}{2} \sin(\omega t) \end{aligned}$$

Larmor precession

The results we found describe a spin that:

- Has zero projection along the vertical axis, i.e. remains in the equatorial plane.
- The x and y projections oscillate 90° out of phase, at frequency ω

→ Larmor precession



Properties of the time evolution operator

$U(t, t_0)$ has some important properties:

- It is a **unitary** operator.

Its eigenvalues are complex numbers with modulus 1, and it conserves the modulus of any vector it operates on.

- It is **time reversible**.

Take for simplicity $t_0 = 0$. If I know $U(t, 0) = e^{-i\frac{H}{\hbar}t}$ then

$$U(-t, 0) = e^{i\frac{H}{\hbar}t} = U^{-1}(t, 0)$$

→ I can track the system back in time by inverting the time evolution operator

These properties have very profound consequences for quantum systems, especially quantum computers. The unitary and time-reversible nature of the time evolution operator imposes that no information can be lost.

Commutator

In general, the product of matrices is not commutative.

Example:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 8 & 7 \\ 20 & 15 \end{pmatrix}$$
$$\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 12 \\ 11 & 16 \end{pmatrix}$$

Definition: the **commutator** of two operators A and B is:

$$[A, B] = AB - BA$$

Note that the commutator is itself an operator.

When $[A, B] = 0$ we say that the operators commute, so they can be multiplied in any order.

Commutators in Quantum Mechanics

Commutation relations have a profound meaning in quantum mechanics. When two observables do not commute, it means that the order in which the observables are measured changes the outcomes!!

Example: S_x and S_z

$$S_z S_x = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$S_x S_z = \frac{\hbar^2}{4} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

The commutator is:

$$[S_z, S_x] = S_z S_x - S_x S_z = \frac{\hbar^2}{4} \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = i\hbar \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = i\hbar S_y$$

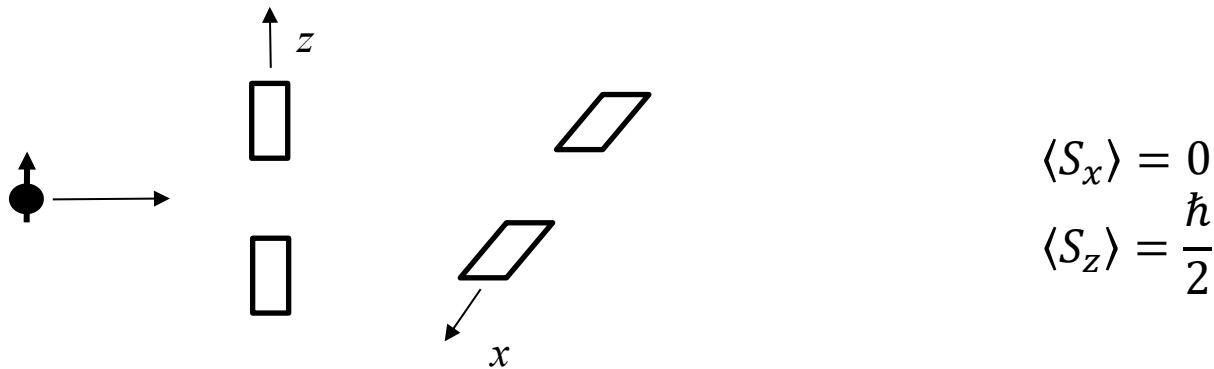
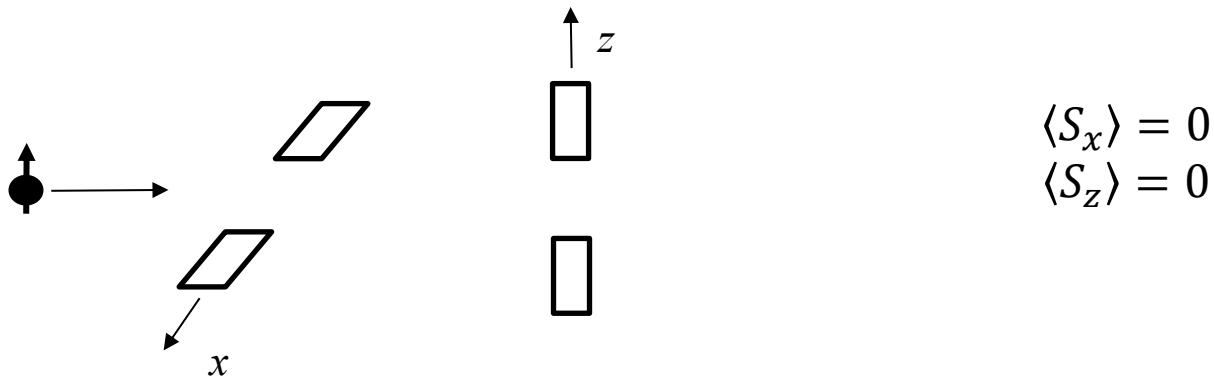
Similarly:

$$[S_x, S_y] = i\hbar S_z$$

$$[S_y, S_z] = i\hbar S_x$$

Sequential measurements

Imagine you want to measure both the x and the z projection of a spin with a pair of Stern-Gerlach machines. The order matters!



Ehrenfest theorem

The expectation value of an observable A at time t is given by:

$$\langle A \rangle(t) = \langle \psi(t) | A | \psi(t) \rangle$$

How does it evolve in time? Assume A is time-independent, and differential with respect to t :

$$\frac{d}{dt} \langle \psi(t) | A | \psi(t) \rangle = \left[\frac{d}{dt} \langle \psi(t) | \right] A | \psi(t) \rangle + \langle \psi(t) | A \left[\frac{d}{dt} | \psi(t) \rangle \right]$$

from the Schroedinger equation:

$$\frac{d}{dt} | \psi(t) \rangle = \frac{1}{i\hbar} H | \psi(t) \rangle \rightarrow \frac{d}{dt} \langle \psi(t) | = \frac{1}{-i\hbar} \langle \psi(t) | H$$

$$\frac{d}{dt} \langle \psi(t) | A | \psi(t) \rangle = \frac{1}{i\hbar} \langle \psi(t) | AH - HA | \psi(t) \rangle$$

$$i\hbar \frac{d}{dt} \langle A \rangle = \langle [A, H] \rangle$$

An observable that commutes with the Hamiltonian is a **constant of the motion**

Pauli matrices

Let us define the following matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

They are called **Pauli matrices**.

Together with the identity matrix $\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ they form a basis of the vector space of 2×2 matrices, i.e. any 2×2 matrix with complex coefficients can be written as:

$$A = a_1 \mathbb{I} + a_x \sigma_x + a_y \sigma_y + a_z \sigma_z$$

Profound consequence:

Any 2-level system can be “mapped” onto a spin. Or, put differently,

All quantum mechanical 2-level systems are equivalent

For a “true” spin, the Pauli matrices are proportional to the operators that describe the spin orientation along the 3 axes in real space. There are other 2-level systems for which there is no such meaning, but it’s still handy to think of them as spins, since we can make a visual picture.

Vector Pauli matrix

The three matrices $\sigma_x, \sigma_y, \sigma_z$ can be grouped into a “vector matrices”

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{pmatrix}$$

This is called **vector Pauli matrix**.

It is useful when calculating the inner product of a “real” vector in 3 dimensions, with the 3 spin operators.

Example: the Hamiltonian of a spin in a magnetic field $\mathbf{B}$ is

$$H = -\gamma \mathbf{B} \cdot \mathbf{S} = -\gamma \frac{\hbar}{2} \mathbf{B} \cdot \boldsymbol{\sigma} = -\gamma \frac{\hbar}{2} (B_x \sigma_x + B_y \sigma_y + B_z \sigma_z)$$

“normal” vector in 3D

vector Pauli matrix

Non-Hermitian operators

All operators we've seen so far were Hermitian (except $U(t, t_0)$), and described physical observables.

Now consider the following operators:

$$\sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \sigma_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

These are clearly non-Hermitian. Therefore, they cannot represent physical observables. But they are useful mathematical tools.

Example:

$$\sigma_+ |\downarrow\rangle = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |\uparrow\rangle$$

σ_+ is a “flip up” operator

$$\sigma_+ |\uparrow\rangle = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

applying σ_+ to the spin-up state results in the zero-vector, which is not a legitimate state vector (it has zero norm).

Section 3

Quantum mechanics in real space

Learning outcomes:

- Describe the motion of a particle using the Schroedinger equation
- Recognize how the particle-like description reemerges in the limit of large quantum numbers ($\rightarrow$ will be covered in the lab)

Note on Schroedinger equation(s)

The Schroedinger equation, given as a postulate of QM, describes the time evolution of quantum state:

$$i\hbar \frac{d|\psi(t)\rangle}{dt} = H|\psi(t)\rangle$$

This is sometimes called **“time-dependent” Schroedinger equation**, for the obvious reason that it describes the time dependence of $|\psi(t)\rangle$.

Note on nomenclature: Especially in older books and papers, people also talk about the **“time-independent” Schroedinger equation**, which is simply the eigenvalues equation for the Hamiltonian:

$$H|u_n\rangle = E_n|u_n\rangle$$

(this is only truly time-independent if the Hamiltonian is time independent too), where $|u_n\rangle$ are the eigenvectors of the Hamiltonian with eigenvalue E_n .

The reason this is called a Schroedinger equation is probably because, in the old days, the only way to solve the “true” Schroedinger equation (the time-dependent one) was to first solve the time-independent one, then change basis so that the time-evolution operator becomes diagonal.

Schroedinger equation in space coordinates

We have seen that the Schroedinger equation for a particle moving in one dimension is basically a wave equation, with the Planck-Einstein relations put in “by hand”:

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x, t) + V(x)\psi(x, t)$$

We wish to find a solution of the form:

$$\psi(x, t) = \varphi(x) \cdot \chi(t)$$

i.e., factorizable into a part that only depends on space, and one that only depends on time. The S.E. becomes:

$$i\hbar \varphi(x) \frac{\partial \chi(t)}{\partial t} = -\frac{\hbar^2}{2m} \chi(t) \frac{\partial^2}{\partial x^2} \varphi(x) + \chi(t) V(x) \varphi(x)$$

divide both sides by $\varphi(x) \cdot \chi(t)$

$$\underbrace{\frac{i\hbar}{\chi(t)} \frac{\partial \chi(t)}{\partial t}}_{t \text{ only}} = \underbrace{\frac{1}{\varphi(x)} \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \varphi(x) \right] + V(x)}_{x \text{ only}}$$

This can only hold if both sides of the equation are a constant, which we set $= \hbar\omega$ (check the dimensions: the terms in the equation above have units of energy)

Schroedinger equation in space coordinates

Left-hand side:

$$i\hbar \frac{\partial \chi(t)}{\partial t} = \hbar\omega \cdot \chi(t) \Rightarrow \chi(t) = A \cdot \exp(-i\omega t)$$

Right-hand side:

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \varphi(x) + V(x)\varphi(x) = \hbar\omega \cdot \varphi(x)$$

This is just the “real space” version of the time-independent Schroedinger equation:

$$H\varphi(x) = E\varphi(x) \rightarrow \left[\underbrace{-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}}_{\text{kinetic}} + \underbrace{V(x)}_{\text{potential}} \right] \varphi(x) = E\varphi(x)$$

Therefore, the function:

$$\psi(x, t) = \varphi(x) \exp(-i\omega t)$$

is a solution of the Schroedinger equation, if $\varphi(x)$ is a solution of the t -independent part.

The function $\psi(x, t)$ written above is a **stationary state**

Interlude: Stationary states

Subtle point:

At face value, the function $\psi(x, t) = \varphi(x) \exp(-i\omega t)$ does contain a time dependence: the phase factor $\exp(-i\omega t)$.

However, *taken in isolation*, this phase factor has no physical meaning.

Consider a ket $|\psi(t)\rangle = |\psi_0\rangle e^{-i\omega t}$ and an observable A . The expectation value of the observable is

$$\langle A \rangle_\psi = \langle \psi(t) | A | \psi(t) \rangle = \langle \psi_0 | e^{i\omega t} A e^{-i\omega t} | \psi_0 \rangle = \langle \psi_0 | A | \psi_0 \rangle$$

which is strictly time-independent. This is why a state like $|\psi(t)\rangle = |\psi_0\rangle e^{-i\omega t}$ is called **stationary**.

Later in the course we will learn how to describe quantum systems using “density matrices”, which are simply the ket and the bra multiplied together: $\rho = |\psi\rangle\langle\psi|$

In this case, the density matrix for $|\psi(t)\rangle$ is:

$$\rho = |\psi(t)\rangle\langle\psi(t)| = |\psi_0\rangle e^{-i\omega t} e^{i\omega t} \langle\psi_0| = |\psi_0\rangle\langle\psi_0|$$

so the phase factor actually disappears.

Combining stationary states

The Schroedinger equation is linear $\Rightarrow$ a linear combination of its solutions is also a solution.

$$\psi(x, t) = \sum_n c_n \varphi_n(x) e^{-i\omega_n t}$$

This means that if I know the stationary states $\varphi_n(x)$ and their energies $E_n = \hbar\omega_n$, I can solve the problem easily if the initial state is written as a linear combination of stationary states: $\psi(x, 0) = \sum_n c_n \varphi_n(x)$

If you think of the wave functions $\varphi_n(x)$ as vectors in an infinite-dimensional space, what we are doing here is like using a diagonal time-evolution operator.

Particle with kinetic energy

Time-independent Schroedinger equation:

$$\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + (V(x) - E) \right] \varphi(x) = 0$$
$$\left[\frac{\partial^2}{\partial x^2} + \frac{2m}{\hbar^2} (E - V(x)) \right] \varphi(x) = 0$$

note: here E is
the variable

Assumption: $E > V$:

The particle's total energy is $>$ than the potential energy $\Rightarrow$ there is a kinetic component

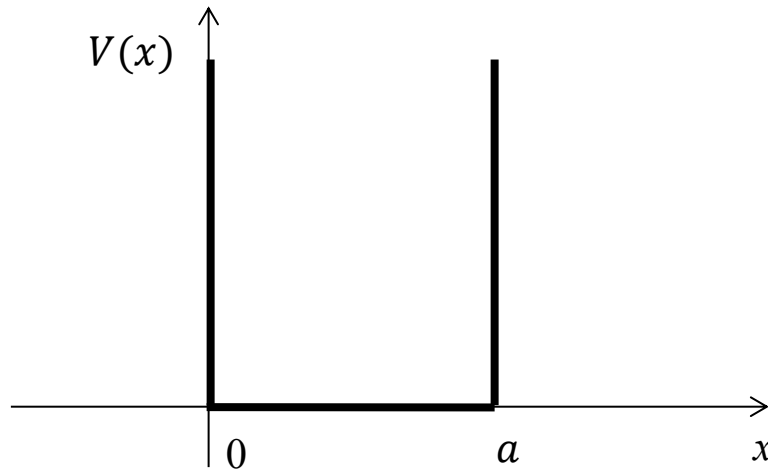
$$E - V = \frac{1}{2} m v^2 = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

General solution of the Schroedinger equation:

$$\varphi(x) = A e^{ikx} + A' e^{-ikx}$$

i.e. a combination of two plane waves, one going left-to-right ($A e^{ikx}$) and one going right-to-left ($A' e^{-ikx}$)

Infinite potential well (a.k.a. “particle in a box”)



$$k = \sqrt{\frac{2mE}{\hbar}} \quad \text{between the walls}$$

$$\varphi(x < 0) = \varphi(x > a) = 0$$

Assume $V(x) = \infty$ for $x < 0$ and $x > a$. The particle cannot access that region.

The infinite potential imposes $\varphi = 0$ at and beyond the walls. $\partial\varphi/\partial x$ needs not be continuous.

General solution between the walls:

$$\varphi(x) = A e^{ikx} + A' e^{-ikx}$$

Infinite potential well (a.k.a. “particle in a box”)

Boundary condition at $x = a$:

$$\varphi(a) = 0 \Rightarrow k_n = \frac{n\pi}{a} \quad (n = 1, 2, \dots)$$

Therefore, the energy eigenvalues $E_n = \frac{\hbar^2 k_n^2}{2m}$ are:

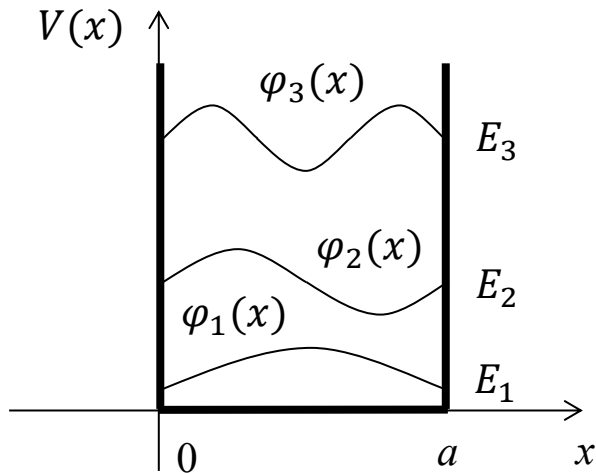
$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

Imposing the normalization of the wavefunction, $\int |\varphi(x)|^2 dx = 1$ one obtains:

$$\varphi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

The boundary conditions have the effect of selecting a discrete set of possible momenta, and therefore possible energies, for the particle in the well. The stationary states are the “standing waves” in the well, analogous to the electromagnetic standing waves in a parallel-plate waveguide.

Stationary states



Subtle point:

Consider e.g. $\varphi_2(x)$. It has energy $E_2 > 0$, and since $V(x) = 0$, that's all kinetic energy. So you would think that the particle must be moving. But, it's not... $\varphi_2(x)$ is a stationary state.

Conceptual exercise:

Think about the meaning of a state like $\varphi_2(x)$ and how it's possible that it describes a stationary system.

Low-energy truncation

So far we have calculated the solutions of the t -independent equation $H\varphi(x) = E\varphi(x)$.

Since H is an observable, its eigenvectors $\varphi_n(x)$ form a basis of the Hilbert space.

In principle, $n \rightarrow \infty$. In practice, one can often “truncate” the problem to a finite set of N low-energy states, e.g. because the temperature is finite, so the very high-energy states are always unoccupied.

Therefore, if the Hilbert space is truncated to a finite dimension N , we can write a finite set of basis kets like:

$$|\varphi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad |\varphi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \quad \dots \quad |\varphi_N\rangle = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

Where the Hamiltonian is simply

$$H = \begin{pmatrix} E_1 & 0 & \dots & 0 \\ 0 & E_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & E_N \end{pmatrix}$$

Each ket $|\varphi_n\rangle$ corresponds to the function of space $\varphi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$.

Non-stationary states

A linear superposition is also a legitimate state:

$$|\psi(t)\rangle = \sum_n c_n(t) |\varphi_n\rangle$$

Example: $|\psi(0)\rangle = \frac{1}{\sqrt{2}} (|\varphi_1\rangle + |\varphi_2\rangle)$

Since we are using the basis of the eigenstates of the Hamiltonian, the time-evolution operator is diagonal:

$$U(t, 0) = \begin{pmatrix} e^{-iE_1 t/\hbar} & 0 & \dots & 0 \\ 0 & e^{-iE_2 t/\hbar} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & e^{-iE_N t/\hbar} \end{pmatrix}$$

The time-evolved state is:

$$|\psi(t)\rangle = U(t, 0) |\psi(0)\rangle = \frac{1}{\sqrt{2}} (e^{-iE_1 t/\hbar} |\varphi_1\rangle + e^{-iE_2 t/\hbar} |\varphi_2\rangle)$$

Non-stationary states

Multiplying by $e^{iE_1 t/\hbar}$ we obtain:

$$|\psi(t)\rangle e^{iE_1 t/\hbar} = \frac{1}{\sqrt{2}} (|\varphi_1\rangle + e^{-i(E_2-E_1)t/\hbar} |\varphi_2\rangle)$$

Key point: the *global* phase factor $e^{iE_1 t/\hbar}$ is irrelevant, but the *relative* phase factor matters!

Call $\omega_{12} = \frac{E_2-E_1}{\hbar}$

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} (|\varphi_1\rangle + e^{-i\omega_{12}t} |\varphi_2\rangle)$$

The corresponding function of real space is:

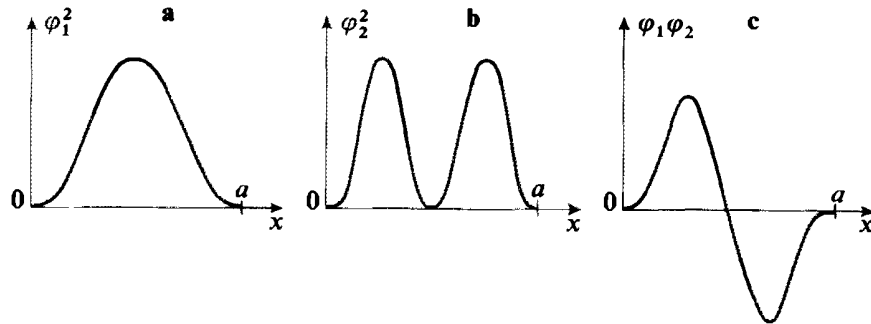
$$\psi(x, t) = \frac{1}{\sqrt{2}} (\varphi_1(x) + e^{-i\omega_{12}t} \varphi_2(x))$$

The probability to find the particle at point x at time t is:

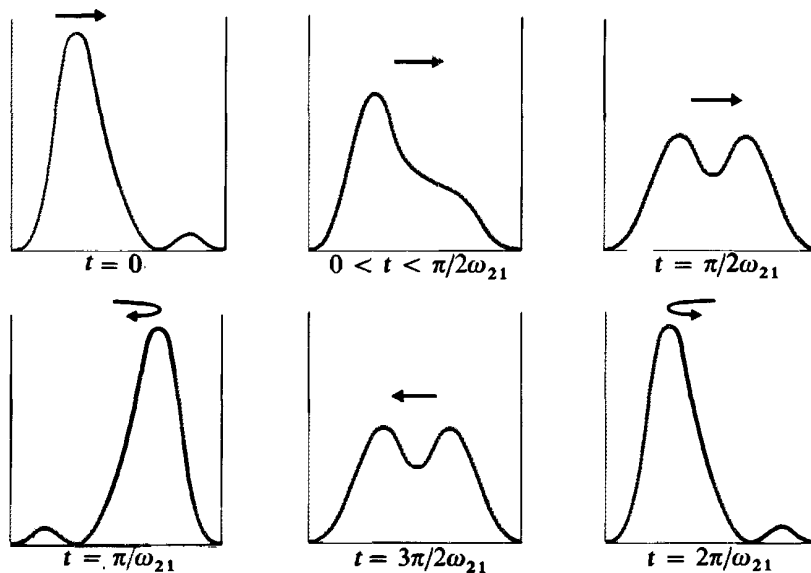
$$|\psi(x, t)|^2 = \frac{1}{2} \varphi_1^2(x) + \frac{1}{2} \varphi_2^2(x) + \varphi_1(x) \varphi_2(x) \cos(\omega_{12}t)$$

Bohr frequencies

The functions $\varphi_1^2(x)$, $\varphi_2^2(x)$, $\varphi_1(x)\varphi_2(x)$ look like this:



Multiply $\varphi_1(x)\varphi_2(x)$ by $\cos(\omega_{12}t)$ and sum them all up:



This describes the motion of a particle bouncing back and forth at a frequency $\omega_{12} \rightarrow$ Bohr frequency

Section 4

Quantum tunnelling

Learning outcomes:

- Understand the phenomenon of tunnelling and how it relates to evanescent waves in electromagnetism
- Describe mathematically a quantum two-level system governed by tunnelling

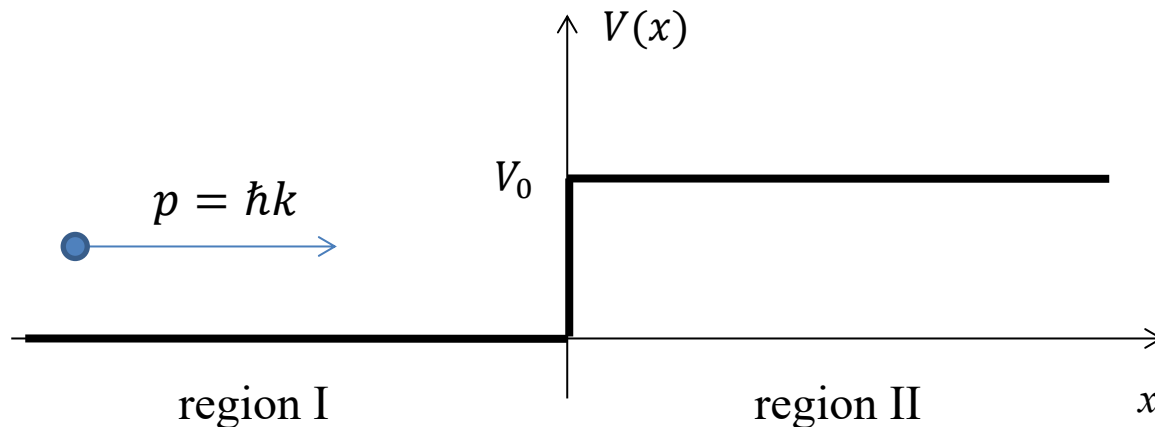
Particle in a step-like potential

We now look at some simple potentials $V(x)$ so that the solution to the time-independent Schroedinger equation is easy to calculate.

Consider a piecewise constant potential:

$$\begin{cases} V(x) = 0, & x < 0 \\ V(x) = V_0, & x \geq 0 \end{cases}$$

and a particle with total energy E and momentum $p = \hbar k$, moving from left to right



Particle in a step-like potential

Time-independent Schroedinger equation:

$$\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + (V(x) - E) \right] \varphi(x) = 0$$

note: here E is
the variable

$$\left[\frac{\partial^2}{\partial x^2} + \frac{2m}{\hbar^2} (E - V(x)) \right] \varphi(x) = 0$$

- Region I: $E > V$:

The particle's total energy is $>$ than the potential energy $\Rightarrow$ there is a kinetic component

$$E - V = \frac{1}{2} m v^2 = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

General solution:

$$\varphi(x) = A e^{ikx} + A' e^{-ikx}$$

i.e. a combination of two plane waves. In this case, since the particle is going left-to-right, we put $A' = 0$.

Evanescent wave

- Region II: $E < V$:

Classically, this case is forbidden (the kinetic energy cannot be negative) $\rightarrow$ the particle would just bounce and reflect against the potential step.

Quantum mechanically, we can formally write:

$$V - E = \frac{\hbar^2 \alpha^2}{2m} \quad \alpha = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

(α takes the place of $-ik$). General solution:

$$\varphi(x) = B e^{-\alpha x} + B' e^{\alpha x}$$

The term $B e^{-\alpha x}$ describes a wavefunction that decays exponentially as the particle tries to go under the step $\rightarrow$ **evanescent wave**.

At the boundary between regions I and II we must have A, A', B, B' chosen so that $\varphi(x)$ and $\partial\varphi(x)/\partial x$ are continuous (unless the step is infinitely high). So the overall problem can be solved by writing the appropriate solution in each constant potential region, then matching them at the boundary.

The maths is identical to the propagation of an electromagnetic wave in a waveguide. The case $E < V$ is like a waveguide below cutoff.

Transmission

- Example: $E > V$ in both regions

$$\begin{aligned}\varphi_I(x) &= A_1 e^{ik_1 x} + A'_1 e^{-ik_1 x} & k_1 &= \frac{\sqrt{2mE}}{\hbar} \\ \varphi_{II}(x) &= A_2 e^{ik_2 x} + A'_2 e^{-ik_2 x} & k_2 &= \frac{\sqrt{2m(E - V_0)}}{\hbar}\end{aligned}$$

with $A'_2 = 0$ since the particle comes from the left. However, I should allow $A'_1 \neq 0$ because it may bounce back at the step.

Matching conditions at $x = 0$:

$$\left[\begin{array}{ll} A_1 + A'_1 = A_2 & \Rightarrow \varphi(x) \text{ is continuous at the boundary} \\ ik_1 A_1 - ik_1 A'_1 = ik_2 A_2 & \Rightarrow \partial\varphi(x)/\partial x \text{ is continuous at the boundary} \end{array} \right.$$

From the above we can calculate the **transmission coefficient** T :

$$T \equiv \frac{k_2}{k_1} \left| \frac{A_2}{A_1} \right|^2 = \frac{4k_1 k_2}{(k_1 + k_2)^2}$$

Reflection

Analogously, the **reflection coefficient** T :

$$R \equiv \left| \frac{A_1'}{A_1} \right|^2 = 1 - T = 1 - \frac{4k_1k_2}{(k_1 + k_2)^2}$$

The condition holds that:

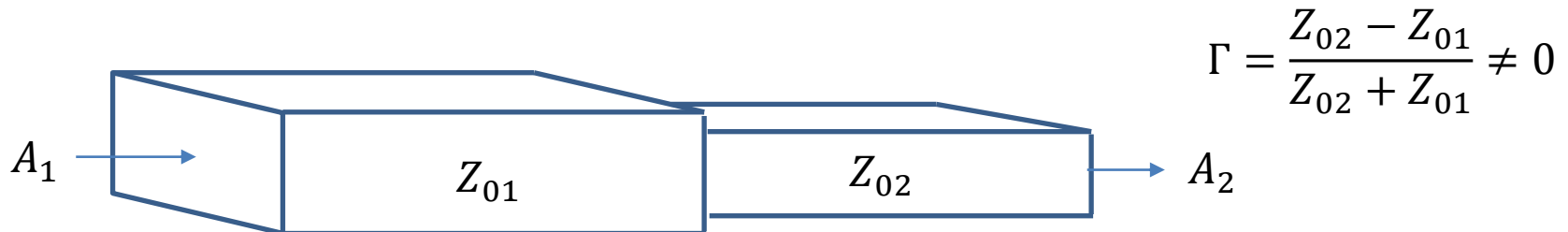
$$T + R = 1$$

“The particle has to be somewhere”

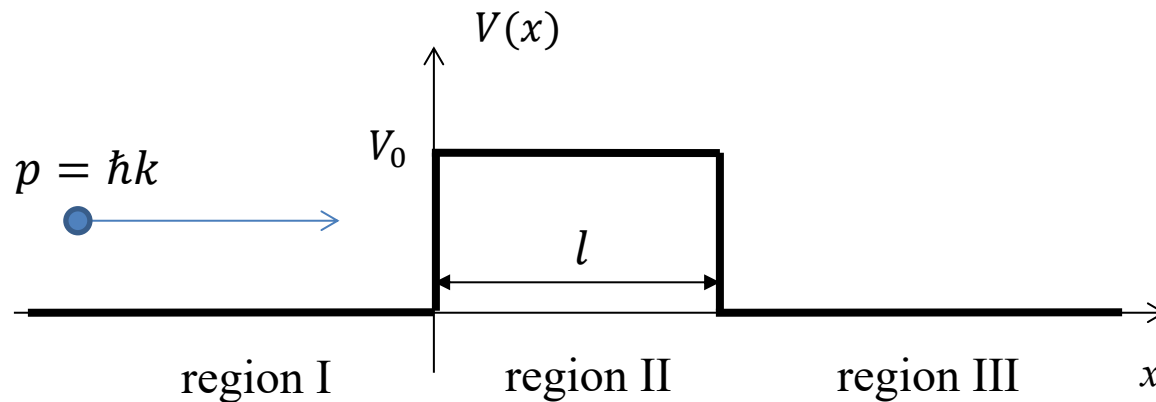
Note: In the classical case, you would expect $R = 0$ because the particle has enough energy to overcome the step – it would just slow down a bit.

Here, instead, there is a nonzero probability of being reflected back!

In terms of waves, however, it makes perfect sense... it's like a transition between two waveguides of different size. Even if both are above cut-off, the wave impedance is different, so there is a nonzero reflection coefficient!



Quantum tunnelling



- Potential step of height V_0 :

$$\varphi_I(x) = A_1 e^{ik_1x} + A'_1 e^{-ik_1x}$$

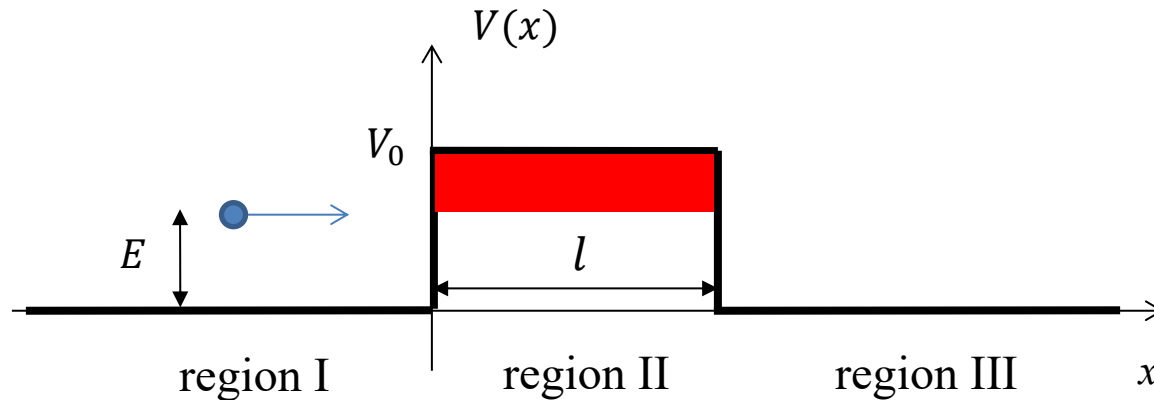
$$\varphi_{II}(x) = B_2 e^{-\alpha_2x} + B'_2 e^{\alpha_2x}$$

$$\varphi_{III}(x) = A_3 e^{ik_1x} + A'_3 e^{-ik_1x}$$

with $A'_3 = 0$. This can be solved by breaking down the problem in two steps, matching the solutions for zones I and II at $x = 0$, and zones II and III at $x = l$. Result:

$$T = \frac{16E(V_0 - E)}{V_0^2} \exp(-\alpha_2 l) = \frac{16E(V_0 - E)}{V_0^2} \exp\left(-l \frac{\sqrt{2m(V_0 - E)}}{\hbar}\right)$$

Quantum tunnelling



$T \neq 0$ because the wave function under the barrier is an evanescent wave, so it has an exponentially decaying “tail” that leaks out to the other side of the barrier

The transmission coefficient is proportional to $\exp(-\alpha_2 l) = \exp\left(-l \frac{\sqrt{2m(V_0 - E)}}{\hbar}\right)$

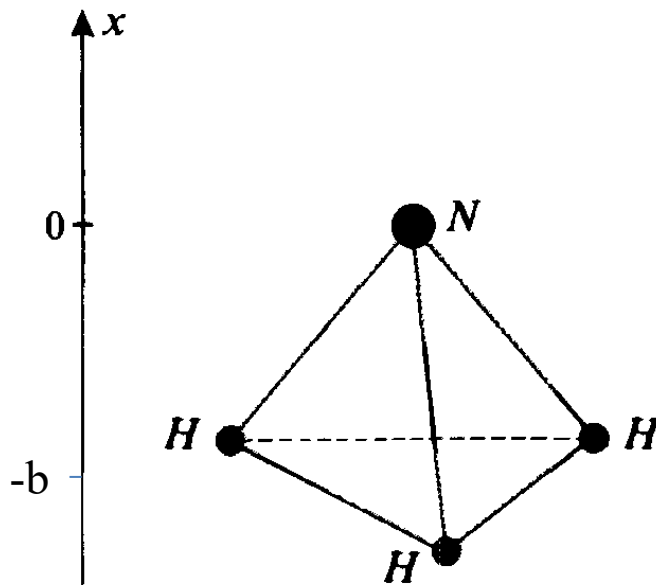
→ the probability of tunnelling through the barrier depends exponentially on the width (l) and on the height of the barrier, measured from the energy of the impinging particle ($V_0 - E$).

Note: the particle can be reflected back with probability $1 - T$, but if it tunnels through the barrier, it comes out with the same energy and momentum as when it arrived!

→ “Elastic tunnelling”

Ammonia molecule

The ammonia molecule (NH_3) has an “umbrella” – like structure:

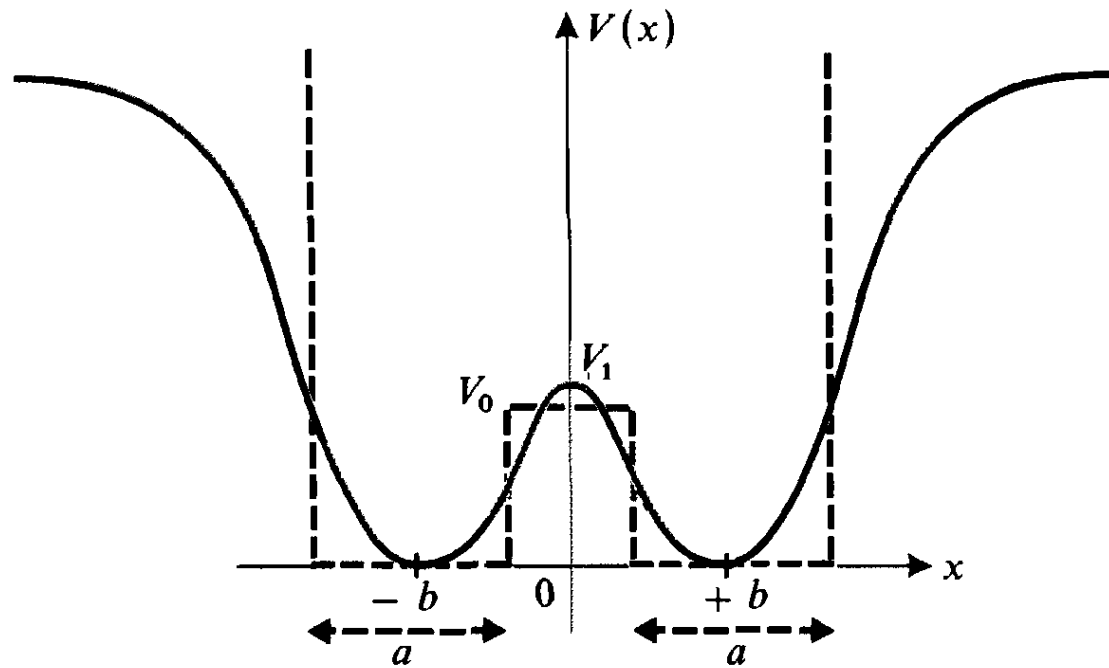


Since the nitrogen is much heavier than the hydrogens, you can think of it as fixed at $x = 0$, i.e. the centre of mass is more or less where the N atom is. The three H atoms, instead, can move. They have two stable positions around $x = \pm b$

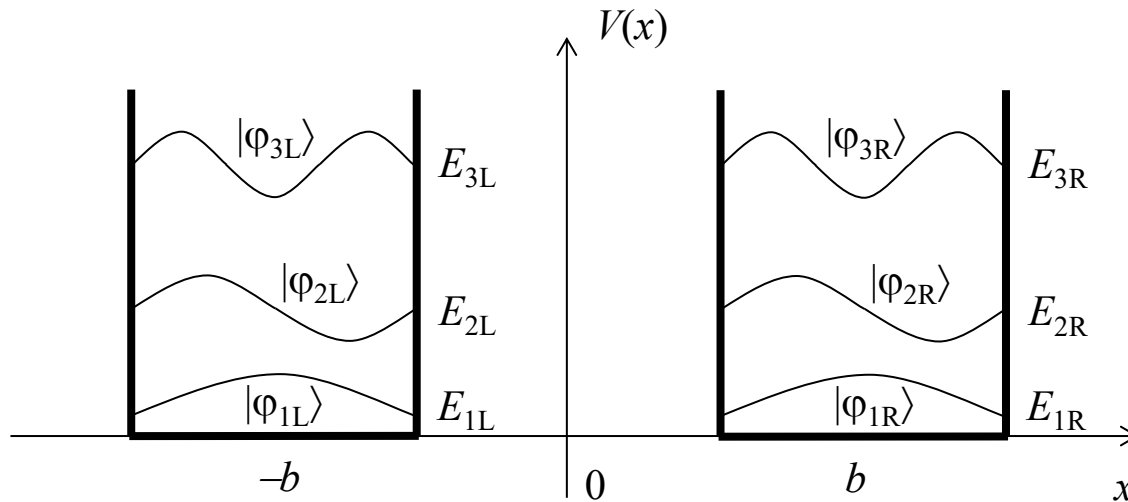
Double-well approximation

If you want to flip the umbrella upside down, the H have to overcome the electrostatic repulsion of the N.

Thus, there is a potential barrier against flipping the umbrella, but its height is finite. For simplicity, we approximate the potential as two rectangular wells, with a finite barrier in the middle and infinite walls outside (dashed line)



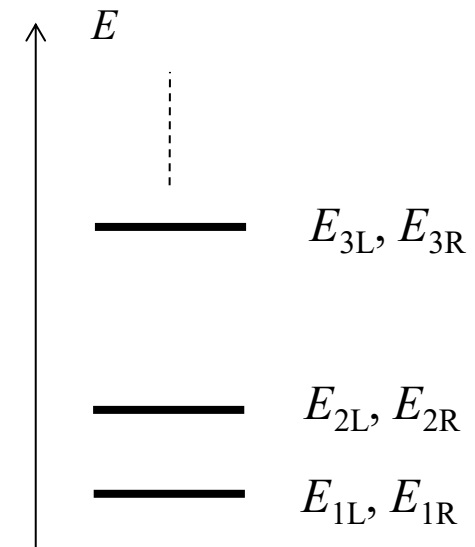
First approximation: infinite barrier



With ∞ potential barrier in the middle, you just have two independent wells.

Each eigenvalue is doubly degenerate:

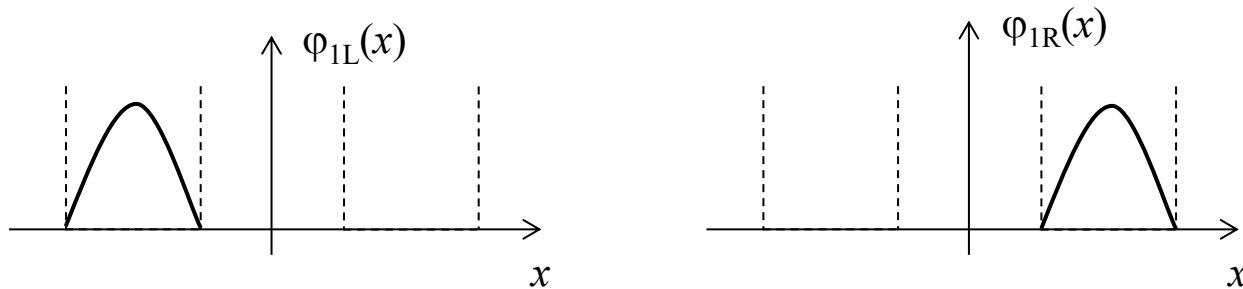
i.e. there are two orthogonal eigenvectors (one for the left well and one for the right) with the same eigenvalue. The energy spectrum is:



Truncation of the Hamiltonian

Assume for simplicity that $k_B T \ll E_2 - E_1$ so only the 2 lowest energy states $|\varphi_{1L}\rangle, |\varphi_{1R}\rangle$ are populated $\rightarrow$ truncate to a 2-dimensional space

The “natural” basis is:



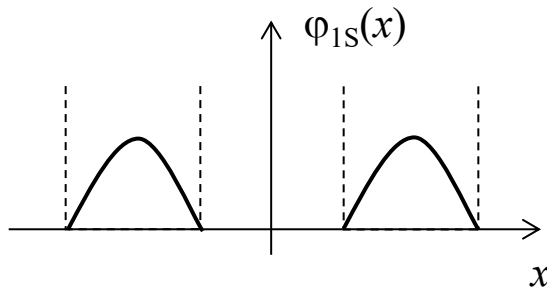
The two basis states have the same energy E_1 , so the Hamiltonian is:

$$H = \begin{pmatrix} E_1 & 0 \\ 0 & E_1 \end{pmatrix} = E_1 \mathbb{I}$$

Since the potential energy is defined with respect to an arbitrary constant, I can set $E_1 = 0$, i.e. measure the energies starting from E_1 upwards.

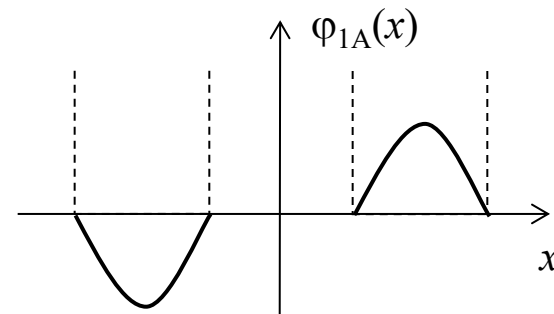
Change of basis

Instead of the “natural” basis, I can also use this one:



$$|\varphi_{1S}\rangle = \frac{1}{\sqrt{2}} (|\varphi_{1R}\rangle + |\varphi_{1L}\rangle)$$

“symmetric state”



$$|\varphi_{1A}\rangle = \frac{1}{\sqrt{2}} (|\varphi_{1R}\rangle - |\varphi_{1L}\rangle)$$

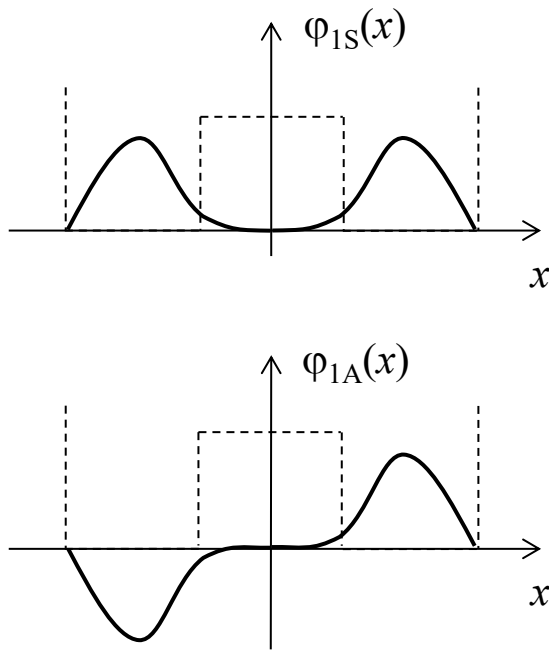
“antisymmetric state”

The states are orthogonal and normalized

Finite barrier

Now take the realistic case, where the barrier between the two wells is not infinite. We know that, in that case, the wavefunction does not go to zero at the edge, but only decays exponentially, so it has a finite probability to “leak” to the other side
→ tunnel effect.

The exact eigenstates of this Hamiltonian now look like:



Both eigenstates have some leakage under the barrier, but in different amount

→ $|\phi_{1S}\rangle$ and $|\phi_{1A}\rangle$ now have different energy

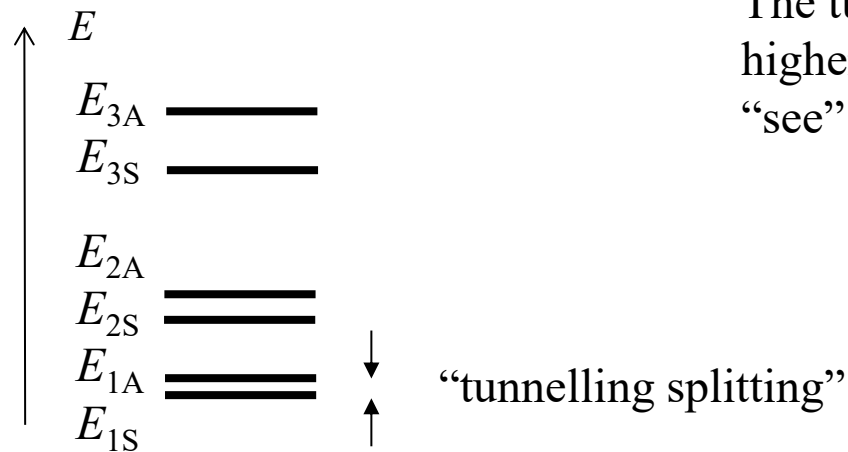
$$E_A - E_S = \text{tunnelling splitting}$$

Tunnelling splitting

When the middle barrier was infinitely high, we had pairs of degenerate states (left/right).

With finite barrier, each pair splits in energy by an amount that depends on how each symmetric/antisymmetric eigenstate leaks under the barrier

→ **tunnelling splitting**



The tunnelling splitting increases for higher-energy doublets, because they “see” a lower barrier

Tunnelling oscillations

Prepare the system in the initial state:

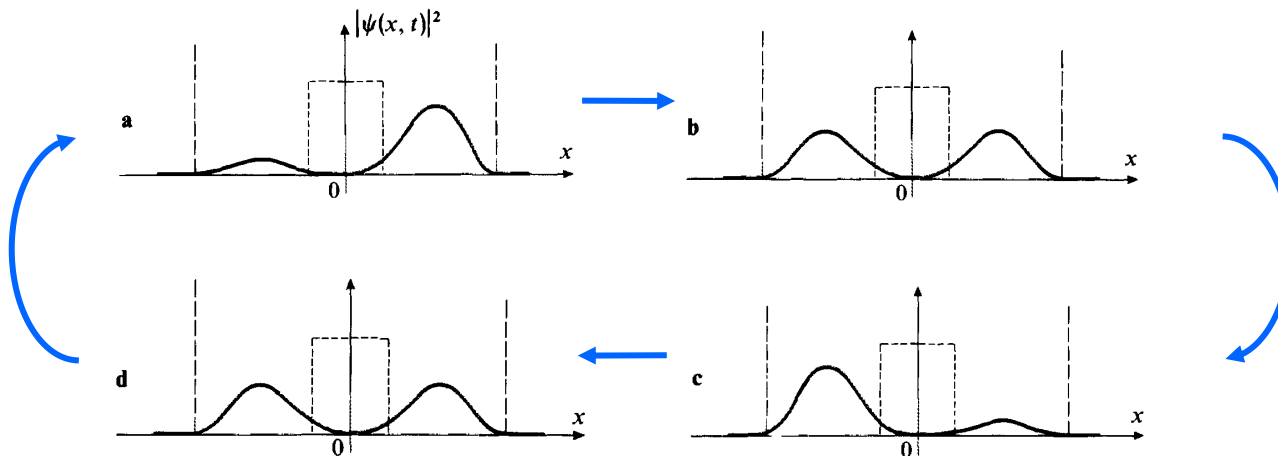
$$|\psi(0)\rangle = \frac{1}{\sqrt{2}} (|\varphi_{1S}\rangle + |\varphi_{1A}\rangle) = |\varphi_{1R}\rangle$$

(this is the “umbrella on the right” state, written in the basis of the symmetric/antisymmetric eigenstates of the Hamiltonian)

At subsequent times we’ll have:

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} (|\varphi_{1S}\rangle + e^{-i(E_{1A}-E_{1S})t/\hbar} |\varphi_{1A}\rangle)$$

The probability of finding the hydrogen atoms evolves like:



Tunnelling oscillations
at frequency

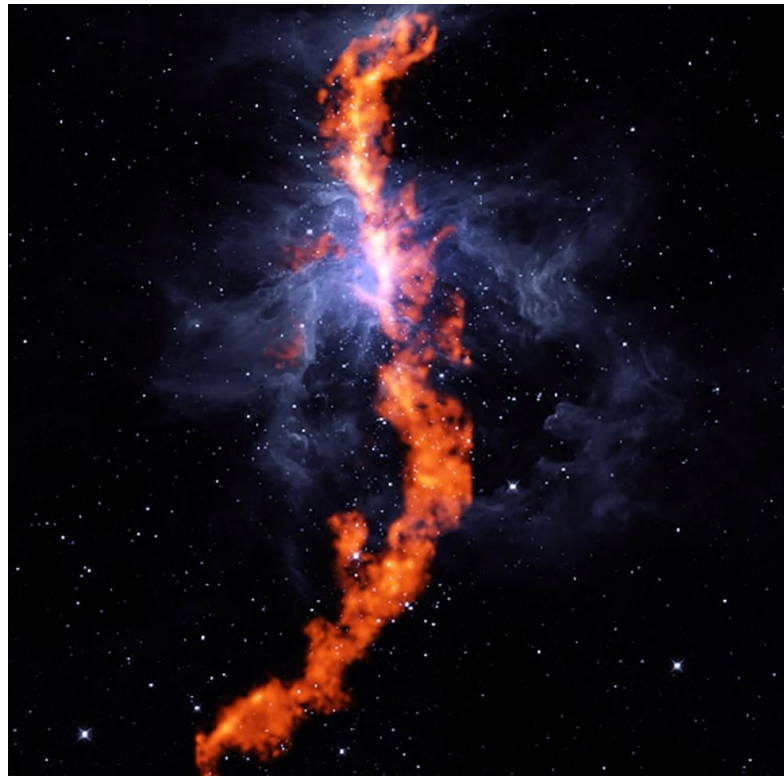
$$\omega_T = \frac{E_{1A} - E_{1S}}{\hbar}$$

Microwave emission from ammonia

The ammonia molecules emits radiation at a well-defined frequency:

$$\omega_T = \frac{E_{1A} - E_{1S}}{\hbar} \approx 2\pi \times 24 \text{ GHz}$$

This is used, for example, to discover that there is ammonia in certain nebulae in the distant universe!



Analogy with a spin

In the presence of the tunnel coupling, the Hamiltonian of the ammonia molecule (truncated to the lowest-energy doublet) becomes:

$$H = \begin{pmatrix} 0 & -\Delta \\ -\Delta & 0 \end{pmatrix}$$

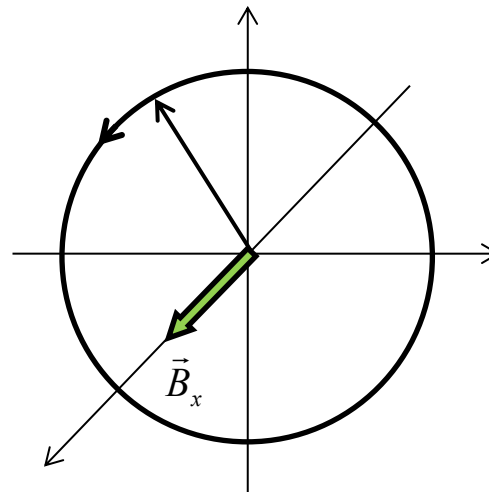
where

$$2\Delta = E_{1A} - E_{1S}$$

→ the tunnel coupling plays the same role as a magnetic field // x on a spin

Note: the eigenstates $|\varphi_{1S}\rangle$ and $|\varphi_{1A}\rangle$ look exactly like $|X\rangle$ and $|-X\rangle$ for a spin!

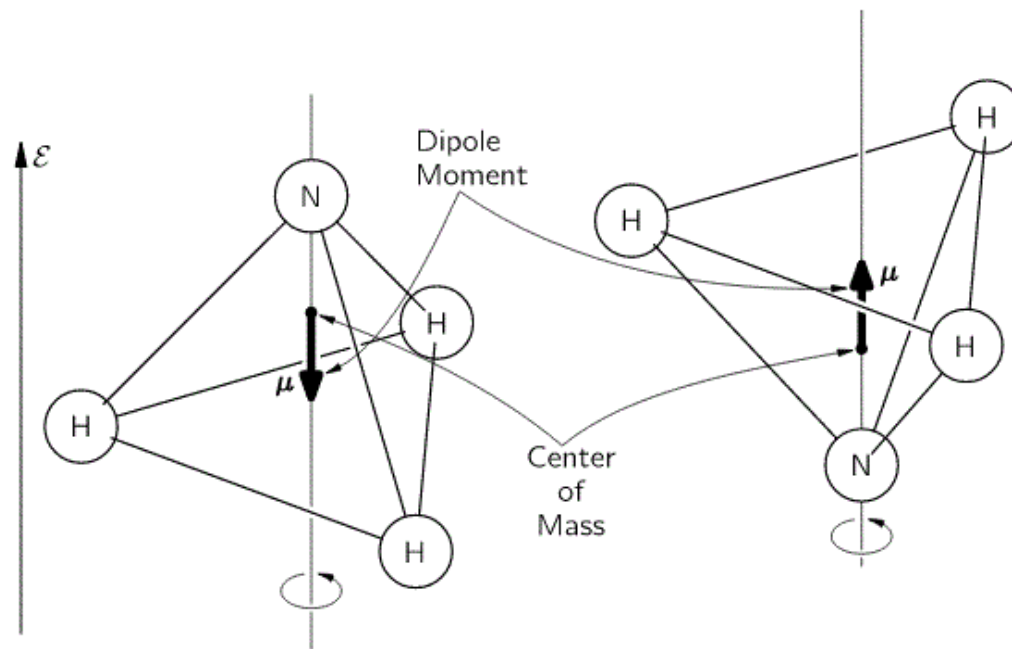
The tunnelling oscillations have the same form as the precession in the yz plane of a spin placed in a magnetic field // x



Analogy with a spin

The ammonia molecule has an electric dipole μ , because of the way the charge is distributed between the N atom and the 3 H $\rightarrow$ an electric field applied along the axis of the molecule “pulls” the umbrella on one side.

Therefore, an electric field plays the same role as a magnetic field $\parallel z$, which splits the energy of the natural basis states (in this case $|\varphi_{1R}\rangle$ and $|\varphi_{1L}\rangle$)



Equivalent Hamiltonian

Overall, we can write the Hamiltonian of the ammonia molecules like:

$$H = \begin{pmatrix} \epsilon & -\Delta \\ -\Delta & -\epsilon \end{pmatrix} = -\Delta\sigma_x + \epsilon\sigma_z$$

where

$$2\Delta = E_{1A} - E_{1S} \quad @ \epsilon = 0 \quad \rightarrow \text{“tunnel coupling”}$$

$$2\epsilon = E_{1R} - E_{1L} \quad @ \Delta = 0 \quad \rightarrow \text{“detuning” by electric field}$$

Profound consequence: under suitable circumstances (low temperature), a complex system like an ammonia molecules is very well described by the same Hamiltonian as that of a spin in a magnetic field that has x and z components.

Section 5

Position and momentum operators

Learning outcomes:

- Write the matrix operators for position and momentum
- Write an arbitrary Hamiltonian of a particle in real space, and solve for its “trajectory” (→ will be covered in the lab)

Position and momentum operators

So far we have treated two examples, where the Hilbert space was constructed out of the eigenstates of the Hamiltonian:

- 1) The spin, where the basis states are $|\uparrow\rangle, |\downarrow\rangle$ and the energy is proportional to the S_z operator
- 2) The particle in a box, where we have obtained the eigenstates $\varphi_n(x)$ of the Hamiltonian directly from the known solution of the Schroedinger equation, and then built the Hilbert space from simple orthogonal kets of the form

$$|\varphi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad |\varphi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \quad \dots \quad |\varphi_N\rangle = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

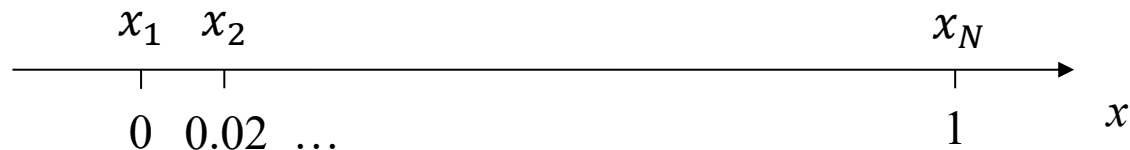
We now want to describe the motion of a generic system, for which we need to write the operators describing the position (called X) and the momentum (called P)

However, the Hamiltonian is not directly proportional to either of them.

Position eigenbasis

Postulate #2: every physical quantity is described by a Hermitian operator.

The position of a particle is no exception! To write it as a finite matrix, we just discretize the space – exactly as you would do in Python. Example:



With this choice, I have discretized the x -axis between 0 and 1 in $N = 51$ points (50 intervals). I can define a “position eigenbasis” with orthonormal vectors in a 51-dimensional space:

$$|\varphi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad |\varphi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \quad \dots \quad |\varphi_{51}\rangle = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

Each ket represents a particle that is maximally concentrated on the i -th point along the x -axis (in the limit where the discretization step is infinitesimal, it represents a δ -function at position x_i)

Position operator

Postulate #3: When measuring the value of the quantity $\mathcal{A}$, we can only obtain as a result one of the eigenvalues a_n of the observable A .

Example: if we measure X while the state of the system is $|\varphi_2\rangle$, we will get $x = 0.02$

So the X operator should look like this:

$$X = \begin{pmatrix} x_1 & 0 & \dots & 0 \\ 0 & x_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & x_N \end{pmatrix} = \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0.02 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & 1 \end{pmatrix}$$

Unlike all the spin and the particle in a box, now the operator that's diagonal in the chosen basis is *not* the Hamiltonian of the system.

This would be a big headache if we were to solve the problem by hand, but... we will use Python instead!

Momentum operator

Next we need to find out how to write the momentum operator.

Recall the time-independent Schroedinger equation, which is the eigenvalue problem for the Hamiltonian:

$$\left[\underbrace{-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}}_{\text{kinetic}} + \underbrace{V(x)}_{\text{potential}} \right] \varphi(x) = E\varphi(x)$$

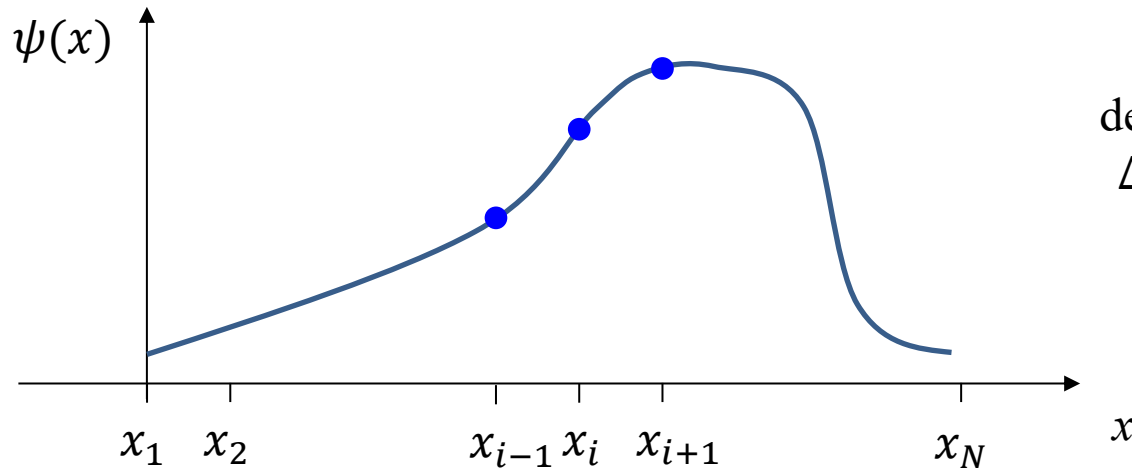
Since the kinetic energy is also given by $p^2/2m$ we can interpret this as meaning that the momentum operator must be:

$$P = -i\hbar \frac{\partial}{\partial x}$$

The derivative is a linear operator, so it must be writable as a matrix. How?

Momentum operator

As usual, imagine we are doing this in Python, with a discretized array of points in space. We just have to calculate the derivative numerically.



define:

$$\Delta x = x_{i+1} - x_i$$

$$\frac{\partial \psi(x_i)}{\partial x} \approx \frac{1}{2} \left(\frac{\psi(x_{i+1}) - \psi(x_i)}{\Delta x} + \frac{\psi(x_i) - \psi(x_{i-1}))}{\Delta x} \right) = \frac{\psi(x_{i+1}) - \psi(x_{i-1}))}{2\Delta x}$$

→ take the average of the slope before and after point x_i

Momentum operator

In total, the matrix representing the momentum operator is given by:

$$P = \frac{-i\hbar}{2\Delta x} \begin{pmatrix} 0 & 1 & \dots & 0 \\ -1 & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 0 & \dots & -1 & 0 \end{pmatrix}$$

With this we can write explicitly the Hamiltonian of a particle as:

$$\mathcal{H} = \frac{P^2}{2m} + V(x)$$

Where P is given by the (non-diagonal!) matrix above, and $V(x)$ is easy to write because it's just a diagonal matrix with the numerical value of the potential at each point x_i .

With this, we can calculate the time evolution operator, etc...

Commutation relations

X and P do not commute!

If we calculate brute-force the commutator using the matrices as written on pages 124 – 127, we find:

$$[X, P] = XP - PX = \frac{i\hbar}{2} \begin{pmatrix} 0 & 1 & \dots & 0 \\ 1 & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 0 & \dots & 1 & 0 \end{pmatrix} \approx i\hbar \mathbb{I}$$

(try to think about why the matrix above is approximately equal to twice the identity)

This represents, in matrix form, the Heisenberg uncertainty principle!

Section 6

Dynamics and control of a quantum bit (qubit)

Learning outcomes:

- Recognize that any quantum mechanical two-level system can be described in a unique way, as a quantum bit
- Understand and apply the concept of rotating reference frame
- Learn how to control a quantum two-level system in a universal way
- Describe some simple time-dependent physical systems

Qubit Hamiltonian

We will often find systems that, under suitable conditions (typically at low temperatures), can be described by a Hamiltonian like:

$$H = \varepsilon\sigma_z + \Delta\sigma_x$$

This is called **qubit Hamiltonian**.

Qubit means “quantum bit”, i.e. a quantum two-level system, that can be used to encode quantum digital information.

- “Unperturbed” case:

$$H_0 = \varepsilon\sigma_z \quad \text{with eigenstates } |\uparrow\rangle, |\downarrow\rangle \quad \text{and eigenvalues } \pm\varepsilon$$

- “Coupled” case:

$$H = \varepsilon\sigma_z + \Delta\sigma_x = \begin{pmatrix} \varepsilon & \Delta \\ \Delta & -\varepsilon \end{pmatrix}$$

to find the eigenvalues:

$$\det(H - \lambda\mathbb{I}) = 0 \quad \rightarrow \quad (\varepsilon - \lambda)(-\varepsilon - \lambda) - \Delta^2 = \lambda^2 - \varepsilon^2 - \Delta^2 = 0$$

$$\lambda_{1,2} = E_{h,l} = \pm\sqrt{\varepsilon^2 + \Delta^2}$$

Energy anticrossing

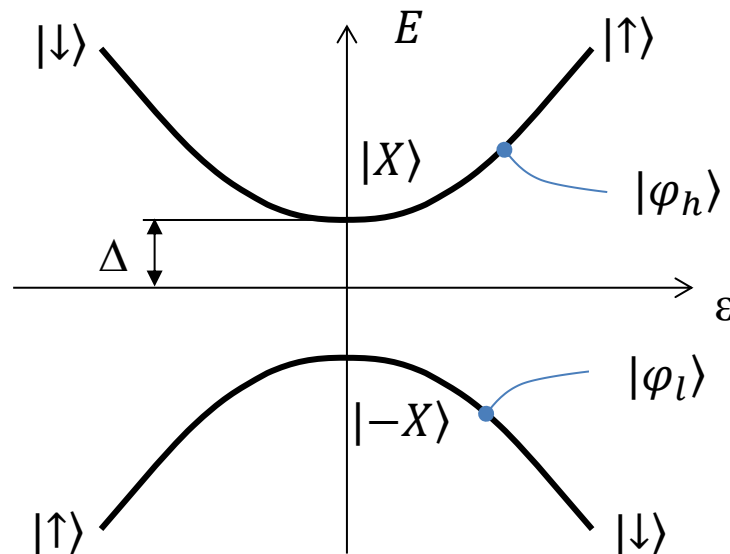
Consider a qubit Hamiltonian where $\Delta = \text{const.}$ and $\varepsilon = -\infty \rightarrow +\infty$

The eigenvectors will be:

$$|\varphi_h\rangle = \sin\frac{\theta}{2} |\downarrow\rangle + \cos\frac{\theta}{2} |\uparrow\rangle \quad (\text{high energy})$$

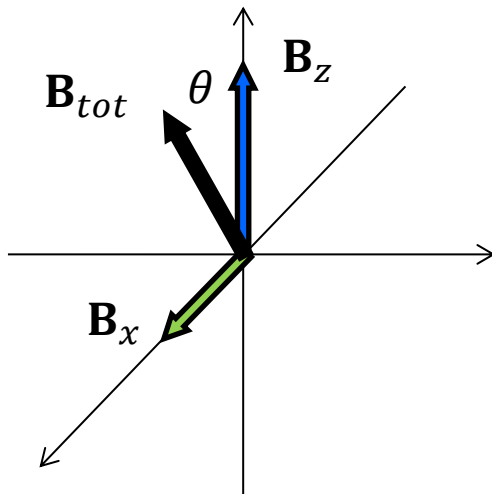
$$|\varphi_l\rangle = \cos\frac{\theta}{2} |\downarrow\rangle - \sin\frac{\theta}{2} |\uparrow\rangle \quad (\text{low energy})$$

where $\tan \theta = \frac{\Delta}{\varepsilon}$ for $0 \leq \theta < \pi$



Spin mapping

This qubit Hamiltonian maps onto that of a spin subject to a variable magnetic field along z, and a constant magnetic field along x, so that the total field makes an angle θ with the z axis.



$$\varepsilon = \gamma \frac{\hbar}{2} B_z$$

$$\Delta = \gamma \frac{\hbar}{2} B_x$$

The energy anticrossing occurs because, since $\Delta \neq 0$, the eigenvalues are never degenerate.

Time evolution: general formula

We have seen that we can write the time evolution operator as $U(t, t_0) = e^{-i\frac{H}{\hbar}(t-t_0)}$ and that it is diagonal if we use the eigenvectors of the Hamiltonian as basis of the

Hilbert space, i.e. $U(t, t_0) = \begin{pmatrix} e^{-i\frac{E_1}{\hbar}(t-t_0)} & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & e^{-i\frac{E_N}{\hbar}(t-t_0)} \end{pmatrix}$

To calculate the time evolution analytically, we must therefore:

- 1) Express the initial state $|\psi(0)\rangle$ in the eigenbasis of the Hamiltonian:

$$|\psi(0)\rangle = \sum_n c_n(0) |\varphi_n\rangle = \begin{pmatrix} c_1(0) \\ \vdots \\ c_N(0) \end{pmatrix}$$

- 2) Apply the time evolution operator and obtain the time-evolved state as:

$$|\psi(t)\rangle = \sum_n c_n(0) e^{-i\frac{E_n}{\hbar}t} |\varphi_n\rangle = \begin{pmatrix} e^{-i\frac{E_1}{\hbar}t} c_1(0) \\ \vdots \\ e^{-i\frac{E_N}{\hbar}t} c_N(0) \end{pmatrix}$$

Example: sudden change

Consider a qubit initially described by the Hamiltonian $H_0 = \varepsilon \sigma_z$ and prepared in its eigenstate $|\uparrow\rangle = |\psi(0^-)\rangle$

At time $t = 0$, switch on the term $\Delta \sigma_x$. Now $|\psi(0^-)\rangle$ is no longer an eigenstate.

1) Express $|\psi(0^-)\rangle = |\uparrow\rangle$ in the new eigenbasis $|\varphi_h\rangle, |\varphi_l\rangle$:

$$|\psi(0^-)\rangle = \cos \frac{\theta}{2} |\varphi_h\rangle - \sin \frac{\theta}{2} |\varphi_l\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ -\sin \frac{\theta}{2} \end{pmatrix}$$

2) Time-evolve the coefficients:

$$|\psi(t)\rangle = e^{-i\frac{E_h}{\hbar}t} \cos \frac{\theta}{2} |\varphi_h\rangle - e^{-i\frac{E_l}{\hbar}t} \sin \frac{\theta}{2} |\varphi_l\rangle = \begin{pmatrix} e^{-i\frac{E_h}{\hbar}t} \cos \frac{\theta}{2} \\ -e^{-i\frac{E_l}{\hbar}t} \sin \frac{\theta}{2} \end{pmatrix}$$

Time evolution

We started in the state $|\uparrow\rangle$. The probability to find the spin $|\downarrow\rangle$ at time t is $|\langle\downarrow|\psi(t)\rangle|^2$

Express $|\downarrow\rangle$ in the new eigenbasis $|\varphi_h\rangle, |\varphi_l\rangle$:

$$\begin{aligned}
 |\downarrow\rangle &= \sin\frac{\theta}{2}|\varphi_h\rangle + \cos\frac{\theta}{2}|\varphi_l\rangle = \begin{pmatrix} \sin\frac{\theta}{2} \\ \cos\frac{\theta}{2} \end{pmatrix} \\
 \langle\downarrow|\psi(t)\rangle &= \begin{pmatrix} \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} \begin{pmatrix} e^{-i\frac{E_h}{\hbar}t} \cos\frac{\theta}{2} \\ -e^{-i\frac{E_l}{\hbar}t} \sin\frac{\theta}{2} \end{pmatrix} = \\
 &= \sin\frac{\theta}{2}\cos\frac{\theta}{2}e^{-i\frac{E_h}{\hbar}t} - \sin\frac{\theta}{2}\cos\frac{\theta}{2}e^{-i\frac{E_l}{\hbar}t} = \frac{1}{2}\sin\theta\left[e^{-i\frac{E_h}{\hbar}t} - e^{-i\frac{E_l}{\hbar}t}\right]
 \end{aligned}$$

Note: $\sin \theta = \frac{\Delta}{\sqrt{\varepsilon^2 + \Delta^2}}$

$$E_h = -E_l = \sqrt{\varepsilon^2 + \Delta^2} \rightarrow e^{-i\frac{E_h}{\hbar}t} - e^{-i\frac{E_l}{\hbar}t} = 2i \sin\left(\frac{\sqrt{\varepsilon^2 + \Delta^2}}{\hbar}t\right)$$

Rabi formula

$$\langle \downarrow | \psi(t) \rangle = i \frac{\Delta}{\sqrt{\varepsilon^2 + \Delta^2}} \sin \left(\frac{\sqrt{\varepsilon^2 + \Delta^2}}{\hbar} t \right)$$

The probability of finding the system in the state $|\downarrow\rangle$ is:

$$\mathcal{P}_{\downarrow} = |\langle \downarrow | \psi(t) \rangle|^2 = \frac{\Delta^2}{\varepsilon^2 + \Delta^2} \sin^2 \left(\frac{\sqrt{\varepsilon^2 + \Delta^2}}{\hbar} t \right)$$

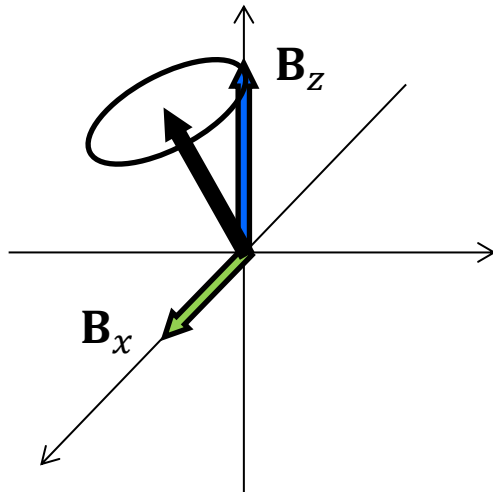
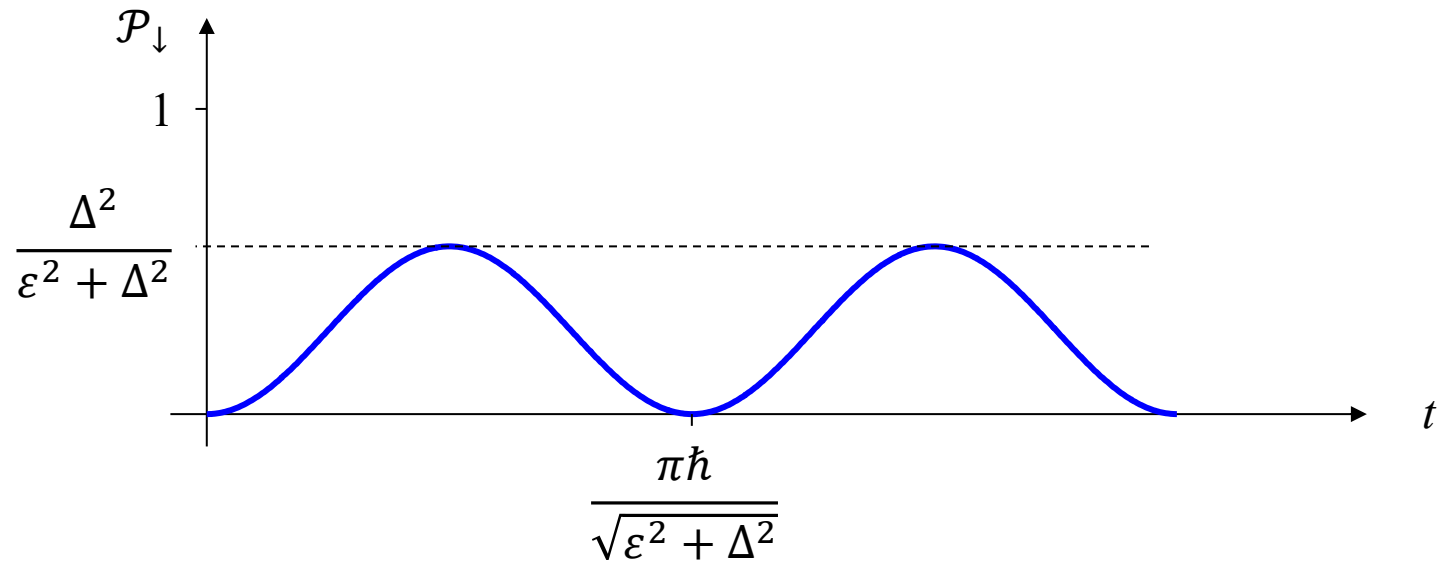
This is known as **Rabi formula**

The probability oscillates at frequency

$$\omega_R = 2 \frac{\sqrt{\varepsilon^2 + \Delta^2}}{\hbar}$$

which is equal to the energy difference $E_h - E_l$ divided by $\hbar$

Rabi oscillations



Physical interpretation for spins:
the spin starts along z , then precesses around the new total field when B_x is switched on

Notice that, in this way, we can never get $\mathcal{P}_{\downarrow} = 1$
(it would require $B_x \rightarrow \infty$)

Time-dependent Hamiltonian

In certain physical systems, the Hamiltonian can be explicitly time-dependent (example: a spin in a time-varying magnetic field). The time-evolution operator is still given by:

$$U(t, t_0) = \int_{t_0}^t \exp\left(-i \frac{H(t')}{\hbar}\right) dt'$$

but the integral may be impossible to solve analytically.

Numerical solution: discretize the time in short intervals Δt , during which the Hamiltonian is approximated as a constant. Calling $t_n = t_0 + n\Delta t$:

$$U(t_{n+1}, t_n) = e^{-i \frac{H(t_n)}{\hbar} \Delta t}$$

$$|\psi(t_{n+1})\rangle = U(t_{n+1}, t_n) |\psi(t_n)\rangle$$

$$|\psi(t_N)\rangle = \left(\prod_{n=0}^{N-1} U(t_{n+1}, t_n) \right) |\psi(t_0)\rangle$$

Note: this a product of matrices! You have to keep them in “time order”

This can be easily coded in Python. However, it means recalculating the exponential of a matrix at every tiny time step, so it's computationally very expensive!

Time-periodic Hamiltonian

Consider a Hamiltonian with a periodic time dependence:

$$H(t) = H_0 + H_1 \cos(\omega t)$$

$T = \frac{2\pi}{\omega}$ is the period. Since $H(t + T) = H(t)$ we just need to calculate the time evolution operator that takes us from the start to the end of the period, splitting the period T in N small intervals, $T = N\Delta t$:

$$U(t + T, t) = \prod_{n=0}^{N-1} U(t_{n+1}, t_n)$$

$U(t + T, t)$ is called “**Floquet operator**”.

Now we can calculate the state after each period as:

$$|\psi(t + kT)\rangle = U^k(t + T, t)|\psi(t)\rangle$$

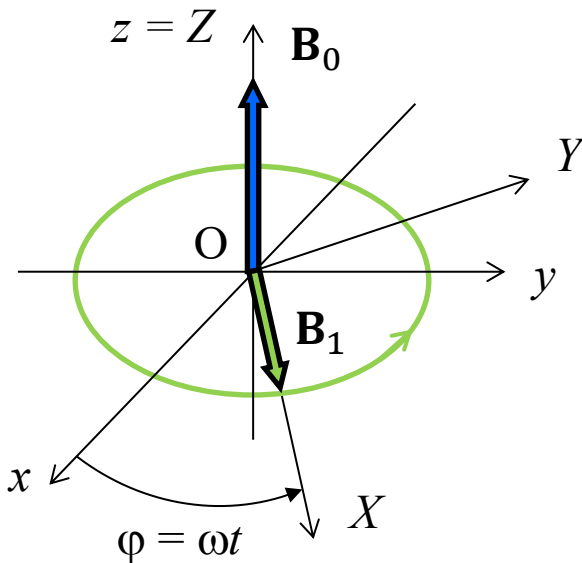
So we only need to do the numerically expensive calculation once, to get $U(t + T, t)$. From then onwards, we can keep multiplying by the same time evolution operator.

Rotating frame

In some cases, a time-dependent Hamiltonian can be turned into a time-independent one, by performing a change of basis. This makes solving the problem much “cheaper”, and more intuitive.

Consider a magnetic dipole $\mathbf{M}$ placed in a static vertical magnetic field $\mathbf{B}_0 = B_0 \mathbf{z}$, plus a rotating magnetic field in the xy plane, $\mathbf{B}_1 = B_1(\cos(\omega t) \mathbf{x} + \sin(\omega t) \mathbf{y})$:

$$H(t) = -\mathbf{M} \cdot \mathbf{B}(t) = -M_z B_z - M_x B_1 \cos(\omega t) - M_y B_1 \sin(\omega t)$$



Trick: define a **rotating reference frame** $OXYZ$ that rotates at frequency ω around the z -axis.

The vector $\mathbf{B}_1$ appears static in the rotating frame!

Quantum treatment

- Define:

$\omega_0 = \gamma B_0 \rightarrow$ Larmor precession frequency of the spin in the static field $\mathbf{B}_0$

$\omega_1 = \gamma B_1 \rightarrow$ Larmor frequency the spin would have if it were placed in a static field B_1

$\Delta\omega = \omega - \omega_0 \rightarrow$ “detuning” between precession frequency and rotating field

- Hamiltonian:

$$\begin{aligned} H(t) &= \gamma B_0 S_z + \gamma B_1 (S_x \cos(\omega t) + S_y \sin(\omega t)) = \\ &= \omega_0 S_z + \omega_1 \left(\frac{e^{i\omega t} + e^{-i\omega t}}{2} S_x + \frac{e^{i\omega t} - e^{-i\omega t}}{2i} S_y \right) = \frac{\hbar}{2} \begin{pmatrix} \omega_0 & \omega_1 e^{-i\omega t} \\ \omega_1 e^{i\omega t} & -\omega_0 \end{pmatrix} \end{aligned}$$

- Generic state at time t , in the basis $|\uparrow\rangle, |\downarrow\rangle$:

$$|\psi(t)\rangle = c_\uparrow(t)|\uparrow\rangle + c_\downarrow(t)|\downarrow\rangle = \begin{pmatrix} c_\uparrow(t) \\ c_\downarrow(t) \end{pmatrix}$$

- Time evolution:

$$\begin{aligned} i\hbar \frac{d|\psi(t)\rangle}{dt} &= H(t)|\psi(t)\rangle \rightarrow i\hbar \frac{d}{dt} \begin{pmatrix} c_\uparrow(t) \\ c_\downarrow(t) \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} \omega_0 & \omega_1 e^{-i\omega t} \\ \omega_1 e^{i\omega t} & -\omega_0 \end{pmatrix} \begin{pmatrix} c_\uparrow(t) \\ c_\downarrow(t) \end{pmatrix} \\ \left\{ \begin{aligned} i \frac{d}{dt} c_\uparrow(t) &= \frac{\omega_0}{2} c_\uparrow(t) + \frac{\omega_1}{2} e^{-i\omega t} c_\downarrow(t) \\ i \frac{d}{dt} c_\downarrow(t) &= \frac{\omega_1}{2} e^{i\omega t} c_\uparrow(t) - \frac{\omega_0}{2} c_\downarrow(t) \end{aligned} \right. \end{aligned}$$

Change of basis

A rotation by an angle φ around an axis $\mathbf{u} = u_x \mathbf{x} + u_y \mathbf{y} + u_z \mathbf{z}$, with $|\mathbf{u}|^2 = 1$, is expressed in the 2-dimensional Hilbert space by the operator

$$R_{\mathbf{u},\varphi} = \exp\left(i \frac{\varphi}{2} \sigma_{\mathbf{u}}\right)$$

where $\sigma_{\mathbf{u}} = \mathbf{u} \cdot \boldsymbol{\sigma}$ is the Pauli matrix along the unit vector $\mathbf{u}$. Note: $R_{\mathbf{u},\varphi}$ is unitary.

We want to rotate our basis by ωt around $\mathbf{z}$, so we need

$$R_{\mathbf{z},\omega t} = \exp\left(i \frac{\omega t}{2} \sigma_z\right) = \begin{pmatrix} e^{i\omega t/2} & 0 \\ 0 & e^{-i\omega t/2} \end{pmatrix}$$

- Define: $|\tilde{\psi}(t)\rangle = R_{\mathbf{z},\omega t} |\psi(t)\rangle = \begin{pmatrix} \tilde{c}_{\uparrow}(t) \\ \tilde{c}_{\downarrow}(t) \end{pmatrix} \rightarrow |\psi(t)\rangle = R_{\mathbf{z},\omega t}^{-1} |\tilde{\psi}(t)\rangle =$

$$\begin{pmatrix} e^{-i\omega t/2} & 0 \\ 0 & e^{i\omega t/2} \end{pmatrix} \begin{pmatrix} \tilde{c}_{\uparrow}(t) \\ \tilde{c}_{\downarrow}(t) \end{pmatrix}$$

Therefore:

$$\begin{cases} c_{\uparrow}(t) = e^{-i\omega t/2} \tilde{c}_{\uparrow}(t) \\ c_{\downarrow}(t) = e^{i\omega t/2} \tilde{c}_{\downarrow}(t) \end{cases} \quad \begin{cases} \frac{d}{dt} c_{\uparrow}(t) = \frac{-i\omega}{2} \tilde{c}_{\uparrow}(t) e^{-\frac{i\omega t}{2}} + e^{-\frac{i\omega t}{2}} \frac{d}{dt} \tilde{c}_{\uparrow}(t) \\ \frac{d}{dt} c_{\downarrow}(t) = \frac{i\omega}{2} \tilde{c}_{\downarrow}(t) e^{\frac{i\omega t}{2}} + e^{\frac{i\omega t}{2}} \frac{d}{dt} \tilde{c}_{\downarrow}(t) \end{cases}$$

Schroedinger equation in the rotating frame

Substitute these expression for $c_{\uparrow\downarrow}(t)$ and $dc_{\uparrow\downarrow}(t)/dt$ into the Schroedinger equation:

$$\begin{cases} i \frac{d}{dt} \tilde{c}_{\uparrow}(t) = -\frac{\Delta\omega}{2} \tilde{c}_{\uparrow}(t) + \frac{\omega_1}{2} \tilde{c}_{\downarrow}(t) \\ i \frac{d}{dt} \tilde{c}_{\downarrow}(t) = \frac{\omega_1}{2} \tilde{c}_{\uparrow}(t) + \frac{\Delta\omega}{2} \tilde{c}_{\downarrow}(t) \end{cases}$$

This is the same as writing:

$$i\hbar \frac{d|\tilde{\psi}(t)\rangle}{dt} = \tilde{H}|\tilde{\psi}(t)\rangle$$

where

$$\tilde{H} = \frac{\hbar}{2} \begin{pmatrix} -\Delta\omega & \omega_1 \\ \omega_1 & \Delta\omega \end{pmatrix} = -\frac{\hbar}{2} \Delta\omega \sigma_z + \frac{\hbar}{2} \omega_1 \sigma_x$$

The Hamiltonian in the rotating frame appears **time-independent!**

The time dependence is moved to the reference frame

Magnetic resonance

The Hamiltonian in the rotating frame looks just like a qubit Hamiltonian with

$$\varepsilon = -\frac{\hbar}{2}\Delta\omega \quad \Delta = \frac{\hbar}{2}\omega_1$$

Assume initial state $|\psi(0)\rangle \equiv |\tilde{\psi}(0)\rangle = |\uparrow\rangle$

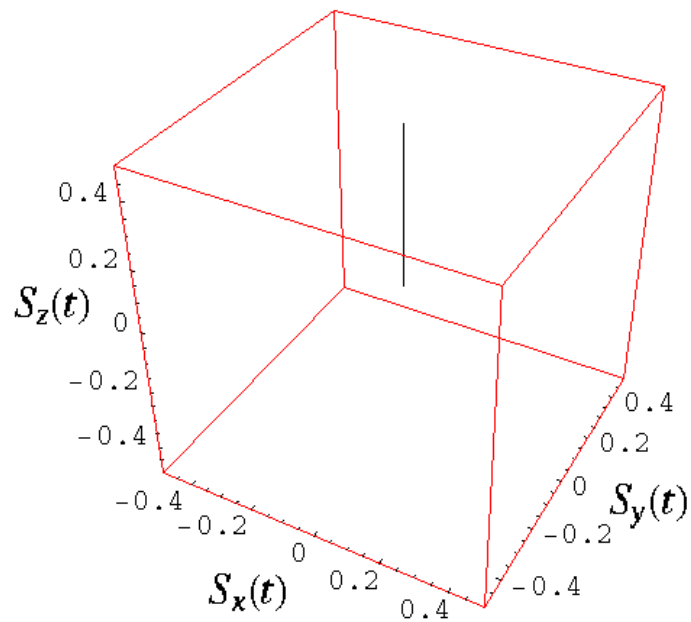
The probability to flip the state to $|\downarrow\rangle$ is obtained from the Rabi formula as:

$$\mathcal{P}_{\downarrow} = |\langle\downarrow|\psi(t)\rangle|^2 = \frac{\omega_1^2}{(\Delta\omega)^2 + \omega_1^2} \sin^2\left(\frac{\sqrt{(\Delta\omega)^2 + \omega_1^2}}{2}t\right)$$

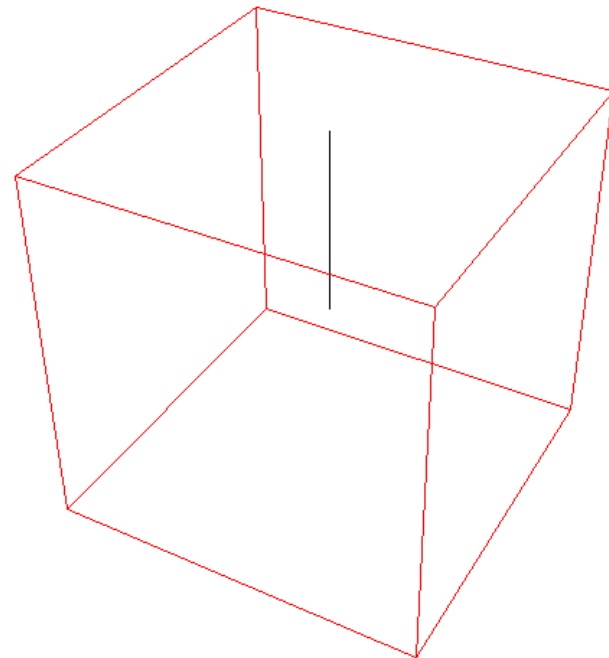
- If $\Delta\omega = 0$, i.e. B_1 rotates at the same frequency as the Larmor precession of the spin in the field B_0 , then $\mathcal{P}_{\downarrow}$ can reach 1 $\rightarrow$ **magnetic resonance**
- If $\Delta\omega \gg \omega_1$, $\mathcal{P}_{\downarrow} \approx 0 \rightarrow$ **off-resonance** the spin state is left untouched
- At resonance, i.e. $\Delta\omega = 0$, the Rabi frequency is equal to $\omega_1 \rightarrow$ the B_0 field is “cancelled out” in the rotating frame, so the spin precesses at frequency ω_1 around the field B_1 .

Spin nutation in fixed vs. rotating frame

Fixed (laboratory) frame



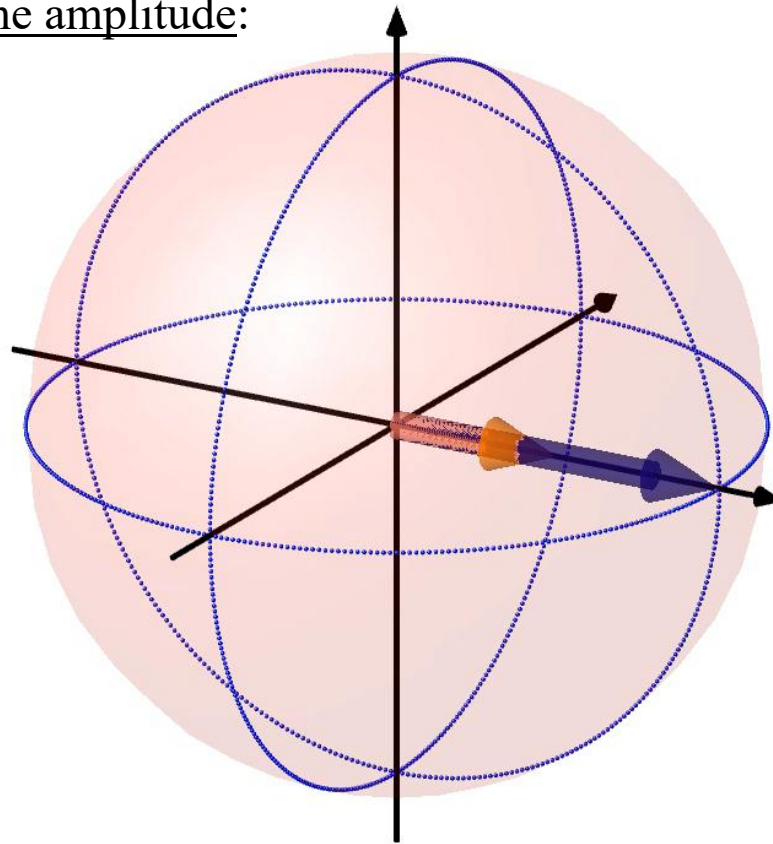
Rotating frame



Rotating wave approximation

Producing a rotating magnetic field requires two coils driven by currents 90° out phase – cumbersome arrangement. It's much easier to have a single coil, which makes a linearly oscillating field when driven by an AC current.

However recall that an oscillating magnetic field can be decomposed into two counter-rotating fields of half the amplitude:

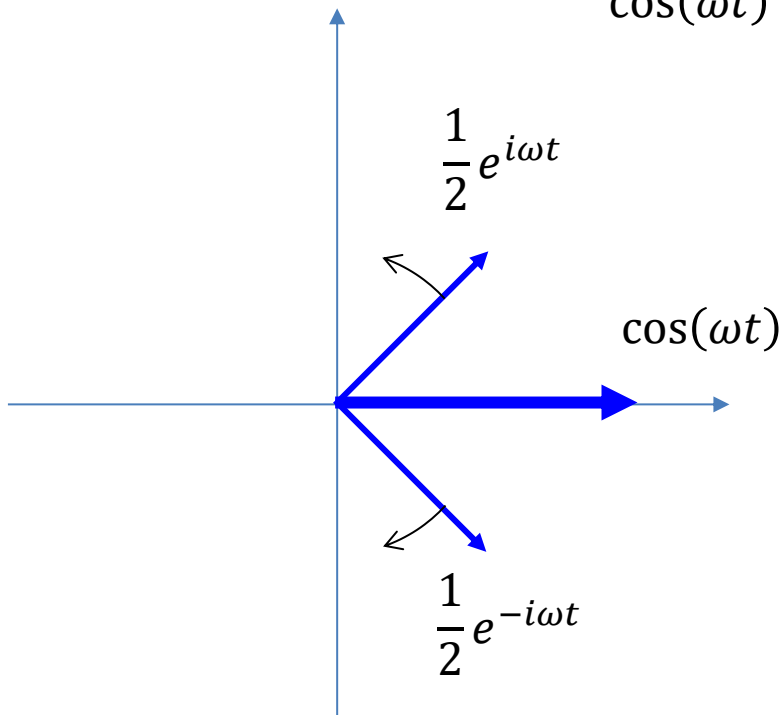


Rotating wave approximation

Producing a rotating magnetic field requires two coils driven by currents 90° out phase – cumbersome arrangement. It's much easier to have a single coil, which makes a linearly oscillating field when driven by an AC current.

However recall that an oscillating magnetic field can be decomposed into two counter-rotating fields of half the amplitude:

$$\cos(\omega t) = \frac{1}{2}(e^{i\omega t} + e^{-i\omega t})$$



rotating field at frequency $-\omega$
→ detuned by 2ω from the resonance!

Only the component rotating at $+\omega$ has an effect
→ The Rabi frequency becomes $\omega_1/2$

Nuclear magnetic resonance (NMR)

Most nuclei have a spin, but they all have different gyromagnetic ratios

Examples:

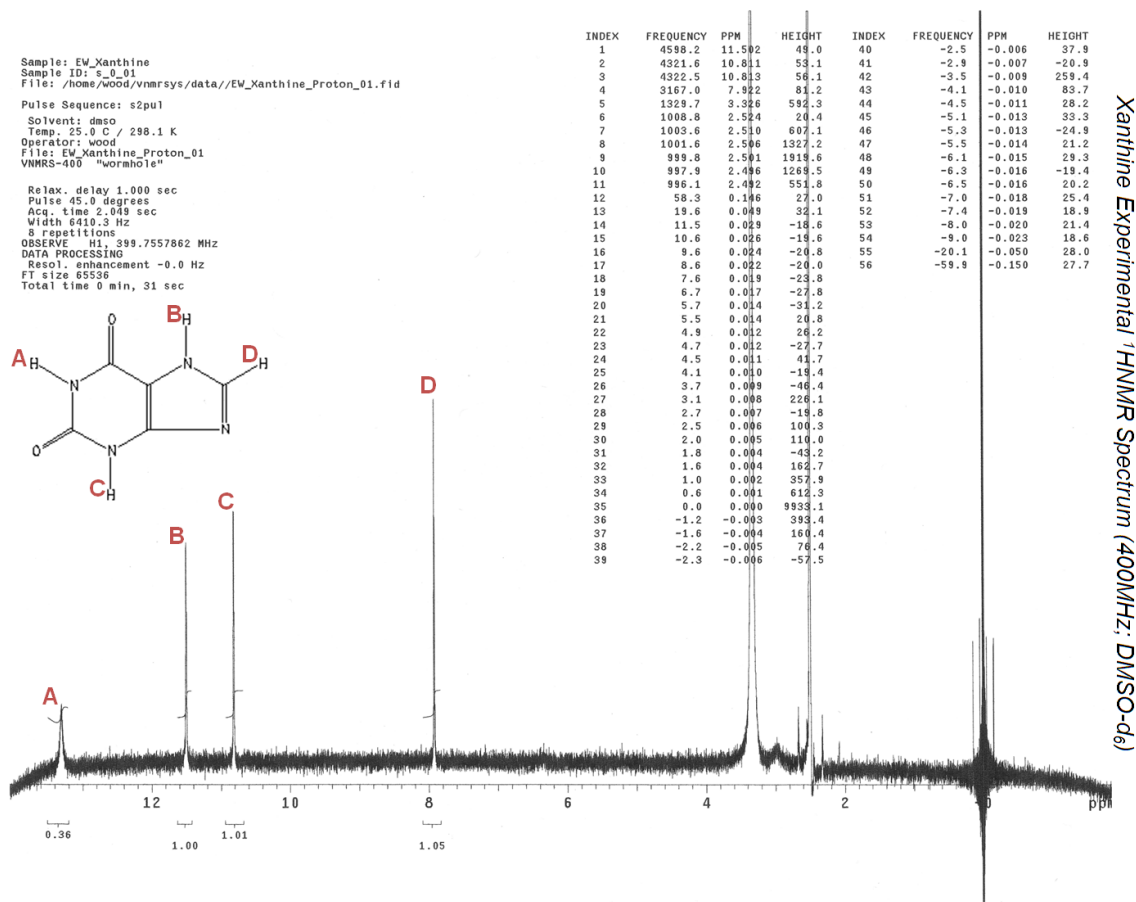
- ^1H $\gamma = 42.58 \text{ MHz/T}$
- ^{13}C $\gamma = 10.7 \text{ MHz/T}$
- ^{19}F $\gamma = 40.06 \text{ MHz/T}$
- ^{31}P $\gamma = 17.23 \text{ MHz/T}$

→ at fixed B_0 they can be distinguished by the frequency at which they respond, or

→ at fixed ω , they can be distinguished by the value of B_0 at which they respond

NMR spectra → chemical analysis

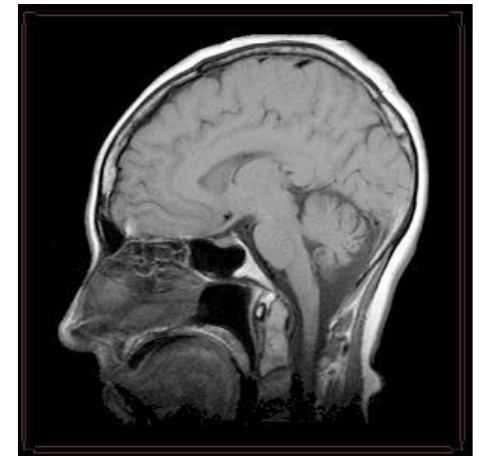
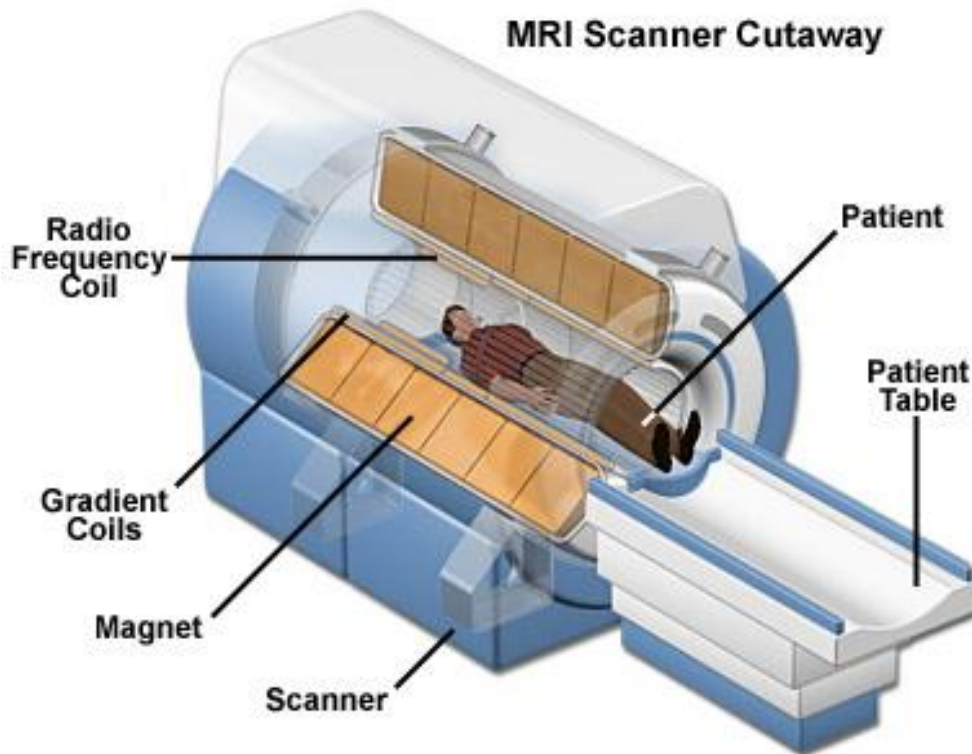
NMR spectroscopy is a key tool in chemistry. The study of NMR spectra is used to determine the elements present in newly synthesized molecules, and to derive their structure



Magnetic resonance imaging (MRI)

A set of “gradient coils” produces a spatially inhomogeneous static B_0 field

Rabi formula: only the spins with $\Delta\omega < \omega_1$ are significantly affected by the radiofrequency $\rightarrow$ the signal comes only from a small volume in space

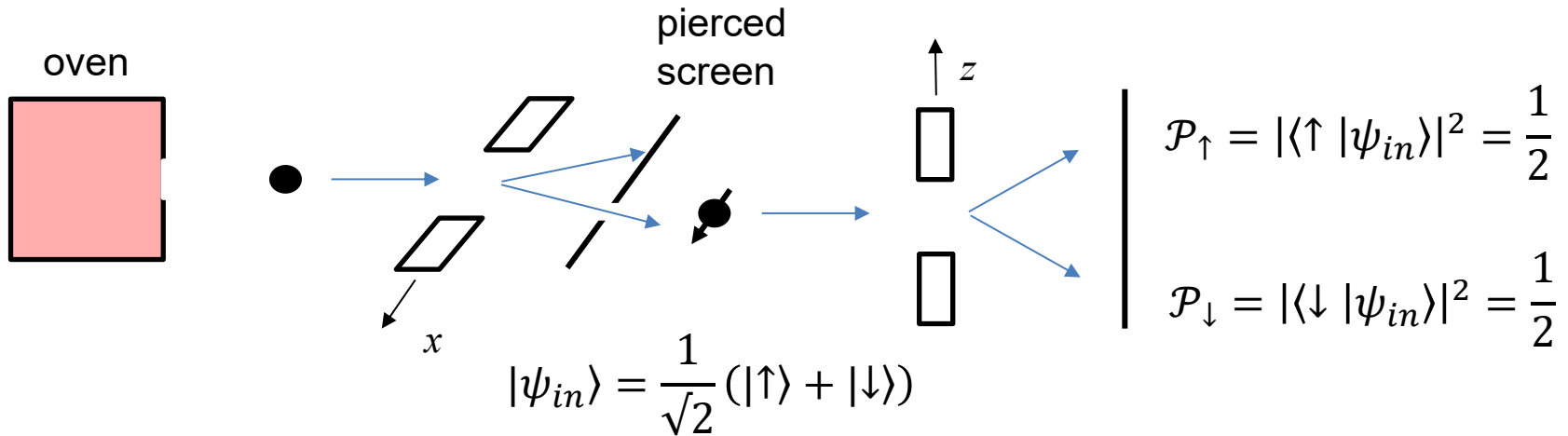


bright $\rightarrow$ high density
of ^1H spins

Pure states

We have seen how to calculate the time evolution, expectation values, etc. for systems in which the quantum state $|\psi(t)\rangle$ is exactly known. This means we have somehow “initialized” the state, usually by performing a measurement.

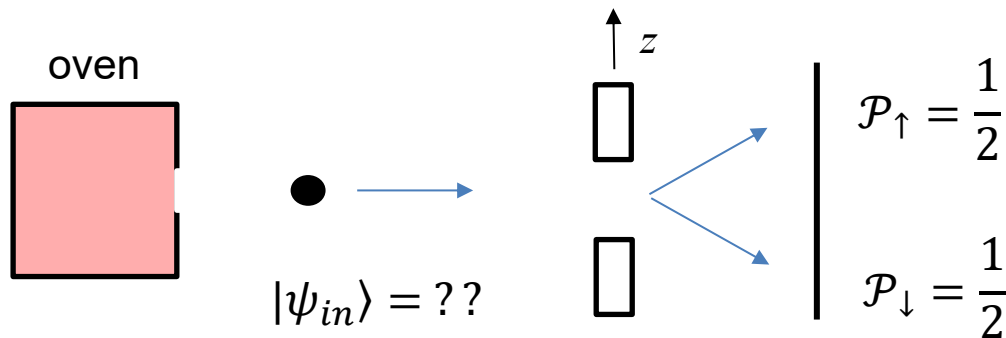
Example:



The pierced screen after the x -oriented Stern-Gerlach prepares the input state $|\psi_{in}\rangle = |X\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle + |\downarrow\rangle)$, which is well-defined quantum state, from which we can calculate everything else $\rightarrow$ **pure state**

Mixed states

Now consider this:



The measurement statistics after the z -oriented Stern-Gerlach are the same as before, but we haven't prepared a pure state as an input. The atoms came out of the oven in a completely random state.

The input state can't be assigned a well-defined ket. The best we can do is say that it is in a mixture of $|\uparrow\rangle$ and $|\downarrow\rangle$ with probabilities $\mathcal{P}_{\uparrow}$ and $\mathcal{P}_{\downarrow} \rightarrow$ **mixed state**

How do we describe the difference between the pure state $\frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ and the random mixture of $|\uparrow\rangle$ and $|\downarrow\rangle$?

Density operator

For pure states, we can define a **density operator**, or density matrix, as:

$$\rho = |\psi\rangle\langle\psi|$$

$$\text{Explicitly: } |\psi\rangle = \begin{pmatrix} c_1 \\ \vdots \\ c_N \end{pmatrix} \rightarrow \rho = \begin{pmatrix} c_1 \\ \vdots \\ c_N \end{pmatrix} (c_1^* \quad \dots \quad c_N^*) = \begin{pmatrix} c_1 c_1^* & \dots & c_1 c_N^* \\ \vdots & \ddots & \vdots \\ c_N c_1^* & \dots & c_N c_N^* \end{pmatrix}$$

or, calling $|u_n\rangle$ the basis vectors of the Hilbert space: $\rho_{pn} = \langle u_p | \rho | u_n \rangle = c_p c_n^*$

- ρ is obviously Hermitian: $\rho^\dagger = \rho$
- ρ is a projector onto the (normalized) state $|\psi\rangle$ and therefore: $\rho = \rho^2$
because $\rho^2 = |\psi\rangle \underbrace{\langle\psi|\psi\rangle}_{=1} \langle\psi| = |\psi\rangle\langle\psi| = \rho$

Properties of the density operator

- Since $|\psi\rangle$ is normalized, i.e. $\sum_n |c_n|^2 = \sum_n c_n c_n^* = 1$, then $\boxed{\text{Tr}(\rho) = 1}$
- Also, since $\rho^2 = \rho$, $\boxed{\text{Tr}(\rho^2) = 1}$
- $\langle A \rangle_\psi = \langle \psi | A | \psi \rangle = \sum_{n,p} c_n^* c_p A_{np} = \sum_{n,p} \langle u_p | \rho | u_n \rangle \langle u_n | A | u_p \rangle = \sum_p \langle u_p | \rho A | u_p \rangle$
 $\boxed{\langle A \rangle_\psi = \text{Tr}(\rho A)}$

- Time evolution:

$$\begin{aligned} \frac{d}{dt} \rho(t) &= \left[\frac{d}{dt} |\psi(t)\rangle \right] \langle \psi(t)| + |\psi(t)\rangle \left[\frac{d}{dt} \langle \psi(t)| \right] = \\ &= \frac{1}{i\hbar} H(t) |\psi(t)\rangle \langle \psi(t)| + \frac{1}{-i\hbar} |\psi(t)\rangle \langle \psi(t)| H(t) \end{aligned}$$

$$\boxed{\frac{d}{dt} \rho(t) = \frac{1}{i\hbar} [H(t), \rho(t)]}$$

For pure states, the density operator contains the same information as the state vector.

Density operator for mixed states

Now consider a mixed state, like that of a silver atom coming out of the oven.

All we know about it is that if we measure it in the z -basis, we get $\mathcal{P}_\uparrow = \frac{1}{2}$, $\mathcal{P}_\downarrow = \frac{1}{2}$.

We can still write $\langle S_z \rangle = \frac{\hbar}{2} \mathcal{P}_\uparrow - \frac{\hbar}{2} \mathcal{P}_\downarrow = 0$, which would be satisfied if we *define*

$$\rho = \begin{pmatrix} \mathcal{P}_\uparrow & 0 \\ 0 & \mathcal{P}_\downarrow \end{pmatrix}$$

since, with this definition, we retain the property that

$$\langle S_z \rangle = \text{Tr}(\rho S_z)$$

In general, we define the density operator for mixed states as

$$\rho = \begin{pmatrix} \mathcal{P}_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \mathcal{P}_N \end{pmatrix}$$

where $\mathcal{P}_n$ is the probability of obtaining the eigenvalue a_n when measuring A . This does not require knowing the precise quantum state at the start of the experiment.

Note: now it no longer holds that $\text{Tr}(\rho^2) = 1$ and $\rho^2 = \rho$

Populations and coherences

The diagonal elements of the density matrix are called **populations**.

They describe the probability $\mathcal{P}_n$ of finding the system in the state $|u_n\rangle$ when performing a measurement. This holds for both pure and mixed states.

The off-diagonal elements of the density matrix are called **coherences**.

They describe the phase relationship between different components of a quantum superposition of states.

- Example:

$$\begin{aligned} |X\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \rightarrow \quad \rho_X = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \\ |Y\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\rangle + i|\downarrow\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \quad \rightarrow \quad \rho_Y = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} \end{aligned}$$

The populations are the same (the probabilities of measuring $|\uparrow\rangle$ or $|\downarrow\rangle$ are $\frac{1}{2}$ in both cases) but the coherences are different (the $|Y\rangle$ state has a different phase from $|X\rangle$)

Decoherence

Consider an NMR experiment, where a large ensemble of identical spins are initially all prepared in the pure state $|X\rangle$. We can describe this ensemble with:

$$\rho_X = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

After preparation, each spin precesses at the Larmor frequency $\omega_0 = \gamma B_0$. In reality, each spin sees a slightly different and unknown $B_0(\mathbf{r})$ field at its location, so after some time we lose track of what exactly is the phase of each spin – they end up pointing randomly in the xy -plane. Now, we can describe the ensemble with

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

since we know that each spin is “somewhere” in the xy -plane.

→ the coherences have decayed to zero.

Decoherence is the phenomenon by which the randomness and lack of knowledge of the precise details of the environment of quantum system results in a loss of knowledge of which pure state the system is in – but we can still describe it with a density matrix!

State purity

Have a pure vs. mixed state is not a black-or-white thing.

Consider the following density matrix

$$\rho_{\psi} = \frac{1}{2} \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.25 \\ 0.25 & 0.5 \end{pmatrix}$$

From which:

$$\rho_{\psi}^2 = \begin{pmatrix} 0.3125 & 0.25 \\ 0.25 & 0.3125 \end{pmatrix}$$

$$\text{Tr}(\rho_{\psi}^2) = 0.625 < 1$$

which shows that ρ_{ψ} does not describe a pure state.

However, it still has nonzero coherences. It means you know “something” about the phase, but there is some uncertainty.

The fully mixed state $\rho = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ would have

$$\text{Tr}(\rho^2) = 0.5$$

which is less than $\text{Tr}(\rho_{\psi}^2)$

$\text{Tr}(\rho^2)$ is called **state purity**

Section 7

Coupling and entanglement of quantum particles

Learning outcomes:

- Learn how to construct the mathematical description of multiple quantum particles
- Understand the fundamental concept of quantum entanglement, which is at the basis of nearly all revolutionary quantum technologies
- Identify physical systems that can exhibit entanglement
- Understand the meaning and significance of Bell's inequality.

Coupled quantum systems

So far we have looked at the quantum description of *individual* physical systems (spins, particles, molecules, etc.) and we have found that we could often recognize the classical analog of their behaviour, as long as we accept wave-particle duality (quantum tunneling is just like evanescent waves, for example...)

The truly striking properties of quantum systems become evident when we look at more than one particle.

Consider a spin: it is described by basis state vectors

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and all its physical properties can be described by suitable combinations of Pauli matrices. The way the Pauli matrices act on the basis states is:

$$\begin{aligned} \sigma_x |\uparrow\rangle &= |\downarrow\rangle, & \sigma_x |\downarrow\rangle &= |\uparrow\rangle & \rightarrow \text{“spin-flip” operator} \\ \sigma_y |\uparrow\rangle &= i |\downarrow\rangle, & \sigma_y |\downarrow\rangle &= -i |\uparrow\rangle \\ \sigma_z |\uparrow\rangle &= |\uparrow\rangle, & \sigma_z |\downarrow\rangle &= -|\downarrow\rangle \end{aligned}$$

Coupled quantum systems

Now I want to describe the combined state of *two spins*. How?

In analogy with binary logic, I can construct a basis for an Hilbert space that describes both spins as the logical combinations of basis states:

$$|\uparrow\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |\uparrow\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |\downarrow\uparrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |\downarrow\downarrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

This basis is obviously orthonormal.

Next, we want to construct the Pauli matrices for each spin. The natural rule to apply is that each matrix should act on its own spin like a normal Pauli matrix, and leave the other spin untouched.

Pauli z matrices

The Pauli z matrices should behave like:

$$\sigma_{z1}|\uparrow\uparrow\rangle = |\uparrow\uparrow\rangle, \quad \sigma_{z1}|\uparrow\downarrow\rangle = |\uparrow\downarrow\rangle, \quad \sigma_{z1}|\downarrow\uparrow\rangle = -|\downarrow\uparrow\rangle, \quad \sigma_{z1}|\downarrow\downarrow\rangle = -|\downarrow\downarrow\rangle$$

therefore:

$$\sigma_{z1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\sigma_{z2}|\uparrow\uparrow\rangle = |\uparrow\uparrow\rangle, \quad \sigma_{z2}|\uparrow\downarrow\rangle = -|\uparrow\downarrow\rangle, \quad \sigma_{z2}|\downarrow\uparrow\rangle = |\downarrow\uparrow\rangle, \quad \sigma_{z2}|\downarrow\downarrow\rangle = -|\downarrow\downarrow\rangle$$

therefore:

$$\sigma_{z2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Pauli x matrices

The Pauli x matrices should behave like:

$$\sigma_{x1}|\uparrow\uparrow\rangle = |\downarrow\uparrow\rangle, \quad \sigma_{x1}|\uparrow\downarrow\rangle = |\downarrow\downarrow\rangle, \quad \sigma_{x1}|\downarrow\uparrow\rangle = |\uparrow\uparrow\rangle, \quad \sigma_{x1}|\downarrow\downarrow\rangle = |\uparrow\downarrow\rangle$$

therefore:

$$\sigma_{x1} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\sigma_{x2}|\uparrow\uparrow\rangle = |\uparrow\downarrow\rangle, \quad \sigma_{x2}|\uparrow\downarrow\rangle = |\uparrow\uparrow\rangle, \quad \sigma_{x2}|\downarrow\uparrow\rangle = |\downarrow\downarrow\rangle, \quad \sigma_{x2}|\downarrow\downarrow\rangle = |\downarrow\uparrow\rangle$$

therefore:

$$\sigma_{x2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Tensor product

Instead of doing it “by hand”, the matrices describing multiple quantum systems can be obtained using the mathematical operation called **tensor product** (or Kronecker product).

- Given two matrices A and B (here 2×2 matrices for simplicity), their tensor product is obtained as the block matrix:

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{pmatrix}$$

- This is valid for any dimension of the matrices being tensor-multiplied:
If A has dimensions $m \times n$ and B has $p \times q$, then $A \otimes B$ has $mp \times nq$.
- The tensor product is associative:

$$(A \otimes B) \otimes C = A \otimes (B \otimes C)$$

so we just write $A \otimes B \otimes C$

- Python (numpy) instruction:**

$$A \otimes B \rightarrow \text{np.kron}(A, B)$$

Construction of a large Hilbert space

Now we can use the tensor product to automatically and systematically construct the Hilbert space of as many particles as we wish.

Consider two spins, each one described in the basis:

$$|\uparrow\rangle_i = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle_i = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad i = 1, 2$$

The basis for their combined Hilbert space is obtained by taking all the combinations of tensor products of individual basis vectors:

$$|\uparrow\uparrow\rangle = |\uparrow\rangle_1 \otimes |\uparrow\rangle_2 = \begin{pmatrix} 1 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |\uparrow\downarrow\rangle = |\uparrow\rangle_1 \otimes |\downarrow\rangle_2 = \begin{pmatrix} 1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$|\downarrow\uparrow\rangle = |\downarrow\rangle_1 \otimes |\uparrow\rangle_2 = \begin{pmatrix} 0 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 1 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |\downarrow\downarrow\rangle = |\downarrow\rangle_1 \otimes |\downarrow\rangle_2 = \begin{pmatrix} 0 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Operators in a tensor product Hilbert space

We have already constructed “by hand” the spin operators for two spins, using the rule that each operator must act only on the spin that “belongs” to it.

With tensor products, the operators are obtained simply by using the desired operator on the spin we wish to address, and the identity matrix where we want the spin to be left untouched. Example:

$$\sigma_{x1} = \sigma_x \otimes \mathbb{I} = \begin{pmatrix} 0 \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & 1 \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ 1 \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & 0 \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\sigma_{x2} = \mathbb{I} \otimes \sigma_x = \begin{pmatrix} 1 \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 0 \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ 0 \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 1 \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

In general, if I have N spins:

$$\sigma_{xi} = \overbrace{\mathbb{I} \otimes \mathbb{I} \otimes \cdots \otimes \sigma_x \otimes \cdots \otimes \mathbb{I}}^{N \text{ products}}$$

↑
 i -th position

Product states

In the Hilbert space describing the two spins, I can use the tensor product to write states that represent combinations of individual spin states.

For example, I want to express the state where spin 1 is pointing along +z, and spin 2 is pointing along +x:

$$|\psi_1\rangle = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |\psi_2\rangle = |X\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The corresponding product state in the combined Hilbert space of the two spins is:

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

States obtained from the tensor product of individual particle states are called **separable**

→ they describe the individual properties of separate systems. For example, you can easily verify that

$$\langle\psi|\sigma_{z1}|\psi\rangle = 1 \quad \text{and} \quad \langle\psi|\sigma_{x2}|\psi\rangle = 1$$

as expected.

Spin-spin interactions

Consider two electron spins, each described by vector Pauli matrices $\boldsymbol{\sigma}_1, \boldsymbol{\sigma}_2$.

When placed in close proximity, they can interact in a variety of ways:

- **dipolar interaction:** each spin produces a magnetic field, which in turn affect the energy of the other. This is analogous to the classical case;
- **exchange interaction:** the orbits of the electrons overlap, and the quantum mechanical eigenstates of the system are symmetric / antisymmetric combinations of orbitals, to which correspond antisymmetric / symmetric spin configurations.
($\rightarrow$ will be discussed in Section 8)

In any case, the two-spin Hamiltonian can be written in a simplified form as:

$$H = J \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$$

where

$J < 0 \rightarrow$ ferromagnetic interaction (favours parallel spin alignment)

$J > 0 \rightarrow$ antiferromagnetic interaction (favours antiparallel spin alignment)

Eigenstates of coupled spins

Write out the product of Pauli matrices:

$$H = J \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 = J(\sigma_{x1}\sigma_{x2} + \sigma_{y1}\sigma_{y2} + \sigma_{z1}\sigma_{z2}) = J \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The eigenstates of this matrix are:

$ T_+\rangle = \uparrow\uparrow\rangle$	}	triplet states, with eigenvalue $E_T = J$ (3-fold degenerate)
$ T_0\rangle = \frac{1}{\sqrt{2}}(\uparrow\downarrow\rangle + \downarrow\uparrow\rangle)$		
$ T_-\rangle = \downarrow\downarrow\rangle$		
$ S\rangle = \frac{1}{\sqrt{2}}(\uparrow\downarrow\rangle - \downarrow\uparrow\rangle)$		singlet state, with eigenvalue $E_S = -3J$ (non-degenerate)

With this definition, the singlet-triplet energy splitting is $4J$. The factor 4 comes because I defined the Hamiltonian with Pauli matrices. If I had used instead spin 1/2 matrices, the splitting would be simply J .

Singlet / triplet states

To understand the meaning of these eigenstates, as usual, it's best to calculate the expectation values of some observable. Consider for example the observable that describes the sum of the z-projections of the two spins:

$$\sigma_{z,tot} = \sigma_{z1} + \sigma_{z2} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

$$\langle T_+ | \sigma_{z,tot} | T_+ \rangle = 2$$

$$\langle T_0 | \sigma_{z,tot} | T_0 \rangle = 0$$

$$\langle T_- | \sigma_{z,tot} | T_- \rangle = -2$$

$$\langle S | \sigma_{z,tot} | S \rangle = 0$$

Both $|T_0\rangle$ and $|S\rangle$ have zero total z projection. How do they differ from each other?

Singlet / triplet states

Consider the square of the total spin:

$$\sigma^2 = (\sigma_{x1} + \sigma_{x2})^2 + (\sigma_{y1} + \sigma_{y2})^2 + (\sigma_{z1} + \sigma_{z2})^2 = \begin{pmatrix} 8 & 0 & 0 & 0 \\ 0 & 4 & 4 & 0 \\ 0 & 4 & 4 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

$$\langle T_+ | \sigma^2 | T_+ \rangle = 8$$

$$\langle T_0 | \sigma^2 | T_0 \rangle = 8$$

$$\langle T_- | \sigma^2 | T_- \rangle = 8$$

$$\langle S | \sigma^2 | S \rangle = 0$$

All triplet states have the same value of the square of the total spin, whereas the singlet has zero total spin...

Entangled states

Consider the singlet state: what is the direction of spin 1 and spin 2 in this state?

Write it out explicitly:

$$|S\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

To understand where each spin points, we need to write it as product state, i.e. find the two 2-dimensional vectors $|\psi_1\rangle$ and $|\psi_2\rangle$ that satisfy:

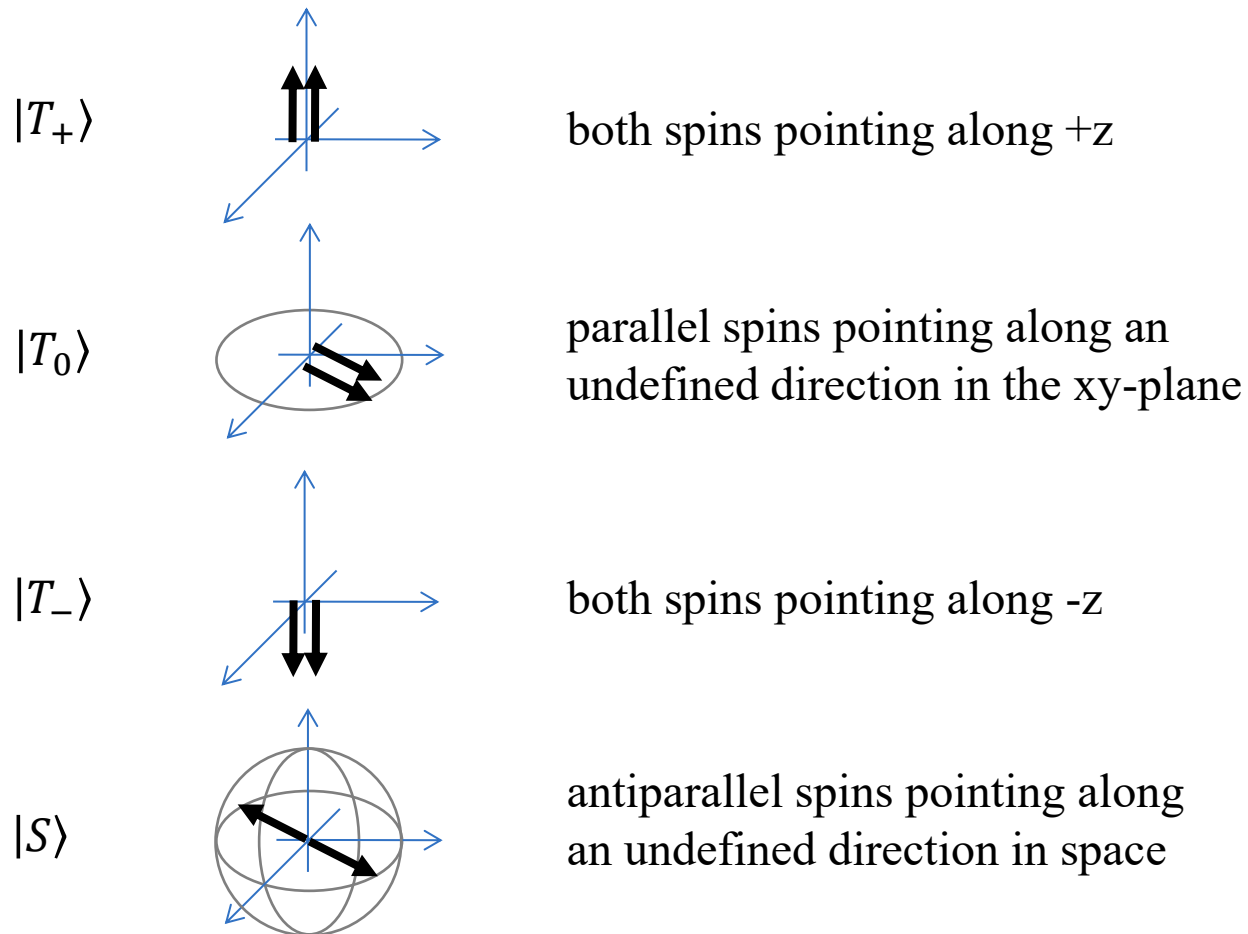
$$|S\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$$

→ this is impossible!

$|S\rangle$ and $|T_0\rangle$ are **entangled states**, that cannot be obtained as the tensor product of two single-particle states. It is not possible to attribute a defined state to the individual constituents.

Now you should understand the deep meaning of the 1st postulate of QM: for instance, in a 4-dimensional Hilbert space constructed as the tensor product of two 2-dimensional spaces, you can have legitimate states that are not tensor products of states in the 2D spaces.

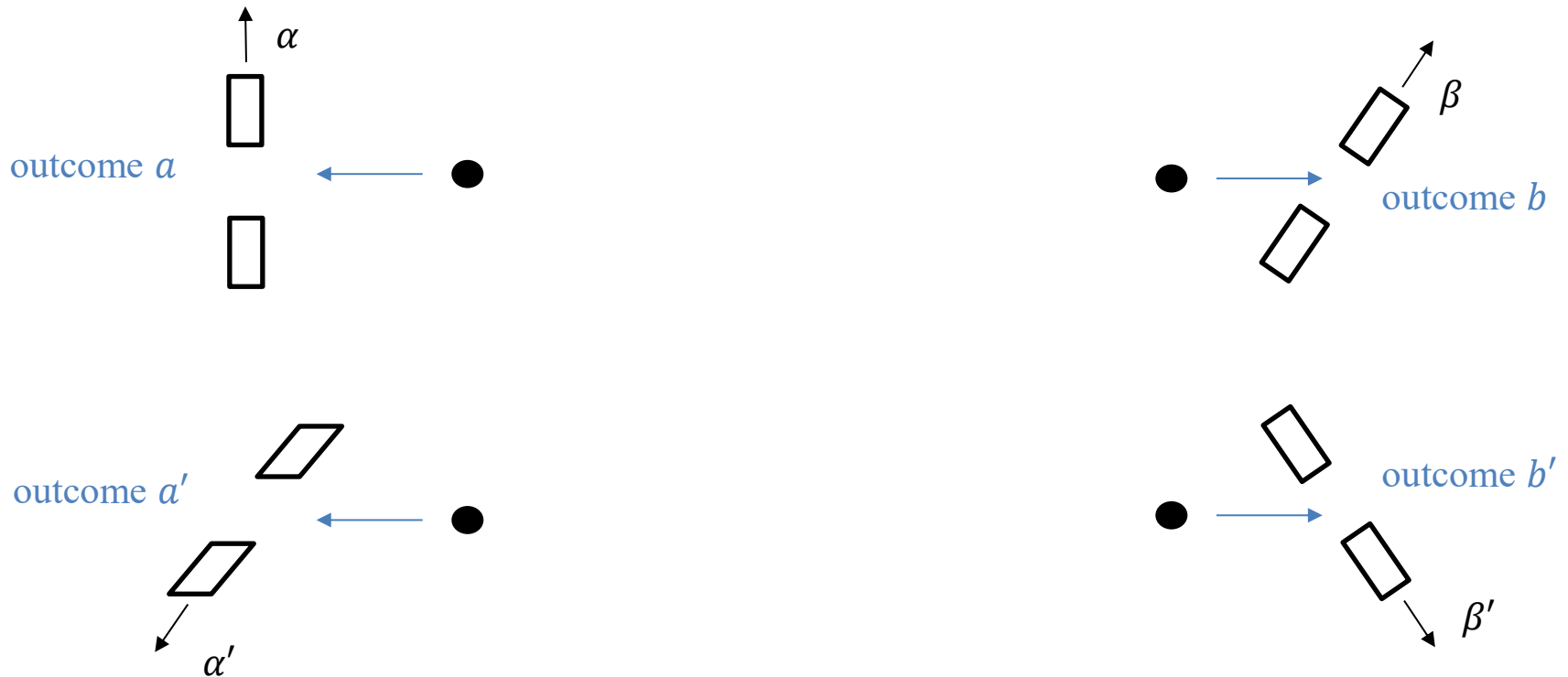
Geometric interpretation



Bell's inequality

Consider two spins, which are prepared initially in some joint state $|\psi\rangle$.

Separate them out, and measure their projection along two axes (defined as the colatitude θ) on each side: α and α' on the left, and β and β' on the right. Express them as Pauli matrices, so that the outcome of the measurement can only be $+1$ or -1 .



... and so on, for all 4 combinations

Bell's inequality

Repeat the experiment multiple times, and calculate each time the quantity:

$$M = ab - ab' + a'b + a'b' = a(b - b') + a'(b + b')$$

Important note: the measurement outcomes can only be ± 1 , so it's obvious to see that M can only take values ± 2 .

Therefore, if I repeat the experiment many times and calculate the average of M , it seems obvious to expect that:

$$-2 \leq \langle M \rangle \leq 2$$

This is known as the Clauser-Horne-Shimony-Holt (CHSH) formulation of Bell's inequality.

Violation of Bell's inequality

Assume the two spins were initially prepared in a singlet state:

$$|S\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

We wish to calculate the probability $\mathcal{P}_{1,1}$ of obtaining the outcome $a = 1, b = 1$ from the measurement of the individual spins along angles α and β .

Recall the expression (page 59) of the eigenstate of the operator S_u making an angle θ with the vertical axis, with positive eigenvalue:

$$|U_\theta\rangle = \cos\frac{\theta}{2}|\uparrow\rangle + \sin\frac{\theta}{2}|\downarrow\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2} \end{pmatrix}$$

Therefore, the probability is:

$$\mathcal{P}_{1,1} = |\langle S|U_\alpha U_\beta\rangle|^2$$

where $|U_\alpha U_\beta\rangle = |U_\alpha\rangle \otimes |U_\beta\rangle$

Violation of Bell's inequality

$$\begin{aligned}\mathcal{P}_{1,1} &= |\langle S | U_\alpha U_\beta \rangle|^2 = \frac{1}{2} \left| \begin{pmatrix} 0 & 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} \cos \frac{\alpha}{2} \cos \frac{\beta}{2} \\ \cos \frac{\alpha}{2} \sin \frac{\beta}{2} \\ \sin \frac{\alpha}{2} \cos \frac{\beta}{2} \\ \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \end{pmatrix} \right|^2 = \\ &= \frac{1}{2} \left| \cos \frac{\alpha}{2} \sin \frac{\beta}{2} - \sin \frac{\alpha}{2} \cos \frac{\beta}{2} \right|^2\end{aligned}$$

Recall the trigonometric identity:

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

Therefore:

$$\mathcal{P}_{1,1} = \frac{1}{2} \sin^2 \left(\frac{\alpha - \beta}{2} \right)$$

Violation of Bell's inequality

Similarly one finds:

$$\mathcal{P}_{-1,-1} = \frac{1}{2} \sin^2 \left(\frac{\alpha - \beta}{2} \right)$$
$$\mathcal{P}_{1,-1} = \mathcal{P}_{-1,1} = \frac{1}{2} \cos^2 \left(\frac{\alpha - \beta}{2} \right)$$

From this we can calculate the expected average of the outcomes of the measurement of the spins along directions α (on the left) and β (on the right):

$$\langle ab \rangle = \mathcal{P}_{1,1} + \mathcal{P}_{-1,-1} - \mathcal{P}_{1,-1} - \mathcal{P}_{-1,1} = -\cos(\alpha - \beta)$$

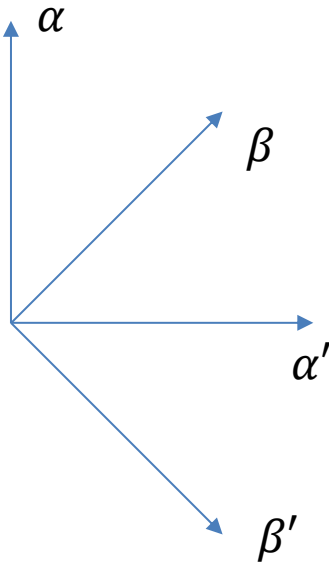
(recall that $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$)

Therefore, the Bell quantity $\langle M \rangle$ can be rewritten as:

$$\langle M \rangle = -\cos(\alpha - \beta) + \cos(\alpha - \beta') - \cos(\alpha' - \beta) - \cos(\alpha' - \beta')$$

Violation of Bell's inequality

Now choose the angles $\alpha, \alpha', \beta, \beta'$ as follows:



$$\begin{aligned} -\cos(\alpha - \beta) &= -\frac{\sqrt{2}}{2} \\ \cos(\alpha - \beta') &= -\frac{\sqrt{2}}{2} \\ -\cos(\alpha' - \beta) &= -\frac{\sqrt{2}}{2} \\ -\cos(\alpha' - \beta') &= -\frac{\sqrt{2}}{2} \end{aligned}$$

In total:

$$\langle M \rangle = -2\sqrt{2}$$

An entangled quantum state can violate Bell's inequality

2022 Nobel Prize for physics



III. Niklas Elmehed © Nobel Prize Outreach

Alain Aspect

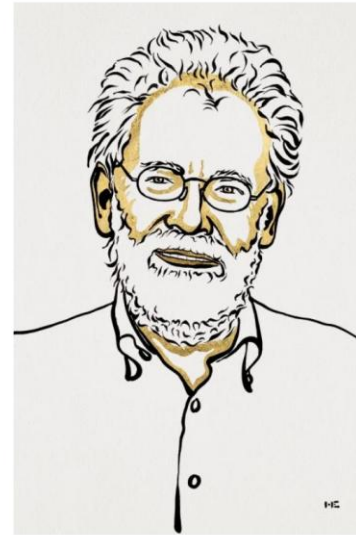
Prize share: 1/3



III. Niklas Elmehed © Nobel Prize Outreach

John F. Clauser

Prize share: 1/3



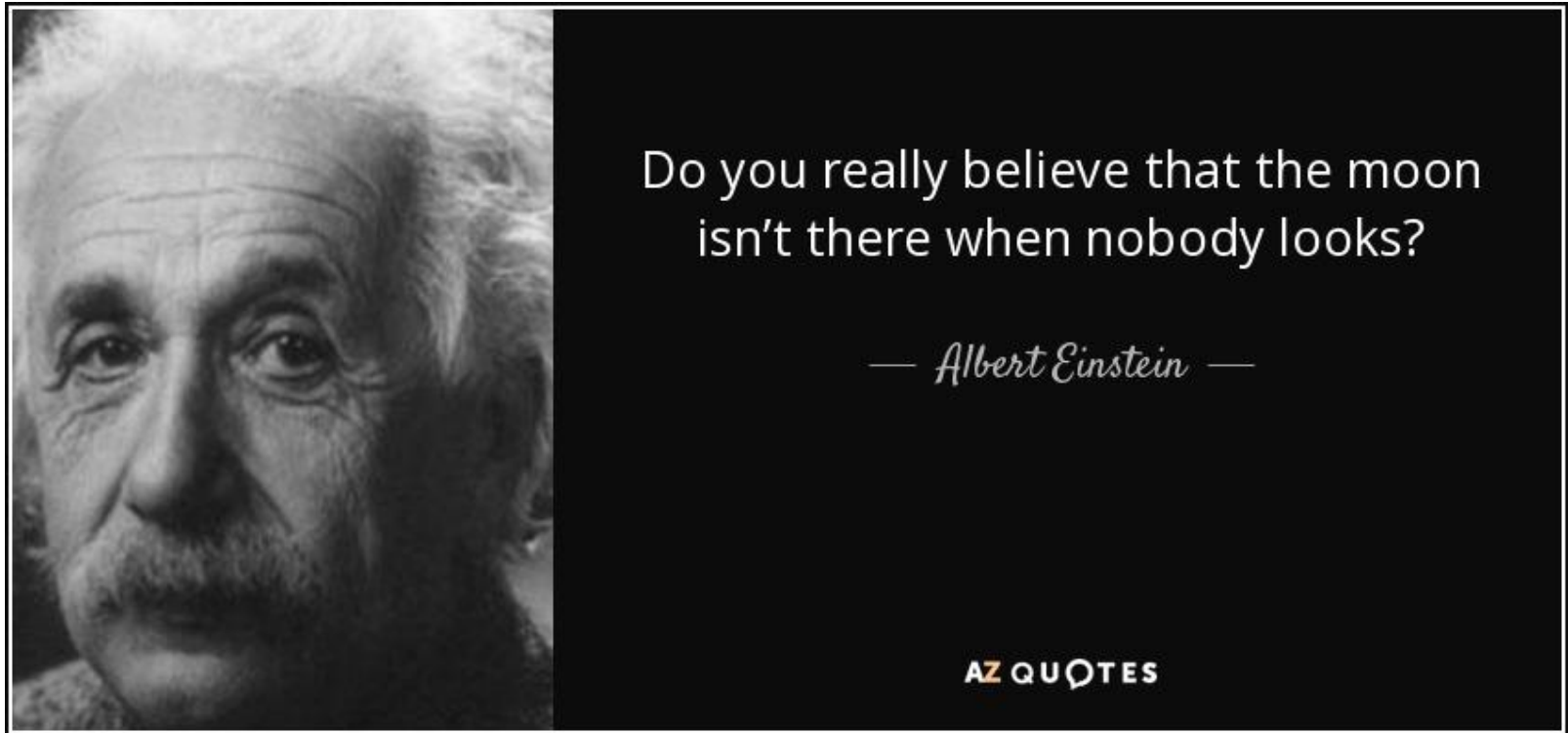
III. Niklas Elmehed © Nobel Prize Outreach

Anton Zeilinger

Prize share: 1/3

The Nobel Prize in Physics 2022 was awarded jointly to Alain Aspect, John F. Clauser and Anton Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science"

Local realism



Bell's inequality, $-2 \leq \langle M \rangle \leq 2$ relies upon three assumptions:

- locality
- realism
- free choice

Local realism

- **locality**: a physical system cannot be instantly affected by distant objects
- **realism**: the physical properties of an object exist already before they are observed.
- **free choice**: we can freely choose the measurement settings

$$\langle M \rangle = \langle ab \rangle - \langle ab' \rangle + \langle a'b \rangle + \langle a'b' \rangle$$

To obtain values > 2 from the equation above, we must admit that either:

- the physical properties of the system which give rise to the values a, a', b, b' do not pre-exist the measurement, and therefore we can't sum the values taken during different experiments (non-realism)
- the choice of measurement axis at one end instantly influences the statistics of the measurement outcomes at the other end (non-locality)
- everything, including us setting up a Bell's inequality test and seemingly randomly choosing measurement settings $\alpha, \alpha', \beta, \beta'$ was already pre-determined (super-determinism).

Multi-particle density matrices

Recall the definition of density matrix: a quantum system described by a state $|\psi\rangle$ can also be described by a density matrix

$$\rho = |\psi\rangle\langle\psi|$$

Now consider two quantum systems, A and B , described by states $|\psi_A\rangle$ and $|\psi_B\rangle$, which are vectors in vector spaces with bases:

$\{|u_m\rangle\}, m = 1 \dots M$ for system A

$\{|v_n\rangle\}, n = 1 \dots N$ for system B

The combined system will be described by the tensor product state $|\psi_C\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$, which is a state defined in a vector space having as basis $\{|w_p\rangle\}$ the tensor products of $\{|u_m\rangle\}$ and $\{|v_n\rangle\}$:

$$\{|w_p\rangle\} = \{|u_m\rangle \otimes |v_n\rangle\}, m = 1 \dots M, n = 1 \dots N \rightarrow p = 1 \dots M \times N$$

(note: we will simply write $|u_m\rangle \otimes |v_n\rangle$ as $|u_m\rangle|v_n\rangle$)

It can be shown that the density matrix describing the combined system is just the tensor product of the individual density matrices:

$$\rho^C = \rho^A \otimes \rho^B = (|\psi_A\rangle\langle\psi_A|) \otimes (|\psi_B\rangle\langle\psi_B|)$$

Partial trace

Given a tensor-product density matrix $\rho^C = \rho^A \otimes \rho^B$, is it possible to reconstruct the individual ones, ρ^A and ρ^B ?

Let us recall how to find the individual matrix elements:

$$\rho_{m,m'}^A = \langle u_m | \rho^A | u_{m'} \rangle$$

$$\rho_{n,n'}^B = \langle v_n | \rho^B | v_{n'} \rangle$$

Similarly, for the combined system:

$$\rho_{p,p'}^C = \langle w_p | \rho^C | w_{p'} \rangle = \langle u_m | \langle v_n | \rho^C | u_{m'} \rangle | v_{n'} \rangle$$

An individual density matrix is recovered by taking the **partial trace** of the combined density matrix over the basis states of the other particle:

$$\rho_{m,m'}^A = \left(\text{Tr}_B(\rho^C) \right)_{m,m'} = \sum_n \langle u_m | \langle v_n | \rho^C | u_{m'} \rangle | v_n \rangle$$

same index

$$\rho_{n,n'}^B = \left(\text{Tr}_A(\rho^C) \right)_{n,n'} = \sum_m \langle u_m | \langle v_n | \rho^C | u_m \rangle | v_{n'} \rangle$$

same index

Partial trace

The meaning of the partial trace is that it gives the state of one particle, averaged over the possible states of the other (with the proper populations).

Example:

$$|\psi_A\rangle = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix} \rightarrow \rho^A = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix} \begin{pmatrix} 0.6 & 0.8 \end{pmatrix} = \begin{pmatrix} 0.36 & 0.48 \\ 0.48 & 0.64 \end{pmatrix}$$

$$|\psi_B\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \rho^B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\rho^C = \rho^A \otimes \rho^B = \begin{pmatrix} 0.36 & 0 & 0.48 & 0 \\ 0 & 0 & 0 & 0 \\ 0.48 & 0 & 0.64 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

From this, using the partial trace:

$$\rho^B = \text{Tr}_A(\rho^C) = \begin{pmatrix} 0.36 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0.64 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

When the joint state is a product state, averaging over the other has no effect – there is no influence between the two.

Partial trace over non-product states

We have seen that in the tensor-product vector space, one can find states that are not the tensor product of single-particle states.

However, if we use the density matrix ρ instead of the state $|\psi\rangle$, we can always calculate a partial trace!

When taking the partial trace over a non-product state, one finds something that is not a pure state $\rightarrow$ mixed state.

Example: singlet state

$$|S\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} \rightarrow \rho^S = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Trace over spin A:

$$\rho^B = \text{Tr}_A(\rho^S) = \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}$$

State purity:

$$\text{Tr}[(\rho^B)^2] = \text{Tr} \begin{pmatrix} 0.25 & 0 \\ 0 & 0.25 \end{pmatrix} = 0.5 < 1$$

Entropy

The concept of entropy originates from thermodynamics and statistical mechanics. It is a measure of the “disorder” of a system.

Consider a physical system that can have N physical states, each with probability p_i of being occupied. The thermodynamic entropy is:

$$S = -k_B \sum_i p_i \log p_i$$

$S = 0$ if only one state $|j\rangle$ is occupied, i.e. $p_i = \delta_{ij}$;

$S = k_B \log N = \text{maximum}$ when $p_i = \frac{1}{N} \quad \forall i$

The concept has been extended to information theory with the Shannon entropy:

$$S = \sum_i p_i \log_2 \left(\frac{1}{p_i} \right)$$

where p_i is the probability of obtaining a piece of information.

- $\log_2(1/p_i)$ is the “information gain” when we receive the message X_i
- p_i is the probability of receiving it.

Entanglement entropy

The diagonal elements ρ_{ii} of a density matrix describe the probability p_i of finding a system in the state $|u_i\rangle$. Therefore it is logical to define the entropy of a quantum state as:

$$S = - \sum_i p_i \log p_i = -\text{Tr}(\rho \log \rho)$$

For a pure state, $S = 0$

For a maximally mixed state, $\rho = \frac{1}{N} \mathbb{I}$, $S = \log N$

Now recall that the partial trace over the density matrix of an entangled state yields a mixed state. We can therefore use the entropy of the partial trace as a measure of entanglement:

$$S = -\text{Tr}(\text{Tr}_A(\rho) \log(\text{Tr}_A(\rho)))$$

Section 8

Identical particles in quantum mechanics

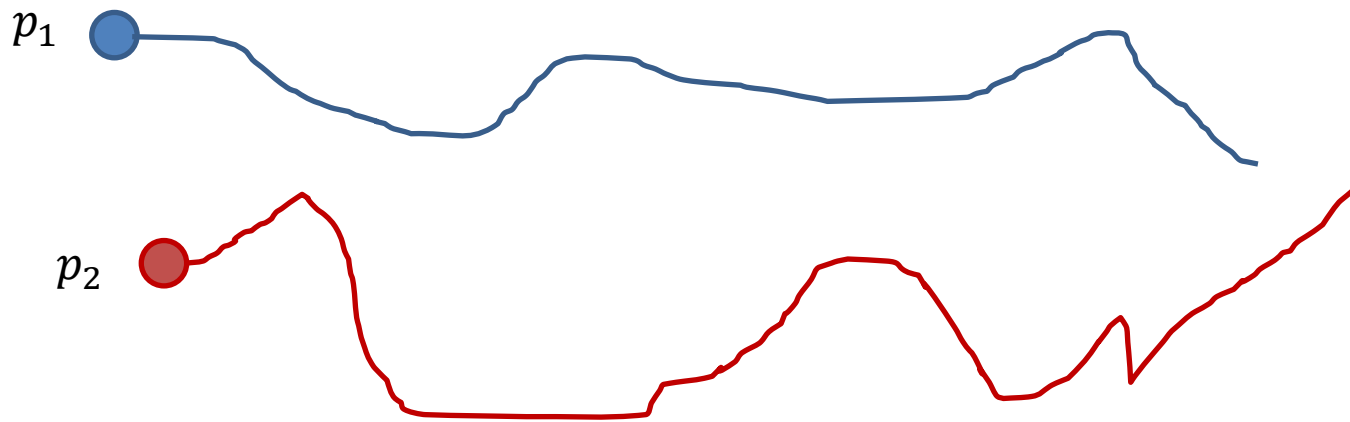
Learning outcomes:

- Understand the origin of the fundamental difference between the quantum behaviour of fermions and bosons
- Understand the physical origin of the exchange interaction

Identical particles in classical mechanics

The world is built up from basic particles (electrons, protons, neutrons, etc) which are formally identical.

In classical physics, we can in principle “label” identical particles and write a “mathematical state” where each particle is identified by some physical property, e.g. their position or velocity.

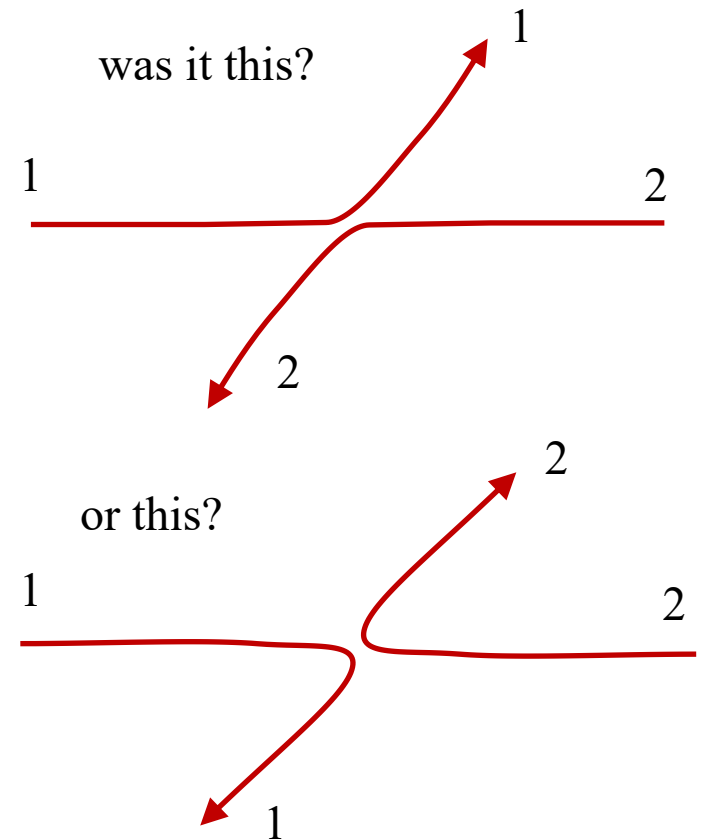
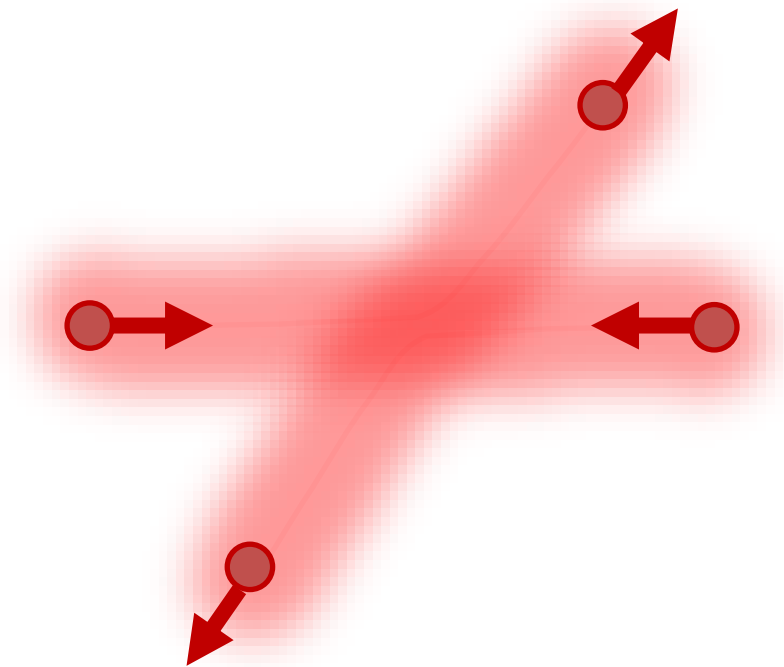


Note, however, that I can also swap the labels, $1 \leftrightarrow 2$, obtaining a different “mathematical state” that corresponds to the same “physical state”, since the particles are identical. The physical predictions won’t depend on the choice of labels.

Identical particles in quantum mechanics

In quantum physics, things are more complicated. There is no such thing as “trajectory”. Moreover, due to Heisenberg’s uncertainty principle, one cannot identify two particles which come in close proximity, closer than their position uncertainty.

Example: collision between two particles



Permutation operator

Consider two spins described by 2×2 Pauli matrices. Since their Hilbert spaces have the same dimension, each one of them can be described in the same basis $\{|u_i\rangle\}, i = 1, 2$.

I will use the notation $|u_i(1)\rangle$ for the basis vector $|u_i\rangle$ in the Hilbert space of spin 1, etc.

The basis for the combined systems is obtained from the tensor product:

$$\underbrace{\{|u_i(1)\rangle \otimes |u_j(2)\rangle\}}_{|u_i(1)u_j(2)\rangle}, i = 1, 2 ; j = 1, 2$$

short notation

The **permutation operator** P_{21} is defined by its action on the basis vectors:

$$P_{21}|u_i(1)u_j(2)\rangle = |u_i(2)u_j(1)\rangle \equiv |u_j(1)u_i(2)\rangle$$

its effect is to swap the state of spin 2 with that of spin 1.

Note: this is true in general, i.e. I have not assumed that the particles are identical (they could be an electron and a proton, for example).

Notes on permutation

In general, the tensor product is not commutative:

$$A \otimes B \neq B \otimes A$$

Therefore, formally, $|u_i(1)u_j(2)\rangle \neq |u_j(2)u_i(1)\rangle$

However, this difference does not correspond to a different physical state: it's just a difference in the order in which I did the tensor product to create the basis of the composite state! Both tensor products describe spin 1 in state $|u_i\rangle$ and spin 2 in state $|u_j\rangle$.

Moreover, if the particles are identical you might think that $|u_i(1)u_j(2)\rangle$ is equivalent to $|u_j(1)u_i(2)\rangle$ because that simply amounts to relabeling the particles, i.e. you still have one particle in state $|u_i\rangle$ and the other in state $|u_j\rangle$.

Conversely, imagine that spin 1 and spin 2 are not identical particles, e.g. spin 1 is an electron and spin 2 is a proton, so they are physically distinguishable, although they both can be described with the same basis vectors. Then, the states $|u_i(1)u_j(2)\rangle$ and $|u_j(1)u_i(2)\rangle$ really are different!

Thus, in general:

$$|u_i(1)u_j(2)\rangle \equiv |u_j(2)u_i(1)\rangle$$

$$|u_i(1)u_j(2)\rangle \neq |u_j(1)u_i(2)\rangle$$

Example of permutation operator

Consider two spins, each described in the usual basis $|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

and the joint basis $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |\uparrow\uparrow\rangle$, $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |\uparrow\downarrow\rangle$, $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |\downarrow\uparrow\rangle$, $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |\downarrow\downarrow\rangle$

The action of the permutation operator will be:

$$P_{21}|\uparrow\uparrow\rangle = |\uparrow\uparrow\rangle ; \quad P_{21}|\uparrow\downarrow\rangle = |\downarrow\uparrow\rangle ; \quad P_{21}|\downarrow\uparrow\rangle = |\uparrow\downarrow\rangle ; \quad P_{21}|\downarrow\downarrow\rangle = |\downarrow\downarrow\rangle ;$$

Therefore:

$$P_{21} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Properties of the permutation operator

- P_{21} is its own inverse, since obviously $P_{21}^2 = \mathbb{I}$
- P_{21} is Hermitian: $P_{21} = P_{21}^\dagger$
- P_{21} is unitary: $P_{21}^\dagger P_{21} = \mathbb{I}$
- The eigenvalues of P_{21} are $+1$ or -1 , since their squares are 1
- There are two types of eigenvectors for P_{21} :
 - symmetric* eigenvectors $P_{21}|\psi_S\rangle = |\psi_S\rangle$
 - antisymmetric* eigenvectors: $P_{21}|\psi_A\rangle = -|\psi_A\rangle$

From the permutation operator, we can construct the following operators:

$$S = \frac{1}{2}(\mathbb{I} + P_{21}) \quad \text{"symmetrizer"}$$

$$A = \frac{1}{2}(\mathbb{I} - P_{21}) \quad \text{"antisymmetrizer"}$$

they have the property that:

$$P_{21}S|\psi\rangle = S|\psi\rangle$$

$$P_{21}A|\psi\rangle = -A|\psi\rangle$$

Symmetrization postulate

When dealing with multiple identical particles, we need to add an extra postulate to quantum mechanics, to account for the fact that we cannot meaningfully describe indistinguishable particles with simple pure states.

Symmetrization postulate:

When a system consists of several identical particles, only certain kets can describe the physical state of the system. The physical kets must be either completely symmetric or completely antisymmetric, depending on the nature of the particle.

- The particles that require symmetric physical kets are called **bosons**;
- The particles that require antisymmetric physical kets are called **fermions**;

Empirical rule:

- bosons have integer spin $(0, 1, 2, \dots)$
- fermions have half-integer spin $(\frac{1}{2}, \frac{3}{2}, \dots)$

Construction of the physical kets

From the symmetrization postulate, and the properties of the permutation operator, we can define a set of rules to construct the physical kets:

- 1) Number the particles in a convenient way, and construct the desired ket using tensor products, etc.
- 2) For bosons: apply the symmetrizer operator S
For fermions: apply the antisymmetrizer operator A
- 3) normalize the ket so obtained

Example: consider two particles, the first described by $|\varphi\rangle$ and the second by $|\chi\rangle$

- 1) tensor product ket: $|\psi\rangle = |\varphi(1)\rangle \otimes |\chi(2)\rangle \equiv |\varphi(1)\chi(2)\rangle$
- 2) For bosons:

$$S|\psi\rangle = \frac{1}{2}(\mathbb{I} + P_{21})|\psi\rangle = \frac{1}{2}(|\varphi(1)\chi(2)\rangle + |\varphi(2)\chi(1)\rangle)$$

For fermions:

$$A|\psi\rangle = \frac{1}{2}(\mathbb{I} - P_{21})|\psi\rangle = \frac{1}{2}(|\varphi(1)\chi(2)\rangle - |\varphi(2)\chi(1)\rangle)$$

- 3) Normalize as necessary

Pauli exclusion principle

Consider the case where the two particles are in the same state, $|\varphi(1)\rangle$ and $|\varphi(2)\rangle$

1) tensor product ket: $|\psi\rangle = |\varphi(1)\rangle \otimes |\varphi(2)\rangle \equiv |\varphi(1)\varphi(2)\rangle$

2) For bosons:

$$S|\psi\rangle = \frac{1}{2}(\mathbb{I} + P_{21})|\psi\rangle = \frac{1}{2}(|\varphi(1)\varphi(2)\rangle + |\varphi(2)\varphi(1)\rangle) = |\varphi(1)\varphi(2)\rangle$$

i.e. we get the simple tensor product of the two states, since it is already symmetric.

However, for fermions:

$$A|\psi\rangle = \frac{1}{2}(\mathbb{I} - P_{21})|\psi\rangle = \frac{1}{2}(|\varphi(1)\varphi(2)\rangle - |\varphi(2)\varphi(1)\rangle) = 0$$

i.e. this leads to an absurd result.

→ two identical fermions cannot be in the same individual quantum state

“Pauli exclusion principle”

Why do we need the symmetrization postulate?

The necessity of introducing the symmetrization postulate comes from the observation that, without it, one would come at some physically absurd results in certain cases.

Example: consider two electron spins in a magnetic field $B_0 \parallel z$. Assume the electrons are very close to each other, so they cannot be distinguished (“labelled”) by their position.

The Hamiltonian is of the form:

$$H = \frac{\gamma B_0}{2} (\sigma_{z1} + \sigma_{z2}) = \gamma B_0 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

in the usual basis. The Hamiltonian is diagonal and its eigenvalues are obviously:

$$\{\gamma B_0, 0, 0, -\gamma B_0\}$$

and the eigenvectors could be:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |\uparrow\uparrow\rangle, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |\uparrow\downarrow\rangle, \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |\downarrow\uparrow\rangle, \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |\downarrow\downarrow\rangle$$

Degenerate eigenvalues

Note: the 0 eigenstate is doubly-degenerate, i.e. there are two eigenvectors associated with it. I have chosen to write them as $|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle$ but that's totally arbitrary. I could have chosen any combination of them, of the form:

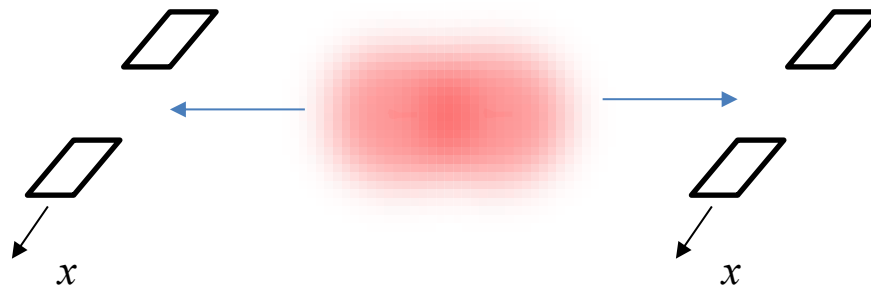
$$|\psi_1\rangle = \alpha_1|\uparrow\downarrow\rangle + \beta_1|\downarrow\uparrow\rangle, \quad |\psi_2\rangle = \alpha_2|\uparrow\downarrow\rangle + \beta_2|\downarrow\uparrow\rangle$$

as long as $|\alpha_i|^2 + |\beta_i|^2 = 1$, and the two vectors are orthogonal.

These two vectors would be a legitimate orthogonal pair of eigenvectors with eigenvalue 0. Therefore, I would be led to think that any of these kets constitute a legitimate mathematical description of the physical state.

Problem: The predictions on the outcome of the measurement of physical observables depend on the choice of coefficients!!

Say I want to pass these two spins through a pair of Stern-Gerlach systems, oriented along $+x$. What is the probability of getting $+\hbar/2$ in both measurements?



Exchange degeneracy

Choosing e.g. $|\psi_1\rangle$, the probability of getting $+\hbar/2$ in both measurements is:

$$P_{++} = |\langle XX|\psi_1\rangle|^2$$

where

$$|XX\rangle = |X\rangle \otimes |X\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Therefore

$$P_{++} = \frac{1}{4} \left| \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ \alpha_1 \\ \beta_1 \\ 0 \end{pmatrix} \right|^2 = \frac{1}{4} |\alpha_1 + \beta_1|^2$$

which can vary between 0 and 1/4 depending on the choice of coefficients!!

Therefore, we cannot freely choose the coefficients α and β – this “**exchange degeneracy**” must be removed, by imposing some specific choice of coefficients.

This is precisely what the symmetrisation postulate does: it imposes a choice of coefficients by requiring that the mathematical state obeys a certain symmetry.

Antisymmetrized state

Since electrons are fermions, we need to make an antisymmetric state. The antisymmetrizer in 4 dimension is:

$$A = \frac{1}{2}(\mathbb{I} - P_{21}) = \frac{1}{2} \left[\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Start for example from the state $|\uparrow\downarrow\rangle$:

$$A|\uparrow\downarrow\rangle = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

This state is not normalized yet, we need to do it by hand (“step 3” on page 197):

$$|\psi_A\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

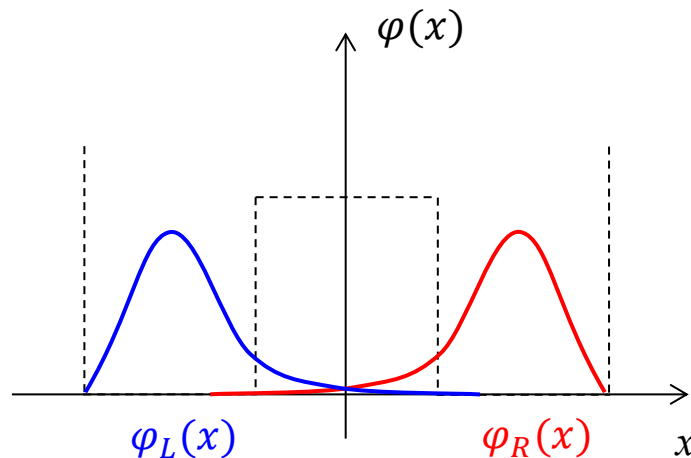
Here you recognize the well-known singlet state.

But, there’s still a problem... we also need *another* antisymmetric state with eigenvalue 0, and orthogonal to this one!

Orbital wavefunction

We have omitted an important (and pretty obvious) fact so far: an electron is not only described by its spin, it's also described by its position (i.e. orbital wavefunction)!

Simple example: electrons held in adjacent potential wells, with a thin, finite barrier in between, so that their wavefunctions overlap. I can use basis functions $|\varphi_L\rangle$, $|\varphi_R\rangle$ to describe the position of each electron



The symmetrization postulate for fermions says that the **total** mathematical state must be antisymmetric upon permutation of particles. This must include the orbital component!

Total antisymmetrized states

From the basis states $|\varphi_L\rangle, |\varphi_R\rangle, |\uparrow\rangle, |\downarrow\rangle$ I can construct 16 basis states for the tensor product Hilbert space of the two electrons, which now will have basis states like:

- $|\varphi_L\rangle \otimes |\varphi_L\rangle \otimes |\uparrow\rangle \otimes |\uparrow\rangle \rightarrow$ “electron 1 on the left with spin up, electron 2 on the left with spin up”, or
- $|\varphi_L\rangle \otimes |\varphi_R\rangle \otimes |\uparrow\rangle \otimes |\downarrow\rangle \rightarrow$ “electron 1 on the left with spin up, electron 2 on the right with spin down”, etc.

Most of these are not legitimate states though! They must be antisymmetrized.

Key point:

If the spin part is antisymmetric, then the total wavefunction will be antisymmetric if the orbital part is symmetric, and the other way around.

Example: if the spins are in a singlet state, we want the orbital wavefunction to be the symmetric combination of left and right (I ignore the normalization factor here):

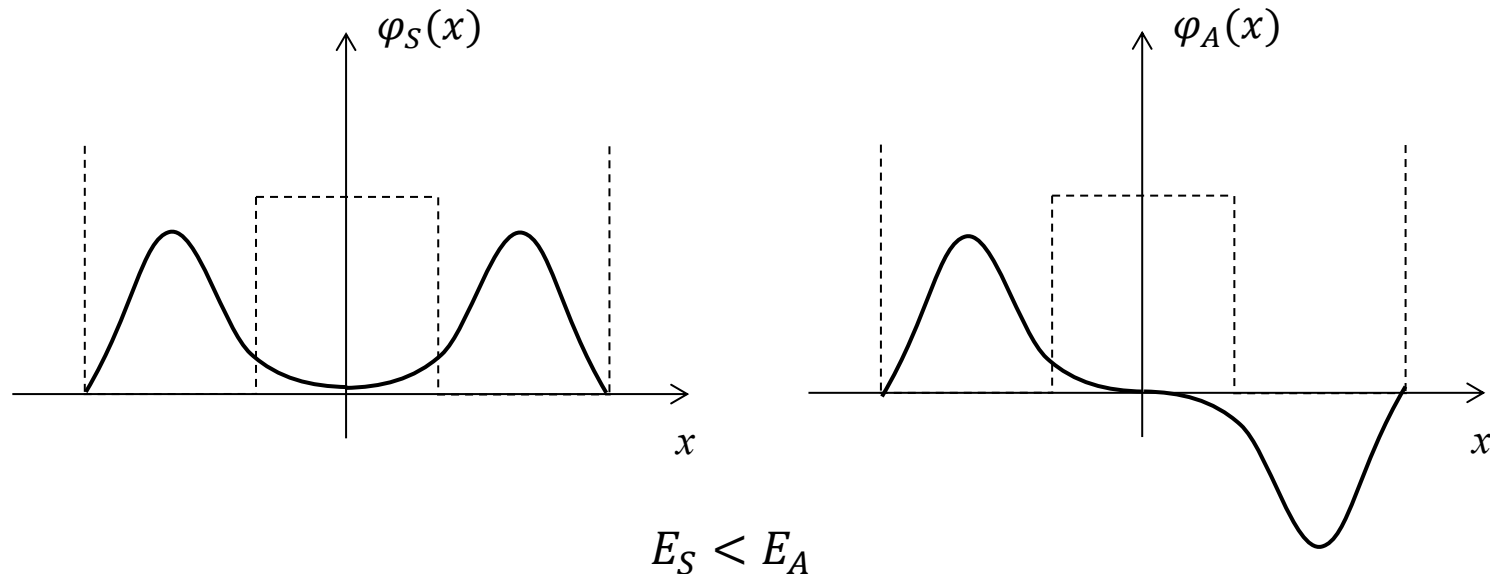
$$|\Psi_S\rangle = (\underbrace{|\varphi_L\varphi_R\rangle + |\varphi_R\varphi_L\rangle}_{|\varphi_S\rangle}) \otimes (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

Other possibility, if the spins are in a triplet state (which is spin-symmetric):

$$|\Psi_T\rangle = (\underbrace{|\varphi_L\varphi_R\rangle - |\varphi_R\varphi_L\rangle}_{|\varphi_A\rangle}) \otimes (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \quad \rightarrow \text{this is the second state we were looking for}$$

Exchange coupling

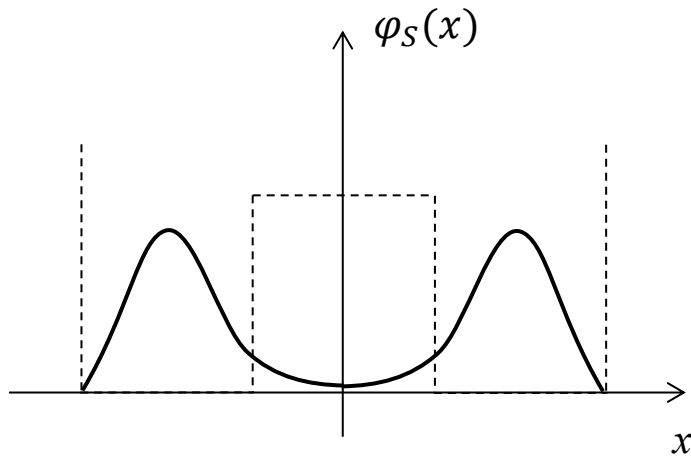
Recall the ammonia molecule: the symmetric and antisymmetric orbital combinations have different energy, due to the different amount of wavefunction going under the barrier, once you impose the continuity condition:



Since the spin part of the quantum states $(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ and $(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$ contributes the same energy (zero), **the total states $|\Psi_S\rangle$ and $|\Psi_T\rangle$ differ in energy by $E_A - E_S$**
→ Exchange coupling

Exchange coupling

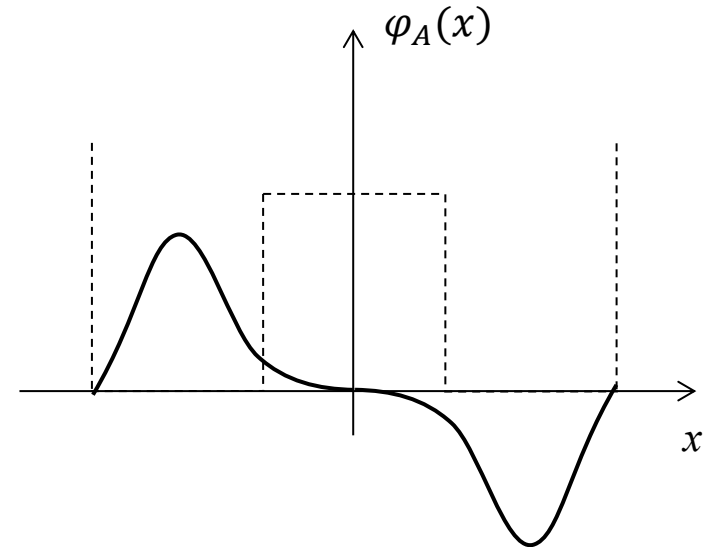
Note: the states $(|\varphi_L\varphi_R\rangle - |\varphi_R\varphi_L\rangle) \otimes |\uparrow\uparrow\rangle$ and $(|\varphi_L\varphi_R\rangle - |\varphi_R\varphi_L\rangle) \otimes |\downarrow\downarrow\rangle$ are also appropriately antisymmetric. In the absence of a magnetic field, we have:



$$(|\varphi_L\varphi_R\rangle + |\varphi_R\varphi_L\rangle) \otimes (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

Energy E_S

Singlet state



$$(|\varphi_L\varphi_R\rangle - |\varphi_R\varphi_L\rangle) \otimes \begin{cases} |\uparrow\uparrow\rangle \\ (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \\ |\downarrow\downarrow\rangle \end{cases}$$

Energy E_A

Triplet states

Exchange Hamiltonian

We now know the physical origin of the exchange Hamiltonian, which we introduced already on page 168:

$$H = J \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$$

Although this Hamiltonian appears to depend only on the spin components, the exchange coupling originates from the orbital component of the electrons.

Since the singlet-triplet energy splitting is $4J$, we simply have

$$J = \frac{E_A - E_S}{4}$$

The reason we can write it this way, without explicitly including the orbital degrees of freedom, is because there is a one-to-one relation between orbital and spin states, imposed by the symmetrization postulate.

Section 9

Atoms, bonds and crystals

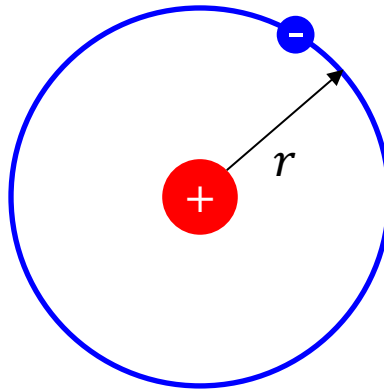
Learning outcomes:

- Understand the quantum origin of the properties of the simplest atomic system, the hydrogen atom.
- Understand the quantum origin of the covalent chemical bond.
- Discover how to derive, with a very simple model, the “band gap” between valence and conduction bands, which constitutes the basic physical property of all semiconductors.

Hydrogen atom

The study of atomic spectra gave a great impetus to the development of quantum mechanics. Many intuitions, later formalized into postulates, come from that (e.g. de Broglie wavelength, Pauli exclusion principle, ...)

The crudest classical picture of a hydrogen atom is a proton, charge $+q$, with an electron, charge $-q$ and mass m_0 , orbiting around it at distance r :



The Coulomb attractive force between electron and proton is:

$$F_q(r) = \frac{q^2}{4\pi\epsilon_0 r^2}$$

Hydrogen atom

Equilibrium orbit: the Coulomb attraction equals the centripetal force $m_0 v^2 / r$

$$\frac{q^2}{4\pi\epsilon_0 r^2} = \frac{m_0 v^2}{r} \Rightarrow v = \sqrt{\frac{q^2}{4\pi\epsilon_0 m_0 r}}$$

The total energy of the atom is the sum of the kinetic energy (we assume the proton doesn't move, since it's so much heavier), and the Coulomb interaction energy:

$$E = \underbrace{\frac{1}{2} m_0 v^2}_{\text{kinetic}} - \underbrace{\frac{q^2}{4\pi\epsilon_0 r}}_{\text{Coulomb}} = -\frac{q^2}{8\pi\epsilon_0 r}$$

Note the – sign,
since the force
is attractive

Problem: the description above is complete nonsense!

It's OK with gravitational forces, but a charged particle moving on a curved orbit emits electromagnetic energy, so the electron would inexorably slow down and spiral in towards the nucleus.

de Broglie hypothesis

In 1924, Louis de Broglie hypothesized that, to every “particle”, is associated a “wave” with properties (frequency ω , wave number $k = 2\pi/\lambda$) that obey the Planck-Einstein relations (which, themselves, originated from the study of the photoelectric effect):

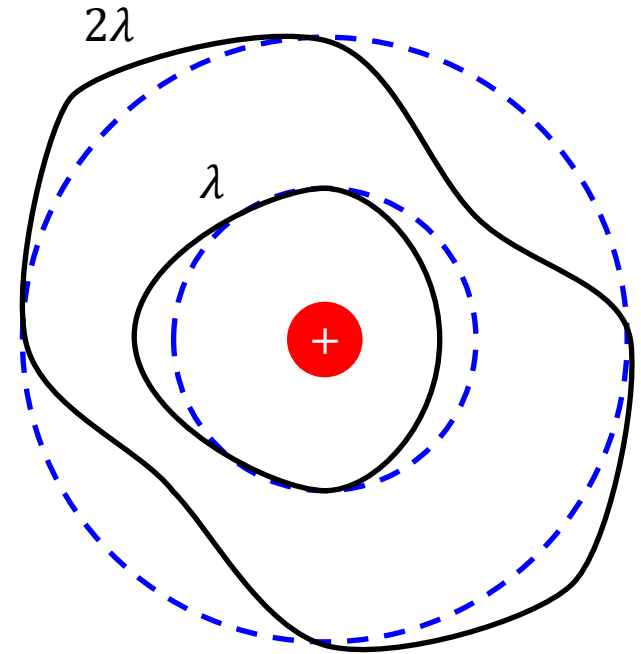
$$\omega = \frac{E}{\hbar}, \quad k = \frac{p}{\hbar}$$

Since the momentum is $p = m_0 v$ and the kinetic energy is $E_k = \frac{1}{2} m_0 v^2$, the wavelength is:

$$\lambda = \frac{h}{m_0 v} = \frac{h}{\sqrt{2m_0 E_k}}$$

Idea: the stable atomic orbits are the ones where the electron does an integer multiple of de Broglie wavelengths around the nucleus!

$$2\pi r = n\lambda = n \frac{h}{m_0 v}$$



Bohr orbits

Combining this quantization condition with the value of the velocity on a stable orbit from page 210, we obtain:

$$2\pi r_n = \frac{nh}{m_0 v} = \frac{nh}{m_0 q} \sqrt{4\pi\epsilon_0 m_0 r_n}$$
$$r_n = n^2 a_0$$

where

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_0 q^2} = 0.529 \text{ \AA}$$

is the **Bohr radius**.

From this, the energy of the n -th orbit is:

$$E_n = -\frac{q^2}{8\pi\epsilon_0 r_n} = -\frac{1}{n^2} E_0$$

where

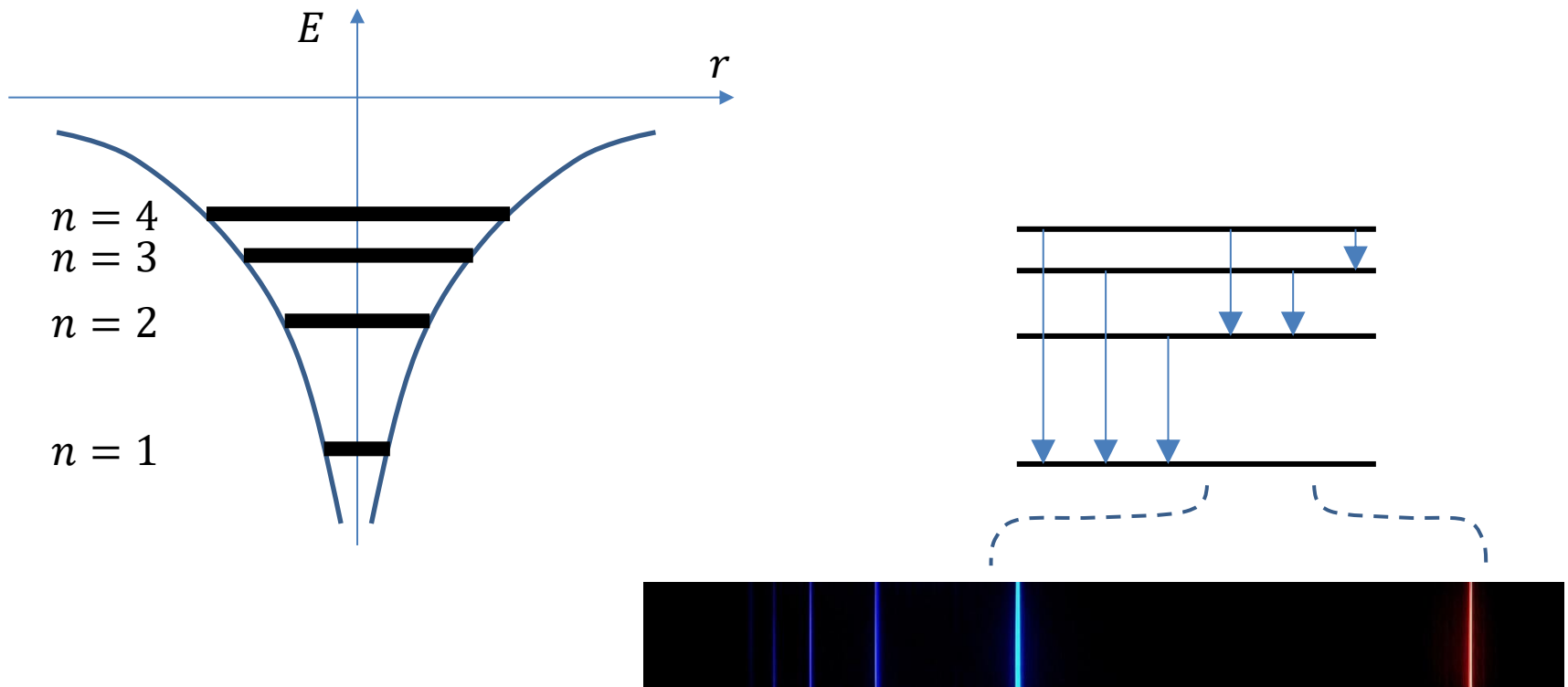
$$E_0 = \frac{q^2}{8\pi\epsilon_0 a_0} = 13.6 \text{ eV}$$

is the **Rydberg constant**.

Photon emission

One of the earliest experimental tests of quantum mechanics was the observation that the spectrum of light emitted by hydrogen atoms has discrete frequencies, given by:

$$\hbar\omega_{mn} = E_m - E_n = E_0 \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$



Example: Balmer spectrum ($n = 2, m > 2$)

Quantum mechanical treatment

The proper way to find the orbital eigenstates of the hydrogen atom is to solve the (time-independent) Schrodinger equation:

$$\left(-\frac{\hbar^2}{2m_0}\nabla^2 + V(r)\right)\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

where $V(r)$ is the Coulomb potential. It's best to write it in spherical coordinates:

$$\nabla^2\psi(r, \theta, \phi) + \frac{2m_0}{\hbar^2}(E - V(r))\psi(r, \theta, \phi) = 0$$

We seek a factorizable solution of the form:

$$\psi(r, \theta, \phi) = R(r) \cdot \Theta(\theta) \cdot \Phi(\phi)$$

Writing the Laplacian in spherical coordinates and expanding the terms we get:

$$\frac{\sin^2 \theta}{R} \cdot \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{1}{\Phi} \cdot \frac{\partial^2 \Phi}{\partial \phi^2} + \frac{\sin \theta}{\Theta} \cdot \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Theta}{\partial \theta} \right) + r^2 \sin^2 \theta \cdot \frac{2m_0}{\hbar^2} (E - V) = 0$$

This equation can be solved by the method of “separation of variables”: since there are terms that depend on some variables but not others, they must all be a constant for the equation to hold.

Quantum numbers

Take for example the second term: it's a function of ϕ only, so it must be a constant, which we set equal to $-m^2$, where m is the “magnetic quantum number”:

$$\frac{1}{\Phi} \cdot \frac{\partial^2 \Phi}{\partial \phi^2} = -m^2$$

Similarly we can separate the equations for r and θ , which will result in two other constant, labelled n and l .

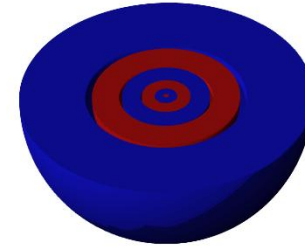
- $n = 1, 2, 3, \dots$ **principal quantum number:**
represents the energy level $E_n = -\frac{1}{n^2} E_0$.
The group of orbitals having the same value of n is called shell
- $l = 0, 1, \dots n - 1$ **azimuthal quantum number:**
represents the angular momentum of the electron orbit, in units of $\hbar$
The group of orbitals having the same value of l is called subshell
- $m = -l, -l + 1, \dots l - 1, l$ **magnetic quantum number:**
represents the z-projection of the angular momentum (hence $-l \leq m \leq l$)

Overall, the solutions of the Schroedinger equation for an electron in a Coulomb potential are complicated, but analytically solvable, functions of r, θ, ϕ identified by the quantum numbers n, l, m .

Atomic orbitals

- $l = 0 \rightarrow$ **s orbitals**

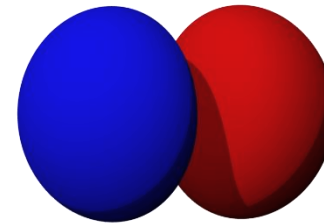
zero angular momentum ($L = 0$), and are therefore spherically symmetric around the nucleus



6s

- $l = 1 \rightarrow$ **p orbitals**

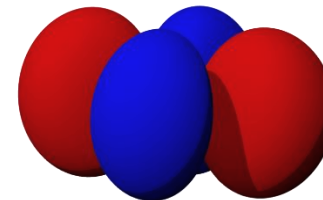
$L = \hbar$, and its projection along z can be $-\hbar, 0, \hbar$ (identified by the m quantum number)



2p

- $l = 2 \rightarrow$ **d orbitals**

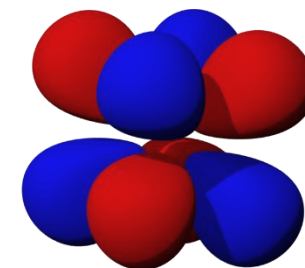
$L = 2\hbar$, and its projection along z can be $2\hbar, -\hbar, 0, \hbar, 2\hbar$



3d

- $l = 3 \rightarrow$ **f orbitals**

$L = 3\hbar$, etc...



4f

Atomic orbitals

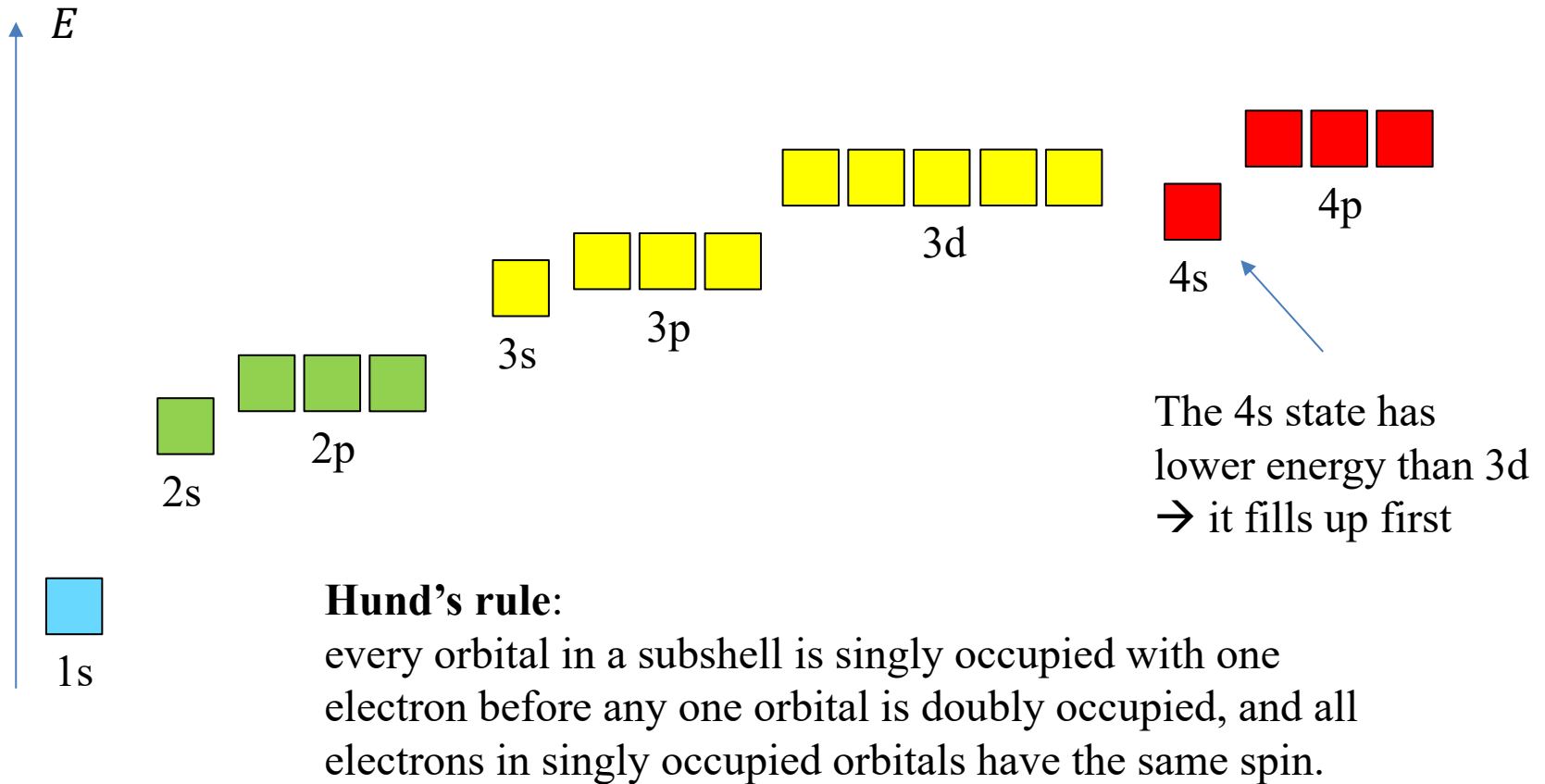
	s ($\ell = 0$)	p ($\ell = 1$)			d ($\ell = 2$)					f ($\ell = 3$)						
	$m = 0$	$m = 0$	$m = \pm 1$		$m = 0$	$m = \pm 1$		$m = \pm 2$		$m = 0$	$m = \pm 1$		$m = \pm 2$		$m = \pm 3$	
	s	p _z	p _x	p _y	d _{z²}	d _{xz}	d _{yz}	d _{xy}	d _{x²-y²}	f _{z³}	f _{xz²}	f _{yz²}	f _{xyz}	f _{z(x²-y²)}	f _{x(x²-3y²)}	f _{y(3x²-y²)}
$n = 1$																
$n = 2$																
$n = 3$																
$n = 4$																
$n = 5$										...	...	...	...	...	...	...
$n = 6$					...	...	...	...	...	...	...	...	...	...	...	...
$n = 7$		...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

Note: each electron also has a spin! This yields one more quantum number:

- $m_s = \pm \frac{1}{2}$ **spin quantum number**

Aufbau principles

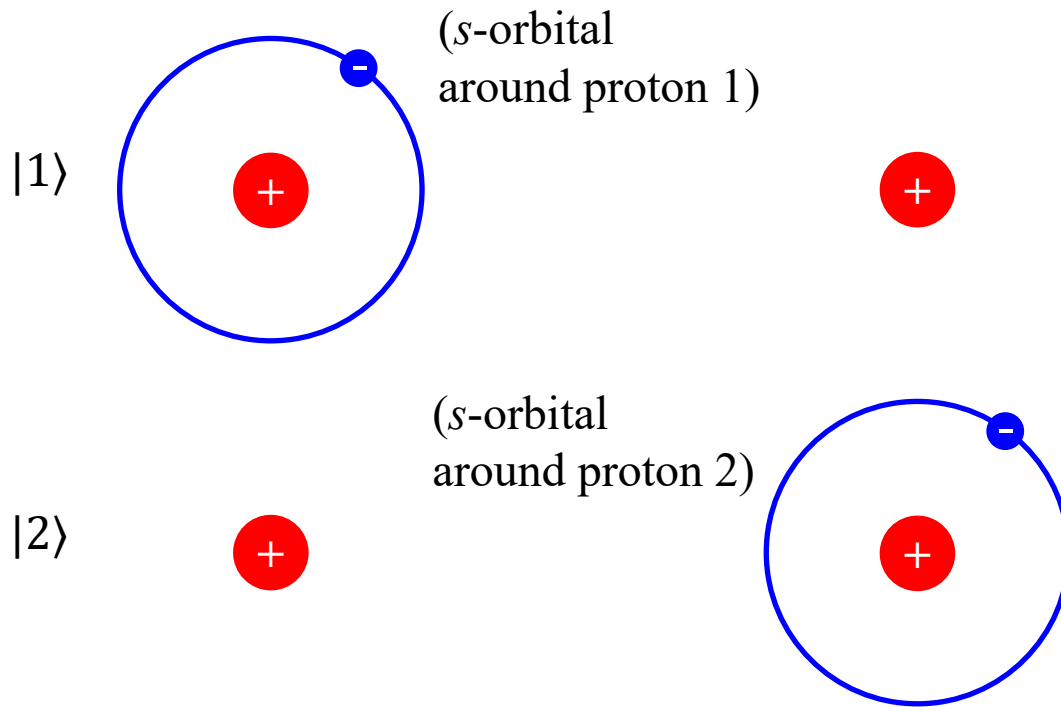
Pauli exclusion principle: no two electrons can occupy the same quantum state. In multi-electron atoms, the orbitals are filled up starting from the lowest energy. Each orbital can take two electrons, in opposite spin state (singlet state!)



H_2^+ molecule

Consider two protons and one electron. The protons are separated by distance R . What are the lowest energy states of this system?

- **Large distance** ($R \gg a_0$)



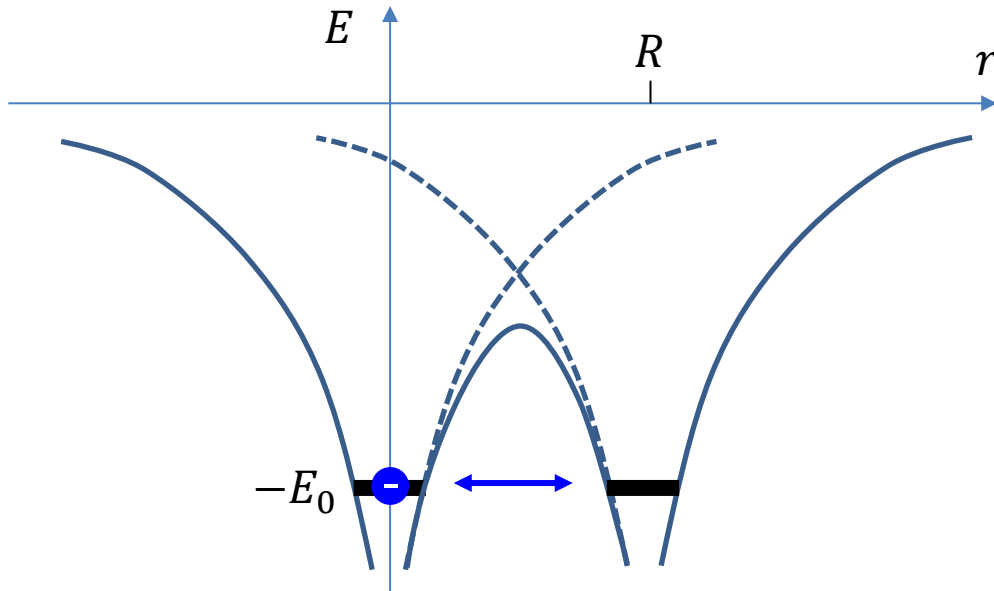
The electron will bind to one of the protons, while the other proton sits by itself.

Total energy:
 $-E_0 = -13.6 \text{ eV}$

States $|1\rangle$ and $|2\rangle$ have the same energy

H₂⁺ molecule

- Short distance ($R \sim a_0$)

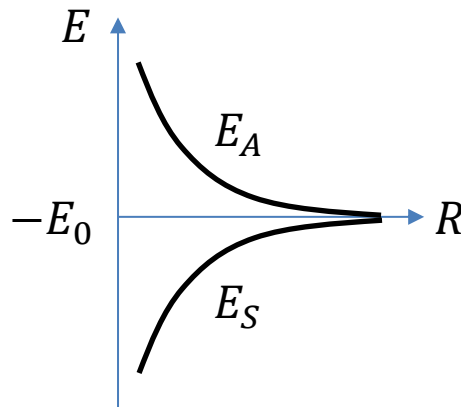


The electron can tunnel from one proton to the other. Let's call the tunnelling frequency

$$\omega_t = \frac{2\Delta}{\hbar}$$

In the basis of the states $|1\rangle$ and $|2\rangle$, the Hamiltonian is:

$$\mathcal{H} = \begin{pmatrix} -E_0 & -\Delta \\ -\Delta & -E_0 \end{pmatrix} = -E_0 \mathbb{I} - \Delta \sigma_x$$



Eigenstates:

$$|\phi_S\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle); \quad |\phi_A\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |2\rangle);$$

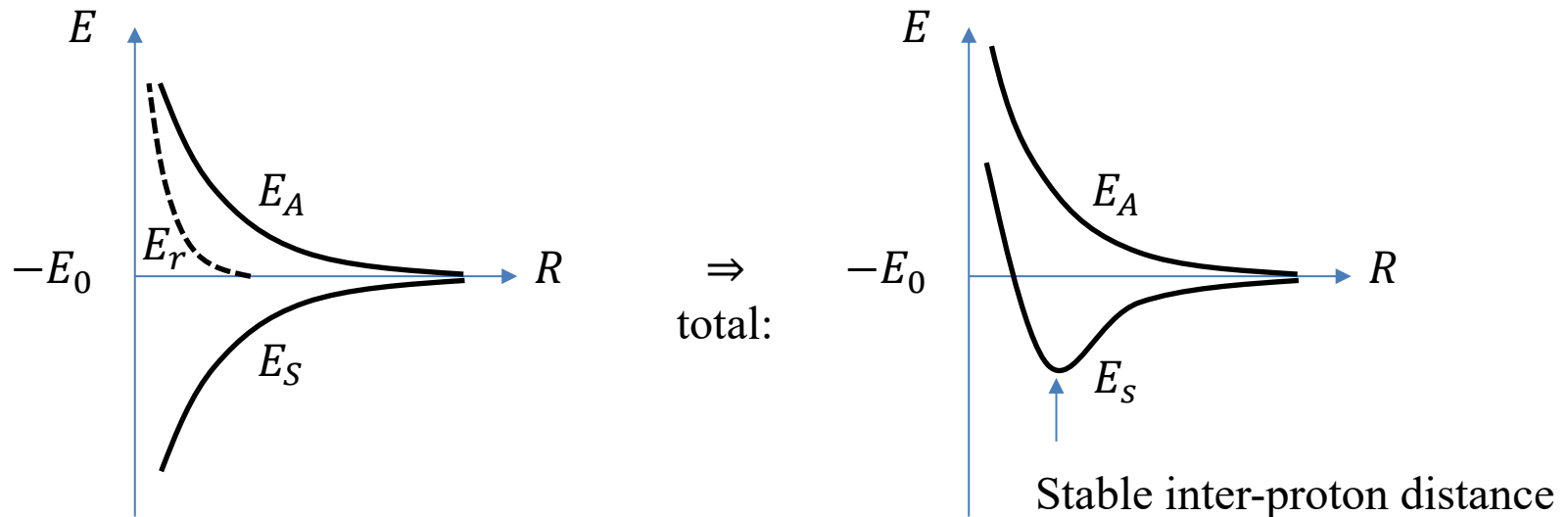
Eigenenergies:

$$E_S = -E_0 - \Delta; \quad E_A = -E_0 + \Delta$$

Chemical bond

In the preceding discussion, we neglected an important term: the Coulomb repulsion between the protons.

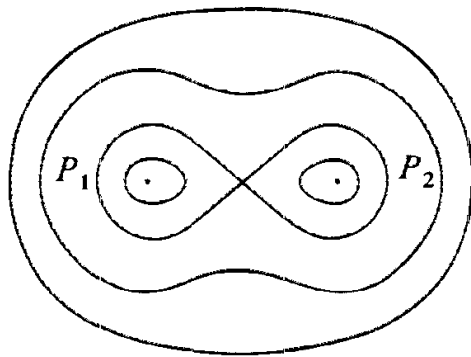
For $R \gg a_0$, the proton with an electron bound to it will appear like a neutral system to the other proton (the electron screens the positive charge), so there's no repulsion at all. But, when $R \sim a_0$, the electron (in a s -orbital of radius a_0) no longer fully screens, so there start to be a repulsion, which contributes a positive energy $E_r(R)$



By sharing an electron, the system can attain a lower energy than when the electron is located around just one of the protons $\rightarrow$ **chemical bond**

Bonding / antibonding orbitals

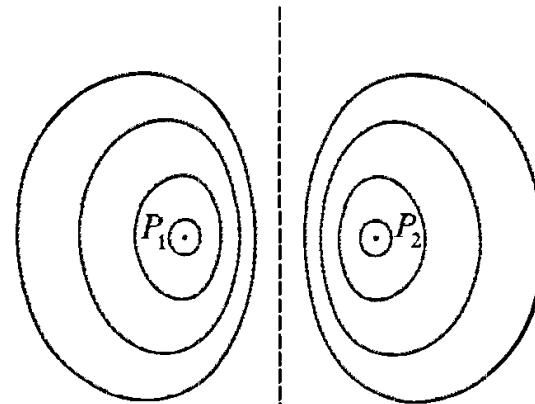
$$|\phi_S\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle)$$



+

“bonding” orbital

$$|\phi_A\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |2\rangle)$$



+

-

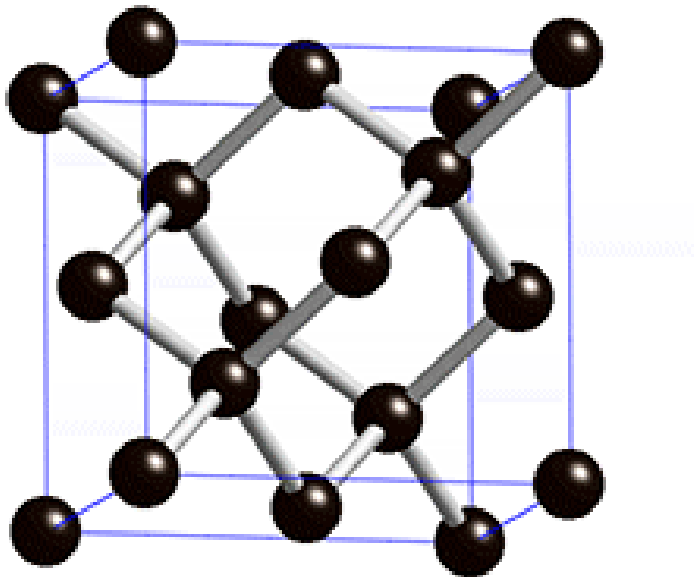
“antibonding” orbital

Quantum description of solids

Solids are formed by placing atoms at regular places in a lattice.

Therefore, their properties need to be understood from the quantum description of atomic orbitals, how they bond and interact, and what happens as a consequence of Pauli exclusion principle and crystal symmetries.

Example: diamond crystal lattice



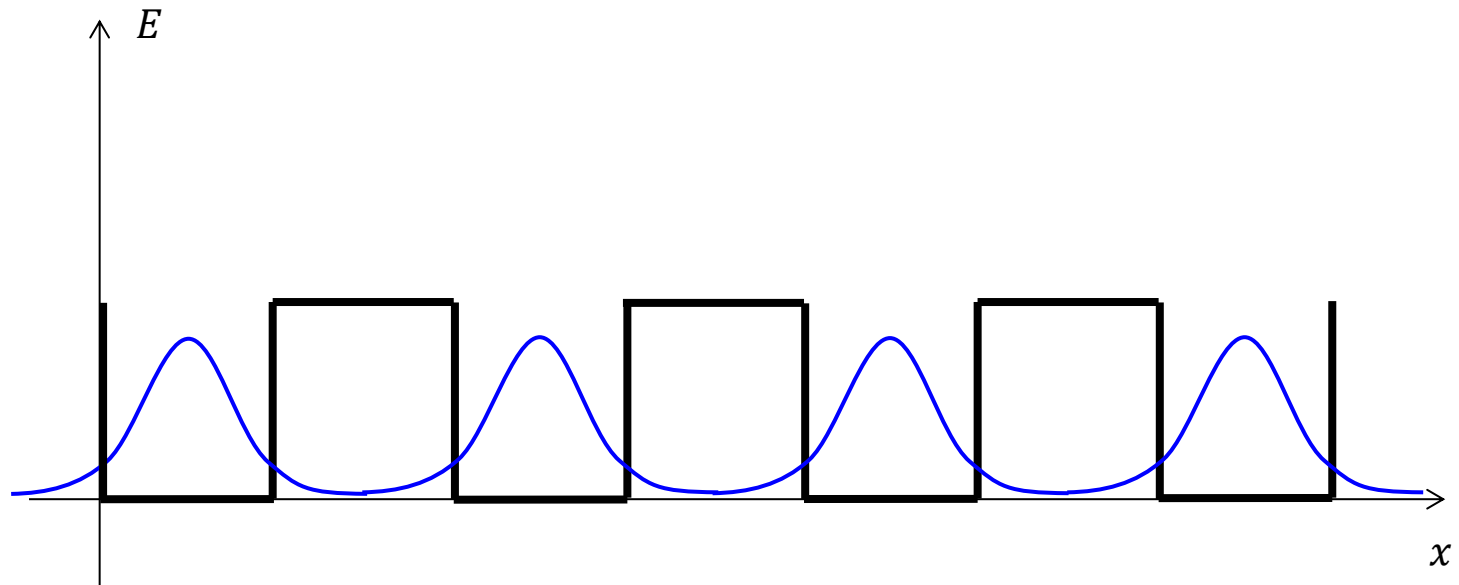
Note:

- diamond is an insulator
 - silicon is a semiconductor
- they have the same crystal structure and the same chemical valence → what's different?

Electron in a periodic potential

Simplest possible model:

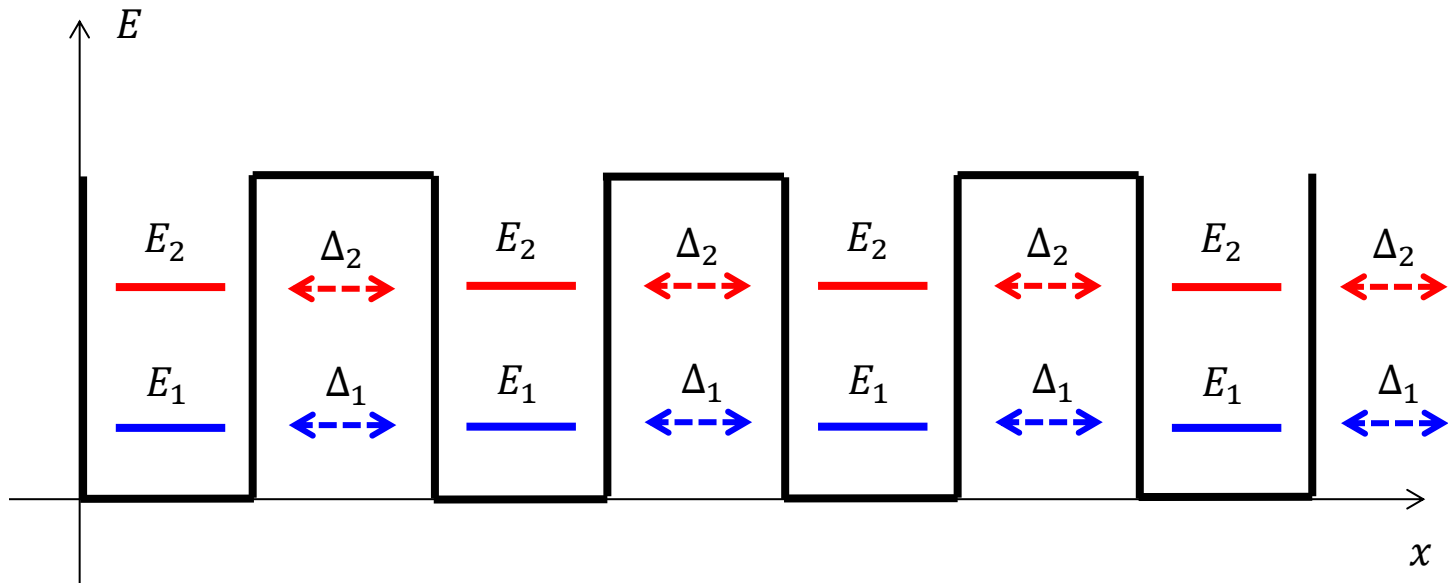
1-dimensional row of potential wells, with finite barriers



This could be solved in Python, but how many potential wells do we need for an accurate description of a solid?

Periodic boundary conditions

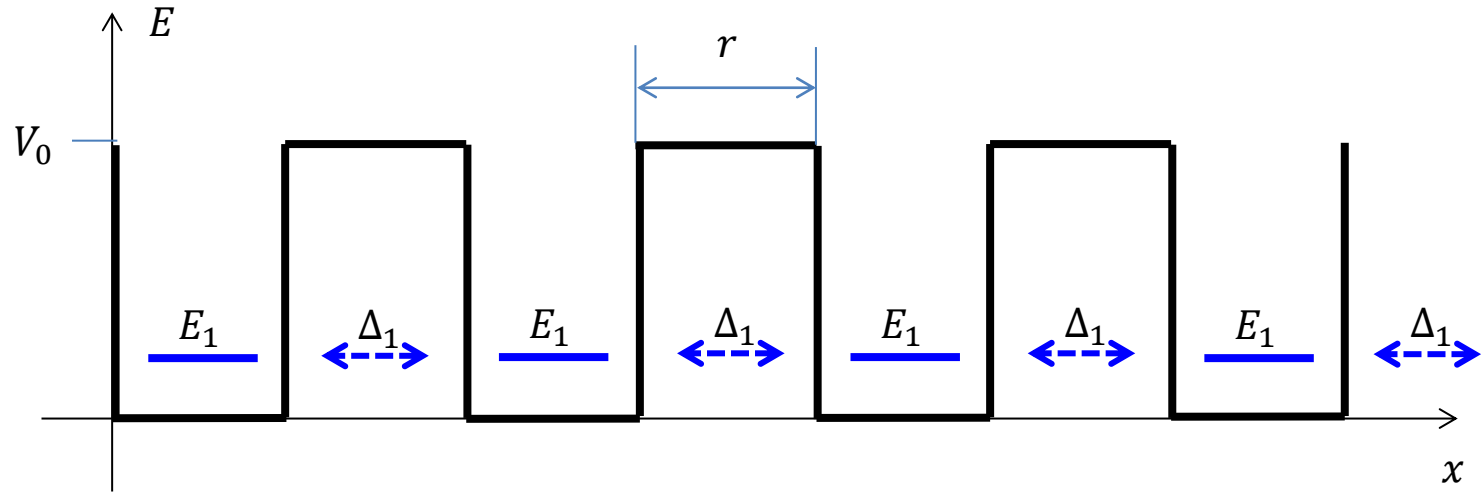
Rather than solving for huge numbers of potential wells, we can “pretend” that the first atom in the row is connected to the last, as in a circle



Each level, of energy E_i , is coupled to the same level in the neighboring well by a tunnel coupling Δ_i , and the last well couples to the first.

1-level case

Consider only the first level in each well:

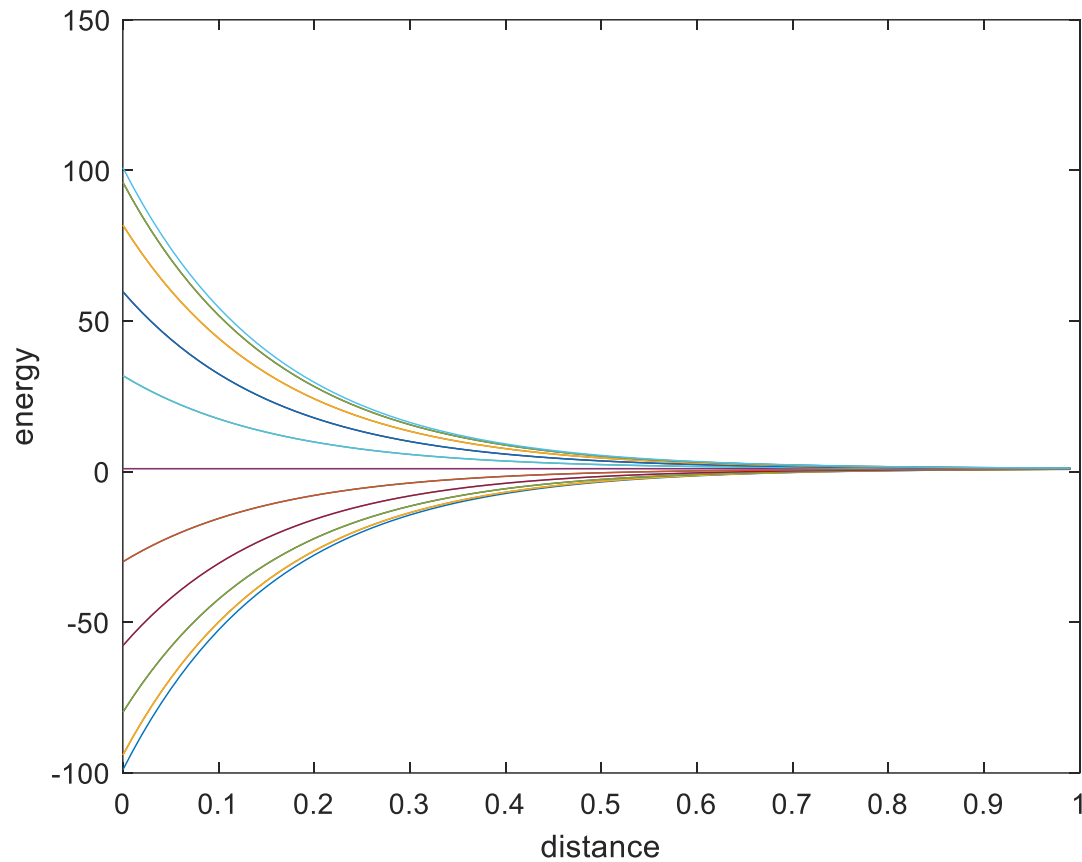


$$H = \begin{pmatrix} E_1 & \Delta_1 & 0 & \Delta_1 \\ \Delta_1 & E_1 & \Delta_1 & 0 \\ 0 & \Delta_1 & \ddots & \Delta_1 \\ \Delta_1 & 0 & \Delta_1 & E_1 \end{pmatrix}$$

The transmission coefficient, and thus the tunnel coupling, depends exponentially on the width r of the barrier, and the height measured from the energy of the particle, $V_0 - E$ (page 107):

$$\Delta_1 = k \exp \left(- \frac{\sqrt{2m(V_0 - E_1)}}{\hbar} \cdot r \right)$$

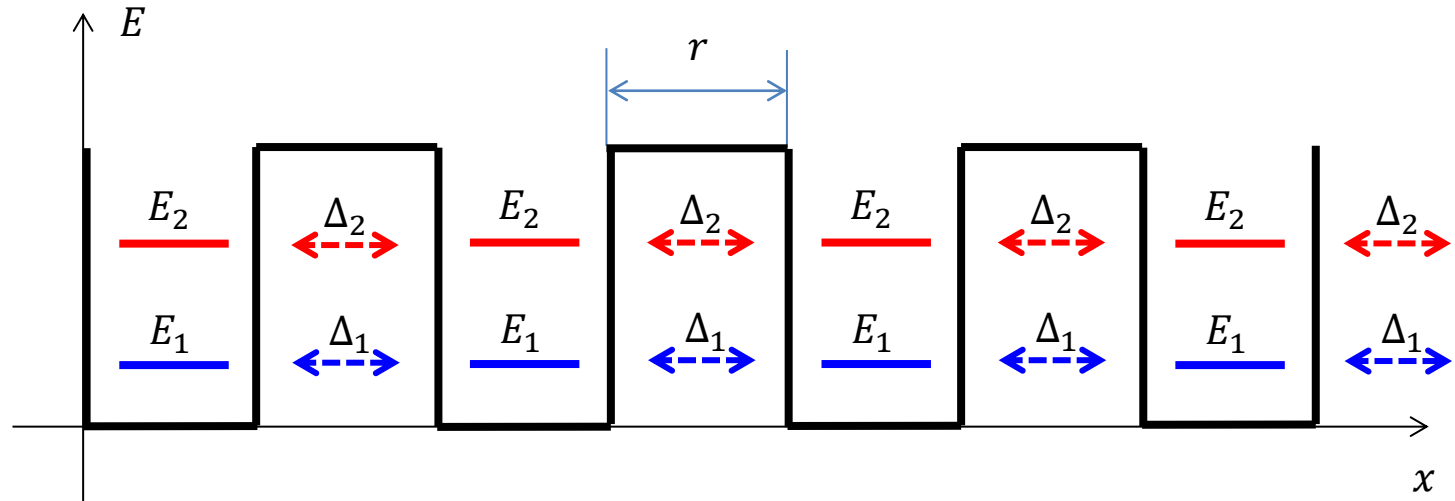
1-level case



The narrower the barrier, the more the electron leaks through, and the levels split due to the tunnel coupling

2-level case

Add a second level in each well:



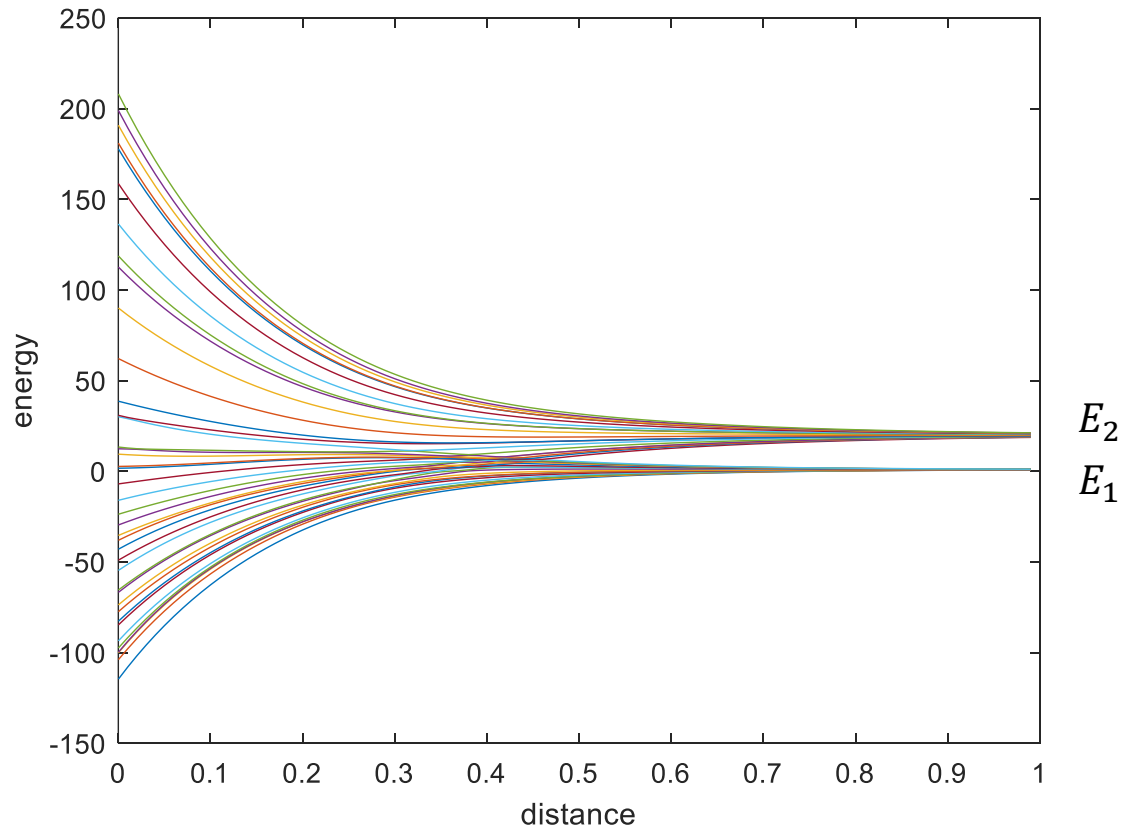
$$H = \begin{pmatrix} E_1 & 0 & \Delta_1 & 0 \\ 0 & E_2 & 0 & \Delta_2 \\ \Delta_1 & 0 & E_1 & 0 \\ 0 & \Delta_2 & 0 & E_2 \end{pmatrix}$$

Since the barrier “seen” from energy E_2 is lower, we have:

$$\Delta_2 = k \exp\left(-\frac{\sqrt{2m(V_0 - E_2)}}{\hbar} \cdot r\right)$$

with $\Delta_2 \gg \Delta_1$ because $(V_0 - E_2) < (V_0 - E_1)$

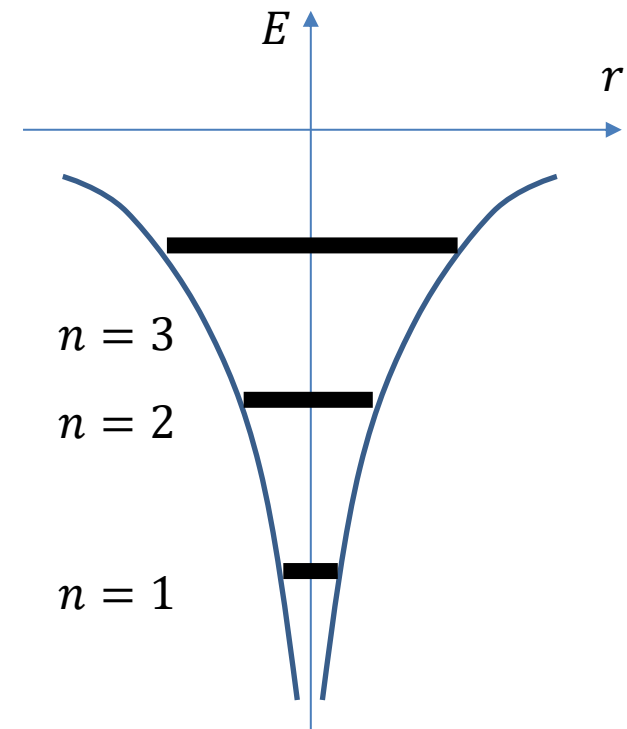
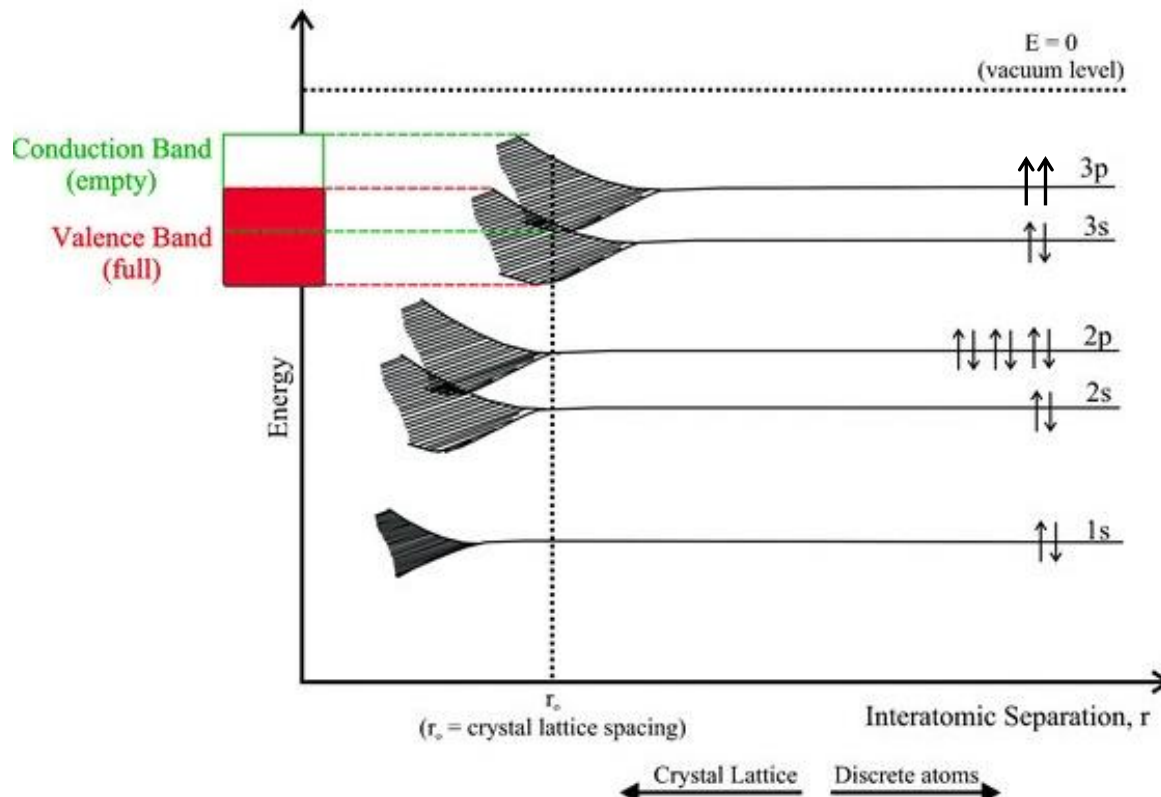
2-level case



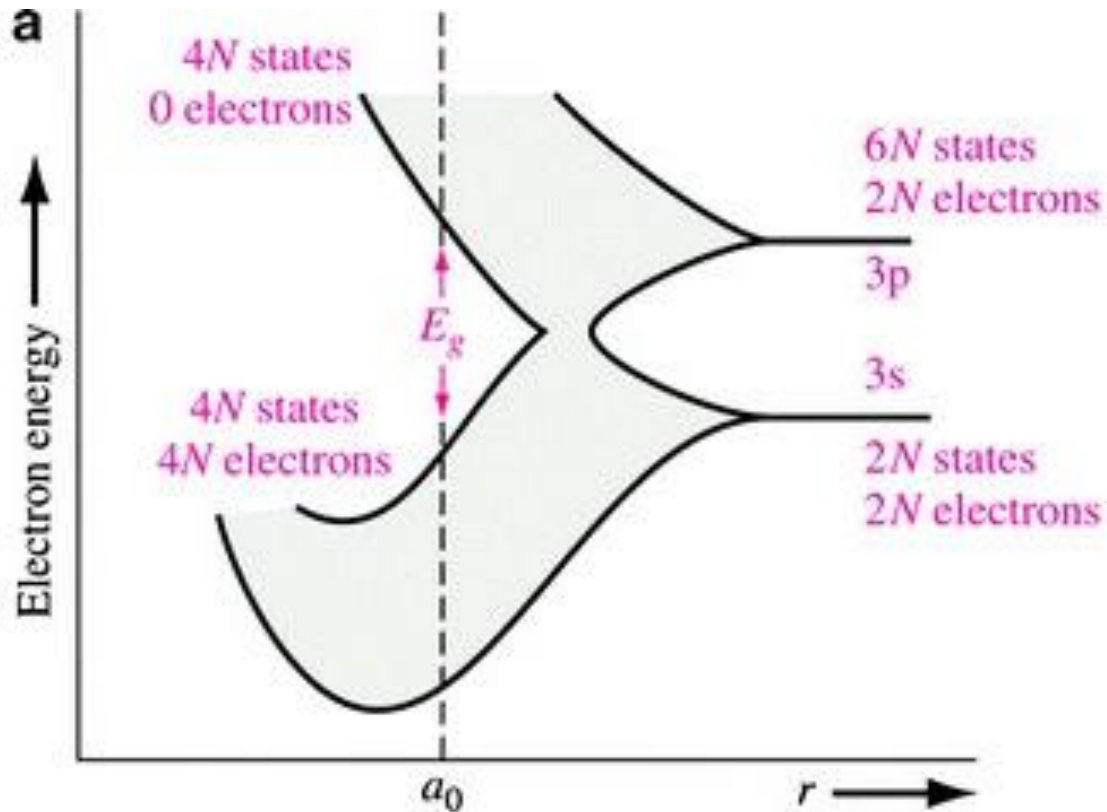
Because of the stronger tunnel coupling, the states that originate from the 2nd level split faster than the ones from the 1st level

Orbital overlap

In addition to the tunnel barrier being lower in energy, higher-energy orbitals in actual atoms (not square potentials) have larger “size”, so they overlap more strongly



Formation of bands

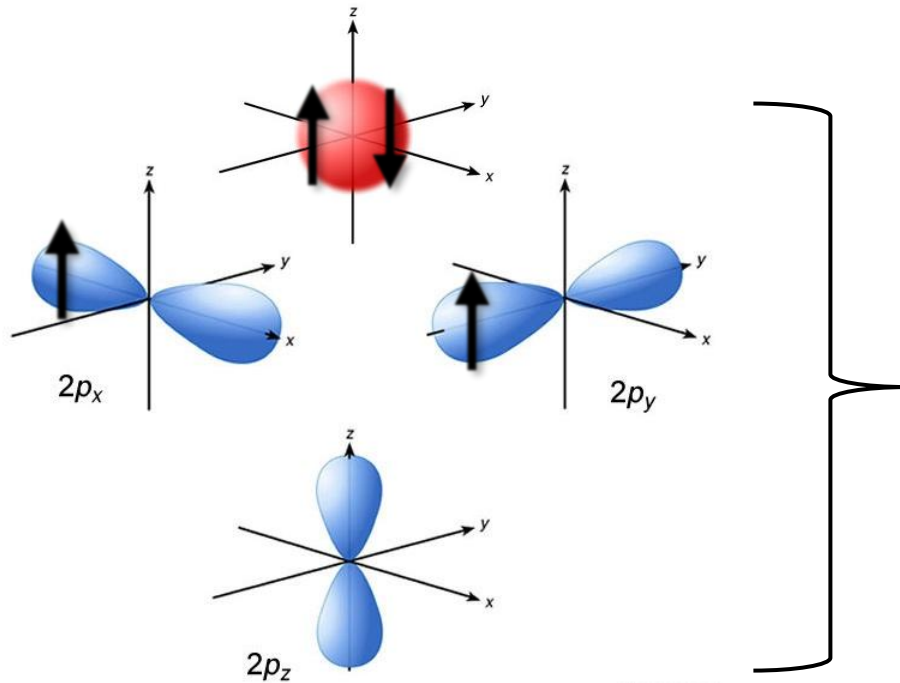


Below a certain value of inter-atomic distance, an energy gap appears again
→ **band gap**

Orbital hybridization

Isolated atom:

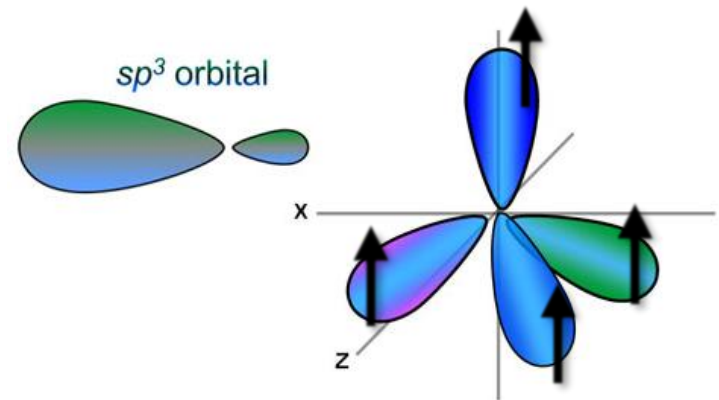
$n s^2, n p^2$ ($n=2,3,4$)



One s-orbital with 2 electrons
Three p-orbitals with 2 electrons

In a crystal:

$n sp^3$ ($n=2,3,4$)



4 hybridized orbitals with
one electron each

Section 10

Quantum transport of electrical charge

Learning outcomes:

- Understand the breakdown of classical Ohm's law when the size of the conductor becomes comparable to the wavelength of the electrons.
- Understand the similarity between quantum transport and waveguide propagation.
- Recognize how the quantized nature of conductance in nanoscale devices can be used to build very sensitive electrometers.

Ohm's law

$$V = R \cdot I \rightarrow I = G \cdot V$$

or, in local form:

$$\vec{j} = \sigma \vec{E}$$

where σ is the conductivity, inverse of the resistivity ρ

At the microscopic level, the conductivity is expressed as:

$$\sigma = \frac{1}{\rho} = \frac{ne^2\tau}{m}$$

n = density of charge carriers

e = electron charge

m = electron or hole (effective) mass

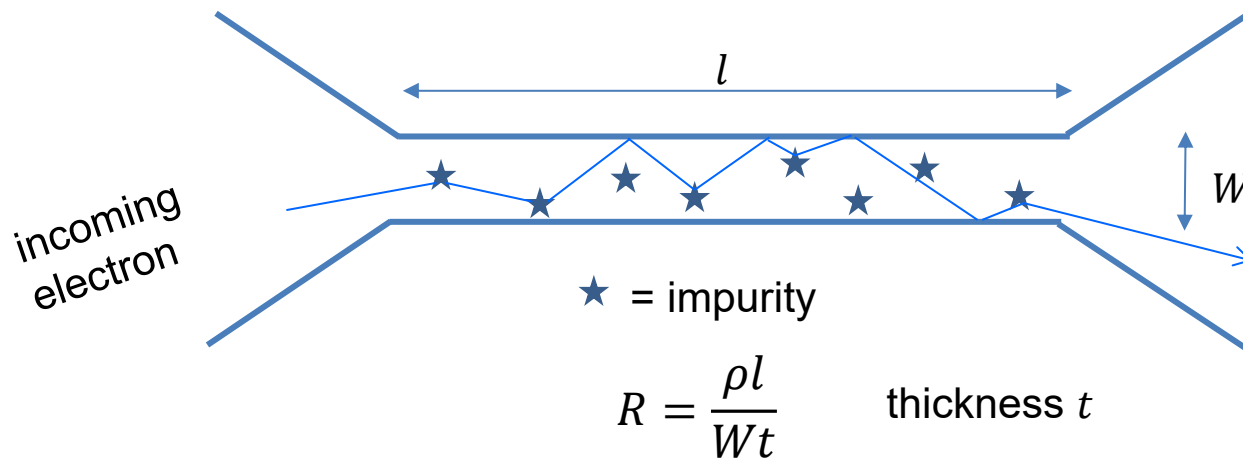
τ = average time between inelastic
scattering events

This is called “Drude model”. It’s based on the idea that an electric field accelerates the charges, then they stop when they hit an obstacle, then accelerate again... so the carrier velocity, and therefore the current, will depend on the average time between scattering events, τ .

Ohm's law

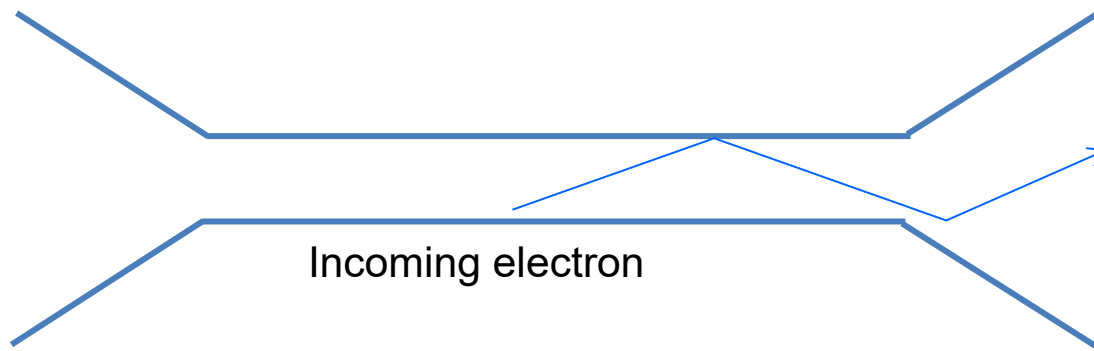
Notice that, unlike what Drude thought, we now know that the electrons (or holes) do not lose their energy by scattering against atoms in a regular crystal lattice. That's because in a regular crystal, the quantum-mechanical wave function of an electron has the same periodicity as the lattice ("Bloch waves"). Scattering occurs when you perturb the periodicity, either with an impurity, or with a lattice displacement (thermal vibrations).

For instance in a wire:



Ballistic transport

If there are no defects in the wire, the electrons bounce elastically against the walls, so they don't lose any momentum in the direction of current flow



Does this mean $R = 0$?

Quantum transport

If the cross-section of the wire is small enough, quantum effects become important, because of wave-particle duality.

The maximum energy of the electrons is the Fermi energy E_F

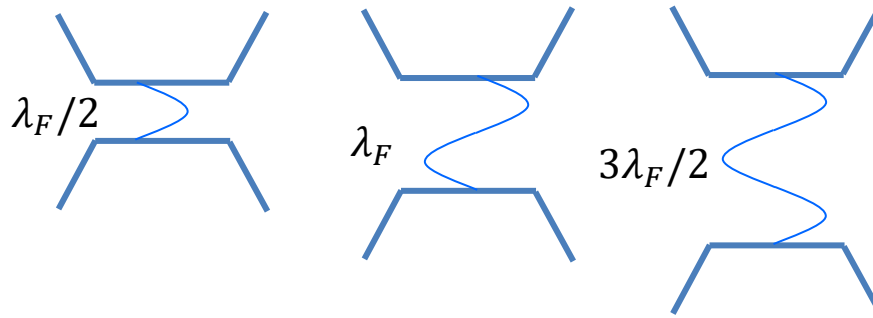
$$E_F = \frac{\hbar^2 k_F^2}{2m} \Rightarrow p_F = \hbar k = \sqrt{2mE_F}$$

$$\lambda_F = \frac{2\pi}{k_F} = \frac{h}{\sqrt{2mE_F}}$$

λ_F is therefore the minimum wavelength of the electrons.

Quantum effects arise when $W \sim \lambda_F$

Quantum transport



The wire acts as a waveguide. As you increase its width, it can carry more modes. Each mode has a quantum of conductance

$$G_0 = \frac{e^2}{h} = \frac{1}{25.8 \text{ k}\Omega}$$

We just said that the electron transport in the “quantum wire” is ballistic, just elastic bouncing off the walls. So where is the power dissipated?

In the contacts! The ballistic wire acts as a waveguide, while the contacts act as non-reflecting termination, because an electron that arrived at the drain never goes back into the wire. So the quantum conductance is like the characteristic impedance of the waveguide. It’s nonzero even if the waveguide has no loss!

Quantum point contact (QPC)

Constrictions of cross-sectional size comparable to the Fermi wavelength, so that quantum effects are visible

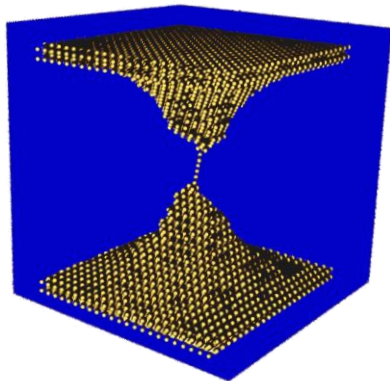
Consider a metallic wire, e.g. gold.

$$E_F = 5.53 \text{ eV}$$

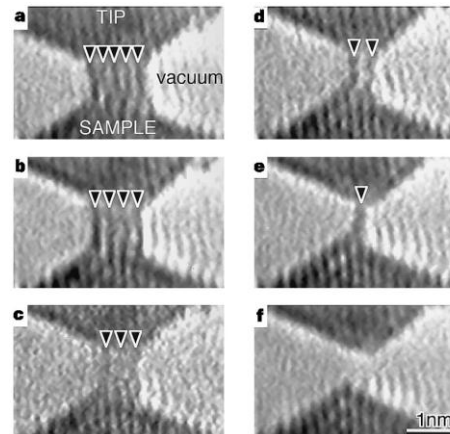
$$k_F = 1.21 \times 10^{10} \text{ m}^{-1} \Rightarrow \lambda_F = 5.2 \text{ \AA}$$

The lattice constant in bulk gold is $a = 4.08 \text{ \AA} \Rightarrow$ to observe quantized conductance, you need make a wire with few-atoms cross section!

When two gold contacts are pulled apart, they spontaneously form a single-atom chain just before breaking



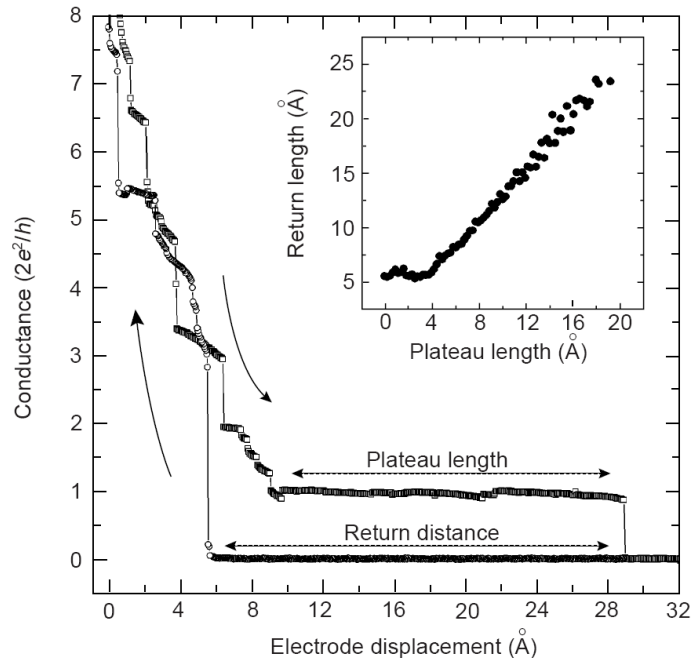
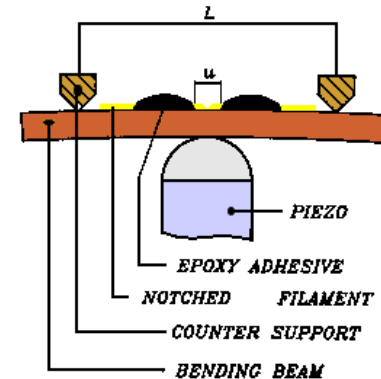
Simulation



Experiment:
Dark = gold wire
Light = vacuum

Quantum point contact (QPC)

A piece of gold wire is clamped and pulled by bending the substrate onto which the ends are glued. The resistance is monitored until it breaks.

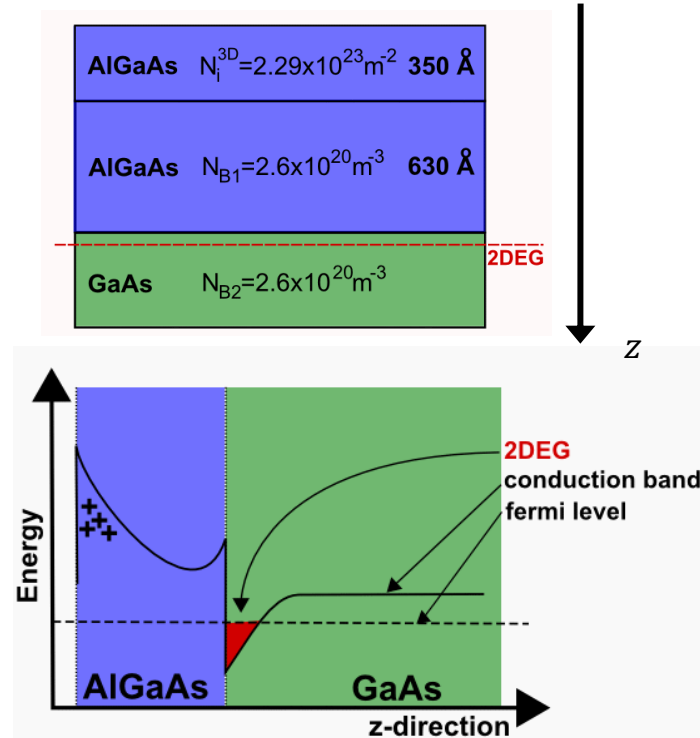


The last resistance plateau, associated with the single-atom chain, has conductance $2e^2/h$.

The factor 2 comes because of spin degeneracy: each conductance channel can carry 2 electrons on opposite spin

2-Dimensional Electron Gas (2DEG)

Different semiconductors have different band gaps $\rightarrow$ this can be used to make heterostructures with peculiar energy profiles.

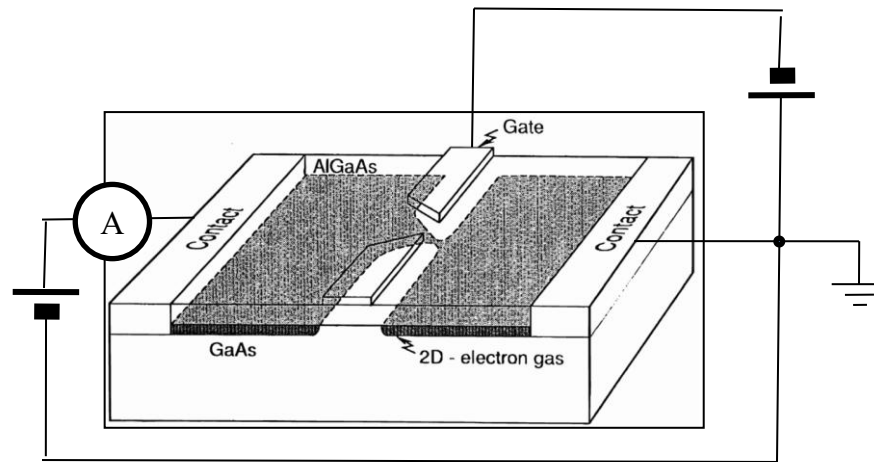


A thin (2-dimensional) layer of electrons is formed at the interface between GaAs and AlGaAs. Metal gates can be fabricated on top of the structure to deplete the 2DEG if desired.

Quantum point contacts in semiconductors

The Fermi energy in a semiconductor is much smaller than in a metal

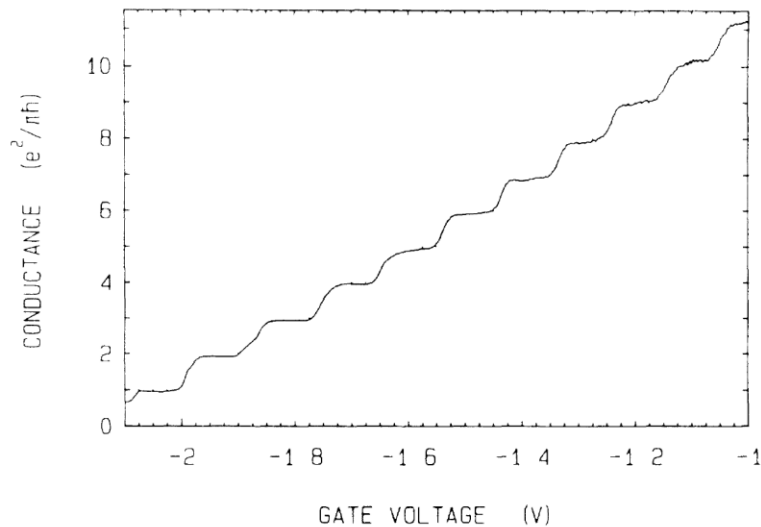
$\Rightarrow \lambda_F$ is larger so, in principle, a quantum point contact can be obtained with a much larger structure. For instance, in a GaAs/AlGaAs heterostructure you can make a 2-dimensional electron gas (2DEG) with $\lambda_F \sim 20 - 40$ nm, comparable to the size of lithographic features made by electron beam lithography



A negatively biased metal gate on top of the 2DEG is used to deplete the electron layer. With a more negative voltage, the QPC can be narrowed until the channel is completely pinched off.

$\rightarrow$ it's the electrostatic version of breaking a wire...

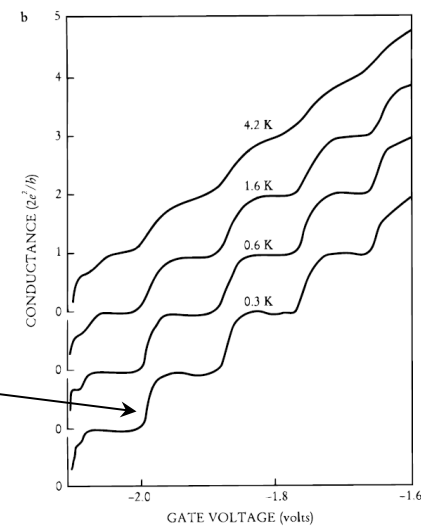
Conductance quantization in QPC



This is of course much easier to control than a broken wire!

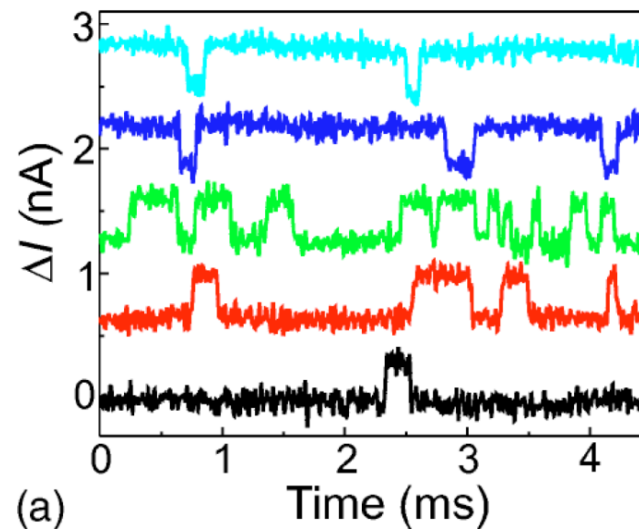
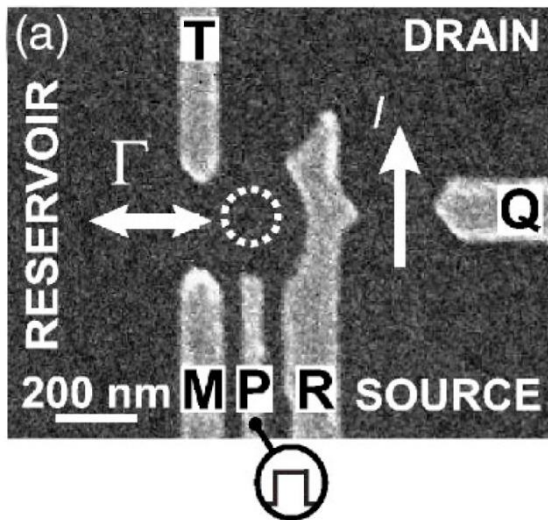
You need low temperatures to observe clear steps, otherwise they get washed out.

Tuning the QPC on the steep slope between two adjacent steps means that the current is very sensitive to the gate voltage, or something equivalent...



The QPC as a charge sensor

When the QPC is tuned to a steep slope, the displacement of even just one electron is sufficient to alter its conductance in a measurable way → it is often incorporated in quantum dot structures



Detect in real time an electron jumping between the quantum dot and the large 2-dimensional electron gas
(quantum dots will be described in the next section)

Section 11

Single-electron devices

Learning outcomes:

- Understand the phenomenon of Coulomb blockade.
- Recognize the applicability of single-electron transistors as charge sensors.
- Learn how to confine electrons in very small semiconductor structures, such that their charge and their spin can be manipulated as quantum degrees of freedom.

Energy-time uncertainty principle

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

This uncertainty relation is a bit different from $\Delta x \Delta p \geq \hbar/2$ (Heisenberg uncertainty principle), because t is a parameter, not a dynamical variable.

Here Δt is defined from:

$$\Delta A = \left| \frac{d\langle A \rangle}{dt} \right| \Delta t$$

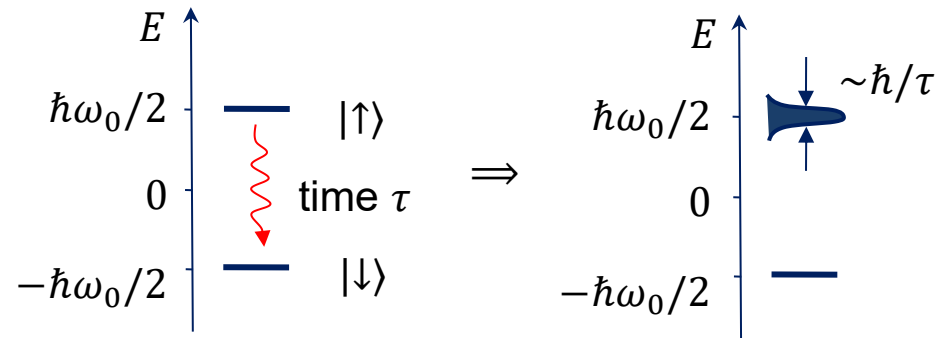
where A is an observable. That is, Δt represents the typical timescale over which the expectation value of the observable changes in a perceptible way.

Lifetime broadening

Example 1: spin in a magnetic field

$$\mathcal{H} = \omega_0 S_z + \mathcal{H}'$$

If we had only the first term $\omega_0 S_z$, $|\uparrow\rangle$ and $|\downarrow\rangle$ would be the eigenstates, the system would stay in these states indefinitely. The extra $\mathcal{H}'$ term describes the fact that, if we prepare the high-energy state $|\uparrow\rangle$, after a while it decays to $|\downarrow\rangle$. Physically, it is coupled to a larger system with many states.

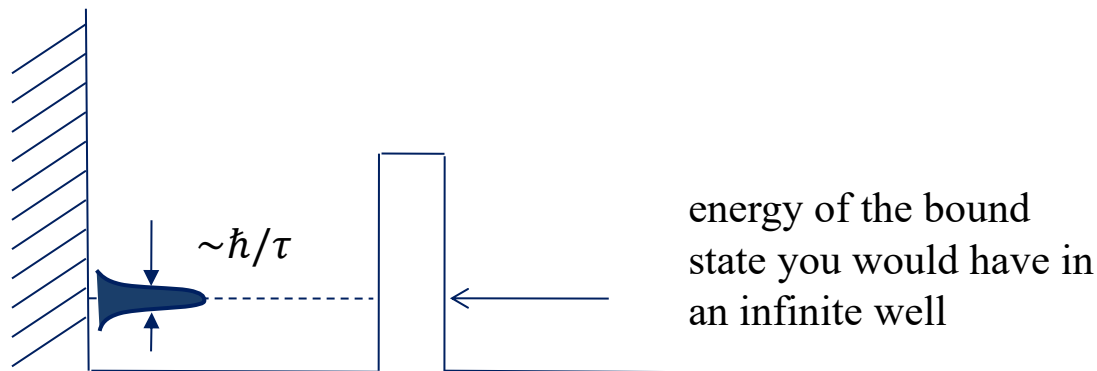


The energy of $|\uparrow\rangle$ is no longer sharply defined, because it's not a true eigenstate anymore. We say it has a “lifetime broadening”, meaning that it is an “approximate eigenstate” with an uncertainty on its energy that relates to its lifetime. Physically, the “broadened state” is a “band” of many states that arise from coupling to the larger system, which is what absorbs the energy upon relaxation of $|\uparrow\rangle$ towards $|\downarrow\rangle$.

Lifetime broadening

Example 2: particle in a non-infinite potential well

Since the particle has a non-zero tunnelling probability, it will leave the well after a characteristic time τ . This broadens the energy level of the bound state.



Again, the “proper” way to describe it would be to write the full Hamiltonian of the particle in the well, coupled to all the states outside the well. This is computationally too heavy, but we can have a very good approximation by treating the single level inside the well, and giving it a broadening $\hbar/\tau$

Constants and conversions

From the Planck-Einstein relations, we already know that energy and frequency can be converted into each other:

$$E = \hbar\omega = hf$$

where $h = 6.64 \times 10^{-34} \text{ J} \cdot \text{s}$

Energy can also be expressed “in temperature units”, using the relation

$$E = k_B T$$

where $k_B = 1.38 \times 10^{-23} \text{ J/K}$ is the Boltzmann constant.

In electronics, we often express energy in units of electronvolt (eV), i.e. the energy of a single electron with charge $e = 1.6 \times 10^{-19} \text{ C}$ subjected to a 1 V potential:

$$E = e \cdot V$$

It is *very useful* to remember the following conversion factors:

$$1 \text{ K} = 20.84 \text{ GHz} = 86 \text{ } \mu\text{eV}$$

Single-electron devices

Consider the below metallic “nanosphere”. What energy is needed to charge this sphere with just a single electron?



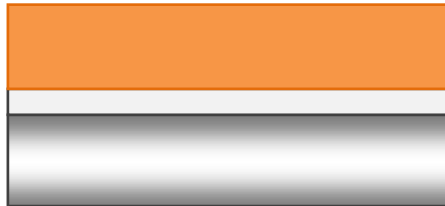
Single electron
charging energy:

$$E_c = \frac{e^2}{2C}$$

R (μm)	C (F)	E (meV)	T (K)
10	1.1E-15	0.072	0.84 (^3He)
1	1.1E-16	0.72	8.4 (LHe)
0.1	1.1E-17	7.20	84 (LN_2)
0.01	1.1E-18	72	840 (Oven)

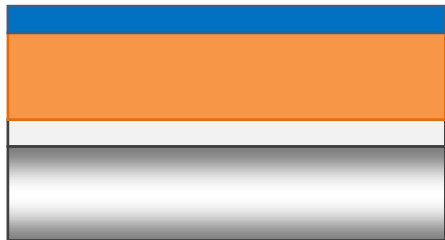
Fabricating small capacitors

We can create small electronic devices in the lab with a process known as shadow evaporation, using electron beam lithography (EBL) and thermal evaporation of metals.



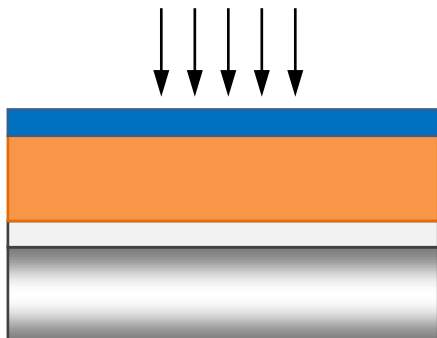
MAA
SiO₂
Si

(a) Spin on a first layer of polymer resist (MAA)



PMMA
MAA
SiO₂
Si

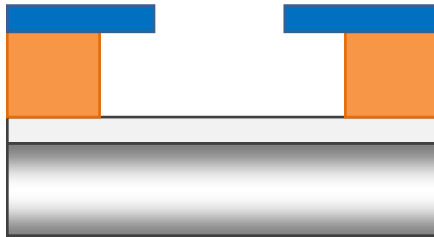
(b) Spin on a “harder” second polymer resist layer (PMMA) on top



PMMA
MAA
SiO₂
Si

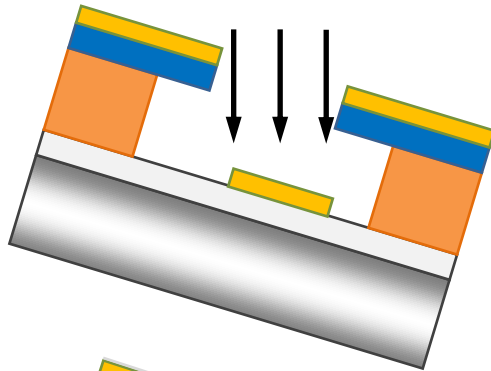
(c) Expose pattern with beam of electrons

Fabricating small capacitors

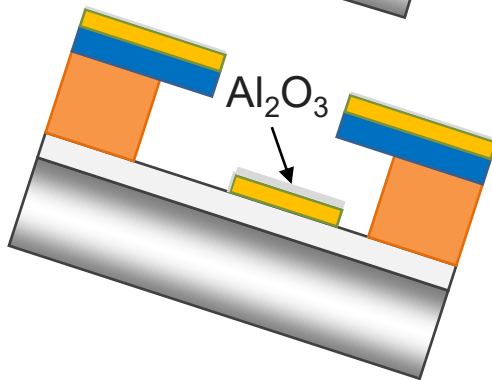


PMMA
MAA
SiO₂
Si

(d) Put device in special chemical developer to dissolve resist. The first layer is more sensitive to the exposure and dissolves quicker (it “undercuts”)

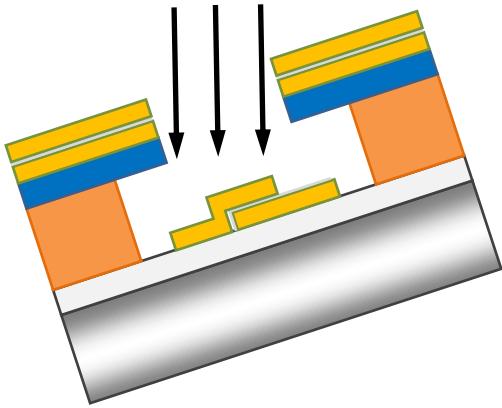


(e) Perform first evaporation of Aluminium, at an angle (e.g. 10 degrees), by tilting the chip
→ “shadow evaporation”

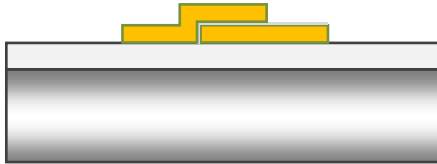


(f) Oxidise first Aluminium layer, resulting in
~1 nm thick Al₂O₃

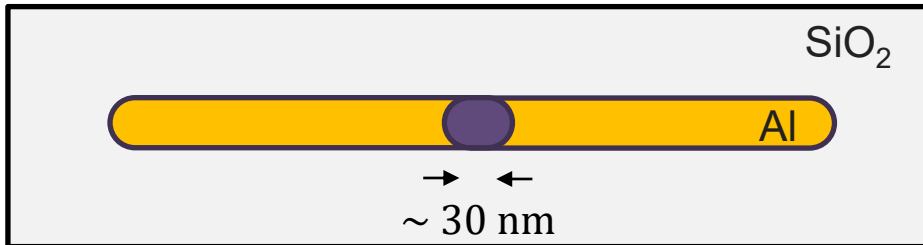
Fabricating small capacitors



(g) Perform a second evaporation at opposite angle



(h) Dissolve the MAA and PMMA resist in acetone



Top view

Charging energy

Let's consider two layers of aluminium forming:

overlap area $A = 30 \times 30 \text{ nm}^2$

oxide thickness $d = 1 \text{ nm}$

If we neglect for now the fact that electrons can tunnel through the oxide, this is just a capacitor...

Junction capacitance:

$$C = \epsilon_0 \epsilon_r \frac{A}{d} = 8.85 \times 10^{-12} \times 10 \times \frac{(3 \times 10^{-8})^2}{10^{-9}} = 80 \text{ aF}$$

The charge on a capacitor is $Q = CV$, and the stored energy is

$$E = \frac{1}{2} \frac{Q^2}{C}$$

Charging energy

Now let's charge the $30 \times 30 \text{ nm}^2$ capacitor with just one electron

→ charge $Q = e = 1.6 \times 10^{-19} \text{ C}$

$$E_C = \frac{e^2}{2C} = \frac{(1.6 \times 10^{-19})^2}{1.6 \times 10^{-16}} = 1.6 \times 10^{-22} \text{ J} \quad \text{“charging energy”}$$

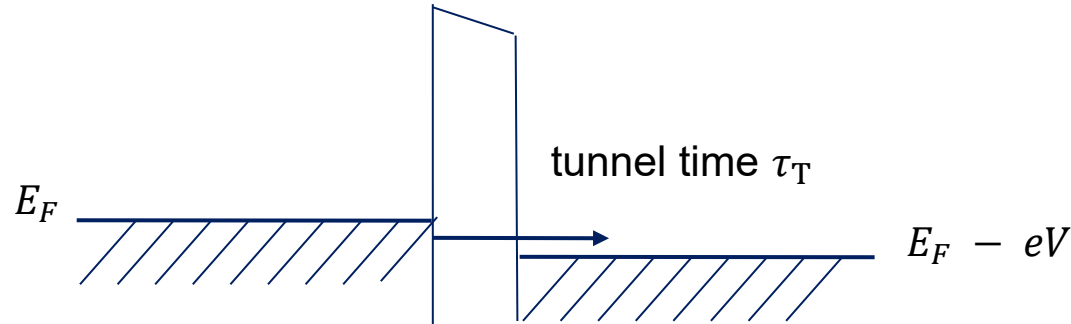
Divide by the Boltzmann constant $k_B = 1.38 \times 10^{-23} \text{ J/K}$ to convert to temperature units

→ $E_C / k_B = 11.5 \text{ K}$

at $T = 1 \text{ K}$, the energy necessary to add even one electron to this capacitor is much larger than the available thermal energy!

Tunnelling conductance

Since the energy barrier created by the Al_2O_3 is not infinitely high and not infinitely thick, there is some tunneling probability $\rightarrow$ nonzero tunnel conductance G_T

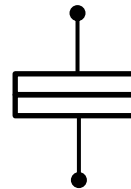


$$G_T = \frac{1}{R_T} = e^2 N(E_F) \Gamma_T$$

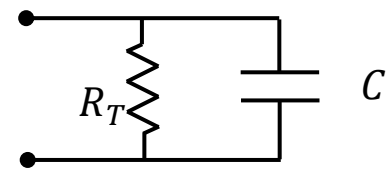
$N(E_F)$ = density of electron states around the Fermi level and $\Gamma_T = 1/\tau_T$ = tunnel rate

Small tunnel junction as a circuit element:

symbol:



equivalent circuit:



Tunnelling conductance

This “capacitor with tunneling” does not hold the charge indefinitely. It will discharge itself with a time constant:

$$\tau = R_T C$$

If I charge the capacitor with one electron, the “lifetime” of the electron on the capacitor is $\tau = R_T C$ (note: this is different from the tunnel time τ_T)

$\Rightarrow$ the energy uncertainty is $\Delta E = \hbar/\tau = \hbar/R_T C$

What happens if $R_T \sim h/2e^2 \approx 12.9 \text{ k}\Omega$, the inverse of the quantum of conductance?

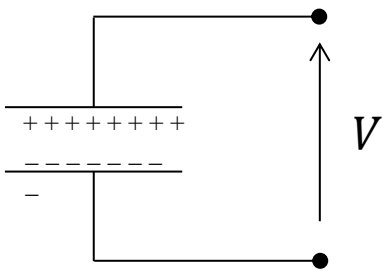
$$\Delta E = \frac{\hbar}{C} \frac{2e^2}{h} = \frac{e^2}{\pi C}$$

which is comparable to the charging energy $e^2/2C$

$\Rightarrow$ if the tunneling resistance is of the order of the quantum of resistance, then the energy uncertainty is of order the charging energy

$\Rightarrow$ the number of electrons stored in the capacitor is not well defined, it has “quantum fluctuations”

Electric displacement vs. “real” charge



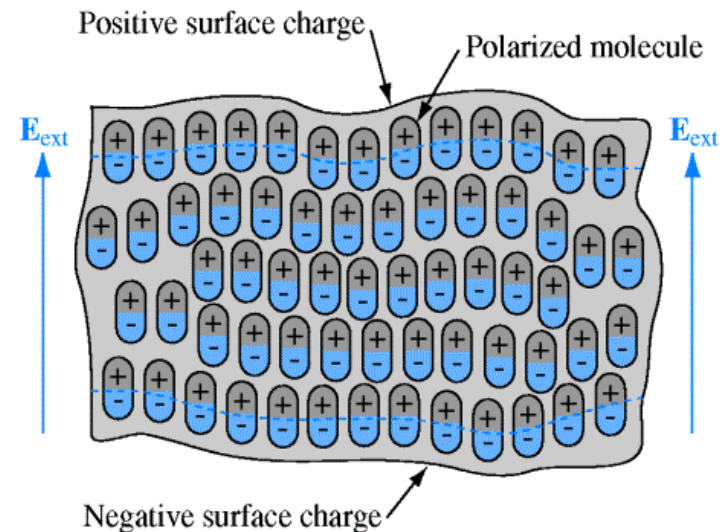
Consider a capacitor with a voltage V applied at its terminals. The charge induced on the plates of the capacitor is therefore

$$Q = CV$$

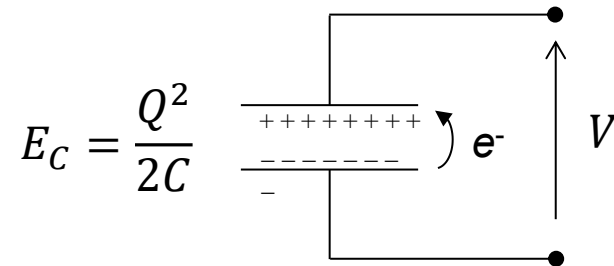
Since V is a continuously variable quantity, so is Q .

Remember that Q here is **not** related to some electron charges being added or removed from the capacitor plates!

Q represents the effect of the electric displacement on the capacitor plates and on the dielectric in between.



Coulomb blockade



If an electron tunnels from the negative electrode to the positive one, the charge goes from Q to $Q - |e|$, i.e. the energy goes from $Q^2/2C$ to $(Q - |e|)^2/2C$

$\Rightarrow$ this constitutes an energy increase, unless the initial charge was $Q \geq |e|/2$, i.e.

$$V \geq |e|/2C$$

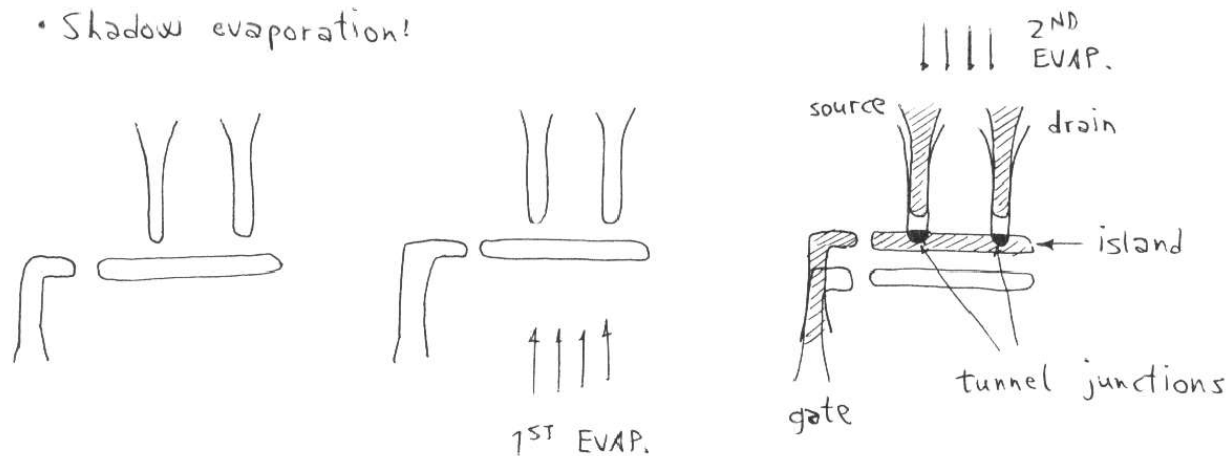
$\Rightarrow$ there is a threshold voltage below which electron tunneling is blocked by electrostatic effects $\rightarrow$ “Coulomb blockade”

e.g. $C = 80 \text{ aF} \Rightarrow V_C = 1.6 \times 10^{-19} / 1.6 \times 10^{-16} = 1 \text{ mV}$

In other words: if $Q \geq \frac{|e|}{2}$, moving an electron to the other side discharges the capacitor. Otherwise, the capacitor gets charged, which is forbidden by energy conservation.

Single-electron transistor (SET)

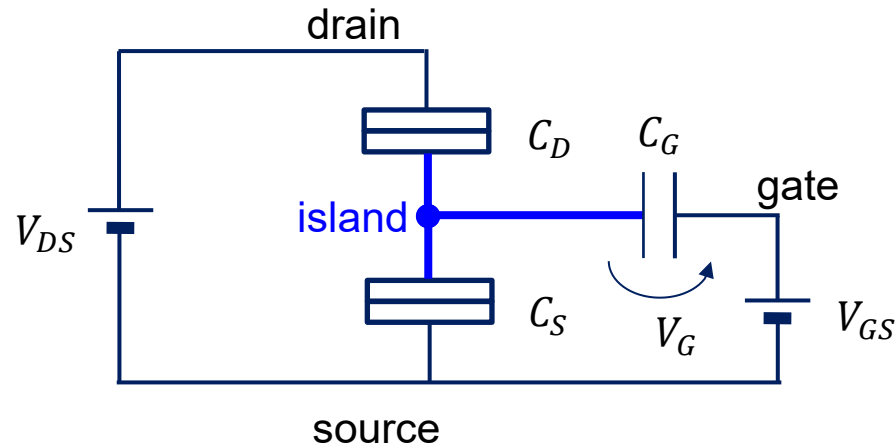
The SET is a metallic island connected to source/drain contacts by low- C tunnel junctions, and capacitively coupled to a control gate. It can be made by shadow evaporation of aluminium:



After the first evaporation, oxygen is introduced to the evaporation chamber to oxidize the Al with a controlled thickness, then the chip is rotated by an angle (typ. from -10° to $+10^\circ$) and metal is evaporated again.

The source and drain electrodes have a small overlap with the island, forming tunnel junctions. The gate is on the side, so has only capacitive coupling and no tunneling.

SET equivalent circuit



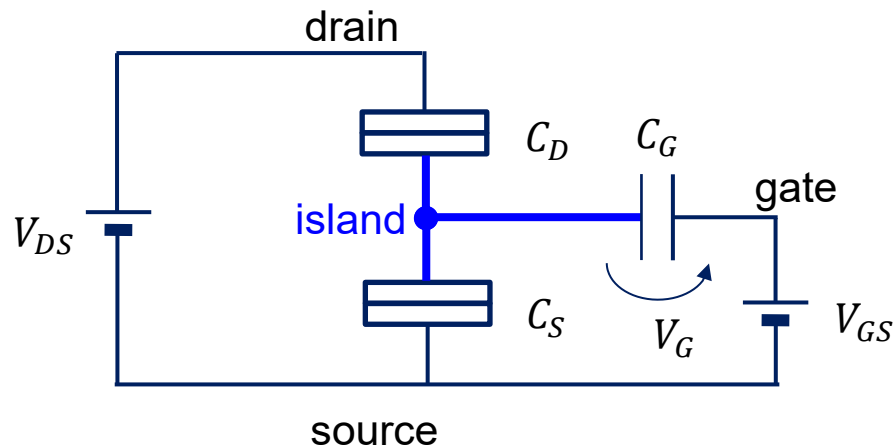
Important note: the charge on C_G doesn't need to be an integer multiple of e . It's a polarization charge that can take any continuous value $Q_G = C_G V_G$.

We shall use a normalized charge $n_G = C_G V_G / e$

$V_G \neq V_{GS}$ because there can be a voltage drop across the source/island junction.

Instead, the charge on the island has a discrete value, because the tunnel barriers are designed to have resistance $R > h/2e^2$

SET equivalent circuit



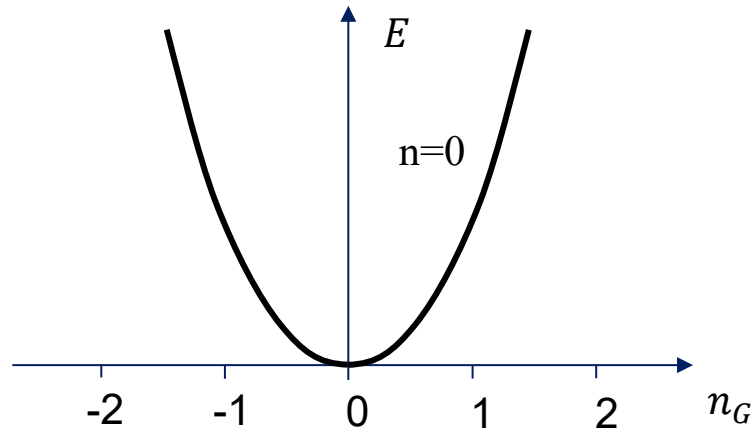
Typical values for the tunneling resistance are $R \sim 50 - 100 \text{ k}\Omega$.

This only works at low temperatures ($\sim 1 \text{ K}$) because you need $k_B T < E_C$

Here, the island is coupled to the rest of the world by 3 capacitors: C_S, C_D, C_G . The charging energy is

$$E_C = e^2/2C_\Sigma \text{ where } C_\Sigma = C_S + C_D + C_G$$

SET energy diagram



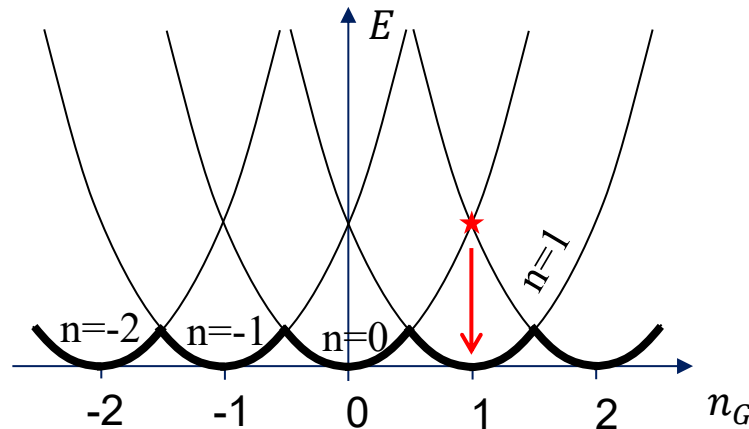
Start with the case where the tunneling resistances are infinite:
→ the electron number on the island can not change

The parabola above then represents $E = Q^2/2C_\Sigma = (en_G)^2/2C_\Sigma$

Even if the electron number is fixed, the capacitor C_G can be charged because of the polarization of the dielectric upon applying a voltage V_G

SET energy diagram

If we allow for electron tunneling on and off the island:



Consider for example the point $\star$: If there are zero electrons on the island, there is a polarization charge on C_G , responsible for the energy increase. If now an electron tunnels onto the island, it exactly compensates the polarization charge, so C_G is discharged and $E = 0$.

For $n_G = n + 1/2$ ($n = \text{number of excess charges}$ on the island), there are two island charge numbers with the same energy: $n = 0$ or $1 \Rightarrow$ an electron can tunnel on and off at no energy cost. As a function of n_G , the energy follows the thick line, if tunneling is allowed to maintain the charge number with the lowest energy.

Electrochemical potential

The total energy of the island with a voltage V_G applied to the gate and n excess electrons is:

$$E(n) = \frac{[-en + C_G V_G]^2}{2C_\Sigma}$$

In this expression we ignore any quantum effects due to confinement of electrons in the potential and we ignore the polarization charge induced by the source-drain voltage. We can see that this energy is quadratic with the applied voltage V_G (or induced charge n_G).

We define the electrochemical potential as the energy required to add the n^{th} electron (with $n - 1$ electrons already on it)

$$\begin{aligned}\mu(n) &= E(n) - E(n - 1) = \frac{[-en + C_G V_G]^2}{2C_\Sigma} - \frac{[-e(n - 1) + C_G V_G]^2}{2C_\Sigma} \\ &= \frac{e^2}{C_\Sigma} \left(n - \frac{1}{2} - n_G \right) \quad [\text{Recall } C_G V_G = en_G]\end{aligned}$$

Electrochemical potential

The electrochemical potential is linear in the induced gate charge n_G . It can be pictured as a ladder of levels that's shifted up or down by applying the gate voltage. The energy spacing between the electrochemical potential levels is constant (independent of both n and n_G)

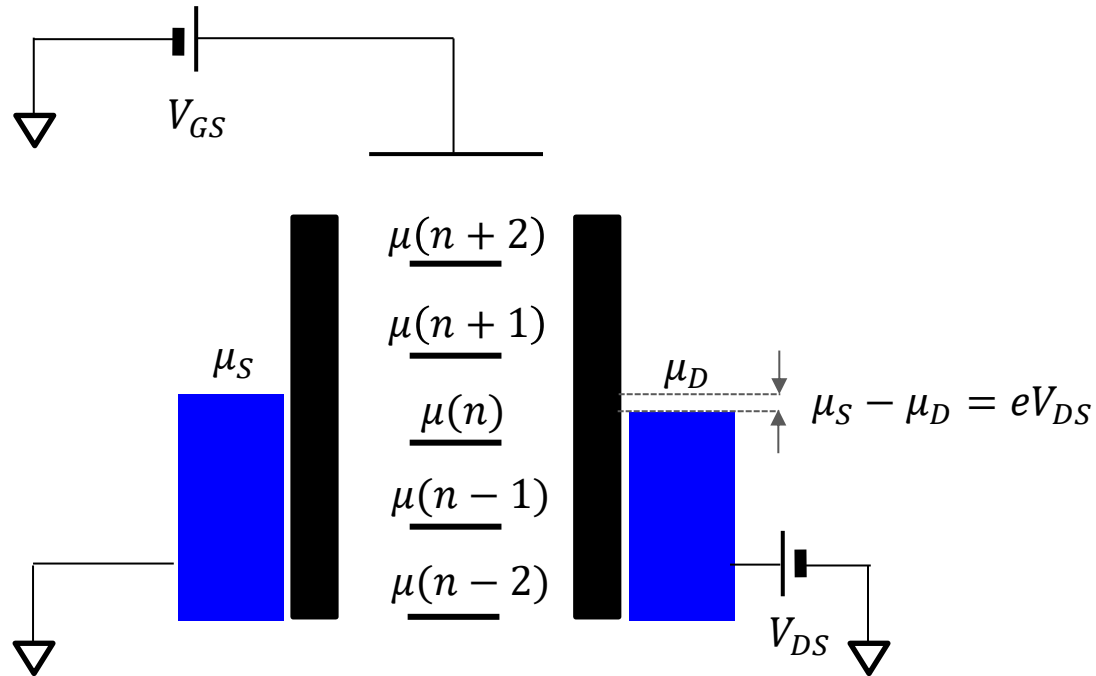
$$E_{\text{add}} = \mu(n) - \mu(n - 1) = \frac{e^2}{C_\Sigma} = 2E_C$$

This is called the “addition energy”, it's the energy required to add **the next electron**.

$$E_{\text{add}} \left\{ \begin{array}{l} \text{---} \mu(n+2) = E_{\text{add}} \left(n + \frac{3}{2} - n_G \right) \\ \text{---} \mu(n+1) = E_{\text{add}} \left(n + \frac{1}{2} - n_G \right) \\ \text{---} \mu(n) = E_{\text{add}} \left(n - \frac{1}{2} - n_G \right) \\ \text{---} \mu(n-1) = E_{\text{add}} \left(n - \frac{3}{2} - n_G \right) \\ \text{---} \mu(n-2) = E_{\text{add}} \left(n - \frac{5}{2} - n_G \right) \end{array} \right.$$

Note, some people define $2E_C$ as the charging energy, as it corresponds to the energy required to charge the island with an additional single electron.

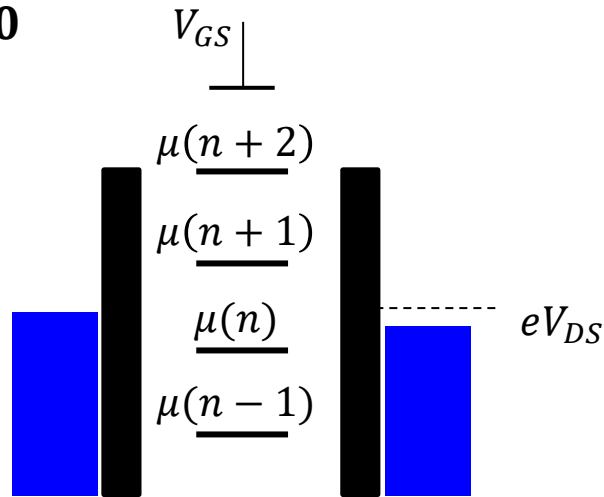
SET charging diagram



The charging diagram shows the relative electrochemical potentials in the source, island and drain. The ladder of black lines represents the electrochemical potential in the island for different charge states. The spacing between the levels is the (classical) Coulomb energy needed to add the next electron E_{add} . This has nothing to do with the quantum-mechanical eigenstates of the electrons in the potential well!

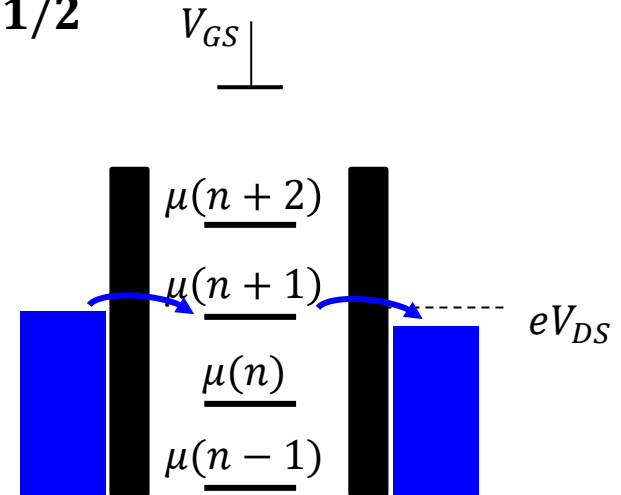
Coulomb blockade

$$n_G = 0$$



Coulomb blockade: The electrons in the source don't have enough energy to tunnel onto the island. Transport is "blockaded" by the Coulomb repulsion of the electrons.

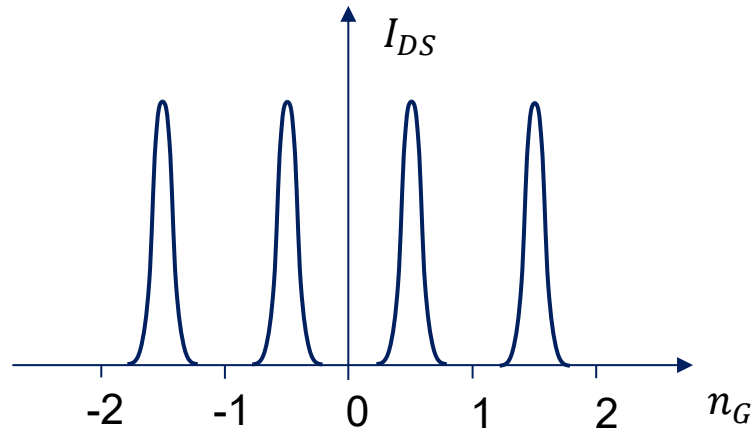
$$n_G = 1/2$$



Blockade lifted: An extra electron can be added at no cost, coming from the source, then tunnels off to the drain.

→ sequential tunneling, the electrons go through one by one

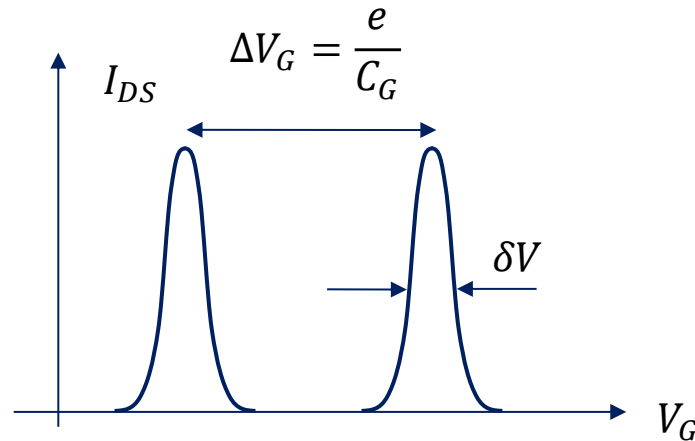
Coulomb peaks



The current between source and drain can only flow at the charge degeneracy points, $n_G = n + 1/2$, where $\mu(n + 1) = 0$
 $\Rightarrow$ as a function of gate voltage $V_G = en_G/C_G$ you find sharp current peaks.

Note: if $V_{DS} \geq e/C_\Sigma$ there is always current flowing, because there is at least one degeneracy point within the source-drain voltage range (“**bias window**”) $\rightarrow$ that’s why you need nanometer-size devices, otherwise C_Σ is large and any tiniest voltage will give you some current.

SET as an electrometer



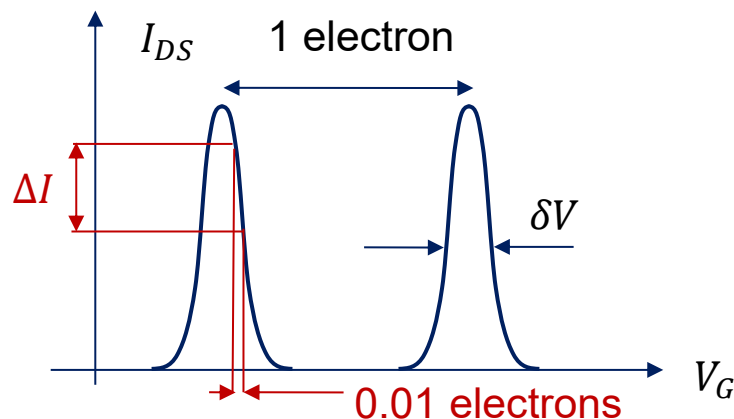
The Coulomb peaks have a finite width, given by:

$$\delta V = \max \left\{ \begin{array}{l} k_B T / e \rightarrow \text{thermal broadening} \\ \hbar \Gamma / e \rightarrow \text{lifetime broadening } (\Gamma = 1/\text{tunnel time}) \\ \sim V_{DS} \rightarrow \text{bias window} \end{array} \right.$$

Like the QPC, the SET is a quantum device whose conductance can be extremely sensitive to the electrostatic potential in its surroundings. The most sensitive point is on the shoulder of a Coulomb peak.

Notice that quantum mechanics here only enters in the tunneling of electrons on and off the island. **Coulomb blockade is a classical effect.**

SET as an electrometer



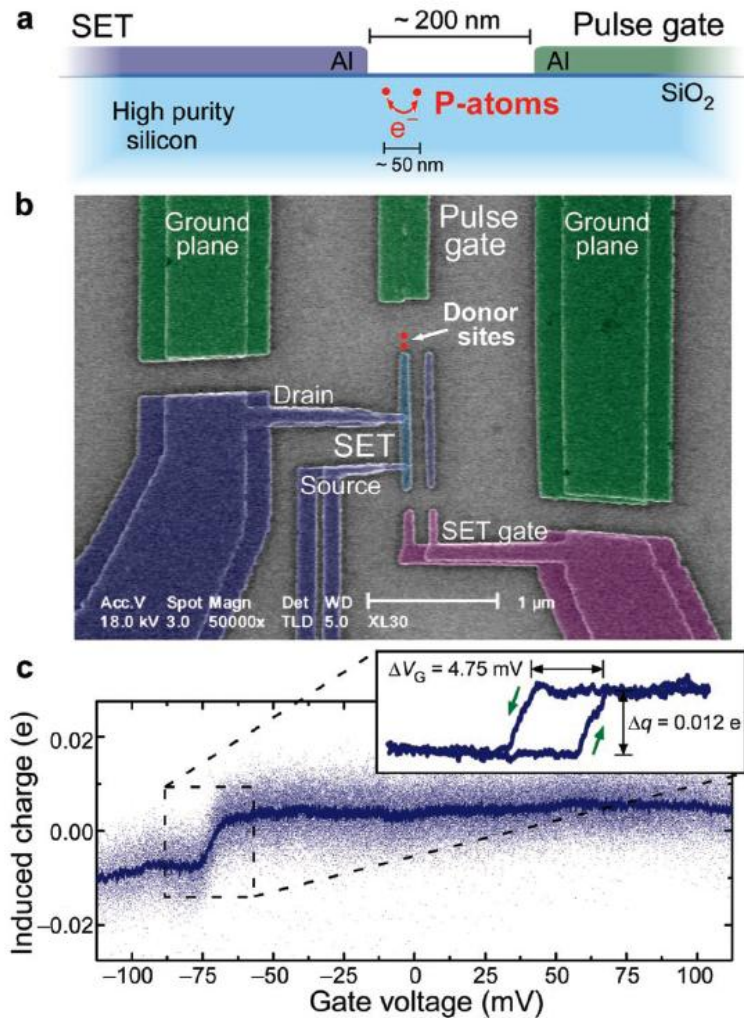
The spacing between the peaks corresponds to taking one electron from infinity and adding it to the island. A much smaller charge displacement can be easily resolved.

The minimum detectable charge ΔQ depends on the noise floor of the measuring electronics, ΔI_N , and on the sharpness of the peaks. The electronics noise depends on the bandwidth, $\Delta I_N \propto \sqrt{BW}$.

Typical values are $\Delta Q \sim 10 \text{ } \mu\text{e}/\sqrt{\text{Hz}}$

$\Rightarrow$ you can detect a charge signal of 0.01e in a time $\sim 1\mu\text{s}$

Detecting the displacement of a single e^-



Two implanted P donors in silicon, with a SET above it.

Changing the “pulse gate” voltage tips an electron from one donor to the other

The displacement of this electron shifts the bias point of the SET and makes its current change.

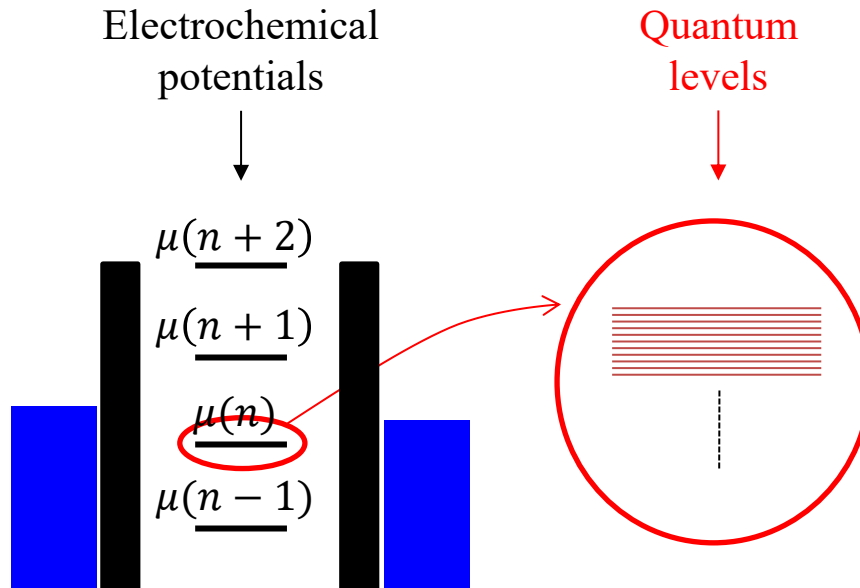
The change in current through the SET is large enough that it can be detected with ~ 100 kHz bandwidth, i.e. within about 10μ s.

SET vs. quantum dots

In the SET, the Coulomb blockade peaks are visible when:

- $E_C \gg k_B T \rightarrow$ low temperatures
- $R_T \gg h/2e^2 \rightarrow$ well-defined electron number

However, the actual number of electrons on the island is macroscopic, and the quantum energy spectrum is a quasi-continuum. Like a “particle in a box” where the box is very wide



In contrast, a quantum dot is a structure where the gap between single-particle energy levels (a **quantum** effect) is large and comparable with the charging energy (a **classical** effect).

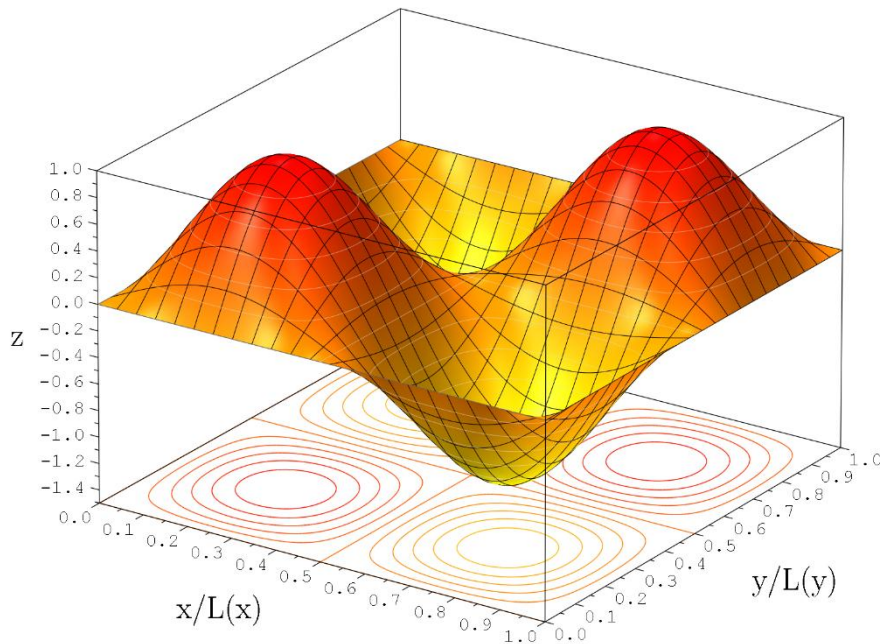
Like a very tight particle in a box

Quantum dots: “artificial atoms”

Recall the hydrogen atom: its orbital states are the solution of the time-independent Schrödinger equation in a central potential (spherically symmetric in 3 dimensions)

Another example we saw is the “particle in a box”, which is a hard-wall potential in 1 dimension.

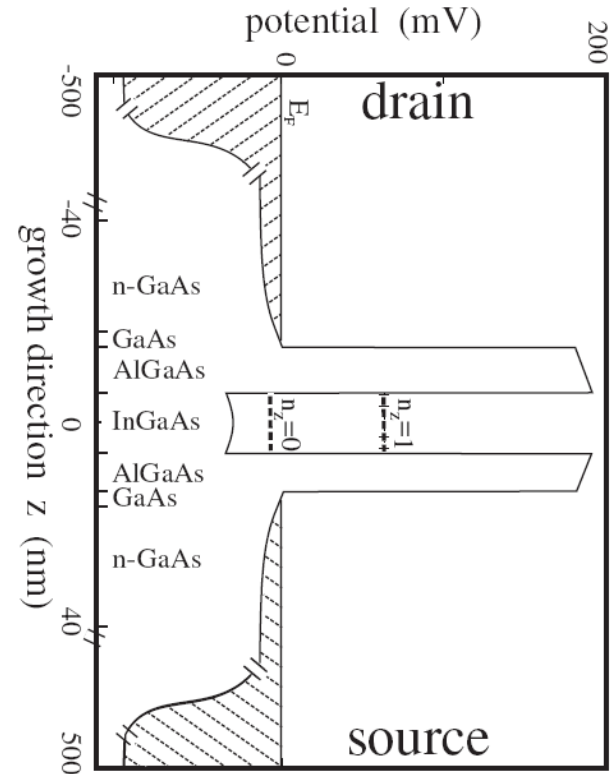
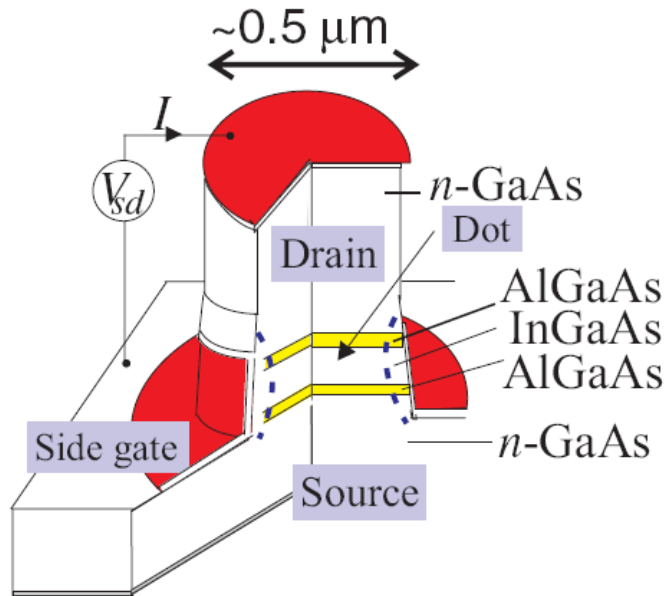
Now suppose you have a confinement potential in 2 dimensions:



This example has two energy minima with a finite barrier in between. You can imagine that, if you placed one electron in there, its wave functions would be the 2-dimensional equivalent of a H_2^+ molecule.

Quantum dots are the 2-dimensional equivalent of atoms and molecules, built artificially in semiconductors!

Vertical few-electron quantum dots



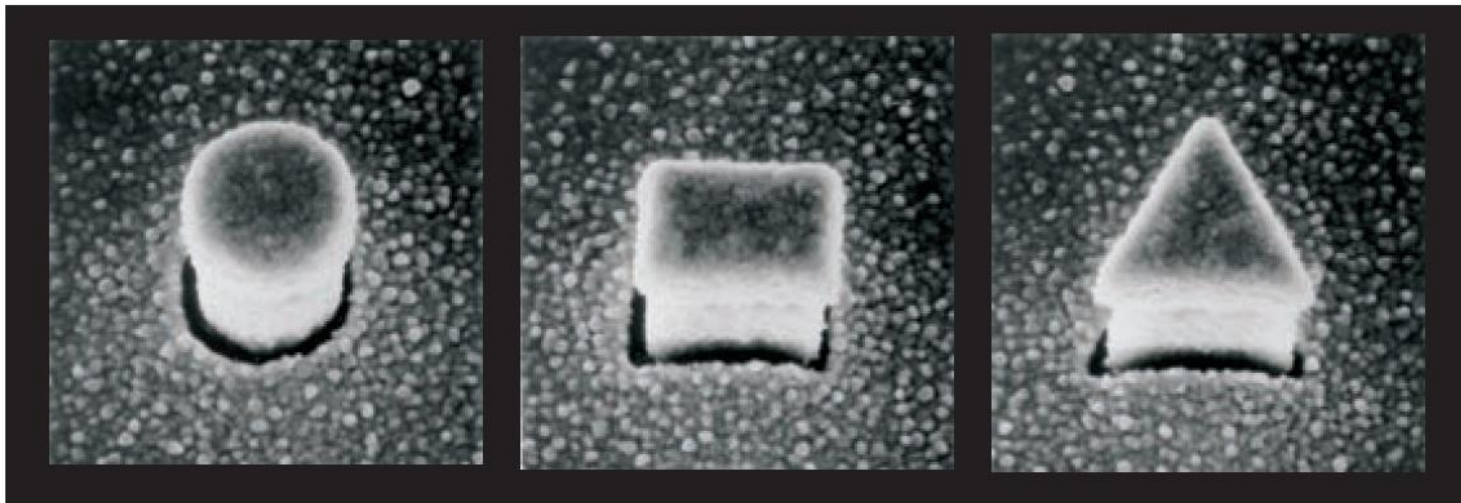
A negative potential on the wrap-around side gate allows one to squeeze out all the electrons, down to the last one!

Here the tunnel barriers are due to the different band gaps of the semiconductors in the stack. They are not as high (in energy) as those in Al_2O_3 – which is a real insulator – so they can be made a bit thicker.

Vertical few-electron quantum dots

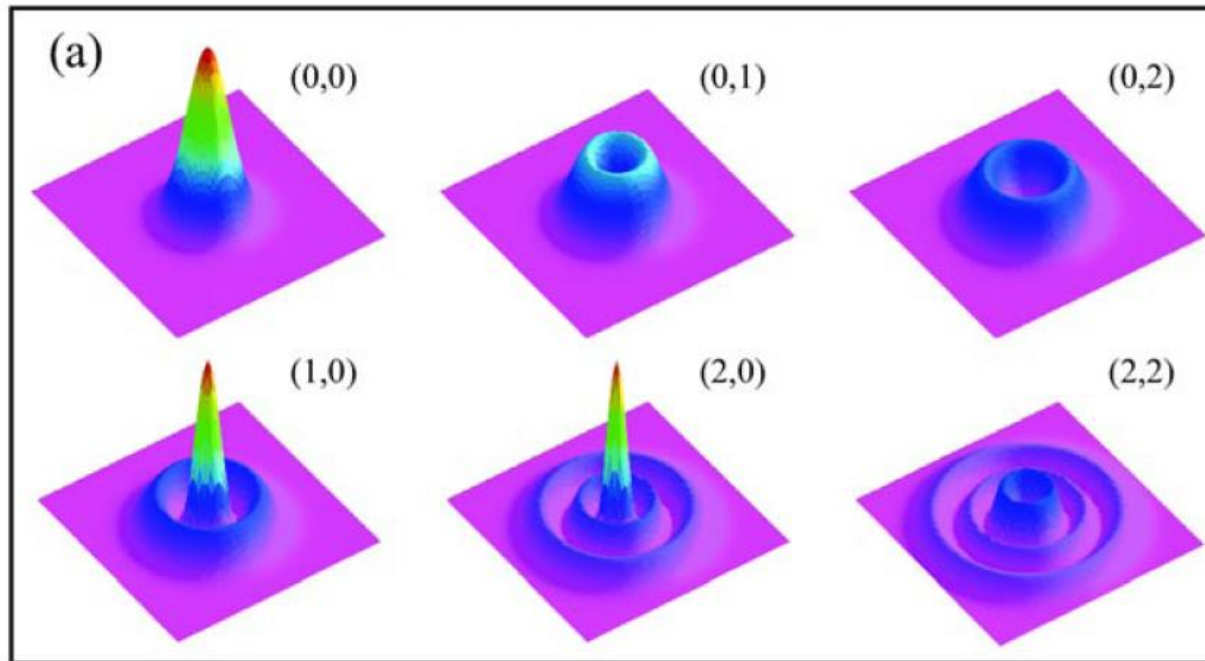
In the semiconductor we can choose whatever shape and size we want, and we can verify that the orbital energies change accordingly.

→ You can't do this with natural atoms!!

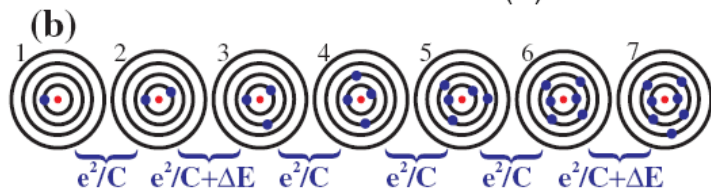
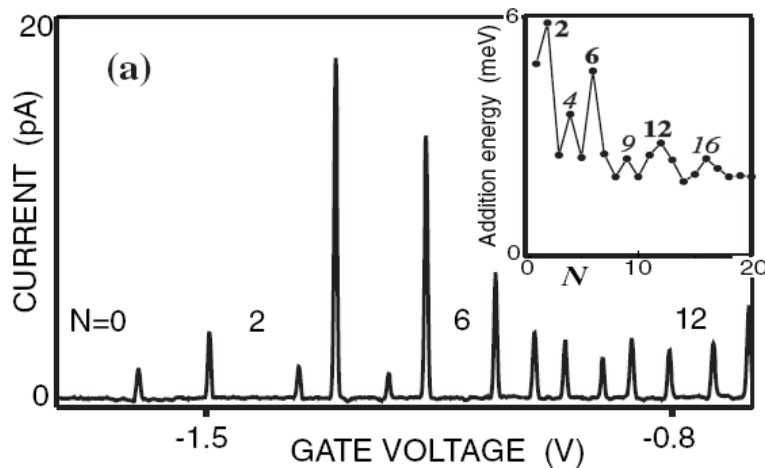


2D orbital eigenstates

The orbital eigenstates of an electron confined in 2D by a parabolic potential are called Fock-Darwin states. n sets the number of nodes in the radial direction whereas l determines the size of the dip in the centre and the radial extent of the wavefunction.



Shell filling



(c) **Periodic Table of 2D Artificial Atoms**

1 Ta							2 Ha	
3 Et	4 Au					5 Ko	6 Oo	
7 Sa	8 To	9 Ho				10 Mi	11 Cr	12 Ja
13	14	15	16 Wi	17 Fr	18 El	19	20 Da	

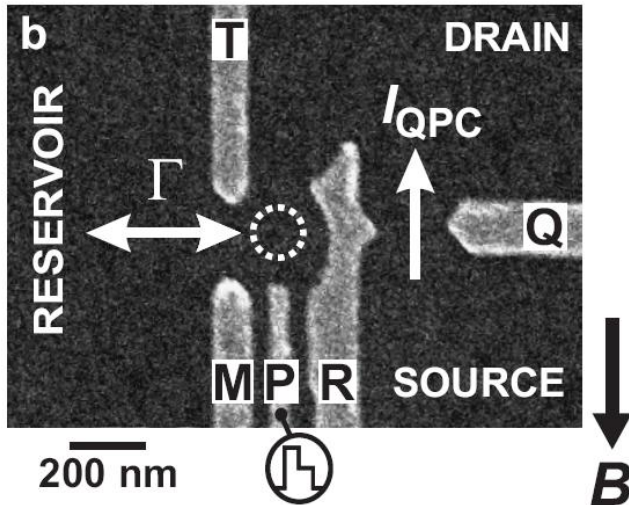
Each time you add an electron, you need to pay the classical Coulomb charging energy (like in the SET). But now the single-particle levels are quantized!

Each orbital can take 2 electrons. For the next electron, you pay the addition energy (e^2/C) plus the extra energy to occupy the next-up quantum orbital.

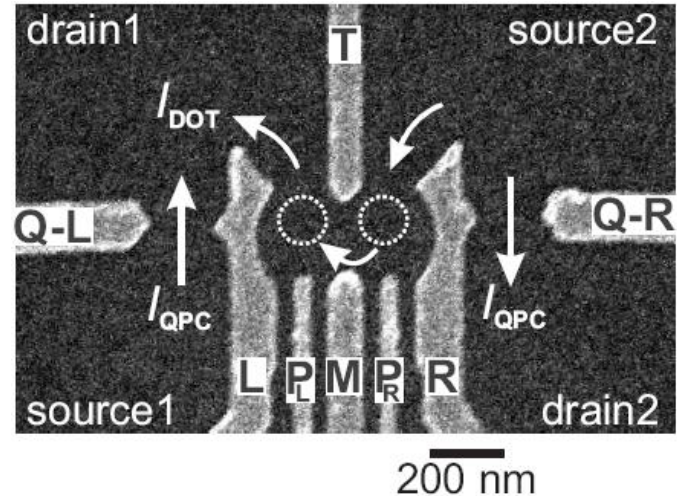
This can be measured via the current through the quantum dot!

You can scan an “artificial periodic table” just by changing a gate voltage!

Lateral quantum dots



Single dot



Double dot

As compared to the vertical dots, here you can:

- Fabricate multiple dots side by side
- tune the height of the tunnel barriers with gate voltages
- incorporate a QPC or SET charge sensor to detect electron motion in real time

Spin-to-charge conversion

The magnetic moment of a single spin is extremely small, and very difficult to detect directly. Instead, we convert it to another degree of freedom (charge), that we are better equipped to measure (e.g. by using a QPC or SET).

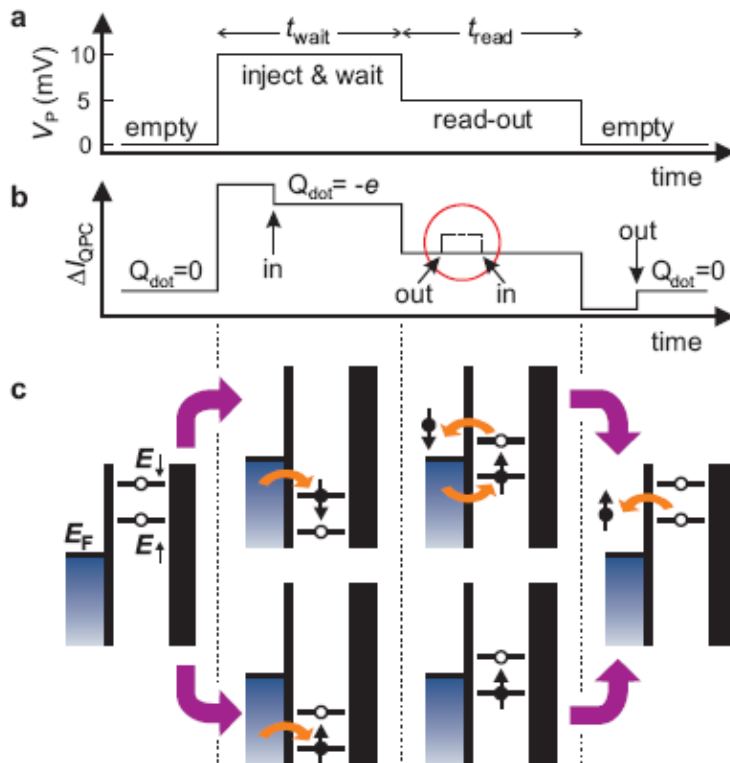
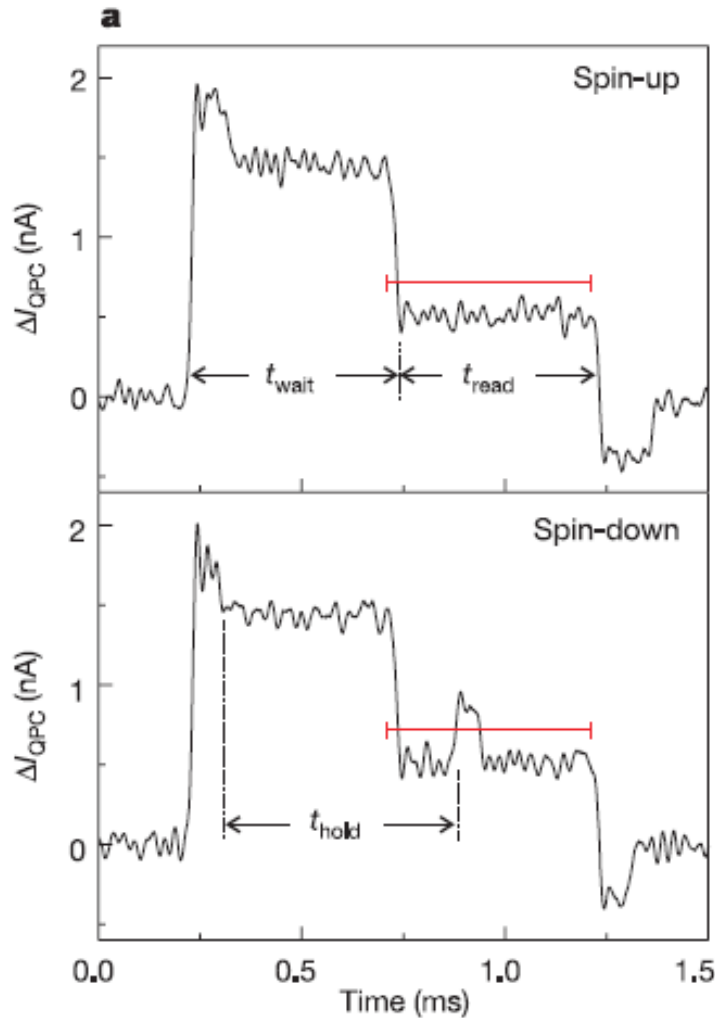


Figure 5.2: Two-level pulse technique used to inject a single electron and measure its spin orientation. (a) Shape of the voltage pulse applied to gate P . The pulse level is 10 mV during t_{wait} and 5 mV during t_{read} (which is 0.5 ms for all measurements). (b) Schematic QPC pulse-response if the injected electron has spin- $\uparrow$ (solid line) or spin- $\downarrow$ (dotted line; the difference with the solid line is only seen during the read-out stage). Arrows indicate the moment an electron tunnels into or out of the quantum dot. (c) Schematic energy diagrams for spin- $\uparrow$ ($E_{\uparrow}$) and spin- $\downarrow$ ($E_{\downarrow}$) during the different stages of the pulse. Black vertical lines indicate the tunnel barriers. The tunnel rate between the dot and the QPC-drain on the right is set to zero. The rate between the dot and the reservoir on the left is tuned to a specific value, Γ . If the spin is $\uparrow$ at the start of the read-out stage, no change in the charge on the dot occurs during t_{read} . In contrast, if the spin is $\downarrow$, the electron can escape and be replaced by a spin- $\uparrow$ electron. This charge transition is detected in the QPC-current (dotted line inside red circle in (b)).

Note: GaAs is a peculiar material where the electron gyromagnetic ratio has positive sign (instead of negative as a free electron would). Therefore, $\uparrow$ is the low-energy state and $\downarrow$ is the high-energy one.

Real-time single-shot spin readout



The “blip” in the QPC current during the readout phase indicated that the spin was in the state $|\downarrow\rangle$. A lack of a blip indicates the spin was $|\uparrow\rangle$. Note, the high energy spin state is $|\downarrow\rangle$ in GaAs, since it has a negative electron g-factor.

This is called **spin-to-charge conversion**, because the displacement of the electron charge is dependent upon its spin state.

It’s like a solid-state version of the Stern-Gerlach apparatus, in a man-made electronic circuit!

Quantum computing with spins in dots

Recall, two spins coupled by an exchange interaction (of strength J) in zero magnetic field form a singlet state and degenerate triplet states:

$$\mathcal{H} = J \mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{J}{4} (\sigma_{x1}\sigma_{x2} + \sigma_{y1}\sigma_{y2} + \sigma_{z1}\sigma_{z2}) = \frac{J}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

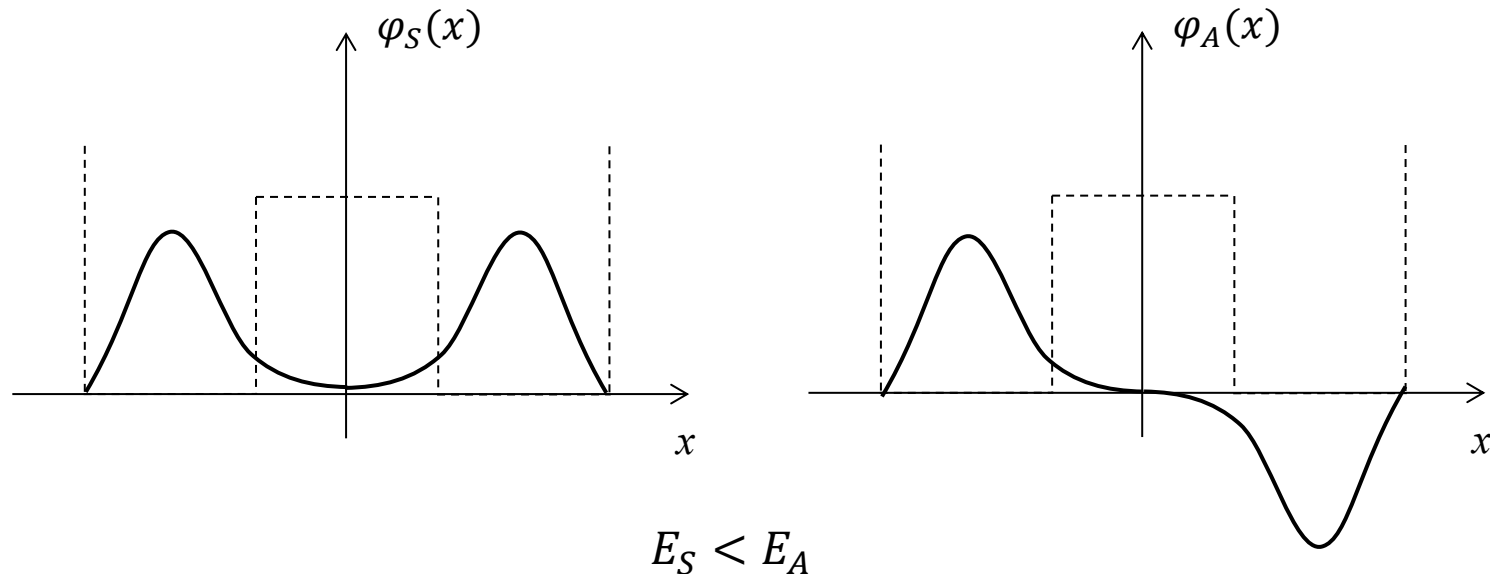
The eigenstates of this matrix are:

$ T_+\rangle = \uparrow\uparrow\rangle$	}	triplet states, with eigenvalue $E_T = J/4$ (3-fold degenerate)
$ T_0\rangle = \frac{1}{\sqrt{2}} (\uparrow\downarrow\rangle + \downarrow\uparrow\rangle)$		
$ T_-\rangle = \downarrow\downarrow\rangle$		
$ S\rangle = \frac{1}{\sqrt{2}} (\uparrow\downarrow\rangle - \downarrow\uparrow\rangle)$		singlet state, with eigenvalue $E_S = -3J/4$ (non-degenerate)

Here we used the S matrices instead of the Pauli σ , so the singlet-triplet energy splitting is J instead of $4J$. We work in units where $\hbar = 1$ so the S matrices are just $\sigma/2$

Exchange coupling

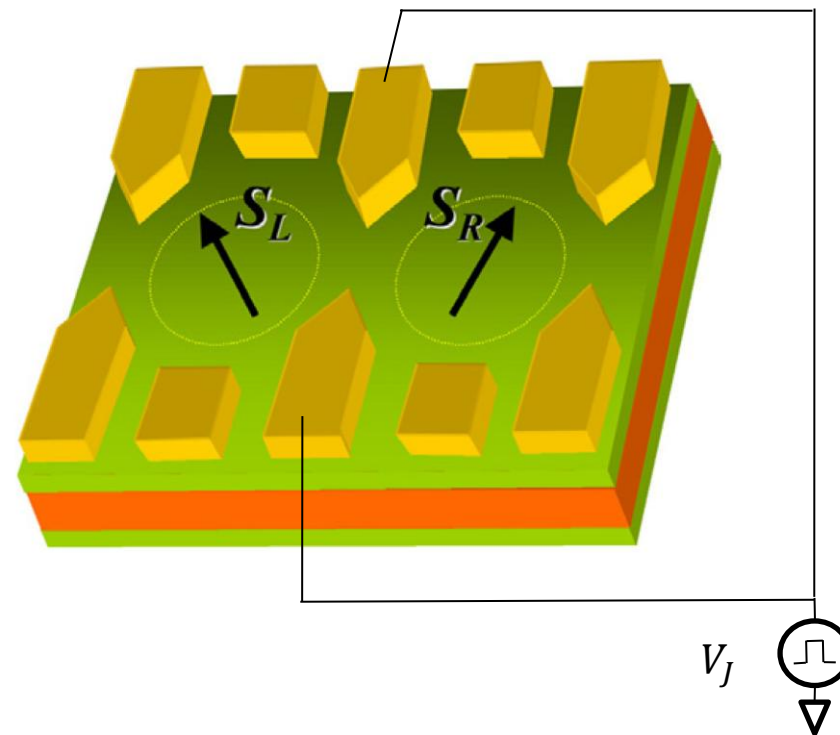
We saw (page 207) that the exchange coupling is a consequence of the symmetrization postulate, and the presence of a finite-height barrier between two potential wells, which makes the symmetric / antisymmetric orbital combinations have different energies:



Is it possible to externally control the exchange coupling?

In a quantum dot, yes! We can change the height of the tunnel barrier using an electrostatic gate.

Controlling the exchange coupling



In a double dot, the exchange interaction J can be controlled by the voltage V_J applied to the middle gates (which raises or lowers the barrier)

$$\mathcal{H}_J = J(t) \mathbf{S}_L \cdot \mathbf{S}_R$$

This means we can “pulse” the exchange interaction J at will.

SWAP gate

If we “turn on” the exchange interaction for a precisely controlled time τ_S , such that $J\tau_S = \pi$, we can implement a SWAP operation on the spins

$$U_S = e^{-i\mathcal{H}_J\tau_S} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The SWAP operation conserves the total spin and its z projection (i.e. $\mathcal{H}_J$ commutes with both $\mathbf{S}^2$ and S_z), so by itself it isn't very useful (it doesn't allow you to explore the full Hilbert space of the coupled system). However, if we stop the SWAP operation halfway through, we perform a $\sqrt{\text{SWAP}}$, which is an entangling operation.

$$U_S^{1/2} = e^{-i\mathcal{H}_J\tau_S/2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1+i}{2} & \frac{1-i}{2} & 0 \\ 0 & \frac{1-i}{2} & \frac{1+i}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

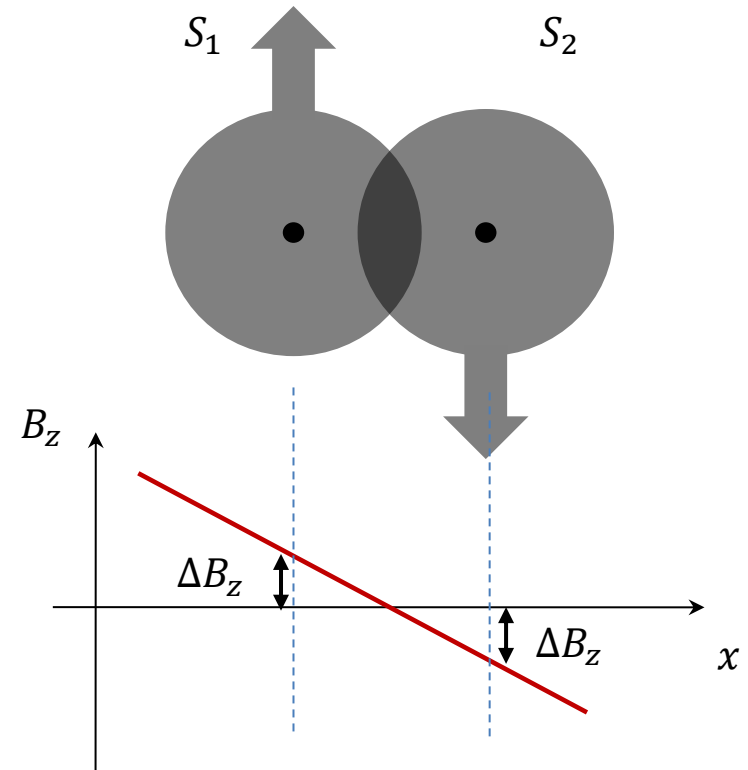
Singlet – triplet qubit

Consider two electron spins, described by the following Hamiltonian:

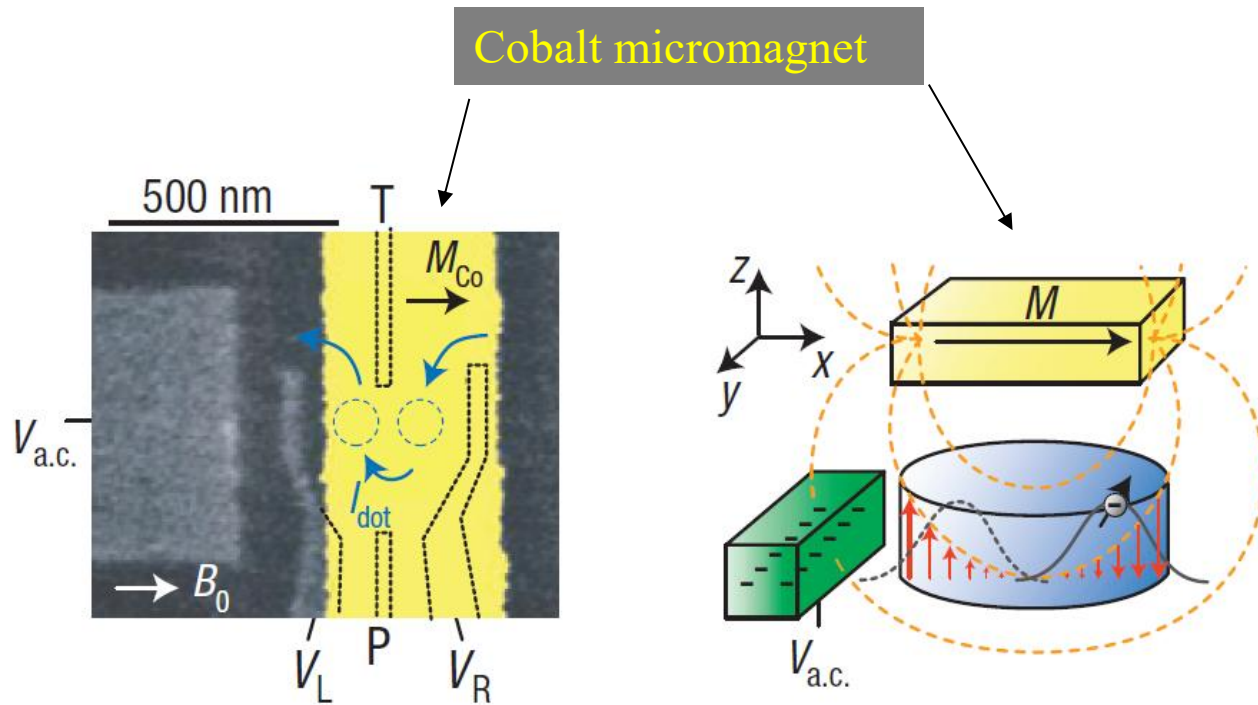
$$\mathcal{H} = J\mathbf{S}_1 \cdot \mathbf{S}_2 + \gamma\Delta B_z(S_{z1} - S_{z2})$$

where J is the exchange interaction strength, and ΔB_z represents a vertically oriented magnetic field, having *opposite* direction at the two spins.

This would be strange in bulk structures (the magnetic field is usually the same everywhere) but becomes feasible in nanoscale devices, where the electrons are close to each other, so they experience some exchange interaction, but far enough apart that a spatially varying magnetic field can have significantly different values at the two locations.



Magnetic field gradient



Deposit a thin strip of a magnetic material (usually cobalt) on top of the semiconductor surface under which you form a double quantum dot:
the magnetic field lines wind around creating a vertical field that points “down” on the right dot and “up” on the left dot.

Singlet – triplet qubit

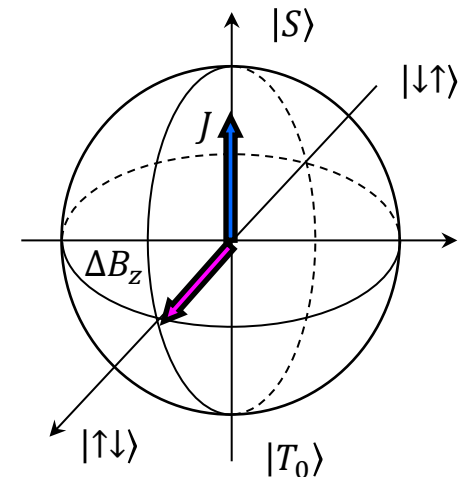
If we prepare a 2-spin system in the $|\uparrow\downarrow\rangle$ state, and subject it to the effect of only an exchange interaction and a magnetic field gradient, the system never occupies the $|T_+\rangle$ or the $|T_-\rangle$ states (see Python code `singlet_triplet.ipynb`).

Even though we have 2 spins, it “effectively” behaves as a two-level system, where:

- the basis states are $|S\rangle$ and $|T_0\rangle$
- J plays the role of a field along z
- ΔB_z plays the role of a field along x

The equivalent Bloch sphere is shown on the right.

(Note: I could also choose the $|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle$ as the basis states – it’s just a rotation of axes in the Hilbert space)



This is a way to make **one** qubit from **two** electrons. The reason for doing it is that it can be controlled with just electrical pulses, without using magnetic resonance.

Section 12

Cryogenic methods

Learning outcomes:

- Understand how to reach the very low temperatures typically necessary to observe quantum phenomena
- Recognize which cooling method is most appropriate for a certain temperature range

Why low temperatures

Almost all quantum phenomena require very low temperatures to become observable. This is just a consequence of the values of the constants of nature.

- Charging energy: $E_C = \frac{e^2}{2C} \rightarrow C = 100 \text{ aF} \Rightarrow \frac{E_C}{k_B} \approx 10 \text{ K}$
- Zeeman energy: $E_Z = \hbar\gamma B \rightarrow B = 1 \text{ T} \Rightarrow \frac{E_Z}{k_B} \approx 1.3 \text{ K}$

The simplest technique to achieve low temperatures is to liquefy an inert gas:

- $\text{N}_2 \rightarrow 77 \text{ K}$
- $\text{H}_2 \rightarrow 20 \text{ K}$
- $\text{He} \rightarrow 4.2 \text{ K}$

These are the boiling temperatures at atmospheric pressure ($P = 1 \text{ bar}$).

Whenever a molecule evaporates from a liquid, it requires a “latent heat of evaporation”. That is, the system “liquid + gas” absorbs heat by increasing its entropy.

Entropy and heat

2nd law of thermodynamics:

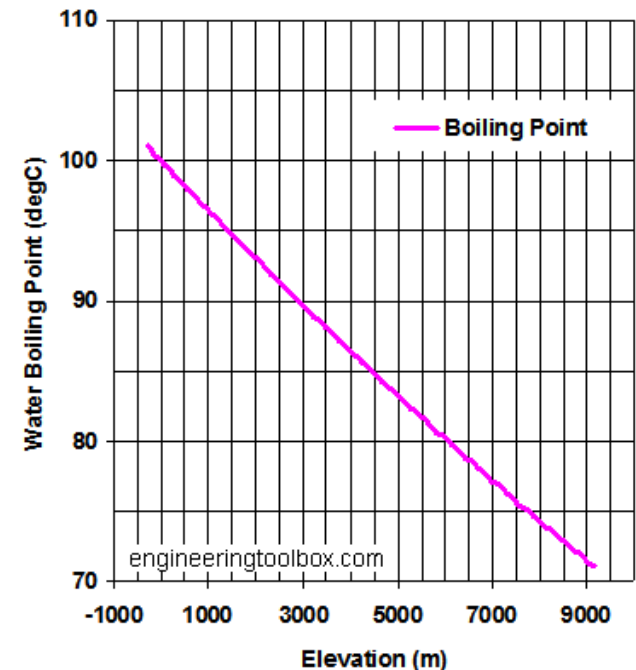
$$dQ = TdS$$

dQ = absorbed heat

dS = increase in entropy (disorder)

Example: If you put a pan of water on the stove, the water molecules tend to move from the liquid (ordered, low-entropy) phase to the liquid phase, i.e. the system becomes more disordered. T remains constant (100 °C) as long as there is liquid, because the vapour pressure is always 1 bar. In a pressure cooker, instead, the water gets hotter than 100 °C because the pressure is higher.

Conversely, reducing the pressure of the gas cools down the liquid (try cooking pasta on a high mountain...)



Cryogenic liquids

Water boils at 100 °C (393 K).

Other, less polar molecules need to be refrigerated to become liquid:

Substance	Boiling point (K)	Freezing point (K)	Lowest practical temperature (K)
H ₂	77.4	63.1	
H ₂	20.3	14	
⁴ He	4.2	-	1.2
³ He	3.3	-	0.3

³He is a light isotope of helium that has only one neutron in the nucleus. It is obtained by the decay of tritium, which is the main ingredient of hydrogen bombs.

1 liter of ³He gas costs \$3000

Cryostats

Cryogenic liquids need to be kept in well-insulated containers, called cryostats, to avoid them evaporating too quickly. Such containers use high-vacuum insulation and radiation shields.

Typical boil-off is a 2-10 liters/day.

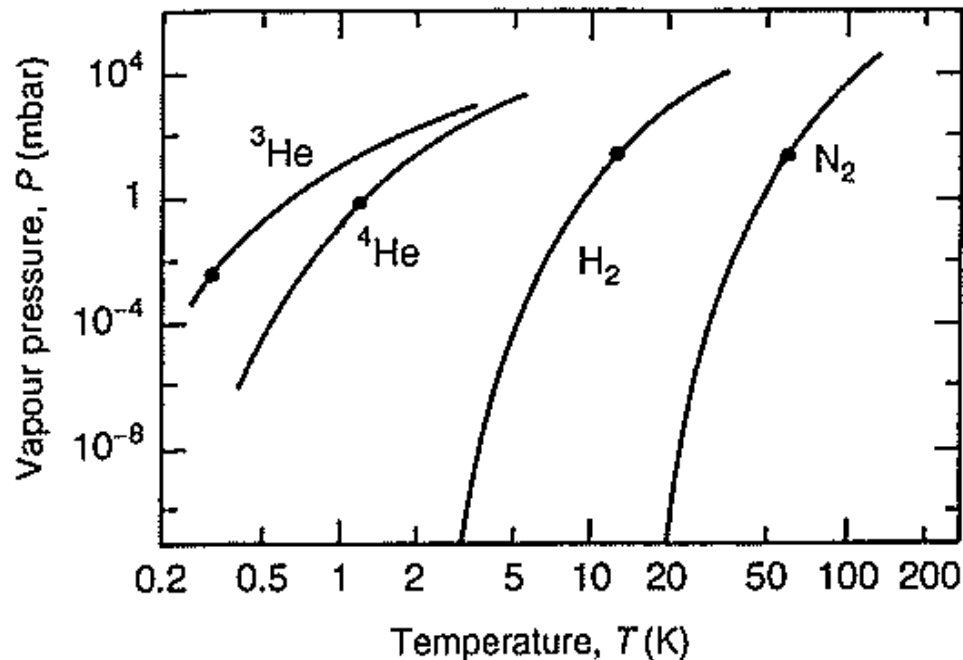
Helium gas is expensive and is usually recovered and reliquefied rather than lost in the air.



Pumped cryostats

The temperature of a liquid/gas system depends on the pressure of the gas. Therefore, connecting a pump to reduce the pressure in the gas phase results in a lower temperature of the liquid.

Using ^3He and a very powerful pump, one can reach 0.3 K



Liquid mixtures

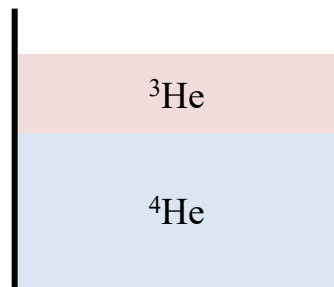
What if we want to go colder (e.g. < 100 mK)? We resort to quantum mechanics!

Consider a liquid mixture of ^3He and ^4He . In general, mixing two liquids increases their entropy (the mixture is a more disordered state).

3rd law of thermodynamics:

$$\lim_{T \rightarrow 0} S = 0$$

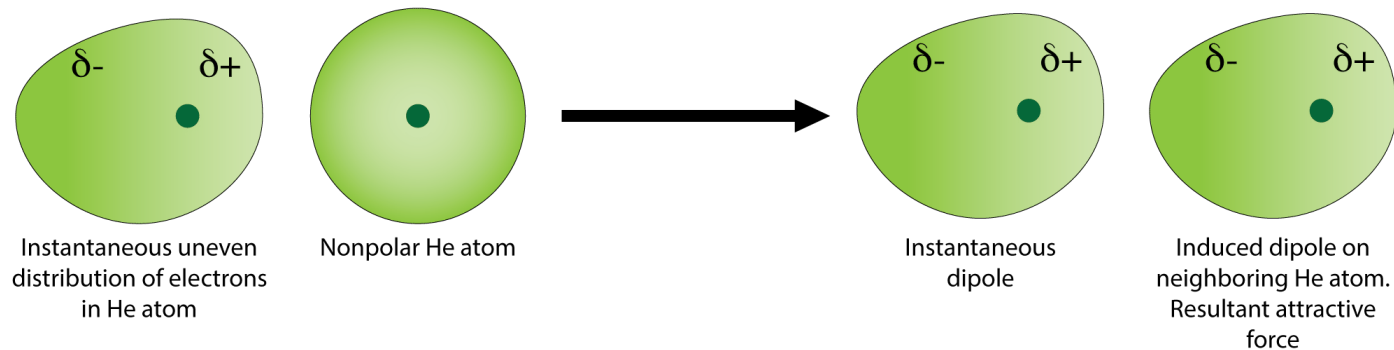
Thus, we would intuitively expect the liquids to completely separate if $T \rightarrow 0$, with the ^3He floating at the top (like oil on water)



In reality this doesn't happen completely, because of quantum effects!

Attractive force between He atoms

There is a very weak type of van der Waals force acting between the helium atoms. It depends only on the structure of the electron shell (i.e. it's identical for ^3He and ^4He) and decays strongly (6th power) with the distance between atoms.



The initial uneven electron charge distribution around the nucleus is caused by quantum fluctuations. This force is entirely quantum-mechanical in nature → it is the result of correlated quantum fluctuations!

Quantum fluctuations

The ^3He atom is 25% lighter than the ^4He atom.

Now think of the Heisenberg uncertainty principle:

$$\Delta x \Delta p > \hbar/2$$

Smaller mass $\rightarrow$ smaller $\Delta p \rightarrow$ larger Δx

This means that, on average, the distance between ^3He and ^4He atoms is smaller than the distance between two ^3He atoms $\rightarrow$ the van der Waals force between ^3He and ^4He is a little stronger $\rightarrow$ ^3He “prefers” to dissolve in the ^4He -rich phase.



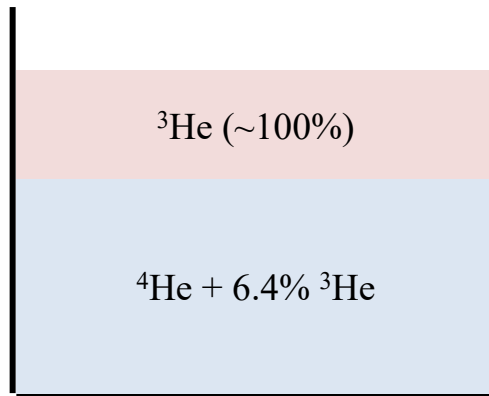
Equilibrium concentration

What is the equilibrium concentration at $T = 0$ K?

^3He has an odd number of protons and neutrons and thus has a spin $I = 1/2$, this makes it a Fermion (just like electrons).

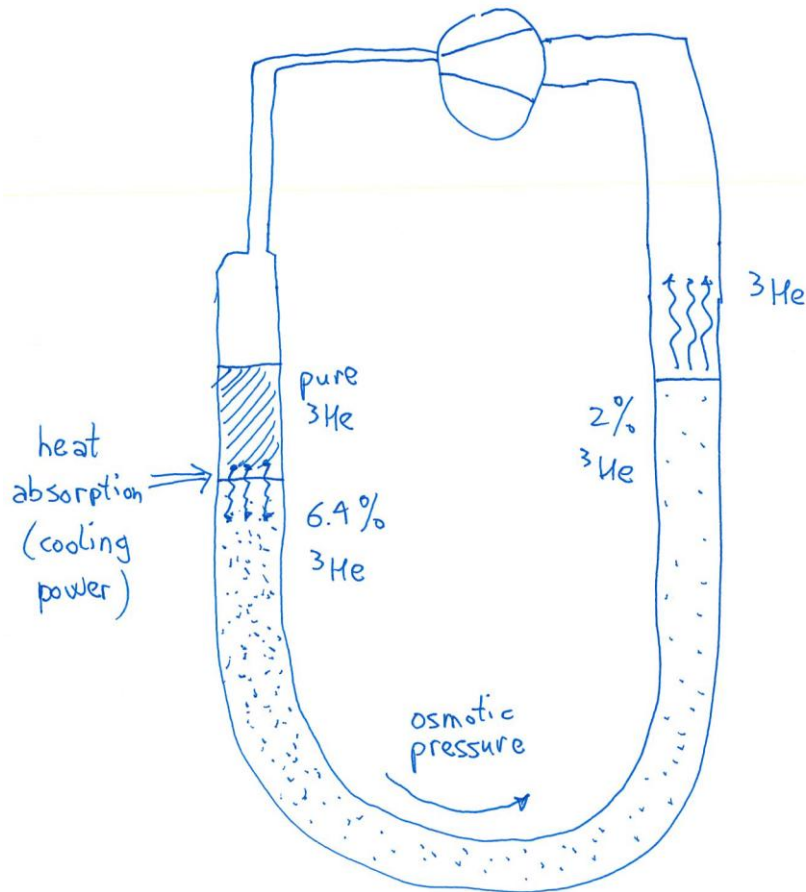
Therefore, you cannot have two ^3He atoms in the same quantum state, they must go into increasing energies (like electrons in a 2DEG or a metal). The equilibrium concentration $n_3 = 6.4\%$ is obtained when the Fermi energy of ^3He in ^4He equals the difference in binding energy ($^3\text{He}-^3\text{He}$)-($^3\text{He}-^4\text{He}$).

Thus, at $T = 0$ K



Dilution refrigerator

Next we place the mixture in a U-tube and connect a pump:

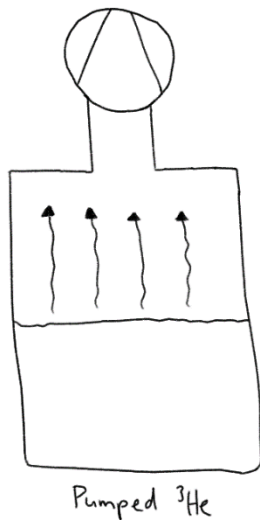


On the pumping side, $n_3 \sim 2\% < 6.4\%$ equilibrium value. A concentration gradient is created that causes a diffusion of ^3He away from the $^3\text{He}/^4\text{He}$ boundary. To restore equilibrium, some ^3He atoms will migrate from the pure phase to the dilute phase, it's like "evaporating ^3He into ^4He ", **there is an associated cooling process.**

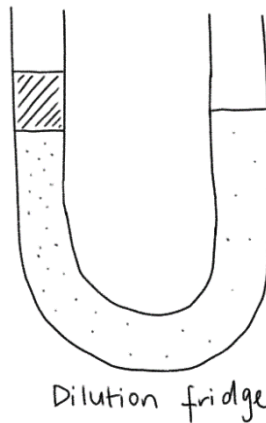
Dilution refrigerator

Why is this better than pumping on a pure ^3He bath? Because the vapour pressure of ^3He goes exponentially to zero for $T \rightarrow 0$ K, while the equilibrium concentration of ^3He in ^4He goes to a constant.

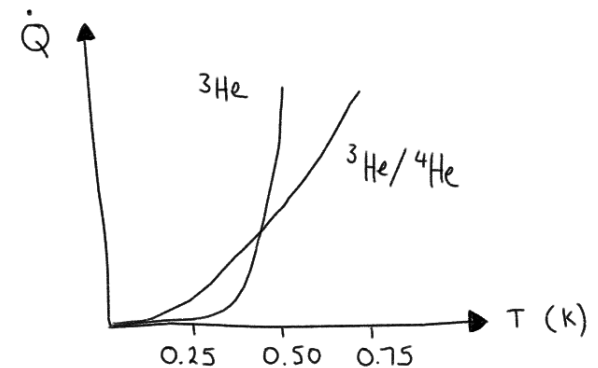
Cooling power: $\dot{Q} = \dot{n}\delta Q$



$$\delta Q \approx \text{constant}$$
$$\dot{n} \propto e^{-K/T}$$



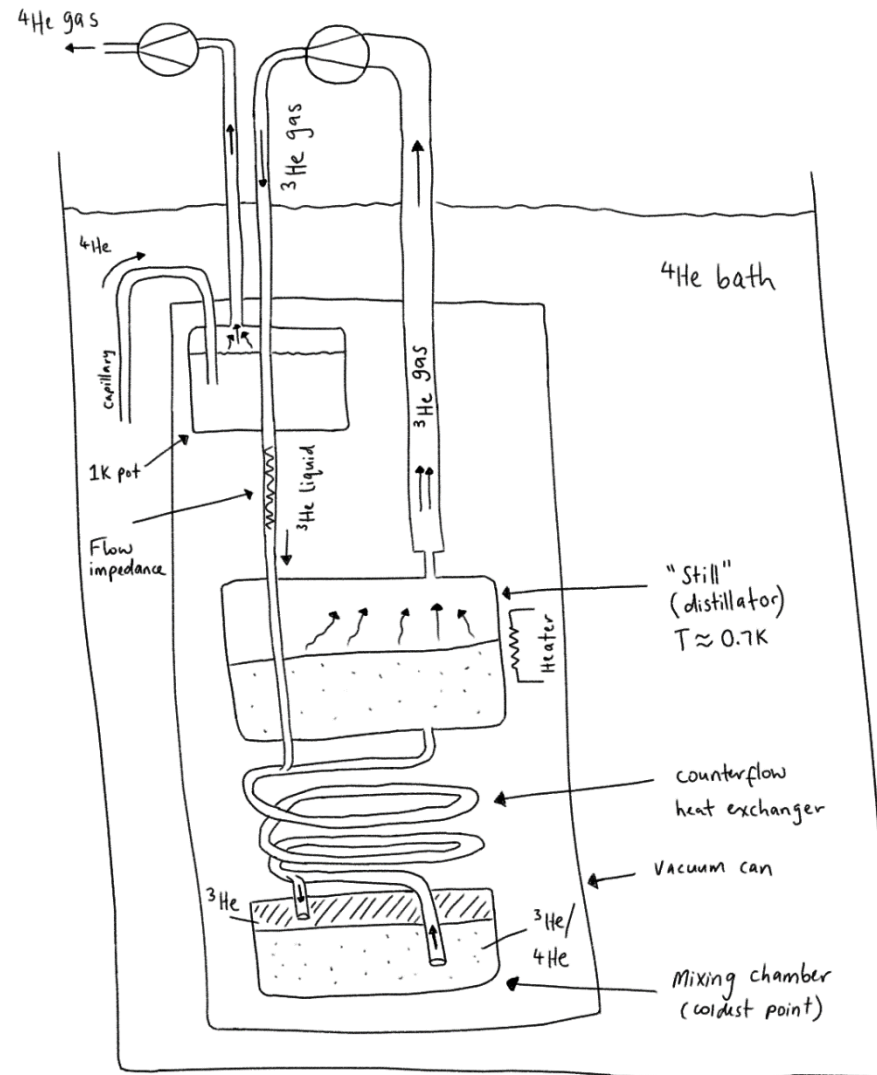
$$\delta Q \propto T^2$$
$$\dot{n} \approx \text{constant}$$



Below $T \sim 0.5$ K the dilution fridge outperforms a pumped ^3He refrigerator. The record for the lowest temperature is 1.9 mK!

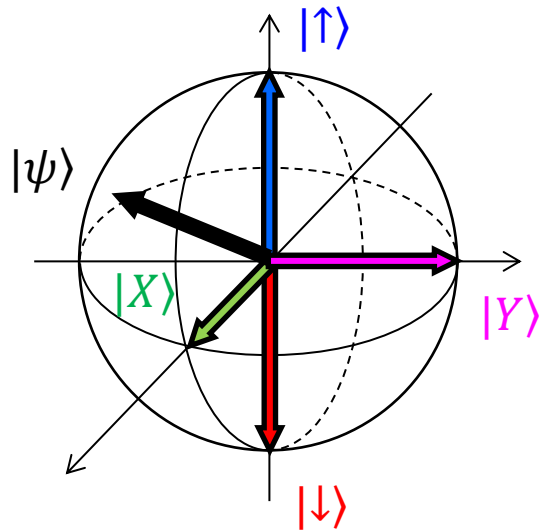
Dilution refrigerator

Construction:



Bloch sphere

Since we work with normalized state vectors, the state of a 2-level system can be represented as a vector on a sphere of radius 1, called **Bloch sphere**



$$|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$$

with $|\alpha|^2 + |\beta|^2 = 1$

Basically, we have mapped the 2-dimensional complex vector space of the states describing the quantum two-level system (TLS) onto a constant-radius sphere in 3 real dimensions.

For a real spin, the 3 dimensions have a physical meaning: they are the directions along which the magnetic moment can be measured. For a general TLS, it's just a mathematical trick.

State overlap

Recall the Born's rule:

$$\mathcal{P}(a_n) = |\langle u_n | \psi \rangle|^2$$

It can be interpreted as follows:

- If I prepare $|\psi_1\rangle = |u_n\rangle$, then $\mathcal{P}(a_n) = 1$
- If I prepare $|\psi_2\rangle \neq |u_n\rangle$, then $\mathcal{P}(a_n) < 1$

If I have machine that prepares at random the states $|\psi_1\rangle$ or $|\psi_2\rangle$, with what confidence am I able to distinguish, *in a single measurement*, which state was prepared?

It depends on the **state overlap** $|\langle \psi_1 | \psi_2 \rangle|^2$

- If $|\langle \psi_1 | \psi_2 \rangle|^2 = 0$ the states are orthogonal. Measuring an outcome completely excludes the other
- If $|\langle \psi_1 | \psi_2 \rangle|^2 \neq 0$ the states are non-orthogonal. It's impossible to tell with certainty which state had been prepared by taking only a single measurement.

Example:

- $|\uparrow\rangle$ and $|\downarrow\rangle$ are orthogonal (in a quantum sense!!)
- $|\uparrow\rangle$ and $|X\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ have a finite overlap

Heisenberg uncertainty for spins

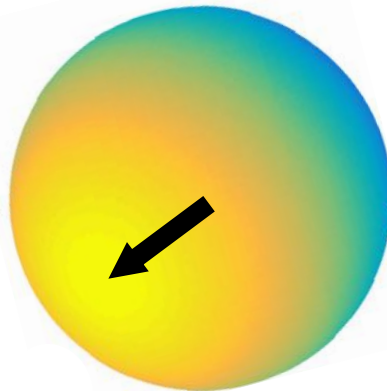
Caution!

It's tempting to say that, for example, the state $|X\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ represents a spin that points along $+x$.

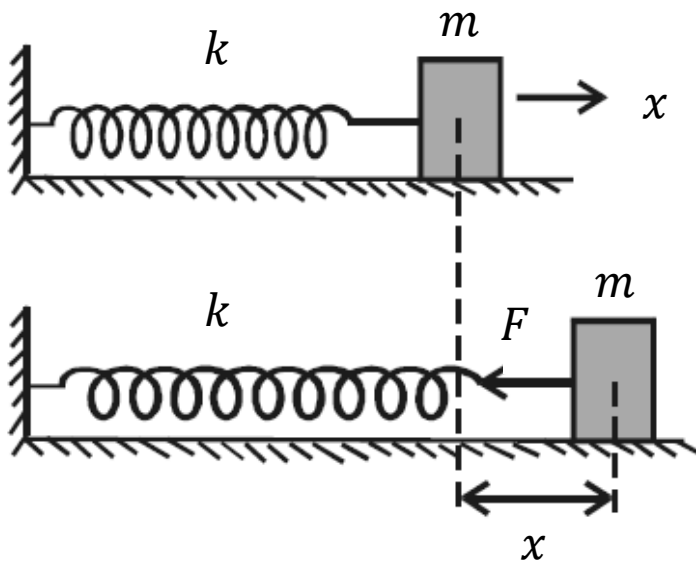
For this to be “true”, you would need to also know that its y and z components are zero. However, if such a spin were measured with a Stern-Gerlach apparatus oriented along z , you would still find it 50% of the time $|\uparrow\rangle$ and 50% of the time $|\downarrow\rangle$.

The fact that the spin operators S_x, S_y, S_z do not commute is a manifestation of the Heisenberg uncertainty principle, for spins!

The “proper” way to think of the state $|X\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ is a state that “maximally points” along $+x$, but has a “smudge” of probability that takes up about half of the Bloch sphere!!



Harmonic oscillator



Recall Hooke's law, which states that a spring (this doesn't have to be an actual spring) will exert a restoring force when it is stretched or compressed:

$$F = -kx$$

Combining this with Newton's second law, we find the second order differential equation

$$F = ma = m \frac{d^2x}{dt^2} = -kx$$

Which has a solution of the form $x = A \cos(\omega t + \phi)$, where A is an amplitude and $\omega = \sqrt{k/m}$ is an angular frequency.

Harmonic oscillator

We can express the restoration force with respect to the more general angular velocity $F = -m\omega^2 x$ and particle mass. The potential energy stored in a compressed or extended harmonic oscillator can be found from the restoring force $F = -dV/dx$

$$V = - \int F dx = \int m\omega^2 x \, dx = \frac{1}{2} m\omega^2 x^2$$

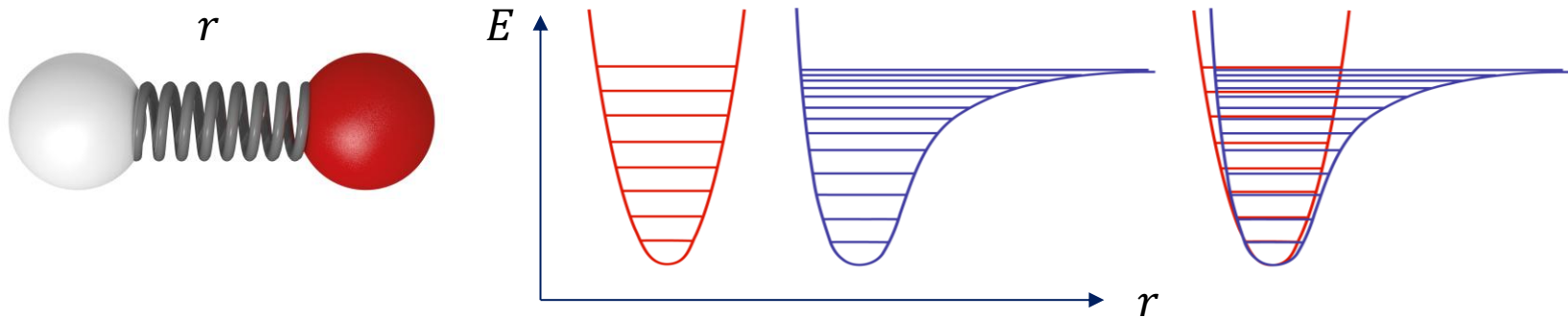
Combining this with the potential energy of the particle $T = mv^2/2 = p^2/2m$ we arrive at the total energy of the Harmonic oscillator

$$\mathcal{H} = T + V = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2$$

Examples of simple Harmonic oscillator systems include pendulums (gravity and torsional), a child on a swing, a tuning fork etc.

Quantum harmonic oscillator

There are many quantum mechanical systems that behave like Harmonic oscillators. Any particle existing in a parabolic-like potential will experience a linear restoration force, e.g.



We can replace the momentum and position variables in the classical Hamiltonian with their quantum mechanical operators to describe the energy of a quantum harmonic oscillator

$$\mathcal{H} = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2$$

Quantum harmonic oscillator

$$\mathcal{H} = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2$$

Our intuition tells us that the relevant energy scale at play likely has something to do with $\hbar\omega$, so let's factor this out

$$\mathcal{H} = \hbar\omega \left[\frac{P^2}{2\hbar m\omega} + \frac{m\omega}{2\hbar} X^2 \right]$$

A dimensional analysis of $2\hbar m\omega$ and $m\omega/2\hbar$ reveals that they possess units of square momentum and inverse-square length. They can be regarded as the characteristic momentum and length of our quantum mechanical harmonic oscillator. We give these characteristic measures variables $p_0 = \sqrt{2\hbar m\omega}$ and $x_0 = \sqrt{2\hbar/m\omega}$

$$\mathcal{H} = \hbar\omega \left[\frac{P^2}{p_0^2} + \frac{X^2}{x_0^2} \right]$$

Commutation relations

$$\mathcal{H} = \hbar\omega \left[\frac{X^2}{x_0^2} + \frac{P^2}{p_0^2} \right]$$

Such a form calls-out for factorisation! For example, for two variables a and b

$$a^2 + b^2 = (a - ib)(a + ib)$$

Can we do the same for $\mathcal{H}$? Careful! X and P are matrices...

$$\begin{aligned} \left[\frac{X}{x_0} - i \frac{P}{p_0} \right] \left[\frac{X}{x_0} + i \frac{P}{p_0} \right] &= \frac{X^2}{x_0^2} + i \frac{XP}{x_0 p_0} - i \frac{PX}{p_0 x_0} + \frac{P^2}{p_0^2} \\ &= \frac{X^2}{x_0^2} + \frac{P^2}{p_0^2} + \frac{i}{x_0 p_0} [X, P] \end{aligned}$$

We see that the result contains the commutator between X and P

Raising and lowering operators

Defining a new operator (which is a linear combination of the momentum and position operators)

$$\hat{a} = \frac{X}{x_0} + i \frac{P}{p_0}$$

and its adjoint

$$\hat{a}^\dagger = \frac{X}{x_0} - i \frac{P}{p_0}$$

allows us to express the Hamiltonian in the simplified form

$$\mathcal{H} = \hbar\omega[\hat{a}^\dagger\hat{a} + 1/2]$$

Commutation relation:

$$[\hat{a}, \hat{a}^\dagger] = \mathbb{I}$$

Raising and lowering operators

What are these new operators $\hat{a}$ and $\hat{a}^\dagger$? Are they observables?

$$\hat{a} = \frac{X}{x_0} + i \frac{P}{p_0}, \quad \hat{a}^\dagger = \frac{X}{x_0} - i \frac{P}{p_0}$$

The following commutation relations will help us to determine the action of $\hat{a}$ on a state $|\psi\rangle$ of our quantum harmonic oscillator

$$\begin{aligned} [\mathcal{H}, \hat{a}] &= \hbar\omega \left[\hat{a}^\dagger \hat{a} \hat{a} + \frac{\hat{a}}{2} - \hat{a} \hat{a}^\dagger \hat{a} - \frac{\hat{a}}{2} \right] \\ &= \hbar\omega [\hat{a}^\dagger \hat{a} \hat{a} - \hat{a} \hat{a}^\dagger \hat{a}] \\ &= \hbar\omega [\hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger] \hat{a} \\ &= \hbar\omega [\hat{a}^\dagger, \hat{a}] \hat{a} \\ &= -\hbar\omega \hat{a} \end{aligned}$$

Similarly, it can be shown that $[\mathcal{H}, \hat{a}^\dagger] = \hbar\omega \hat{a}^\dagger$.

Quantum harmonic oscillator

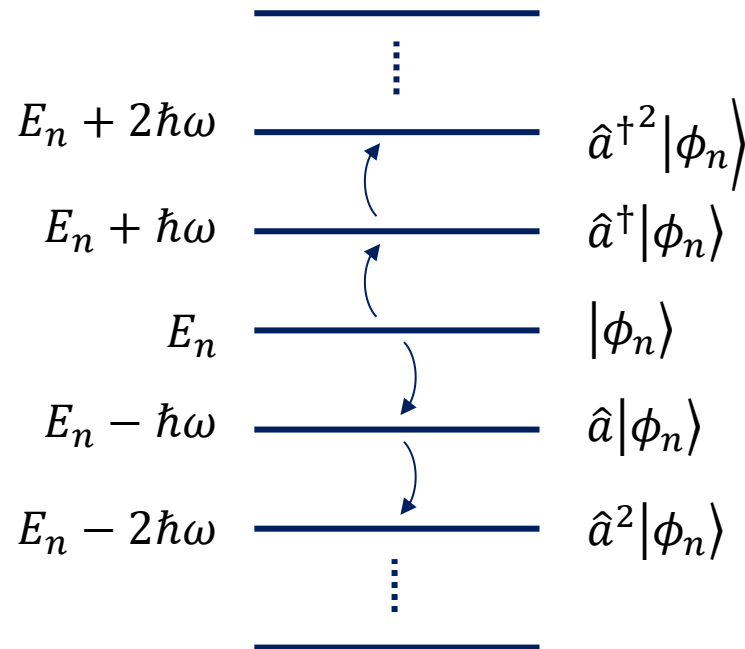
To see what action $\hat{a}$ has, let's apply it to an energy eigenstate $|\phi_n\rangle$ of our system $\mathcal{H}|\phi_n\rangle = E_n|\phi_n\rangle$ and look at the energy of this new state $|\psi\rangle = \hat{a}|\phi_n\rangle$

$$\begin{aligned}\mathcal{H}|\psi\rangle &= \mathcal{H}\hat{a}|\phi_n\rangle \\ &= [\mathcal{H}\hat{a} + \hat{a}\mathcal{H} - \hat{a}\mathcal{H}]\phi_n\rangle \\ &= [\mathcal{H}\hat{a} - \hat{a}\mathcal{H} + \hat{a}\mathcal{H}]\phi_n\rangle \\ &= [[\mathcal{H}, \hat{a}] + \hat{a}\mathcal{H}]\phi_n\rangle \\ &= [-\hbar\omega\hat{a} + \hat{a}\mathcal{H}]\phi_n\rangle \\ &= [E_n - \hbar\omega]\hat{a}|\phi_n\rangle \\ &= [E_n - \hbar\omega]|\psi\rangle\end{aligned}$$

Similarly, $\mathcal{H}\hat{a}^\dagger|\phi_n\rangle = [E_n + \hbar\omega]\hat{a}^\dagger|\phi_n\rangle$. The operator $\hat{a}$ has the effect of “lowering” the energy of a state by a constant amount $\hbar\omega$, whilst $\hat{a}^\dagger$ “raises” the energy by the same amount. We therefore call $\hat{a}$ and $\hat{a}^\dagger$ the **lowering and raising operators**, respectively.

Quantum harmonic oscillator

The Harmonic oscillator describes a ladder of states, with adjacent states separated by an energy $\hbar\omega$



We can show that the energies $\langle \mathcal{H} \rangle$ of these states must be positive. This means there is a ground state $|\phi_0\rangle$. In order to enforce that there are no states lower in energy than the ground state, we set $\hat{a}|\phi_0\rangle = 0$.

Quantum harmonic oscillator

What is the ground state energy? If $\hat{a}|\phi_0\rangle = 0$, then

$$\begin{aligned}\mathcal{H}|\phi_0\rangle &= \hbar\omega[\hat{a}^\dagger\hat{a} + 1/2]|\phi_0\rangle \\ &= \hbar\omega[0 + 1/2]|\phi_0\rangle \\ &= \frac{\hbar\omega}{2}|\phi_0\rangle\end{aligned}$$

The ground state energy is $\hbar\omega/2$. Why isn't the ground state energy zero?

We make one further simplification to the Hamiltonian, by renaming the operator $\hat{a}^\dagger\hat{a}$

$$N = \hat{a}^\dagger\hat{a}$$

This is called the “number operator”, for reasons that will become apparent soon. The final form of the Hamiltonian for the quantum harmonic oscillator is then

$$\mathcal{H} = \hbar\omega[N + 1/2]$$

Quantum harmonic oscillator

$$\mathcal{H} = \hbar\omega[N + 1/2]$$

It is clear that the eigenstates of $\mathcal{H}$ (which we previously labelled $|\phi_n\rangle$) are also eigenstates of the number operator N . We give these states a new label $|n\rangle$, with eigenvalue n , i.e.

$$N|n\rangle = n|n\rangle$$

The energy of the state $|n\rangle$ is therefore

$$\mathcal{H}|n\rangle = \hbar\omega[N + 1/2]|n\rangle = \hbar\omega[n + 1/2]|n\rangle$$

Since the energies of the quantum harmonic oscillator eigenstates are all positive, n must be positive. This is also clear from the definition of the number operator N

$$\begin{aligned}\langle n|N|n\rangle &= n\langle n|n\rangle = n \\ n &= \langle n|N|n\rangle = \langle n|\hat{a}^\dagger\hat{a}|n\rangle = (\hat{a}|n\rangle)^\dagger(\hat{a}|n\rangle) = |\hat{a}|n\rangle|^2 \geq 0\end{aligned}$$

Quantum harmonic oscillator

We saw that for the ground state $|0\rangle$,

$$\mathcal{H}|0\rangle = \frac{\hbar\omega}{2}|0\rangle$$

which implies $n = 0$ for this state. As we also discovered, applying the raising operator $\hat{a}^\dagger$ to $|0\rangle$ creates a new eigenstate of the Hamiltonian with an energy raised by an amount $\hbar\omega$

$$\mathcal{H}\hat{a}^\dagger|0\rangle = \hbar\omega[1 + 1/2]\hat{a}^\dagger|0\rangle$$

This new state $\hat{a}^\dagger|0\rangle$ has a number eigenvalue of $n = 1$ and we label the normalised version of this state $|1\rangle$. The n are thus integers that range from zero to infinity. We call the $|n\rangle$ states “number states”, or **Fock states**, after the Soviet theorist Vladimir Fock.

Unfortunately, direct application of $\hat{a}^\dagger$ to a state $|n\rangle$ does not produce a normalised eigenstate

$$\hat{a}^\dagger|n\rangle = \lambda_{n+1}|n+1\rangle$$

Quantum harmonic oscillator

If we look at the norm of this state

$$\begin{aligned}\langle n|\hat{a}\hat{a}^\dagger|n\rangle &= \langle n|\hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} + \hat{a}^\dagger\hat{a}|n\rangle \\ &= \langle n|[\hat{a}, \hat{a}^\dagger] + \hat{a}^\dagger\hat{a}|n\rangle \\ &= \langle n|1 + N|n\rangle \\ &= n + 1\end{aligned}$$

$$\begin{aligned}\langle n|\hat{a}\hat{a}^\dagger|n\rangle &= |\lambda_{n+1}|^2 \\ \lambda_{n+1} &= \sqrt{n+1}\end{aligned}$$

So we have

$$\begin{aligned}\hat{a}^\dagger|n\rangle &= \sqrt{n+1}|n+1\rangle \\ \hat{a}^\dagger|n-1\rangle &= \sqrt{n}|n\rangle \quad (n \rightarrow n-1)\end{aligned}$$

$$|n\rangle = \frac{\hat{a}^\dagger|n-1\rangle}{\sqrt{n}}$$

Quantum harmonic oscillator

We can recursively define each state in terms of the previous, all the way to the ground state

$$|n\rangle = \frac{\hat{a}^\dagger |n-1\rangle}{\sqrt{n}} = \frac{(\hat{a}^\dagger)^2 |n-2\rangle}{\sqrt{n(n-1)}} = \dots = \frac{(\hat{a}^\dagger)^n |0\rangle}{\sqrt{n!}}$$

Is N an observable? I.e., can we measure these number states $|n\rangle$?

What do their wavefunctions look like? Consider the ground state wavefunction $\phi_0(x) = \langle x|0\rangle$

$$\hat{a}|0\rangle = 0$$

$$\left[\frac{X}{x_0} + i \frac{P}{p_0} \right] |0\rangle = 0$$

$$\langle x| \left[\frac{X}{x_0} + i \frac{P}{p_0} \right] |0\rangle = 0$$

Quantum harmonic oscillator

$$\langle x | \left[\frac{X}{x_0} + i \frac{P}{p_0} \right] | 0 \rangle = 0$$

$$\frac{\langle x | X | 0 \rangle}{x_0} + i \frac{\langle x | P | 0 \rangle}{p_0} = 0$$

$$\frac{x}{x_0} \phi_0(x) + \frac{i}{p_0} \frac{\hbar}{i} \frac{d}{dx} \phi_0(x) = 0$$

$$\frac{p_0}{x_0 \hbar} x \phi_0(x) + \frac{d}{dx} \phi_0(x) = 0$$

$$\frac{m\omega}{\hbar} x \phi_0(x) + \frac{d}{dx} \phi_0(x) = 0$$

$$\frac{2}{x_0^2} x \phi_0(x) + \frac{d}{dx} \phi_0(x) = 0$$

This has a solution $\phi_0(x) = \mathcal{N} e^{-(x/x_0)^2}$, with $\mathcal{N} = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4}$ a normalisation constant. Write down wavefunctions for the next two states $|1\rangle$ and $|2\rangle$ (hint: use the recursive formula).

Quantum harmonic oscillator

$$\begin{aligned}\phi_1(x) &= \frac{\langle x | \hat{a}^\dagger | 0 \rangle}{\sqrt{1!}} \\ &= \langle x | \left[\frac{X}{x_0} - i \frac{P}{p_0} \right] | 0 \rangle \\ &= \frac{x \phi_0(x)}{x_0} - \frac{i \hbar}{p_0 i} \frac{d}{dx} \phi_0(x) \\ &= \frac{2x}{x_0} \left(\frac{m\omega}{\pi \hbar} \right)^{\frac{1}{4}} e^{-(x/x_0)^2}\end{aligned}$$

$$\begin{aligned}\phi_2(x) &= \frac{\langle x | \hat{a}^\dagger | 2 \rangle}{\sqrt{2!}} \\ &= \dots\end{aligned}$$

Quantum harmonic oscillator

The exact wavefunctions of the quantum harmonic oscillator eigenfunctions are provided by following expression

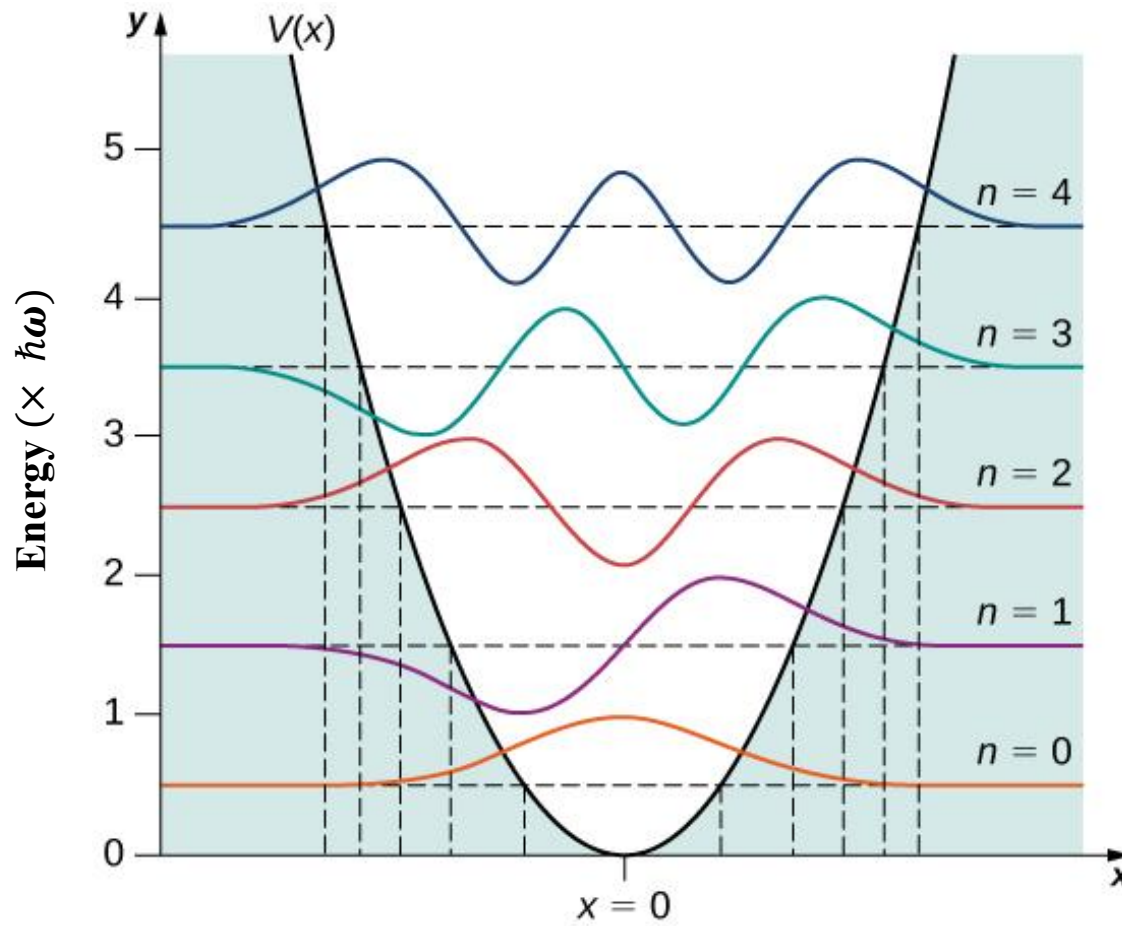
$$\phi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi \hbar} \right)^{\frac{1}{4}} H_n \left(\frac{x}{x_0} \right) e^{-\frac{x^2}{2x_0^2}}$$

Where $H_n \left(\frac{x}{x_0} \right)$ represents the Hermite polynomials

$$H_n(x) = (-1)^n e^{x^2} \left(\frac{d^n}{dx^n} e^{-x^2} \right)$$

Fock/number states

The first five Fock state wavefunctions are shown on the plot below.



Number states

What is the expectation of the position and momentum for the Fock states? Recall that

$$\hat{a} = \frac{X}{x_0} + i \frac{P}{p_0}, \quad \hat{a}^\dagger = \frac{X}{x_0} - i \frac{P}{p_0}$$

$$\text{So, } X = \frac{x_0}{2} (\hat{a} + \hat{a}^\dagger) \text{ and } P = \frac{p_0}{2i} (\hat{a} - \hat{a}^\dagger)$$

$$\langle n | X | n \rangle = \frac{x_0}{2} \langle n | \hat{a} + \hat{a}^\dagger | n \rangle = \frac{x_0}{2} (\langle n | \hat{a} | n \rangle + \langle n | \hat{a}^\dagger | n \rangle) = 0$$

$$\langle n | P | n \rangle = \frac{p_0}{2i} \langle n | \hat{a} - \hat{a}^\dagger | n \rangle = \frac{p_0}{2i} (\langle n | \hat{a} | n \rangle - \langle n | \hat{a}^\dagger | n \rangle) = 0$$

Having a zero average position and momentum is **not** the same thing as having zero position and momentum!

Number state uncertainty

We know that the number states have zero average position and momentum, but are certainly not motionless, Heisenberg's uncertainty principle won't allow it. Particles in these states will have some spread in their position and momentum:

$$\begin{aligned}\langle n|X^2|n\rangle &= \frac{x_0^2}{4} \langle n|(\hat{a} + \hat{a}^\dagger)^2|n\rangle = \frac{x_0^2}{4} \langle n|\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + \hat{a}^{\dagger 2}|n\rangle \\&= \frac{x_0^2}{4} \langle n|\hat{a}\hat{a}^\dagger + N|n\rangle = \frac{x_0^2}{4} \langle n|[\hat{a}, \hat{a}^\dagger] + N + N|n\rangle \\&= \frac{x_0^2}{4} \langle n|2N + 1|n\rangle = \frac{x_0^2}{2} \left(n + \frac{1}{2} \right) \\ \langle n|P^2|n\rangle &= -\frac{p_0^2}{4} \langle n|(\hat{a} - \hat{a}^\dagger)^2|n\rangle \\&= \frac{p_0^2}{2} \left(n + \frac{1}{2} \right)\end{aligned}$$

Number state uncertainty

Remembering the uncertainty of an operator A is defined as $\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$

$$\Delta X = \sqrt{\frac{x_0^2}{2} \left(n + \frac{1}{2} \right)}$$

$$\Delta P = \sqrt{\frac{p_0^2}{2} \left(n + \frac{1}{2} \right)}$$

$$\Delta X \Delta P = \frac{x_0 p_0}{2} \left(n + \frac{1}{2} \right) = \hbar \left(n + \frac{1}{2} \right)$$

For $n = 0$, $\Delta X \Delta P = \hbar/2$, the minimum uncertainty possible.

Correspondence principle

The correspondence principle states that for large quantum numbers, the quantum-mechanical description of a system reproduces classical physics. This is indeed the case for large number states, where the probability density concentrates at the extremes of the potential and the oscillations become more rapid (and unresolvable due to the increasing uncertainty).

